

内分泌经典文献翻译

制作：云梦山

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Endothelium-specific depletion of LRP1 improves glucose homeostasis through inducing osteocalcin

LRP1的内皮特异性缺失通过诱导骨钙素改善葡萄糖稳态

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Abstract

Abstract The **vascular endothelium** is present within metabolic organs and actively regulates energy **metabolism**. Here we show **osteocalcin**, recognized as a bone-secreted metabolic hormone, is expressed in mouse primary **endothelial cells** isolated from heart, lung and liver. In human **osteocalcin** promoter-driven green fluorescent protein transgenic mice, green fluorescent protein signals are enriched in **endothelial cells** lining **aorta**, small vessels and **capillaries** and abundant in **aorta**, **skeletal muscle** and eye of adult mice. The depletion of lipoprotein receptor-related protein 1 induces **osteocalcin** through a Forkhead box O -dependent pathway in **endothelial cells**. Whereas depletion of **osteocalcin** abolishes the glucose-lowering effect of low-density lipoprotein receptor-related protein 1 depletion, **osteocalcin** treatment normalizes **hyperglycemia** in multiple mouse models. Mechanistically, **osteocalcin** receptor-G protein-coupled receptor family C group 6 member A and insulin-like-growth-factor-1 receptor are in the same complex with **osteocalcin** and required for **osteocalcin**-promoted **insulin signaling pathway**. Therefore, our results reveal an **endocrine/paracrine** role of **endothelial cells** in regulating insulin sensitivity, which may have therapeutic implications in treating **diabetes** and **insulin resistance** through manipulating **vascular endothelium**.

中文翻译

摘要 血管内皮存在于代谢器官中，并积极调节能量代谢。本文显示，骨钙素作为一种已知的骨源性代谢激素，在小鼠心脏、肺和肝脏分离的原代内皮细胞中表达。在人类骨钙素启动子驱动的绿色荧光蛋白转基因小鼠中，绿色荧光蛋白信号富集于主动脉、小血管和毛细血管的内皮细胞，并在成年小鼠的主动脉、骨骼肌和眼睛中大量存在。脂蛋白受体相关蛋白1的缺失通过Forkhead box O依赖途径诱导内皮细胞中的骨钙素表达。虽然骨钙素的缺失消除了低密度脂蛋白受体相关蛋白1缺失的降糖效应，但骨钙素处理在多种小鼠模型中使高血糖恢复正常。机制上，骨钙素受体-G蛋白偶联受体家族C组6成员A和胰岛素样生长因子-1受体与骨钙素处于同一复合物中，并且是骨钙素促进胰岛素信号通路所必需的。因此，我们的研究结果揭示了内皮细胞在调节胰岛素敏感性中的内分泌/旁分泌作用，这可能为通过调控血管内皮治疗糖尿病和胰岛素抵抗提供治疗意义。

核心词汇

vascular endothelium /ˈvæskjələr ˌendouˈθi:liəm/ : 血管内皮

osteocalcin /ˌɒstiouˈkælsɪn/ : 骨钙素

aorta /eɪˈɔ:rtə/ : 主动脉

hyperglycemia /ˌhaɪpərglaɪˈsi:miə/ : 高血糖

insulin resistance /ˈɪnsəlɪn rɪˈzɪstəns/ : 胰岛素抵抗

endocrine /ˈendoukrɪn/ : 内分泌

skeletal muscle /ˈskeɪtlɪˈmʌsl/ : 骨骼肌

metabolism /məˈtæbəˌlɪzəm/ : 代谢

endothelial cells /ˌendouˈθi:liəl sɛlz/ : 内皮细胞

capillaries /ˈkæpɪˌleɪrɪz/ : 毛细血管

diabetes /ˈdaɪəˈbi:ti:z/ : 糖尿病

insulin signaling pathway /ˈɪnsəlɪn ˈsɪgnəlɪŋ ˈpæθweɪ/ : 胰岛素信号通路

paracrine /ˈpærəkrɪn/ : 旁分泌

The role of osteocalcin in human glucose metabolism: Marker or mediator?

骨钙素在人体葡萄糖代谢中的作用：标记物还是中介物？

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Abstract

Increasing evidence supports an association between the **skeleton** and **energy metabolism**. These interactions are mediated by a variety of hormones, cytokines and nutrients. Here, the evidence for a role of **osteocalcin** in the regulation of **glucose metabolism** in humans is reviewed. **osteocalcin** is a **bone matrix** protein that regulates **hydroxyapatite** size and shape through its vitamin-K-dependent, γ -carboxylated form. The concentration of **osteocalcin** in the circulation is a measure of **bone formation**. The **undercarboxylated** form of **osteocalcin** is active in **glucose metabolism** in mice. Total **serum osteocalcin** concentrations in humans are inversely associated with measures of **glucose metabolism**; however, human data are inconclusive with regard to the role of uncarboxylated **osteocalcin** in **glucose metabolism** because most studies do not account for the influence of **vitamin K** on the proportion of **undercarboxylated osteocalcin** or differentiate between the total and uncarboxylated forms of **osteocalcin**. Furthermore, most human studies do not concomitantly measure other **bone turnover** markers to isolate the role of **osteocalcin** as a measure of **bone formation** from its effect on **glucose metabolism**. Carefully designed studies are required to define the role of **osteocalcin** and its carboxylated or **undercarboxylated** forms in the regulation of **glucose metabolism** in humans.

中文翻译

越来越多的证据支持骨骼与能量代谢之间存在关联，这些相互作用由多种激素、细胞因子和营养物质介导。本文综述了骨钙素在调节人体葡萄糖代谢中作用的证据。骨钙素是一种骨基质蛋白，通过其依赖维生素K的 γ -羧化形式调节羟基磷灰石的大小和形状。循环中骨钙素的浓度是衡量骨形成的指标。在小鼠中，未羧化形式的骨钙素在葡萄糖代谢中具有活性。人体血清总骨钙素浓度与葡萄糖代谢指标呈负相关；然而，关于未羧化骨钙素在葡萄糖代谢中的作用，人类数据尚无定论，因为大多数研究未考虑维生素K对未羧化骨钙素比例的影响，也未区分骨钙素的总形式和未羧化形式。此外，大多数人类研究未同时测量其他骨转换标志物，以将骨钙素作为骨形成指标的作用与其对葡萄糖代谢的影响区分开来。需要精心设计的研究来明确骨钙素及其羧化或未羧化形式在调节人体葡萄糖代谢中的作用。

核心词汇

osteocalcin /ˌɒstiəʊˈkælsɪn/ : 骨钙素

bone formation /bəʊn fɔːˈmeɪʃən/ : 骨形成

vitamin K /ˈvɪtəˌmɪn keɪ/ : 维生素K

undercarboxylated /ˌʌndəˈkɑːbɒksɪleɪtɪd/ : 羧化不足的

serum /ˈsɪərəm/ : 血清

energy metabolism /ˈenədʒi məˈtæbəlaɪzəm/ : 能量代谢

glucose metabolism /ˈɡluːkəʊs məˈtæbəlaɪzəm/ : 葡萄糖代谢

hydroxyapatite /haɪˌdrɒksiˈæpətait/ : 羟基磷灰石

carboxylation /kɑːˈbɒksɪˈleɪʃən/ : 羧化

bone matrix /bəʊn ˈmeɪtrɪks/ : 骨基质

bone turnover /bəʊn ˈtʃːnəʊvə/ : 骨转换

skeleton /ˈskelɪtən/ : 骨骼系统

Endocrine toxicities of immune checkpoint inhibitors

免疫检查点抑制剂的内分泌毒性

期刊: Nature Reviews Endocrinology 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

immune checkpoint inhibitors (ICIs) are **monoclonal antibodies** that target two key signalling pathways related to **T cell** activation and exhaustion, by binding to and inhibiting cytotoxic T lymphocyte antigen 4 (CTLA4) or PD1 and its ligand PDL1. ICIs, such as nivolumab, pembrolizumab and ipilimumab, are approved for the **treatment** of numerous and diverse **cancer** types, in various combination regimens, and are now an established cornerstone of **cancer** therapeutics. Toxicities induced by ICIs are autoimmune in nature and are referred to as immune-related adverse events (irAEs); these events can affect any organ system in an unpredictable fashion. Importantly, irAEs can manifest as **endocrinopathies** involving the thyroid (**hypothyroidism** or **thyrotoxicosis**), pituitary (**hypophysitis**), adrenal glands (**adrenal insufficiency**) and pancreas (**diabetes mellitus**). These events are a frequent source of acute and persistent morbidity in patients treated with ICIs and can even be fatal. Over the past few years, there has been a growing understanding of the underlying **pathogenesis** of irAEs that has led to the development of more effective management strategies. Herein, we review the current understanding of the pathobiology, **clinical manifestations** and **treatment** approaches to endocrine toxicities arising from ICIs.

中文翻译

免疫检查点抑制剂 (ICIs) 是一类单克隆抗体, 通过结合并抑制细胞毒性T淋巴细胞抗原4 (CTLA4) 或PD1及其配体PDL1, 靶向与T细胞活化和耗竭相关的两个关键信号通路。诸如纳武利尤单抗、帕博利珠单抗和伊匹木单抗等ICIs已获批用于治疗多种不同类型的癌症, 并采用多种联合治疗方案, 现已成为癌症治疗的基石。ICIs引发的毒性本质上是自身免疫性的, 被称为免疫相关不良事件 (irAEs); 这些事件可能以不可预测的方式影响任何器官系统。重要的是, irAEs可表现为累及甲状腺 (甲状腺功能减退或甲状腺毒症)、垂体 (垂体炎)、肾上腺 (肾上腺功能不全) 和胰腺 (糖尿病) 的内分泌疾病。这些事件是接受ICIs治疗患者急性和持续性发病的常见原因, 甚至可能致命。过去几年中, 人们对irAEs的潜在发病机制有了越来越深入的了解, 从而制定了更有效的管理策略。在此, 我们综述了目前对ICIs引起的内分泌毒性的病理生物学、临床表现和治疗方法的理解。

核心词汇

immune checkpoint inhibitors /ɪˈmjuːn ˈtʃekpɔɪnt ɪnˈhɪbɪtəz/ : 免疫检查点抑制剂

T cell /tiː sel/ : T细胞

endocrinopathies /ˌendəʊkrɪˈnɒpəθiːz/ : 内分泌疾病

thyrotoxicosis /ˌθaɪrəʊˈtɒksɪˈkeʊsɪs/ : 甲状腺毒症

adrenal insufficiency /əˈdriːnəl ˌɪnsəˈfɪʃənsi/ : 肾上腺功能不全

pathogenesis /ˌpæθəˈdʒenɪsɪs/ : 发病机制

treatment /ˈtriːtmənt/ : 治疗

monoclonal antibodies /ˌmɒnəˈkləʊnəl ˈæntɪˌbɒdiz/ : 单克隆抗体

cancer /ˈkænsər/ : 癌症

hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症

hypophysitis /haɪˌpɒfɪˈsaɪtɪs/ : 垂体炎

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈleɪtəs/ : 糖尿病

clinical manifestations /ˈklɪnɪkəl ˌmænɪfɛstˈeɪʃənz/ : 临床表现

Hypopituitarism

垂体功能减退症

期刊: The Lancet 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

Partial or complete deficiency of anterior or posterior **pituitary** hormone production leads to central **hypoadrenalism**, central **hypothyroidism**, **hypogonadotropic hypogonadism**, **growth hormone deficiency**, or **arginine vasopressin deficiency** depending on the hormones affected. **hypopituitarism** is rare and likely to be underdiagnosed, with an unknown but rising incidence and prevalence. The most common cause is compressive growth or ablation of a **pituitary** or **hypothalamic** mass. Less common causes include genetic mutations, **hypophysitis** (especially in the context of cancer immunotherapy), infiltrative and infectious disease, and traumatic brain injury. Clinical features vary with timing of onset, cause, and number of **pituitary** axes disrupted. Diagnosis requires measurement of basal circulating hormone concentrations and confirmatory **hormone stimulation testing** as needed. Treatment is aimed at replacement of deficient hormones. Increased mortality might persist despite treatment, particularly in younger patients, females, and those with **arginine vasopressin deficiency**. Patients with complex diagnoses, pregnant patients, and adolescent **pituitary**-deficient patients transitioning to adulthood should ideally be managed at a **pituitary tumour centre** of excellence.

中文翻译

垂体前叶或后叶激素的部分或完全缺乏，根据受影响激素的不同，可导致中枢性肾上腺功能减退、中枢性甲状腺功能减退、低促性腺激素性性腺功能减退、生长激素缺乏或精氨酸加压素缺乏。垂体功能减退症较为罕见，且可能存在诊断不足的情况，其发病率和患病率虽不明确但呈上升趋势。最常见的原因是垂体或下丘脑占位性病变的压迫性生长或切除。较少见的原因包括基因突变、垂体炎（尤其在癌症免疫治疗背景下）、浸润性和感染性疾病以及创伤性脑损伤。临床表现因发病时间、病因及受累垂体轴的数量而异。诊断需测量基础循环激素浓度，并根据需要进行激素激发试验确认。治疗旨在替代缺乏的激素。尽管接受治疗，死亡率仍可能持续升高，尤其是在年轻患者、女性患者及精氨酸加压素缺乏患者中。对于诊断复杂的患者、妊娠期患者以及从青春期过渡至成年期的垂体功能减退青少年患者，理想情况下应在垂体肿瘤卓越中心进行管理。

核心词汇

hypopituitarism /ˌhaɪpəʊpɪˈtuːɪtəˌrɪzəm/ : 垂体功能减退症	hypoadrenalism /ˌhaɪpəʊəˈdriːnəlɪzəm/ : 肾上腺功能减退症
hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症	hypogonadotropic hypogonadism /ˌhaɪpəʊˌɡoʊnədəʊˈtrɒpɪk ˌhaɪpəʊˌɡoʊnədɪzəm/ : 低促性腺激素性性腺功能减退症
growth hormone deficiency /ɡroʊθ ˈhɔːrmoun dɪˈfɪjənsi/ : 生长激素缺乏症	arginine vasopressin deficiency /ˈɑːrdʒɪniːn ˌvæsəʊsˈpreɪsɪn dɪˈfɪjənsi/ : 精氨酸加压素缺乏症
pituitary /pɪˈtuːɪteri/ : 垂体	hypothalamic /ˌhaɪpəʊθəˈlæmɪk/ : 下丘脑的
hypophysitis /ˌhaɪpəʊfəɪˈsaɪtɪs/ : 垂体炎	hormone replacement /ˈhɔːrmoun rɪˈpleɪsmənt/ : 激素替代治疗
hormone stimulation testing /ˈhɔːrmoun ˌstɪmjʊˈleɪʃən ˈtestɪŋ/ : 激素激发试验	pituitary tumour centre /pɪˈtuːɪteri ˈtuːmə ˈsɛntər/ : 垂体肿瘤中心

Pangenomic classification of pituitary neuroendocrine tumors

垂体神经内分泌肿瘤的全基因组分类

期刊: Cancer Cell 日期: 2020 JCR Q1 中科院升级版 医学1区 IF 44.5

Abstract

Pituitary neuroendocrine tumors (PitNETs) are common, with five main histological subtypes: **lactotroph**, **somatotroph**, and **thyrotroph** (POU1F1/PIT1 lineage); **corticotroph** (TBX19/TPIT lineage); and **gonadotroph** (NR5A1/SF1 lineage). We report a comprehensive pangenomic classification of PitNETs. PitNETs from POU1F1/PIT1 lineage showed an **epigenetic signature** of diffuse **DNA hypomethylation**, with transposable elements expression and **chromosomal instability** (except for GNAS-mutated somatotrophs). In TPIT lineage, corticotrophs were divided into three classes: the USP8-mutated with overt secretion, the USP8-wild-type with increased **invasiveness** and increased epithelial-mesenchymal transition, and the large silent tumors with **gonadotroph transdifferentiation**. Unexpected expression of **gonadotroph** markers was also found in GNAS-wild-type somatotrophs (SF1 expression), challenging the current definition of SF1/**gonadotroph** lineage. This classification improves our understanding and affects the **clinical stratification** of patients with PitNETs.

中文翻译

垂体神经内分泌肿瘤 (PitNETs) 较为常见, 主要有五种组织学亚型: 泌乳素细胞瘤、生长激素细胞瘤和促甲状腺激素细胞瘤 (POU1F1/PIT1谱系); 促肾上腺皮质激素细胞瘤 (TBX19/TPIT谱系); 以及促性腺激素细胞瘤 (NR5A1/SF1谱系)。我们报告了PitNETs的全面泛基因组分类。来自POU1F1/PIT1谱系的PitNETs表现出弥漫性DNA低甲基化的表观遗传特征, 伴有转座元件表达和染色体不稳定性 (GNAS突变的生长激素细胞瘤除外)。在TPIT谱系中, 促肾上腺皮质激素细胞瘤被分为三类: USP8突变型伴有明显分泌功能, USP8野生型侵袭性增强且上皮-间质转化增加, 以及大型无功能肿瘤伴有促性腺激素细胞转分化。在GNAS野生型生长激素细胞瘤中也意外发现了促性腺激素标志物的表达 (SF1表达), 这对当前SF1/促性腺激素谱系的定义提出了挑战。这一分类提高了我们对PitNETs的理解, 并影响了患者的临床分层。

核心词汇

pituitary neuroendocrine tumor /piˈtjuːɪtəriˌnjuːərouˌendəˈkraɪnˈtjuːmə/ : 垂体神经内分泌肿瘤

lactotroph /ˈlæktətrof/ : 泌乳素细胞

corticotroph /ˈkɔːrtɪkətrof/ : 促肾上腺皮质激素细胞

epigenetic signature /ˌɛpɪdʒəˈnɛtɪkˈsɪɡnətʃər/ : 表观遗传特征

chromosomal instability /ˌkrouməˈsouməlˌɪnstəˈbɪlɪti/ : 染色体不稳定性

transdifferentiation /ˌtrænsdɪˈfɛrənʃiˈeɪʃən/ : 转分化

histological subtype /ˌhɪstəˈlɒdʒɪkəlˈsʌbtaɪp/ : 组织学亚型

somatotroph /ˈsoumətətrof/ : 生长激素细胞

gonadotroph /ˈɡounədətrof/ : 促性腺激素细胞

DNA hypomethylation /ˌdiːɛnˈeɪˌhaɪprouˌmɛθɪˈleɪʃən/ : DNA低甲基化

invasiveness /ɪnˈveɪsɪvnəs/ : 侵袭性

clinical stratification /ˈklɪnɪkəlˌstrætɪfɪˈkeɪʃən/ : 临床分层

Single-cell and spatial transcriptome analyses reveal tumor heterogeneity and immune remodeling involved in pituitary neuroendocrine tumor progression

单细胞与空间转录组分析揭示垂体神经内分泌肿瘤进展中的肿瘤异质性与免疫重塑

期刊: Nature Communications 日期: 2025 JCR Q1 中科院升级版 综合性期刊1区 IF 15.7

Abstract

Pituitary neuroendocrine tumors (PitNETs) can be invasive or **aggressive**, yet the mechanisms behind these behaviors remain poorly understood, impeding treatment advancements. Here, we integrat single-cell RNA sequencing and **spatial transcriptomics**, analyzing over 177,000 cells and 35,000 spots across 57 tissue samples. This comprehensive approach facilitates the identification of PitNETs tumor populations and characterizes the reconfiguration of the **tumor microenvironment** (TME) as PitNETs progress and invade. We trace the trajectory of TPIT-lineage PitNETs and identify an **aggressive** tumor cluster marked by elevated p53-mediated **proliferation** and a higher Trouillas classification, both associated with **tumor progression**. Additionally, we document the **heterogeneity** of immune stromal cells within PitNETs, particularly noting the enrichment of SPP1+ tumor associated macrophages (TAMs) in invasive tumors. These TAMs facilitate tumor **invasion** through the SPP1-ITGAV/ITGB1 signaling pathway. Our in-depth single-cell and spatial analysis of PitNETs uncovers the molecular dynamics within the TME, suggesting potential targets for **therapeutic intervention**.

中文翻译

垂体神经内分泌肿瘤（PitNETs）可能具有侵袭性或侵袭性，但这些行为背后的机制仍知之甚少，阻碍了治疗进展。本研究整合单细胞RNA测序与空间转录组学技术，分析了57个组织样本中超过17.7万个细胞和3.5万个位点。这种综合方法有助于识别PitNETs肿瘤群体，并描述肿瘤微环境（TME）在PitNETs进展和侵袭过程中的重构特征。我们追踪了TPIT谱系PitNETs的发展轨迹，发现一个以p53介导的增殖增强和更高Trouillas分级为标志的侵袭性肿瘤簇，这两者均与肿瘤进展相关。此外，我们记录了PitNETs内免疫基质细胞的异质性，特别注意到在侵袭性肿瘤中SPP1+肿瘤相关巨噬细胞（TAMs）的富集。这些TAMs通过SPP1-ITGAV/ITGB1信号通路促进肿瘤侵袭。我们对PitNETs的深入单细胞及空间分析揭示了TME内的分子动态，为治疗干预提供了潜在靶点。

核心词汇

pituitary neuroendocrine tumor /piˈtjuː.ɪteriː ˌnjuːərouː.ɛndeː
'kraɪn ˈtjuː.mər/ : 垂体神经内分泌肿瘤

aggressive /əˈɡresɪv/ : 侵袭性

proliferation /prəˌlɪfəˈreɪʃən/ : 增殖

macrophage /ˈmækrəfeɪdʒ/ : 巨噬细胞

tumor progression /ˈtjuː.mər prəˈɡreɪʃən/ : 肿瘤进展

tumor associated macrophage /ˈtjuː.mər əˈsəʊsiətiːd
'mækrəfeɪdʒ/ : 肿瘤相关巨噬细胞

invasion /ɪnˈveɪʒən/ : 侵袭

tumor microenvironment /ˈtjuː.mər ˌmaɪkrouɪnˈvaɪrənmənt/
: 肿瘤微环境

heterogeneity /ˌhetərəʊdʒəˈniːti/ : 异质性

spatial transcriptomics /ˈspeɪʃəl trænˈskriptəmɪks/ : 空间转录组学

therapeutic intervention /ˌθerəˈpiː.tɪk ˌɪntərˈvenʃən/ : 治疗干预

Novel genetic causes of pituitary adenomas

垂体腺瘤的新遗传原因

期刊: Clinical Cancer Research 日期: 2016 JCR Q1 中科院升级版 医学1区 IF 10.2

Abstract

Recently, a number of novel genetic alterations have been identified that predispose individuals to pituitary adenomas. Clinically relevant pituitary adenomas are relatively common, present in 0.1% of the general population. They are mostly benign monoclonal neoplasms that arise from any of the five **hormone**-secreting cell types of the anterior lobe of the **pituitary gland**, and cause disease due to hormonal alterations and local space-occupying effects. The pathomechanism of pituitary adenomas includes alterations in cell-cycle regulation and growth factor signaling, which are mostly due to **epigenetic** changes; somatic and especially germline mutations occur more rarely. A significant proportion of growth **hormone**- and adrenocorticotrophin-secreting adenomas have activating somatic mutations in the GNAS and USP8 genes, respectively. Rarely, germline mutations predispose to pituitary tumorigenesis, often in a **familial** setting. Classical tumor predisposition syndromes include multiple endocrine neoplasia type 1 (MEN1) and type 4 (MEN4) syndromes, Carney complex, and McCune-Albright syndrome. Pituitary tumors have also been described in association with neurofibromatosis type 1, DICER1 syndrome, and SDHx mutations. Pituitary adenomas with no other associated tumors have been described as **familial** isolated pituitary adenomas. Patients with AIP or GPR101 mutations often present with pituitary **gigantism** either in a **familial** or simplex setting. GNAS and GPR101 mutations that arise in early embryonic age can lead to somatic **mosaicism** involving the **pituitary gland** and resulting in growth **hormone** excess. **senescence** has been suggested as the key mechanism protecting pituitary adenomas turning **malignant** in the overwhelming majority of cases. Here we briefly summarize the genetic background of pituitary adenomas, with an emphasis on the recent developments in this field. Clin Cancer Res; 22(20); 5030-42. ©2016 AACR SEE ALL ARTICLES IN THIS CCR FOCUS SECTION, "ENDOCRINE CANCERS REVISING PARADIGMS".

中文翻译

近期研究发现多种新型基因变异可增加垂体腺瘤的易感性。具有临床意义的垂体腺瘤较为常见，在普通人群中发病率为0.1%。这类肿瘤多为良性单克隆新生物，源自垂体前叶五种激素分泌细胞中的任意一种，通过激素异常分泌及局部占位效应引发疾病。其病理机制涉及细胞周期调控与生长因子信号通路的改变，主要源于表观遗传学变化，体细胞突变尤其是胚系突变较为罕见。在生长激素和促肾上腺皮质激素分泌型腺瘤中，分别存在较高比例的GNAS和USP8基因激活突变。胚系突变虽罕见，但常以家族性形式诱发垂体肿瘤发生。经典肿瘤易感综合征包括多发性内分泌腺瘤病1型（MEN1）和4型（MEN4）、Carney综合征及McCune-Albright综合征。此外，垂体肿瘤还与神经纤维瘤病1型、DICER1综合征及SDHx基因突变相关。不伴其他肿瘤的垂体腺瘤被定义为家族性孤立性垂体腺瘤。携带AIP或GPR101基因突变的患者常表现为家族性或散发性垂体性巨人症。胚胎早期发生的GNAS和GPR101突变可导致累及垂体的体细胞嵌合体，进而引发生长激素过量。研究提示细胞衰老是绝大多数垂体腺瘤避免恶性转化的关键保护机制。本文简要综述垂体腺瘤的遗传背景，重点阐述该领域最新研究进展。Clin Cancer Res; 22(20); 5030-42. ©2016 AACR 详见本刊CCR焦点栏目"内分泌肿瘤：范式重构"全部文章。

核心词汇

pituitary adenoma /piˈtjuːɪtəri ˌædɪˈnoumə/ : 垂体腺瘤

hormone /ˈhɔːrmoun/ : 激素

germline mutation /ˈdʒɜːrmlaɪn mjuːˈteɪʃən/ : 种系突变

epigenetic /ˌepɪdʒəˈnetɪk/ : 表观遗传

senescence /sɪˈnesəns/ : 衰老

familial /fəˈmɪliəl/ : 家族性

neoplasm /ˈniːou.plæzəm/ : 肿瘤

pituitary gland /piˈtjuːɪtəri glænd/ : 垂体

somatic mutation /souˈmætɪk mjuːˈteɪʃən/ : 体细胞突变

gigantism /dʒaɪˈɡæntɪzəm/ : 巨人症

mosaicism /mouˈzeɪɪsɪzəm/ : 嵌合体

malignant /məˈlɪɡnənt/ : 恶性

Diagnosis and management of pituitary adenomas

垂体腺瘤的诊断与治疗

期刊: JAMA 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 55.0

Abstract

ImportancePituitary adenomas are neoplasms of the pituitary adenohypophyseal cell lineage and include functioning tumors, characterized by the secretion of pituitary hormones, and nonfunctioning tumors. Clinically evident pituitary adenomas occur in approximately 1 in 1100 persons.ObservationsPituitary adenomas are classified as either macroadenomas (≥ 10 mm) (48% of tumors) or microadenomas (< 10 mm). Macroadenomas may cause mass effect, such as visual field defects, headache, and/or hypopituitarism, which occur in about 18% to 78%, 17% to 75%, and 34% to 89% of patients, respectively. Thirty percent of pituitary adenomas are nonfunctioning adenomas, which do not produce hormones. Functioning tumors are those that produce an excess of normally produced hormones and include prolactinomas, somatotropinomas, corticotropinomas, and thyrotropinomas, which produce prolactin, growth hormone, corticotropin, and thyrotropin, respectively. Approximately 53% of pituitary adenomas are prolactinomas, which can cause hypogonadism, infertility, and/or galactorrhea. Twelve percent are somatotropinomas, which cause acromegaly in adults and gigantism in children, and 4% are corticotropinomas, which secrete corticotropin autonomously, resulting in hypercortisolemia and Cushing disease. All patients with pituitary tumors require endocrine evaluation for hormone hypersecretion. Patients with macroadenomas additionally require evaluation for hypopituitarism, and patients with tumors compressing the optic chiasm should be referred to an ophthalmologist for formal visual field testing. For those requiring treatment, first-line therapy is usually transsphenoidal pituitary surgery, except for prolactinomas, for which medical therapy, either bromocriptine or cabergoline, is usually first line.Conclusions and RelevanceClinically manifest pituitary adenomas affect approximately 1 in 1100 people and can be complicated by syndromes of hormone excess as well as visual field defects and hypopituitarism from mass effect in larger tumors. First-line therapy for prolactinomas consists of bromocriptine or cabergoline, and transsphenoidal pituitary surgery is first-line therapy for other pituitary adenomas requiring treatment.

中文翻译

重要性：垂体腺瘤是垂体腺垂体细胞谱系的肿瘤，包括功能性肿瘤（以分泌垂体激素为特征）和非功能性肿瘤。临床上明显的垂体腺瘤发生率约为每1100人中1例。观察结果：垂体腺瘤分为大腺瘤（ ≥ 10 毫米）（占肿瘤的48%）和微腺瘤（ < 10 毫米）。大腺瘤可能引起占位效应，如视野缺损、头痛和/或垂体功能减退，这些症状分别出现在约18%至78%、17%至75%和34%至89%的患者中。30%的垂体腺瘤为非功能性腺瘤，不产生激素。功能性肿瘤则过量分泌正常产生的激素，包括泌乳素瘤、生长激素瘤、促肾上腺皮质激素瘤和促甲状腺激素瘤，分别分泌泌乳素、生长激素、促肾上腺皮质激素和促甲状腺激素。约53%的垂体腺瘤为泌乳素瘤，可能导致性腺功能减退、不孕和/或溢乳。12%为生长激素瘤，在成人中引起肢端肥大症，在儿童中引起巨人症；4%为促肾上腺皮质激素瘤，自主分泌促肾上腺皮质激素，导致高皮质醇血症和库欣病。所有垂体肿瘤患者均需进行内分泌评估以检测激素分泌是否过量。大腺瘤患者还需评估垂体功能减退情况，而肿瘤压迫视交叉的患者应转诊至眼科医生进行正式视野检查。对于需要治疗的患者，除泌乳素瘤外，一线治疗通常为经蝶垂体手术；泌乳素瘤的一线治疗通常为药物治疗，即溴隐亭或卡麦角林。结论与相关性：临床上明显的垂体腺瘤影响约每1100人中的1例，可能并发激素过量综合征，以及较大肿瘤占位效应引起的视野缺损和垂体功能减退。泌乳素瘤的一线治疗包括溴隐亭或卡麦角林，而其他需要治疗的垂体腺瘤的一线治疗为经蝶垂体手术。

核心词汇

pituitary adenoma /piˈtjuːɪteriː ˌædiˈnoumə/ : 垂体腺瘤	macroadenoma /ˌmækrouˌædiˈnoumə/ : 大腺瘤
microadenoma /ˌmaɪkrouˌædiˈnoumə/ : 微腺瘤	prolactinoma /prɒʊˌlæktiˈnoumə/ : 泌乳素瘤
somatotropinoma /souˌmæteˌtrɒʊpiˈnoumə/ : 生长激素瘤	corticotropinoma /ˌkɔːrtɪkɒʊˌtrɒʊpiˈnoumə/ : 促肾上腺皮质激素瘤
hypopituitarism /ˌhaɪpɒʊpiˈtjuːɪterɪzəm/ : 垂体功能减退症	acromegaly /ˌækrouˈmegəli/ : 肢端肥大症
Cushing disease /ˈkʊʃɪŋ diˈziːz/ : 库欣病	visual field defect /ˈvɪʒuəl fiːld ˈdiːfekt/ : 视野缺损
transsphenoidal surgery /trænsˌsfiːˈnoɪdəl ˈsɜːrdʒəri/ : 经蝶窦手术	endocrine evaluation /ˈendoukrɪn ɪˌvæljuˈeɪʃən/ : 内分泌评估

Current management of cushing's disease

库欣病的当前管理

期刊: Journal of Internal Medicine 日期: 2019 JCR Q1 中科院升级版 医学1区 IF 9.2

Abstract

Abstract Cushing's disease (CD) is caused by a **pituitary tumour** that secretes **adrenocorticotropin** (ACTH) autonomously, leading to excess **cortisol** secretion from the **adrenal glands**. The condition is associated with increased morbidity and mortality that can be mitigated by treatments that result in sustained **endocrine remission**. **transsphenoidal pituitary surgery** (TSS) remains the mainstay of treatment for CD but requires considerable neurosurgical expertise and experience in order to optimize patient outcomes. Up to 90% of patients with microadenomas (tumour below 1 cm in largest diameter) and 65% of patients with macroadenomas (tumour at or above 1 cm in greatest diameter) achieve **endocrine remission** after TSS by an experienced surgeon. Patients who are not in remission postoperatively or those who relapse may benefit from undergoing a second pituitary operation. Alternatively, **radiation therapy** to the sella with interim medical therapy, or **bilateral adrenalectomy**, can be effective as definitive treatments of CD. Medical therapy is currently adjunctive in most patients with CD and is generally prescribed to patients who are about to receive **radiation therapy** and will be awaiting its salutary effects to occur. Available treatment options include **steroidogenesis inhibitors**, centrally acting agents and **glucocorticoid receptor antagonists**. Several novel agents are in clinical trials and may eventually constitute additional treatment options for this serious condition.

中文翻译

摘要库欣病 (CD) 是由垂体肿瘤自主分泌促肾上腺皮质激素 (ACTH) 引起, 导致肾上腺皮质醇分泌过量。该疾病与发病率和死亡率增加相关, 通过实现持续内分泌缓解的治疗可减轻这些风险。经蝶垂体手术 (TSS) 仍是CD的主要治疗方法, 但需要丰富的神经外科专业知识和经验以优化患者预后。在经验丰富的外科医生操作下, 高达90%的微腺瘤 (最大直径小于1厘米的肿瘤) 患者和65%的大腺瘤 (最大直径等于或大于1厘米的肿瘤) 患者在接受TSS后能达到内分泌缓解。术后未缓解或复发的患者可能受益于第二次垂体手术。此外, 针对蝶鞍区的放射治疗配合过渡期药物治疗, 或双侧肾上腺切除术, 均可作为CD的有效根治性治疗方案。目前药物治疗对大多数CD患者而言属于辅助手段, 通常适用于即将接受放射治疗并等待其有益效果显现的患者。现有治疗选择包括类固醇生成抑制剂、中枢作用药物和糖皮质激素受体拮抗剂。若干新型药物正处于临床试验阶段, 未来可能为这一严重疾病提供更多治疗选择。

核心词汇

Cushing's disease /'kʊʃɪŋz dɪ'zi:z/ : 库欣病

adrenocorticotropin /ə'dri:nou,kɔ:'rtɪkəʊ'trɒpɪn/ : 促肾上腺皮质激素

adrenal glands /ə'dri:nəl glændz/ : 肾上腺

transsphenoidal pituitary surgery /træns'sfi:nɔɪdəl pi'tju:ɪtəri 'sɜ:rdʒəri/ : 经蝶窦垂体手术

macroadenoma /,mækrou,ædi'noumə/ : 大腺瘤

bilateral adrenalectomy /baɪ'lætərəl ə'dri:nə'lektəmi/ : 双侧肾上腺切除术

glucocorticoid receptor antagonists /glu:kou'kɔ:'rtɪkɔɪd rɪ'septər æn'tæɡənɪsts/ : 糖皮质激素受体拮抗剂

pituitary tumour /pi'tju:ɪtəri 'tju:mə/ : 垂体肿瘤

cortisol /'kɔ:'rtɪsɒl/ : 皮质醇

endocrine remission /'endəukraɪn rɪ'mɪʃən/ : 内分泌缓解

microadenoma /,maɪkrou,ædi'noumə/ : 微腺瘤

radiation therapy /,reɪdi'eɪʃən 'θerəpi/ : 放射治疗

steroidogenesis inhibitors /strɪ'ɔɪdɒʊ'dʒenɪsɪs ɪn'hɪbɪtəz/ : 类固醇生成抑制剂

Proteogenomic landscape and clinical characterization of GH-producing pituitary adenomas/somatotroph pituitary neuroendocrine tumors

生长激素分泌型垂体腺瘤/生长激素垂体神经内分泌肿瘤的蛋白质基因组学图谱与临床特征

期刊: Communications Biology 日期: 2022 JCR Q1 中科院升级版 生物学1区 IF 5.1

Abstract

The clinical characteristics of growth hormone (GH)-producing pituitary adenomas/somatotroph pituitary neuroendocrine tumors (GHomas/somatotroph PitNETs) vary across patients. In this study, we aimed to integrate the genetic alterations, protein expression profiles, transcriptomes, and clinical characteristics of GHomas/somatotroph PitNETs to identify molecules associated with acromegaly characteristics. Targeted capture sequencing and copy number analysis of 36 genes and nontargeted proteomics analysis were performed on fresh-frozen samples from 121 sporadic GHomas/somatotroph PitNETs. Targeted capture sequencing revealed GNAS as the only driver gene, as previously reported. Classification by consensus clustering using both RNA sequencing and proteomics revealed many similarities between the proteome and the transcriptome. gene ontology analysis was performed for differentially expressed proteins between wild-type and mutant GNAS samples identified by nontargeted proteomics and involved in G protein-coupled receptor (GPCR) pathways. The results suggested that GNAS mutations impact endocrinological features in acromegaly through GPCR pathway induction. ATP2A2 and ARID5B correlated with the GH change rate in the octreotide loading test, and WWC3, SERINC1, and ZFAND3 correlated with the tumor volume change rate after somatostatin analog treatment. These results identified a biological connection between GNAS mutations and the clinical and biochemical characteristics of acromegaly, revealing molecules associated with acromegaly that may affect medical treatment efficacy.

中文翻译

摘要生长激素（GH）分泌型垂体腺瘤/生长激素细胞垂体神经内分泌肿瘤（GHomas/somatotroph PitNETs）的临床特征因患者而异。本研究旨在整合GHomas/somatotroph PitNETs的基因变异、蛋白表达谱、转录组及临床特征，以识别与肢端肥大症特征相关的分子。我们对121例散发性GHomas/somatotroph PitNETs的新鲜冷冻样本进行了36个基因的靶向捕获测序与拷贝数分析，以及非靶向蛋白质组学分析。靶向捕获测序结果显示，正如先前报道，GNAS是唯一的驱动基因。通过RNA测序和蛋白质组学的共识聚类分类显示，蛋白质组与转录组之间存在高度相似性。针对非靶向蛋白质组学鉴定的野生型与突变型GNAS样本间差异表达蛋白进行了基因本体分析，这些蛋白参与G蛋白偶联受体（GPCR）通路。结果表明，GNAS突变通过诱导GPCR通路影响肢端肥大症的内分泌特征。ATP2A2和ARID5B与奥曲肽负荷试验中的GH变化率相关，而WWC3、SERINC1和ZFAND3与生长抑素类似物治疗后的肿瘤体积变化率相关。这些结果揭示了GNAS突变与肢端肥大症临床及生化特征之间的生物学联系，并识别出可能影响药物治疗效果的肢端肥大症相关分子。

核心词汇

acromegaly /ˌækrouˈmegəli/ : 肢端肥大症

somatotroph /soʊˈmætoˌtroʊf/ : 生长激素细胞

transcriptome /trænˈskriptəʊm/ : 转录组

mutation /mjuˈteɪʃən/ : 突变

endocrinological /ˌendəkrɪnəˈlɒdʒɪkəl/ : 内分泌学的

transplantation /ˌtrænsplænˈteɪʃən/ : 移植

pituitary adenoma /pɪˈtjuːɪtəriˌædɪˈnoʊmə/ : 垂体腺瘤

neuroendocrine tumor /ˌnuːroʊˈendəkrɪn ˈtjuːmə/ : 神经内分泌肿瘤

proteomics /ˌprəʊtiˈoʊmɪks/ : 蛋白质组学

gene ontology /dʒiːn ˌɒntɪˈblədʒi/ : 基因本体论

biopsy /ˈbaɪɒpsi/ : 活检

immunotherapy /ˌɪmjʊnoʊˈθerəpi/ : 免疫治疗

Growth hormone deficiency: Health and longevity

生长激素缺乏症：健康与长寿

期刊: Endocrine Reviews 日期: 2019 JCR Q1 中科院升级版 医学1区 IF 22.0

Abstract

Abstract The important role of GH in the control of mammalian **longevity** was first deduced from extended **longevity** of mice with genetic GH **deficiency** (GHD) or GH resistance. Mice with isolated GHD (IGHD) due to GHRH or GHRH **receptor** mutations, combined **deficiency** of GH, prolactin, and TSH, or global deletion of GH receptors live longer than do their normal siblings. They also exhibit multiple features of delayed and/or slower **aging**, accompanied by extension of **healthspan**. The unexpected, remarkable **longevity** benefit of severe endocrine defects in these animals presumably represents evolutionarily conserved trade-offs among **aging**, growth, maturation, fecundity, and the underlying anabolic processes. Importantly, the negative association of GH signaling with **longevity** extends to other mammalian species, apparently including humans. Data obtained in humans with IGHD type 1B, owing to a **mutation** of the GHRH **receptor** gene, in the Itabaianinha County, Brazil, provide a unique opportunity to study the impact of severe reduction in GH signaling on age-related characteristics, health, and functionality. Individuals with IGHD are characterized by proportional short **stature**, doll facies, high-pitched voices, and central **obesity**. They have delayed **puberty** but are **fertile** and generally healthy. Moreover, these IGHD individuals are partially protected from **cancer** and some of the common effects of **aging** and can attain extreme **longevity**, 103 years of age in one case. We think that low, but detectable, residual GH **secretion** combined with life-long reduction of circulating IGF-1 and with some tissue levels of IGF-1 and/or IGF-2 preserved may account for the normal **longevity** and apparent extension of **healthspan** in these individuals.

中文翻译

摘要：生长激素（GH）在调控哺乳动物寿命中的重要作用，最初是从遗传性GH缺乏（GHD）或GH抵抗小鼠寿命延长现象中推断出来的。由于GHRH或GHRH受体突变导致孤立性GHD（IGHD）、GH/催乳素/TSH联合缺乏或GH受体整体缺失的小鼠，其寿命均超过正常同窝小鼠。这些小鼠还表现出多种衰老延迟和/或减缓的特征，并伴随着健康寿命的延长。这些动物因严重内分泌缺陷带来的意外显著长寿效应，可能反映了进化过程中衰老、生长、成熟、繁殖力及合成代谢过程之间的权衡机制。重要的是，GH信号与寿命的负相关关系在其他哺乳动物乃至人类中同样存在。巴西伊塔巴尼亚尼亚县因GHRH受体基因突变导致的1B型IGHD人群，为研究GH信号严重减弱对年龄相关特征、健康状况及功能的影响提供了独特样本。IGHD个体具有比例匀称的矮小身材、娃娃面容、高频嗓音和中心性肥胖特征。他们青春期延迟但具备生育能力且总体健康。值得注意的是，这些IGHD人群对癌症和某些常见衰老效应具有部分抵抗力，其中一例甚至达到103岁极端长寿。我们认为，持续可检测的低水平残余GH分泌、伴随终生的循环IGF-1水平降低，以及部分组织中保留的IGF-1和/或IGF-2水平，可能共同促成了这些个体的正常寿命及显著延长的健康生存期。

核心词汇

longevity /lɒnˈdʒevəti/ : 长寿

mutation /mjuːˈteɪʃən/ : 突变

puberty /ˈpjuːbəti/ : 青春期

cancer /ˈkænsər/ : 癌症

stature /ˈstætʃər/ : 身材

aging /ˈeɪdʒɪŋ/ : 衰老

deficiency /dɪˈfɪjənsi/ : 缺乏症

receptor /rɪˈseptər/ : 受体

obesity /əʊˈbiːsəti/ : 肥胖症

secretion /sɪˈkriːʃən/ : 分泌

fertile /ˈfɜːtaɪl/ : 有生育能力

healthspan /ˈhelθspæn/ : 健康寿命

Acromegaly: Pathogenesis, diagnosis, and management

肢端肥大症：发病机制、诊断与治疗

期刊: The Lancet Diabetes & Endocrinology 日期: 2022

Abstract

growth hormone-secreting pituitary adenomas that cause **acromegaly** arise as monoclonal expansions of differentiated **somatotroph** cells and are usually sporadic. They are almost invariably benign, yet they can be locally invasive and show progressive growth despite treatment. Persistent excess of both **growth hormone** and its target hormone insulin-like growth factor 1 (IGF-1) results in a wide array of cardiovascular, respiratory, metabolic, musculoskeletal, neurological, and neoplastic comorbidities that might not be reversible with disease control. Normalisation of IGF-1 and **growth hormone** are the primary therapeutic aims; additional treatment goals include tumour shrinkage, relieving symptoms, managing complications, reducing excess morbidity, and improving quality of life. A multimodal approach with **surgery**, **medical therapy**, and (more rarely) **radiation therapy** is required to achieve these goals. In this Review, we examine the **epidemiology**, **pathogenesis**, **diagnosis**, complications, and treatment of **acromegaly**, with an emphasis on the importance of tailoring management strategies to each patient to optimise outcomes.

中文翻译

导致肢端肥大症的生长激素分泌型垂体腺瘤源于分化型生长激素细胞的单克隆扩增，通常为散发性。这类肿瘤几乎总是良性的，但可能具有局部侵袭性，并在治疗后仍可能持续生长。生长激素及其靶激素胰岛素样生长因子1 (IGF-1) 的持续过量会导致广泛的心血管、呼吸系统、代谢、骨骼肌肉、神经系统及肿瘤性共病，这些病症在疾病控制后可能无法逆转。治疗的主要目标是使IGF-1和生长激素水平恢复正常；其他治疗目标包括缩小肿瘤体积、缓解症状、处理并发症、降低过度发病率以及提高生活质量。为实现这些目标，需要采用手术、药物治疗和（较少使用的）放射治疗相结合的多模式方法。在本综述中，我们探讨了肢端肥大症的流行病学、发病机制、诊断、并发症及治疗，重点强调了个体化管理策略对优化治疗效果的重要性。

核心词汇

acromegaly /ˌækrouˈmegəli/ : 肢端肥大症

somatotroph /souˈmæteˌtrɒf/ : 生长激素细胞

comorbidity /ˌkoumɔːrˈbɪdɪti/ : 共病

pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制

medical therapy /ˈmedɪkəl ˈθerəpi/ : 药物治疗

diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断

pituitary adenoma /piˈtjuːɪtəri ˌædɪˈnoumə/ : 垂体腺瘤

growth hormone /ɡrouθ ˈhɔːrmoun/ : 生长激素

epidemiology /ˌepɪˌdiːmiˈɔːlədʒi/ : 流行病学

surgery /ˈsɜːrdʒəri/ : 手术

radiation therapy /ˌreɪdiˈeɪʃən ˈθerəpi/ : 放射治疗

Subclinical hypothyroidism

亚临床甲状腺功能减退症

期刊: JAMA 日期: 2019 JCR Q1 中科院升级版 医学1区 IF 55.0

Abstract

IMPORTANCE: **subclinical hypothyroidism**, defined as an elevated serum **thyrotropin** (often referred to as thyroid-stimulating hormone, or TSH) level with normal levels of free **thyroxine** (FT4) affects up to 10% of the adult population. **OBSERVATIONS:** **subclinical hypothyroidism** is most often caused by autoimmune (Hashimoto) thyroiditis. However, serum **thyrotropin** levels rise as people without thyroid disease age; serum **thyrotropin** concentrations may surpass the upper limit of the traditional reference range of 4 to 5 mU/L among elderly patients. This phenomenon has likely led to an overestimation of the true prevalence of **subclinical hypothyroidism** in persons older than 70 years. In patients who have circulating **thyroid peroxidase antibodies**, there is a greater risk of progression from subclinical to **overt hypothyroidism**. **subclinical hypothyroidism** may be associated with an increased risk of **heart failure**, **coronary artery disease** events, and mortality from coronary heart disease. In addition, middle-aged patients with **subclinical hypothyroidism** may have **cognitive impairment**, nonspecific symptoms such as **fatigue**, and altered mood. In the absence of large randomized trials showing benefit from **levothyroxine therapy**, the rationale for treatment is based on the potential for decreasing the risk of adverse cardiovascular events and the possibility of preventing progression to **overt hypothyroidism**. However, **levothyroxine therapy** may be associated with **iatrogenic thyrotoxicosis**, especially in elderly patients, and there is no evidence that it is beneficial in persons aged 65 years or older. **CONCLUSIONS AND RELEVANCE:** **subclinical hypothyroidism** is common and most individuals can be observed without treatment. Treatment might be indicated for patients with **subclinical hypothyroidism** and serum **thyrotropin** levels of 10 mU/L or higher or for young and middle-aged individuals with **subclinical hypothyroidism** and symptoms consistent with mild hypothyroidism.

中文翻译

重要性: 亚临床甲状腺功能减退症, 定义为血清促甲状腺激素 (常称为TSH) 水平升高而游离甲状腺素 (FT4) 水平正常, 影响高达10%的成年人口。观察结果: 亚临床甲状腺功能减退症最常见的原因是自身免疫性 (桥本氏) 甲状腺炎。然而, 无甲状腺疾病的人群随着年龄增长, 血清促甲状腺激素水平会上升; 老年患者的血清促甲状腺激素浓度可能超过传统参考范围上限的4至5 mU/L。这一现象可能导致对70岁以上人群亚临床甲状腺功能减退症真实患病率的高估。对于循环中存在甲状腺过氧化物酶抗体的患者, 从亚临床进展为明显甲状腺功能减退症的风险更大。亚临床甲状腺功能减退症可能与心力衰竭、冠状动脉疾病事件以及冠心病死亡风险增加有关。此外, 患有亚临床甲状腺功能减退症的中年患者可能出现认知障碍、疲劳等非特异性症状以及情绪变化。由于缺乏大型随机试验显示左旋甲状腺素治疗的益处, 治疗的理由基于降低不良心血管事件风险的潜力以及防止进展为明显甲状腺功能减退症的可能性。然而, 左旋甲状腺素治疗可能与医源性甲状腺毒症相关, 尤其是在老年患者中, 并且没有证据表明它对65岁或以上人群有益。结论与相关性: 亚临床甲状腺功能减退症很常见, 大多数个体可以在不治疗的情况下进行观察。对于亚临床甲状腺功能减退症且血清促甲状腺激素水平达到10 mU/L或更高的患者, 或者对于有亚临床甲状腺功能减退症且症状与轻度甲状腺功能减退症一致的中青年个体, 可能需要治疗。

核心词汇

subclinical hypothyroidism /sʌbˈklaɪnɪkəl ˌhaɪpəˈθaɪroɪdɪzəm/ : 亚临床甲状腺功能减退症	thyrotropin /ˌθaɪrəˈtrɒpɪn/ : 促甲状腺激素
thyroxine /θaɪˈrɒksɪːn/ : 甲状腺素	autoimmune thyroiditis /ˌɔːtəʊˈɪmjʊn ˌθaɪrəɪˈdaɪtɪs/ : 自身免疫性甲状腺炎
thyroid peroxidase antibodies /ˈθaɪroɪd pəˈrɒksɪdeɪz ˈæntɪˌbɒdiz/ : 甲状腺过氧化物酶抗体	overt hypothyroidism /oʊˈvɜːrt ˌhaɪpəˈθaɪroɪdɪzəm/ : 显性甲状腺功能减退症
heart failure /hɑːrt ˈfeɪljər/ : 心力衰竭	coronary artery disease /ˈkɒrənəri ˈɑːtəri dɪˈziːz/ : 冠状动脉疾病
cognitive impairment /ˈkɒɡnɪtɪv ɪmˈpeəmənt/ : 认知功能障碍	levothyroxine therapy /ˌliːvəʊˈθaɪˈrɒksɪːn ˈθerəpi/ : 左甲状腺素治疗
iatrogenic thyrotoxicosis /aɪˌætrəˈdʒenɪk ˌθaɪrəʊtɒk ˈsɪkʊsɪs/ : 医源性甲状腺毒症	fatigue /fəˈtiːɡ/ : 疲劳

Diagnosis and management of graves disease: A global overview

格雷夫斯病的诊断与管理：全球概览

期刊: Nature Reviews Endocrinology 日期: 2013 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

Graves disease is an autoimmune disorder characterized by goitre, hyperthyroidism and, in 25% of patients, Graves ophthalmopathy. The hyperthyroidism is caused by thyroid hypertrophy and stimulation of function, resulting from interaction of anti-TSH-receptor antibodies (TRAb) with the TSH receptor on thyroid follicular cells. Measurements of serum levels of TRAb and thyroid ultrasonography represent the most important diagnostic tests for Graves disease. Management of the condition currently relies on antithyroid drugs, which mainly inhibit thyroid hormone synthesis, or ablative treatments ((¹³¹I)-radiotherapy or thyroidectomy) that remove or decrease thyroid tissue. None of these treatments targets the disease process, and patients with treated Graves disease consequently experience either a high rate of recurrence, if receiving antithyroid drugs, or lifelong hypothyroidism, after ablative therapy. Geographical differences in the use of these therapies exist, partially owing to the availability of skilled thyroid surgeons and suitable nuclear medicine units. Novel agents that might act on the disease process are currently under evaluation in preclinical or clinical studies, but evidence of their efficacy and safety is lacking.

中文翻译

格雷夫斯病是一种自身免疫性疾病，其特征为甲状腺肿、甲状腺功能亢进，以及25%的患者伴有格雷夫斯眼病。其甲状腺功能亢进源于抗促甲状腺激素受体抗体（TRAb）与甲状腺滤泡细胞上的促甲状腺激素受体相互作用，导致甲状腺肥大和功能亢进。血清TRAb水平检测和甲状腺超声检查是诊断格雷夫斯病最重要的检测手段。目前该病的治疗主要依赖抗甲状腺药物（主要抑制甲状腺激素合成）或消融治疗（¹³¹I放射治疗或甲状腺切除术）以去除或减少甲状腺组织。这些治疗方法均未针对疾病进程本身，因此接受抗甲状腺药物治疗的患者复发率较高，而接受消融治疗的患者则需终身面临甲状腺功能减退的风险。这些治疗方法的使用存在地域差异，部分原因在于熟练甲状腺外科医生和合适核医学单位的可获得性。目前正在临床前或临床研究中评估可能作用于疾病进程的新型药物，但尚缺乏其有效性和安全性的证据。

核心词汇

Graves disease /greɪvz dɪ'zi:z/ : 格雷夫斯病

goitre /'gɔɪtə/ : 甲状腺肿

ophthalmopathy /'ɒfθəl'mɒpəθi/ : 眼病

thyroid follicular cells /'θaɪrɔɪd fə'ɪkjʊlər sɛlz/ : 甲状腺滤泡细胞

antithyroid drugs /,ænti'θaɪrɔɪd drʌgz/ : 抗甲状腺药物

hypothyroidism /,haɪpə'θaɪrɔɪdɪzəm/ : 甲状腺功能减退症

autoimmune disorder /,ɔ:təʊ'mju:n dɪs'ɔ:rdər/ : 自身免疫性疾病

hyperthyroidism /,haɪpə'θaɪrɔɪdɪzəm/ : 甲状腺功能亢进症

thyroid hypertrophy /'θaɪrɔɪd haɪ'pɜ:rtroʊfi/ : 甲状腺肥大

ultrasonography /,ʌltrəsə'nɒgrəfi/ : 超声检查

thyroidectomy /,θaɪrɔɪ'dektəmi/ : 甲状腺切除术

recurrence /rɪ'kɜ:rəns/ : 复发

Lymphocyte infiltration and thyrocyte destruction are driven by stromal and immune cell components in hashimoto’ s thyroiditis

淋巴细胞浸润与甲状腺细胞破坏由桥本甲状腺炎中的基质细胞和免疫细胞成分驱动。

期刊: Nature Communications 日期: 2022 JCR Q1 中科院升级版 综合性期刊1区 IF 15.7

Abstract

AbstractHashimoto’s thyroiditis (HT) is the most common **autoimmune disease** characterized by lymphocytic infiltration and **thyrocyte** destruction. Dissection of the interaction between the thyroidal stromal microenvironment and the infiltrating immune cells might lead to a better understanding of HT **pathogenesis**. Here we show, using single-cell RNA-sequencing, that three thyroidal **stromal cell** subsets, ACKR1+endothelial cells and CCL21+myofibroblasts and CCL21+fibroblasts, contribute to the thyroidal tissue microenvironment in HT. These cell types occupy distinct histological locations within the **thyroid gland**. Our experiments suggest that they might facilitate **lymphocyte** trafficking from the blood to thyroid tissues, and T cell zone CCL21+fibroblasts may also promote the formation of tertiary lymphoid organs characteristic to HT. Our study also demonstrates the presence of inflammatory macrophages and dendritic cells expressing high levels of IL-1β in the thyroid, which may contribute to **thyrocyte** destruction in HT patients. Our findings thus provide a deeper insight into the cellular interactions that might prompt the **pathogenesis** of HT.

中文翻译

摘要：桥本甲状腺炎（HT）是最常见的自身免疫性疾病，其特征为淋巴细胞浸润和甲状腺细胞破坏。剖析甲状腺基质微环境与浸润免疫细胞之间的相互作用，可能有助于更好地理解HT的发病机制。本文通过单细胞RNA测序技术揭示，三种甲状腺基质细胞亚群——ACKR1+内皮细胞、CCL21+肌成纤维细胞和CCL21+成纤维细胞——共同构成了HT中的甲状腺组织微环境。这些细胞类型在甲状腺内占据不同的组织学位置。实验表明它们可能促进淋巴细胞从血液向甲状腺组织迁移，而T细胞区的CCL21+成纤维细胞还可能促进HT特征性三级淋巴器官的形成。研究还证实甲状腺中存在高表达IL-1β的炎性巨噬细胞和树突状细胞，这可能是导致HT患者甲状腺细胞破坏的重要因素。这些发现为深入理解可能驱动HT发病机制的细胞间相互作用提供了新的视角。

核心词汇

Hashimoto's thyroiditis /hɑːʃiˈmɔʊtəʊz ˌθaɪrɔɪˈdaɪtɪs/ : 桥本甲状腺炎	autoimmune disease /ˌɔːtəʊˈmjuːn dɪˈziːz/ : 自身免疫性疾病
lymphocyte /ˈlɪmfəsaɪt/ : 淋巴细胞	thyrocyte /ˈθaɪrəsaɪt/ : 甲状腺细胞
pathogenesis /ˌpæθəˈdʒenɪsɪs/ : 发病机制	stromal cell /ˈstroʊməl sɛl/ : 基质细胞
endothelial cell /ˌɛndəʊˈθiːliəl sɛl/ : 内皮细胞	fibroblast /ˈfaɪbrəʊblæst/ : 成纤维细胞
thyroid gland /ˈθaɪrɔɪd glænd/ : 甲状腺	macrophage /ˈmækɹəfeɪdʒ/ : 巨噬细胞
dendritic cell /denˈdɹɪtɪk sɛl/ : 树突状细胞	tertiary lymphoid organ /ˈtɜːrʃiˌɛri ˈlɪmfɔɪd ˈɔːrgən/ : 三级淋巴器官

Hashimoto's thyroiditis: An update on pathogenic mechanisms, diagnostic protocols, therapeutic strategies, and potential malignant transformation

桥本甲状腺炎：发病机制、诊断方案、治疗策略及潜在恶性转化的最新进展

期刊: Autoimmunity Reviews 日期: 2020

JCR Q1

中科院升级版 医学1区

IF 8.3

Abstract

Hashimoto's thyroiditis, characterized by thyroid-specific autoantibodies, is one of the commonest autoimmune disorders. Although the exact etiology has not been fully elucidated, Hashimoto's thyroiditis is related to an interaction among genetic elements, environmental factors and epigenetic influences. Cellular and humoral immunity play a key role in the development of the disease; thus, a T and B cells inflammatory infiltration is frequently found. Histopathologic features of the disease include lymphoplasmacytic infiltration, lymphoid follicle formation with germinal centers, and parenchymal atrophy. Moreover, the occurrence of large follicular cells and oxyphilic or Askanazy cells is frequently associated to Hashimoto's thyroiditis. Clinically, Hashimoto's thyroiditis is characterized mainly by systemic manifestations due to the damage of the thyroid gland, developing a primary hypothyroidism. Diagnosis of Hashimoto's thyroiditis is clinical and based on clinical characteristics, positivity to serum antibodies against thyroid antigens (thyroid peroxidase and thyroglobulin), and lymphocytic infiltration on cytological examination. The mainstream of treatment is based on the management of the hypothyroidism with a substitution therapy. A relationship between Hashimoto's thyroiditis and a possible malignant transformation has been proposed in several studies and involves immunological/hormonal pathogenic links although specific correlation is still debated and needs to be further investigated with prospective studies.

中文翻译

桥本甲状腺炎，以甲状腺特异性自身抗体为特征，是最常见的自身免疫性疾病之一。尽管其确切病因尚未完全阐明，但桥本甲状腺炎与遗传因素、环境因素及表观遗传影响之间的相互作用有关。细胞免疫和体液免疫在该疾病的发展中起着关键作用；因此，常可发现T细胞和B细胞的炎性浸润。该疾病的组织病理学特征包括淋巴浆细胞浸润、伴有生发中心的淋巴滤泡形成以及实质萎缩。此外，大型滤泡细胞和嗜酸性或Askanazy细胞的出现常与桥本甲状腺炎相关。临床上，桥本甲状腺炎主要表现为由于甲状腺损伤导致的全身性症状，发展成原发性甲状腺功能减退症。桥本甲状腺炎的诊断基于临床特征、血清抗甲状腺抗原（甲状腺过氧化物酶和甲状腺球蛋白）抗体阳性以及细胞学检查中的淋巴细胞浸润。治疗的主流是通过替代疗法管理甲状腺功能减退症。多项研究提出了桥本甲状腺炎与可能的恶性转化之间的关系，涉及免疫/激素致病联系，尽管具体的相关性仍有争议，需要通过前瞻性研究进一步探讨。

核心词汇

Hashimoto's thyroiditis /həˈʃiː moʊtəʊz ˌθaɪrɔɪˈdaɪtɪs/ : 桥本甲状腺炎

autoantibodies /ˌɔːtəʊˈæntɪˌbɒdiz/ : 自身抗体

thyroid gland /ˈθaɪrɔɪd ɡlænd/ : 甲状腺

parenchymal atrophy /pəˈrɛŋkɪməl ˈætəfɪ/ : 实质萎缩

cytological examination /ˌsaɪtəˈlɒdʒɪkəl ɪɡˌzæmɪˈneɪʃən/ : 细胞学检查

malignant transformation /məˈlɪɡnənt ˌtrænsfərˈmeɪʃən/ : 恶性转化

thyroglobulin /ˌθaɪrəʊˈɡlɒbjʊlɪn/ : 甲状腺球蛋白

autoimmune disorders /ˌɔːtəʊˈmjuːn dɪsˈɔːrðəz/ : 自身免疫性疾病

hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症

lymphoplasmacytic infiltration /ˌlɪmfəʊplæzəˈsɪtɪk ˌɪnfɪˈtreɪʃən/ : 淋巴浆细胞浸润

lymphoid follicle /ˌlɪmfɔɪd ˈfɒlɪkəl/ : 淋巴滤泡

substitution therapy /ˌsʌbstɪˈtjuːʃən ˈθerəpi/ : 替代治疗

thyroid peroxidase /ˈθaɪrɔɪd pəˈrɒksɪdeɪz/ : 甲状腺过氧化物酶

Hashimoto thyroiditis: Clinical and diagnostic criteria

桥本甲状腺炎：临床与诊断标准

期刊: Autoimmunity Reviews 日期: 2014 JCR Q1 中科院升级版 医学1区 IF 8.3

Abstract

Hashimoto thyroiditis (HT), now considered the most common **autoimmune disease**, was described over a century ago as a pronounced lymphoid **goiter** affecting predominantly women. In addition to this classic form, several other clinico-pathologic entities are now included under the term HT: fibrous variant, IgG4-related variant, juvenile form, Hashitoxicosis, and painless thyroiditis (sporadic or post-partum). All forms are characterized pathologically by the infiltration of hematopoietic mononuclear cells, mainly lymphocytes, in the interstitium among the thyroid follicles, although specific features can be recognized in each variant. Thyroid cells undergo atrophy or transform into a bolder type of follicular cell rich in mitochondria called **Hürthle cell**. Most HT forms ultimately evolve into **hypothyroidism**, although at presentation patients can be **euthyroid** or even hyperthyroid. The diagnosis of HT relies on the demonstration of circulating antibodies to thyroid antigens (mainly **thyroperoxidase** and **thyroglobulin**) and reduced **echogenicity** on thyroid **sonogram** in a patient with proper clinical features. The treatment remains symptomatic and based on the administration of synthetic thyroid hormones to correct the **hypothyroidism** as needed. Surgery is performed when the **goiter** is large enough to cause significant compression of the surrounding cervical structures, or when some areas of the thyroid gland mimic the features of a nodule whose **cytology** cannot be ascertained as benign. HT remains a complex and ever expanding disease of unknown pathogenesis that awaits prevention or novel forms of treatment.

中文翻译

桥本甲状腺炎 (HT) 现被视为最常见的自身免疫性疾病，一个多世纪前被描述为一种主要影响女性的显著淋巴性甲状腺肿。除了这种经典形式外，目前HT还包括其他几种临床病理类型：纤维化变异型、IgG4相关变异型、青少年型、桥本毒症以及无痛性甲状腺炎（散发型或产后型）。所有类型在病理学上的共同特征是甲状腺滤泡间质中造血单核细胞（主要是淋巴细胞）的浸润，尽管每种变异型都有其特定的病理特征。甲状腺细胞会发生萎缩，或转化为一种富含线粒体、更为显著的滤泡细胞，称为许特莱细胞。大多数HT最终会发展为甲状腺功能减退，尽管患者在就诊时可能甲状腺功能正常甚至亢进。HT的诊断依赖于在具有相应临床特征的患者中，证实存在循环甲状腺抗原抗体（主要是甲状腺过氧化物酶和甲状腺球蛋白），以及甲状腺超声显示回声减低。治疗仍以对症为主，根据需要给予合成甲状腺激素以纠正甲状腺功能减退。当甲状腺肿大到足以对周围颈部结构造成显著压迫，或甲状腺某些区域呈现结节特征且细胞学无法确定为良性时，则需进行手术。HT仍然是一种复杂且不断扩展的疾病，其发病机制尚不明确，有待预防或新型治疗方法的出现。

核心词汇

Hashimoto thyroiditis /həˈʃiːˈmoʊtaɪˌθaɪrɔɪˈdaɪtɪs/ : 桥本甲状腺炎

goiter /ˈɡoɪtər/ : 甲状腺肿

thyroid follicle /ˈθaɪrɔɪd ˈfɒlɪkəl/ : 甲状腺滤泡

hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症

euthyroid /juːˈθaɪrɔɪd/ : 甲状腺功能正常

thyroglobulin /ˌθaɪrəʊˈɡlɒbjʊlɪn/ : 甲状腺球蛋白

sonogram /ˈsɒnəɡræm/ : 声像图

autoimmune disease /ˌɔːtəʊˈɪmjʊn dɪˈziːz/ : 自身免疫性疾病

lymphocyte /ˈlɪmfəsaɪt/ : 淋巴细胞

Hürthle cell /ˈhɜːrtl̩ seɪl/ : 许特莱细胞

hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症

thyroperoxidase /ˌθaɪrəʊpərˈɒksɪdeɪs/ : 甲状腺过氧化物酶

echogenicity /ˌɛkəʊdʒəˈnɪsɪti/ : 回声性

cytology /saɪˈtɒlədʒi/ : 细胞学

Hyperthyroidism

甲状腺功能亢进症

期刊: The Lancet 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

thyrotoxicosis causes a variety of symptoms and adverse health outcomes. **hyperthyroidism** refers to increased thyroid hormone synthesis and secretion, most commonly from Graves' disease or toxic nodular **goitre**, whereas **thyroiditis** (typically autoimmune, viral, or drug induced) causes **thyrotoxicosis** without **hyperthyroidism**. The diagnosis is based on suppressed serum concentrations of thyroid-stimulating hormone (TSH), accompanied by free **thyroxine** and total or free triiodothyronine concentrations, which are raised (overt **hyperthyroidism**) or within range (subclinical **hyperthyroidism**). The underlying cause is determined by clinical assessment, detection of TSH-receptor antibodies and, if necessary, radionuclide thyroid **scintigraphy**. Treatment options for **hyperthyroidism** include **antithyroid drugs**, **radioactive iodine**, and **thyroidectomy**, whereas **thyroiditis** is managed symptomatically or with **glucocorticoid** therapy. In Graves' disease, first-line treatment is a 12-18-month course of **antithyroid drugs**, whereas for **goitre**, **radioactive iodine** or surgery are preferred for toxic nodules or goitres. Evidence also supports long-term treatment with **antithyroid drugs** as an option for patients with Graves' disease and toxic nodular **goitre**.

中文翻译

甲状腺毒症引发多种症状及不良健康后果。甲状腺功能亢进症指甲状腺激素合成与分泌增加，最常见于格雷夫斯病或毒性结节性甲状腺肿；而甲状腺炎（通常为自身免疫性、病毒性或药物诱发）则导致甲状腺毒症但不伴甲亢。诊断依据为血清促甲状腺激素浓度受抑制，同时游离甲状腺素及总或游离三碘甲状腺原氨酸浓度升高（显性甲亢）或处于正常范围（亚临床甲亢）。病因需通过临床评估、促甲状腺激素受体抗体检测及必要时放射性核素甲状腺显像来确定。甲亢治疗方案包括抗甲状腺药物、放射性碘和甲状腺切除术，而甲状腺炎则采用对症治疗或糖皮质激素疗法。格雷夫斯病的一线治疗为12-18个月疗程的抗甲状腺药物，毒性结节或甲状腺肿则首选放射性碘或手术治疗。证据亦支持对格雷夫斯病和毒性结节性甲状腺肿患者采用长期抗甲状腺药物治疗作为可选方案。

核心词汇

thyrotoxicosis /ˌθaɪrouˈtɒksɪˈkəʊsɪs/ : 甲状腺毒症	hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症
thyroiditis /ˌθaɪrɔɪˈdaɪtɪs/ : 甲状腺炎	goitre /ˈɡɔɪtər/ : 甲状腺肿
thyroidectomy /ˌθaɪrɔɪˈdektəmi/ : 甲状腺切除术	thyroxine /θaɪˈrɒksɪn/ : 甲状腺素
scintigraphy /sɪnˈtɪɡrəfi/ : 闪烁扫描	antithyroid drugs /ˌæntiˈθaɪrɔɪd drʌɡz/ : 抗甲状腺药物
radioactive iodine /ˌreɪdɪəʊˈæktɪv ˈaɪədaɪn/ : 放射性碘	glucocorticoid /ˌɡluːkɒsˈkɔːrtɪkɔɪd/ : 糖皮质激素
nodule /ˈnɒdjuːl/ : 结节	

Hyperthyroidism

甲状腺功能亢进症

期刊: The Lancet 日期: 2016 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

hyperthyroidism is characterised by increased thyroid hormone synthesis and secretion from the **thyroid gland**, whereas **thyrotoxicosis** refers to the clinical syndrome of excess circulating thyroid hormones, irrespective of the source. The most common cause of **hyperthyroidism** is **Graves' disease**, followed by **toxic nodular goitre**. Other important causes of **thyrotoxicosis** include **thyroiditis**, iodine-induced and drug-induced thyroid dysfunction, and factitious ingestion of excess thyroid hormones. Treatment options for **Graves' disease** include **antithyroid drugs**, **radioactive iodine therapy**, and **surgery**, whereas **antithyroid drugs** are not generally used long term in **toxic nodular goitre**, because of the high relapse rate of **thyrotoxicosis** after discontinuation. β blockers are used in symptomatic **thyrotoxicosis**, and might be the only treatment needed for **thyrotoxicosis** not caused by excessive production and release of the thyroid hormones. **thyroid storm** and **hyperthyroidism** in **pregnancy** and during the post-partum period are special circumstances that need careful assessment and treatment.

中文翻译

甲状腺功能亢进症的特征是甲状腺激素合成和分泌增加，而甲状腺毒症则指循环中甲状腺激素过多的临床综合征，不论其来源如何。甲状腺功能亢进症最常见的原因是格雷夫斯病，其次是毒性结节性甲状腺肿。甲状腺毒症的其他重要原因包括甲状腺炎、碘诱导和药物诱导的甲状腺功能障碍，以及人为过量摄入甲状腺激素。格雷夫斯病的治疗选择包括抗甲状腺药物、放射性碘治疗和手术，而抗甲状腺药物通常不长期用于毒性结节性甲状腺肿，因为停药后甲状腺毒症的复发率很高。 β 受体阻滞剂用于有症状的甲状腺毒症，并且可能是非甲状腺激素过量产生和释放引起的甲状腺毒症的唯一治疗方法。甲状腺危象以及孕期和产后的甲状腺功能亢进症是需要仔细评估和治疗的特殊情况。

核心词汇

hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症**thyroid gland** /ˈθaɪrɔɪd glænd/ : 甲状腺**toxic nodular goitre** /ˈtɒksɪk ˈnɒdjʊlər ˈɡɔɪtə/ : 毒性结节性甲状腺肿**antithyroid drugs** /ˌæntiˈθaɪrɔɪd drʌgz/ : 抗甲状腺药物**surgery** /ˈsɜːrdʒəri/ : 手术**pregnancy** /ˈpreɡnənsi/ : 妊娠**thyrotoxicosis** /ˌθaɪrɔʊˌtɒksɪˈkəʊsɪs/ : 甲状腺毒症**Graves' disease** /ɡreɪvz dɪˈziːz/ : 格雷夫斯病**thyroiditis** /ˌθaɪrɔɪˈdaɪtɪs/ : 甲状腺炎**radioactive iodine therapy** /ˌreɪdɪəʊˈæktɪv ˈaɪədəɪn ˈθerəpi/ : 放射性碘治疗**thyroid storm** /ˈθaɪrɔɪd stɔːrm/ : 甲状腺危象

Hyperthyroidism: A Review

甲状腺功能亢进症：综述

期刊: JAMA 日期: 2023

Abstract

hyperthyroidism affects 2.5% of adults worldwide and is associated with osteoporosis, heart disease, and increased mortality. First-line treatments are antithyroid drugs, thyroid surgery, and radioactive iodine treatment. treatment choices should be individualized and patient centered.

中文翻译

甲状腺功能亢进症影响着全球2.5%的成年人，并与骨质疏松症、心脏病和死亡率增加相关。一线治疗方法包括抗甲状腺药物、甲状腺手术和放射性碘治疗。治疗方案的选择应个体化，并以患者为中心。

核心词汇

hyperthyroidism /ˌhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症	osteoporosis /ˌɒstiəʊpəˈreʊsɪs/ : 骨质疏松症
heart disease /hɑːrt dɪˈziːz/ : 心脏病	mortality /mɔːrˈtæləti/ : 死亡率
treatment /ˈtriːtmənt/ : 治疗	antithyroid drugs /ˌæntiˈθaɪrɔɪd drʌgz/ : 抗甲状腺药物
thyroid surgery /ˈθaɪrɔɪd ˈsɜːrdʒəri/ : 甲状腺手术	radioactive iodine treatment /ˌreɪdiəʊˈæktɪv ˈaɪədəɪn ˈtriːtmənt/ : 放射性碘治疗
individualized /ˌɪndɪˈvɪdʒʊəlaɪzd/ : 个体化	patient centered /ˈpeɪʃənt ˈsentəd/ : 以患者为中心

Hyperthyroidism: Aetiology, pathogenesis, diagnosis, management, complications, and prognosis

甲状腺功能亢进症：病因、发病机制、诊断、治疗、并发症及预后

期刊: The Lancet Diabetes & Endocrinology 日期: 2023

Abstract

hyperthyroidism is a common condition with a global **prevalence** of 0.2-1.3%. When **clinical suspicion** of **hyperthyroidism** arises, it should be confirmed by **biochemical tests** (eg, low TSH, high free thyroxine [FT(4)], or high free tri-iodothyronine [FT(3)]). If **hyperthyroidism** is confirmed by **biochemical tests**, a nosological **diagnosis** should be done to find out which disease is causing the **hyperthyroidism**. Helpful tools are TSH-receptor antibodies, **thyroid** peroxidase antibodies, **thyroid** ultrasonography, and scintigraphy. **hyperthyroidism** is mostly caused by Graves' **hyperthyroidism** (70%) or toxic nodular goitre (16%). **hyperthyroidism** can also be caused by subacute granulomatous thyroiditis (3%) and drugs (9%) such as amiodarone, tyrosine kinase inhibitors, and immune checkpoint inhibitors. Disease-specific recommendations are given. Currently, Graves' **hyperthyroidism** is preferably treated with antithyroid drugs. However, recurrence of **hyperthyroidism** after a 12-18 month course of antithyroid drugs occurs in approximately 50% of patients. Being younger than 40 years, having FT(4) concentrations that are 40 pmol/L or higher, having TSH-binding inhibitory immunoglobulins that are higher than 6 U/L, and having a goitre size that is equivalent to or larger than WHO grade 2 before the start of treatment with antithyroid drugs increase risk of recurrence. Long-term treatment with antithyroid drugs (ie, 5-10 years of treatment) is feasible and associated with fewer recurrences (15%) than short-term treatment (ie, 12-18 months of treatment). Toxic nodular goitre is mostly treated with radioiodine (¹³¹I) or thyroidectomy and is rarely treated with radiofrequency ablation. Destructive thyrotoxicosis is usually mild and transient, requiring steroids only in severe cases. Specific attention is given to patients with **hyperthyroidism** who are pregnant, have COVID-19, or have other complications (eg, atrial fibrillation, thyrotoxic periodic paralysis, and **thyroid** storm). **hyperthyroidism** is associated with increased mortality. Prognosis might be improved by rapid and sustained control of **hyperthyroidism**. Innovative new treatments are expected for Graves' disease, by targeting B cells or TSH receptors.

中文翻译

甲状腺功能亢进症是一种常见疾病，全球患病率为0.2-1.3%。当临床怀疑甲亢时，应通过生化检测（如低TSH、高游离甲状腺素[FT(4)]或高游离三碘甲状腺原氨酸[FT(3)]）确诊。若生化检测确诊甲亢，需进行病因学诊断以明确导致甲亢的疾病。辅助检查包括TSH受体抗体、甲状腺过氧化物酶抗体、甲状腺超声及核素显像。甲亢主要由格雷夫斯病（70%）或毒性结节性甲状腺肿（16%）引起，也可由亚急性肉芽肿性甲状腺炎（3%）及药物（9%，如胺碘酮、酪氨酸激酶抑制剂和免疫检查点抑制剂）导致。本文针对不同病因提出特异性治疗建议。目前格雷夫斯病甲亢首选抗甲状腺药物治疗，但经12-18个月疗程后约50%患者出现复发。年龄<40岁、FT(4)浓度≥40 pmol/L、TSH结合抑制性免疫球蛋白>6 U/L、治疗前甲状腺肿大达WHO 2级及以上者复发风险增高。长期抗甲状腺药物治疗（5-10年）可行且复发率（15%）低于短期治疗（12-18个月）。毒性结节性甲状腺肿多采用放射性碘（¹³¹I）或甲状腺切除术治疗，射频消融应用较少。破坏性甲状腺毒症通常轻微且短暂，仅重症需使用糖皮质激素。需特别关注妊娠合并甲亢、COVID-19感染或伴发其他并发症（如房颤、甲亢性周期性麻痹及甲状腺危象）的患者。甲亢与死亡率升高相关，快速持续控制甲亢可能改善预后。针对B细胞或TSH受体的创新疗法有望用于格雷夫斯病治疗。

核心词汇

hyperthyroidism /ˈhaɪpərˈθaɪrɔɪdɪzəm/ : 甲状腺功能亢进症	prevalence /ˈpreɪləns/ : 患病率
clinical suspicion /ˈklɪnɪkəl səˈspɪʃən/ : 临床怀疑	biochemical tests /ˌbaɪəʊˈkɛmɪkəl ˈtɛsts/ : 生化检测
thyroid /ˈθaɪrɔɪd/ : 甲状腺	hormone /ˈhɔːrmoun/ : 激素
diagnosis /ˌdaɪəɡˈnəʊsɪs/ : 诊断	endocrine /ˈendəʊkrɪn/ : 内分泌
metabolism /məˈtæbəlaɪzəm/ : 新陈代谢	symptoms /ˈsɪmptəmz/ : 症状
laboratory /ˈlæbrəˌtɔːri/ : 实验室	screening /ˈskriːnɪŋ/ : 筛查

Hypothyroidism

甲状腺功能减退症

期刊: The Lancet 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

hypothyroidism, the deficiency of **thyroid hormone**, is a common condition worldwide. It affects almost all body systems and has a wide variety of clinical presentations from being asymptomatic to, in rare cases, life threatening. The classic symptoms of **hypothyroidism** include **fatigue**, **lethargy**, weight gain, and cold intolerance; however, these symptoms are non-specific and the diagnosis is typically made on biochemical grounds through serum **thyroid function tests**. The most common cause of **hypothyroidism** is chronic **autoimmune thyroiditis** (**Hashimoto's thyroiditis**), although other causes, including drugs (such as amiodarone, lithium, and immune checkpoint inhibitors), radioactive-iodine treatment, and **thyroid surgery**, are frequent. Historically, severe **iodine deficiency** was the most common cause. Reference ranges for **thyroid function tests** are based on fixed percentiles of the population distribution, but there is increasing awareness of the need for more individualised reference intervals based on key factors such as age, sex, and special circumstances such as pregnancy. **levothyroxine monotherapy** is the standard treatment for **hypothyroidism**; it is safe and inexpensive, restores **thyroid function tests** to within the reference range, and improves symptoms in the majority of patients. However, 10% of patients have persistent symptoms of ill health despite normalisation of **thyroid function tests** biochemically and a substantial proportion of patients on **levothyroxine** have thyroid-stimulating hormone concentrations outside the reference range. Ongoing symptoms despite **levothyroxine** treatment has led to some patients using liothyronine or desiccated thyroid extract. Taken together, these factors have led to intense debate around the treatment thresholds and treatment strategies for **hypothyroidism**. In this Seminar, we review the **epidemiology**, **genetic determinants**, causes, and presentation of **hypothyroidism**; highlight key considerations and controversies in its diagnosis and management; and provide future directions for research.

中文翻译

甲状腺功能减退症，即甲状腺激素缺乏，是全球范围内一种常见疾病。它几乎影响全身所有系统，临床表现多样，从无症状到罕见情况下危及生命不等。甲减的典型症状包括疲劳、嗜睡、体重增加和畏寒；然而这些症状并不具特异性，诊断通常需通过血清甲状腺功能检测进行生化确认。甲减最常见的原因是慢性自身免疫性甲状腺炎（桥本甲状腺炎），但其他原因如药物（如胺碘酮、锂剂和免疫检查点抑制剂）、放射性碘治疗及甲状腺手术也较为常见。历史上，严重碘缺乏曾是主要病因。甲状腺功能检测的参考范围基于人群分布的固定百分位数，但人们日益认识到需要根据年龄、性别及妊娠等特殊状况建立更个体化的参考区间。左甲状腺素单药治疗是甲减的标准疗法，该方案安全经济，能使多数患者的甲状腺功能检测指标恢复至参考范围内并改善症状。然而，10%的患者在生化指标正常化后仍存在持续的健康症状，且相当比例接受左甲状腺素治疗的患者促甲状腺激素浓度超出参考范围。部分患者因左甲状腺素治疗后症状持续存在而转向使用碘塞罗宁或干燥甲状腺提取物。这些因素共同引发了关于甲减治疗阈值与治疗策略的激烈讨论。本研讨会将综述甲减的流行病学特征、遗传决定因素、病因及临床表现，重点探讨诊断与治疗中的关键考量与争议，并为未来研究方向提供指引。

核心词汇

hypothyroidism /ˌhaɪpəʊˈθaɪrɔɪdɪzəm/ : 甲状腺功能减退症	thyroid hormone /ˈθaɪrɔɪd ˈhɔːrmoun/ : 甲状腺激素
autoimmune thyroiditis /ˌɔːtəʊˈɪmjʊːn ˌθaɪrɔɪˈdaɪtɪs/ : 自身免疫性甲状腺炎	Hashimoto's thyroiditis /həˈʃɪˈmoutəʊz ˌθaɪrɔɪˈdaɪtɪs/ : 桥本甲状腺炎
fatigue /feˈtiːɡ/ : 疲劳	lethargy /ˈleθərdʒi/ : 嗜睡
thyroid function tests /ˈθaɪrɔɪd ˈfʌŋkʃən tests/ : 甲状腺功能检查	levothyroxine /ˌliːvəʊθaɪˈrɒksɪn/ : 左甲状腺素
monotherapy /ˌmɒnəʊˈθerəpi/ : 单药治疗	thyroid surgery /ˈθaɪrɔɪd ˈsɜːrdʒəri/ : 甲状腺手术
iodine deficiency /ˈaɪədaɪn dɪˈfɪʃənsi/ : 碘缺乏	epidemiology /ˌepɪˌdɪːmiˈɒlədʒi/ : 流行病学
genetic determinants /dʒəˈnetɪk dɪˈtɜːrmɪnənts/ : 遗传决定因素	

Thyroid Cancer

甲状腺癌

期刊: JAMA 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 55.0

Abstract

Importance Approximately 43 720 new cases of **thyroid carcinoma** are expected to be diagnosed in 2023 in the US. Five-year relative survival is approximately 98.5%. This review summarizes current evidence regarding pathophysiology, diagnosis, and management of early-stage and advanced thyroid cancer. Observations Papillary thyroid cancer accounts for approximately 84% of all thyroid cancers. Papillary, follicular (≈4%), and oncocytic (≈2%) forms arise from thyroid follicular cells and are termed well-differentiated thyroid cancer. Aggressive forms of follicular cell-derived thyroid cancer are poorly differentiated thyroid cancer (≈5%) and **anaplastic thyroid cancer** (≈1%). **medullary thyroid cancer** (≈4%) arises from parafollicular C cells. Most cases of well-differentiated thyroid cancer are asymptomatic and detected during physical examination or incidentally found on **diagnostic imaging** studies. For microcarcinomas (≤1 cm), observation without **surgical resection** can be considered. For tumors larger than 1 cm with or without **lymph node metastases**, surgery with or without **radioactive iodine** is curative in most cases. **surgical resection** is the preferred approach for patients with recurrent locoregional disease. For **metastatic disease**, **surgical resection** or **stereotactic body irradiation** is favored over systemic therapy (eg, lenvatinib, dabrafenib). Antiangiogenic multikinase inhibitors (eg, sorafenib, lenvatinib, cabozantinib) are approved for thyroid cancer that does not respond to **radioactive iodine**, with response rates 12% to 65%. Targeted therapies such as dabrafenib and selipercatinib are directed to genetic mutations (BRAF, RET, NTRK, MEK) that give rise to thyroid cancer and are used in patients with advanced **thyroid carcinoma**. Conclusions Approximately 44 000 new cases of thyroid cancer are diagnosed each year in the US, with a 5-year relative survival of 98.5%. Surgery is curative in most cases of well-differentiated thyroid cancer. **radioactive iodine** treatment after surgery improves overall survival in patients at high risk of recurrence. Antiangiogenic multikinase inhibitors and targeted therapies to genetic mutations that give rise to thyroid cancer are increasingly used in the treatment of **metastatic disease**.

中文翻译

重要性：预计2023年美国将诊断出约43,720例新发甲状腺癌病例。五年相对生存率约为98.5%。本综述总结了当前关于早期和晚期甲状腺癌的病理生理学、诊断和管理的证据。观察结果：乳头状甲状腺癌约占所有甲状腺癌的84%。乳头状、滤泡状（约4%）和嗜酸细胞性（约2%）甲状腺癌起源于甲状腺滤泡细胞，被称为分化型甲状腺癌。滤泡细胞来源的甲状腺癌的侵袭性形式包括低分化甲状腺癌（约5%）和未分化甲状腺癌（约1%）。髓样甲状腺癌（约4%）起源于滤泡旁C细胞。大多数分化型甲状腺癌病例无症状，在体检中发现或在诊断性影像学检查中偶然发现。对于微小癌（≤1厘米），可以考虑不进行手术切除的观察。对于大于1厘米的肿瘤，无论是否伴有淋巴结转移，手术联合或不联合放射性碘治疗在大多数情况下可治愈。对于局部复发性疾病，手术切除是首选方法。对于转移性疾病，手术切除或立体定向体部放疗优于全身治疗（如乐伐替尼、达拉非尼）。抗血管生成多激酶抑制剂（如索拉非尼、乐伐替尼、卡博替尼）已被批准用于对放射性碘无反应的甲状腺癌，反应率为12%至65%。达拉非尼和塞尔帕替尼等靶向疗法针对导致甲状腺癌的基因突变（BRAF、RET、NTRK、MEK），用于晚期甲状腺癌患者。结论：美国每年诊断出约44,000例新发甲状腺癌病例，五年相对生存率为98.5%。在大多数分化型甲状腺癌病例中，手术可治愈。术后放射性碘治疗可提高复发高风险患者的总生存率。抗血管生成多激酶抑制剂和针对导致甲状腺癌的基因突变的靶向疗法越来越多地用于转移性疾病的治疗。

核心词汇

thyroid carcinoma /ˈθaɪrɔɪd ˌkɑːrsɪˈnoʊmə/：甲状腺癌	papillary thyroid cancer /ˈpæpɪˌlɪəri ˈθaɪrɔɪd ˈkænsər/：乳头状甲状腺癌
follicular thyroid cancer /fəˈlɪkjʊlər ˈθaɪrɔɪd ˈkænsər/：滤泡状甲状腺癌	anaplastic thyroid cancer /ænəˈplæstɪk ˈθaɪrɔɪd ˈkænsər/：未分化甲状腺癌
medullary thyroid cancer /ˈmedəˌlɪəri ˈθaɪrɔɪd ˈkænsər/：髓样甲状腺癌	lymph node metastases /lɪmf nɒd məˈtæstəˌsɪz/：淋巴结转移
metastatic disease /ˌmɛtəˈstætɪk dɪˈzɪz/：转移性疾病	surgical resection /ˈsɜːrdʒɪkəl rɪˈsekʃən/：手术切除
radioactive iodine /ˌreɪdɪʊˈæktɪv ˈaɪəˌdaɪn/：放射性碘	stereotactic body irradiation /ˌsterɪʊˈtæktɪk ˈbɒdi ɪˌreɪdɪˈeɪʃən/：立体定向体部放疗
antiangiogenic therapy /ˌæntiˌændʒɪʊˈdʒɛnɪk ˈθerəpi/：抗血管生成治疗	targeted therapy /ˈtɑːrgɪtɪd ˈθerəpi/：靶向治疗

biopsy /'baɪɒpsi/ : 活检

diagnostic imaging /,daɪəɡ'nɒstɪk 'ɪmɪdʒɪŋ/ : 诊断影像学

云梦山-侵权必究

Single-cell RNA sequencing depicts the local cell landscape in thyroid-associated ophthalmopathy

单细胞RNA测序描绘甲状腺相关眼病的局部细胞景观

期刊: Cell Reports Medicine 日期: 2022 JCR Q1 中科院升级版 医学1区 IF 10.6

Abstract

There is a specific reactivity and characteristic remodeling of the periocular tissue in thyroid-associated ophthalmopathy (TAO). However, local cell changes responsible for these pathological processes have not been sufficiently identified. Here, single-cell RNA sequencing is performed to characterize the transcriptional changes of cellular components in the orbital connective tissue in individuals with TAO. Our study shows that lipofibroblasts with RASD1 expression are highly involved in inflammation and adipogenesis during TAO. ACKR1(+) endothelial cells and adipose tissue macrophages may engage in TAO pathogenesis. We find CD8(+)CD57(+) cytotoxic T lymphocytes with the terminal differentiation phenotype to be another source of interferon-gamma, a molecule actively engaging in TAO pathogenesis. Cell-cell communication analysis reveals increased activity of CXCL8/ACKR1 and TNFSF4/TNFRSF4 interactions in TAO. This study provides a comprehensive local cell landscape of TAO and may be valuable for future therapy investigation.

中文翻译

甲状腺相关眼病（TAO）中眼周组织存在特异性反应和特征性重塑，但导致这些病理过程的局部细胞变化尚未被充分阐明。本研究通过单细胞RNA测序技术，描绘了TAO患者眼眶结缔组织中细胞成分的转录变化。结果显示，表达RASD1的脂肪成纤维细胞在TAO炎症反应和脂肪生成过程中高度活跃；ACKR1(+)内皮细胞和脂肪组织巨噬细胞可能参与TAO发病机制。我们发现具有终末分化表型的CD8(+)CD57(+)细胞毒性T淋巴细胞是干扰素-γ的另一来源，该分子在TAO发病中发挥关键作用。细胞间通讯分析显示，TAO中CXCL8/ACKR1和TNFSF4/TNFRSF4的相互作用增强。本研究提供了TAO的全面局部细胞图谱，对未来的治疗研究具有重要价值。

核心词汇

periocular tissue /ˌpɛrɪˈɒkjʊlər ˈtɪʃuː/ : 眼周组织	remodeling /ˈrɪːˈmɒdəlɪŋ/ : 重塑
pathological processes /ˌpæθəˈlɒdʒɪkəl ˈprəʊsesɪz/ : 病理过程	transcriptional changes /ˈtræŋˈskrɪpʃənəl ˈtʃeɪndʒɪz/ : 转录变化
cellular components /ˈseljʊlər kəmˈpəʊnənts/ : 细胞成分	orbital connective tissue /ˈɔːrbɪtəl kəˈnektɪv ˈtɪʃuː/ : 眼眶结缔组织
lipofibroblasts /ˌlɪpɒʊ ˈfaɪbrəʊblæsts/ : 脂成纤维细胞	inflammation /ˌɪnfləˈmeɪʃən/ : 炎症
adipogenesis /ˌædɪpəʊˈdʒenɪsɪs/ : 脂肪生成	sequencing /ˈsiːkwənsɪŋ/ : 测序

An autoimmune stem-like CD8 T cell population drives type 1 diabetes

自身免疫性干细胞样CD8 T细胞群驱动1型糖尿病

期刊: Nature 日期: 2022 JCR Q1 中科院升级版 综合性期刊1区 IF 48.5

Abstract

CD8 T cell-mediated autoimmune diseases result from the **breakdown** of self-**tolerance** mechanisms in **autoreactive** CD8 T cells(1). How autoimmune T cell populations arise and are sustained, and the molecular programmes defining the autoimmune T cell state, are unknown. In type 1 diabetes, beta-cell-specific CD8 T cells destroy insulin-producing beta-cells. Here we followed the fate of beta-cell-specific CD8 T cells in non-obese diabetic mice throughout the course of type 1 diabetes. We identified a stem-like autoimmune progenitor population in the pancreatic draining lymph node (pLN), which self-renews and gives rise to pLN autoimmune mediators. pLN autoimmune mediators migrate to the pancreas, where they differentiate further and destroy beta-cells. Whereas transplantation of as few as 20 autoimmune progenitors induced type 1 diabetes, as many as 100,000 pancreatic autoimmune mediators did not. Pancreatic autoimmune mediators are short-lived, and stem-like autoimmune progenitors must continuously seed the pancreas to sustain beta-cell destruction. Single-cell RNA sequencing and clonal analysis revealed that autoimmune CD8 T cells represent unique T cell differentiation states and identified features driving the transition from autoimmune progenitor to autoimmune mediator. Strategies aimed at targeting the stem-like autoimmune progenitor pool could emerge as novel and powerful immunotherapeutic interventions for type 1 diabetes.

中文翻译

CD8 T细胞介导的自身免疫性疾病源于自身反应性CD8 T细胞中自我耐受机制的破坏(1)。自身免疫性T细胞群体如何产生和维持，以及定义自身免疫性T细胞状态的分子程序尚不清楚。在1型糖尿病中，β细胞特异性CD8 T细胞会破坏产生胰岛素的β细胞。本研究追踪了非肥胖糖尿病小鼠中β细胞特异性CD8 T细胞在整个1型糖尿病病程中的命运。我们在胰腺引流淋巴结（pLN）中发现了一个类似干细胞的自身免疫祖细胞群体，该群体能够自我更新并产生pLN自身免疫介质。pLN自身免疫介质迁移至胰腺，在此进一步分化并破坏β细胞。移植仅20个自身免疫祖细胞即可诱导1型糖尿病，而移植多达10万个胰腺自身免疫介质却无法实现。胰腺自身免疫介质寿命短暂，必须由类似干细胞的自身免疫祖细胞持续向胰腺输送，才能维持β细胞破坏。单细胞RNA测序和克隆分析揭示，自身免疫性CD8 T细胞代表独特的T细胞分化状态，并识别出驱动自身免疫祖细胞向自身免疫介质转化的特征。针对类似干细胞的自身免疫祖细胞库的治疗策略，可能成为1型糖尿病新型且强效的免疫干预手段。

核心词汇

autoimmune disease /ˌɔːtəʊɪ'mjuːn dɪ'ziːz/ : 自身免疫性疾病	autoreactive /ˌɔːtəʊɪ'æktɪv/ : 自身反应性
breakdown /'breɪkdaʊn/ : 破坏/崩溃	mechanism /'mekənɪzəm/ : 机制
tolerance /'tɒlərəns/ : 耐受	autoimmunity /ˌɔːtəʊɪ'mjuːnəti/ : 自身免疫
lymphocyte /'lɪmfəsaɪt/ : 淋巴细胞	cytotoxic /ˌsaɪtə'tɒksɪk/ : 细胞毒性
immune system /ɪ'mjuːn 'sɪstəm/ : 免疫系统	pathogenesis /ˌpæθə'dʒenɪsɪs/ : 发病机制

Single-cell RNA sequencing of murine islets shows high cellular complexity at all stages of autoimmune diabetes

小鼠胰岛单细胞RNA测序揭示自身免疫性糖尿病各阶段高度细胞复杂性

期刊: Journal of Experimental Medicine 日期: 2020 JCR Q1 中科院升级版 医学1区 IF 10.6

Abstract

Tissue-specific **autoimmune diseases** are driven by activation of diverse **immune cells** in the target organs. However, the molecular signatures of immune cell populations over time in an autoimmune process remain poorly defined. Using single-cell RNA sequencing, we performed an unbiased examination of diverse islet-infiltrating cells during **autoimmune diabetes** in the nonobese diabetic mouse. The data revealed a landscape of transcriptional heterogeneity across the **lymphoid** and **myeloid** compartments. Memory CD4 and cytotoxic CD8 T cells appeared early in **islets**, accompanied by **regulatory cells** with distinct phenotypes. Surprisingly, we observed a dramatic remodeling in the islet **microenvironment**, in which the resident **macrophages** underwent a stepwise activation program. This process resulted in **polarization** of the macrophage subpopulations into a terminal **proinflammatory** state. This study provides a single-cell atlas defining the staging of **autoimmune diabetes** and reveals that diabetic autoimmunity is driven by transcriptionally distinct cell populations specialized in divergent biological functions.

中文翻译

组织特异性自身免疫疾病是由靶器官中多种免疫细胞的激活所驱动的。然而，在自身免疫过程中，免疫细胞群体随时间变化的分子特征仍不明确。利用单细胞RNA测序技术，我们对非肥胖糖尿病小鼠在自身免疫性糖尿病期间的各种胰岛浸润细胞进行了无偏倚的检测。数据显示了淋巴和髓系细胞区室中转录异质性的全景。记忆性CD4和细胞毒性CD8 T细胞早期出现在胰岛中，伴随着具有不同表型的调节细胞。令人惊讶的是，我们观察到胰岛微环境发生了剧烈的重塑，其中驻留的巨噬细胞经历了一个逐步的激活程序。这一过程导致巨噬细胞亚群极化为终末促炎状态。本研究提供了一个定义自身免疫性糖尿病分期的单细胞图谱，并揭示了糖尿病自身免疫是由专门执行不同生物学功能的转录上不同的细胞群体所驱动的。

核心词汇

autoimmune diseases /ˌɔːtəʊˈɪmjʊn dɪˈziːzɪz/ : 自身免疫性疾病

immune cells /ɪˈmjʊn sɛlz/ : 免疫细胞

myeloid /ˈmaɪəloɪd/ : 髓样

macrophages /ˈmækroʊfaɪdʒɪz/ : 巨噬细胞

proinflammatory /ˌprəʊɪnˈflæmətɔːri/ : 促炎性

microenvironment /ˌmaɪkrəʊɪnˈvaɪrənmənt/ : 微环境

autoimmune diabetes /ˌɔːtəʊˈɪmjʊn ˌdaɪəˈbiːtɪz/ : 自身免疫性糖尿病

lymphoid /ˈlɪmfɔɪd/ : 淋巴样

regulatory cells /ˌregjʊlətɔːri sɛlz/ : 调节细胞

polarization /ˌpəʊləraɪˈzeɪʃən/ : 极化

islets /ˈaɪləts/ : 胰岛

The relationship between SARS-CoV-2 infection and type 1 diabetes mellitus

SARS-CoV-2感染与1型糖尿病之间的关系

期刊: Nature Reviews Endocrinology 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 42.3

Abstract

Environmental factors, in particular viral infections, are thought to have an important role in the pathogenesis of type 1 diabetes mellitus (T1DM). The COVID-19 pandemic reinforced this hypothesis as many observational studies and meta-analyses reported a notable increase in the incidence of T1DM following infection with SARS-CoV-2 as well as an association between SARS-CoV-2 infection and the risk of new-onset T1DM. Experimental evidence suggests that human β -cells express SARS-CoV-2 receptors and that SARS-CoV-2 can infect and replicate in β -cells, resulting in structural or functional alterations of these cells. These alterations include reduced numbers of insulin-secreting granules, impaired pro-insulin (or insulin) secretion, and β -cell transdifferentiation or dedifferentiation. The inflammatory environment induced by local or systemic SARS-CoV-2 infection might result in a set of signals (such as pro-inflammatory cytokines) that lead to β -cell alteration or apoptosis or to a bystander activation of T cells and disruption of peripheral tolerance that triggers autoimmunity. Other mechanisms, such as viral persistence, molecular mimicry and activation of endogenous human retroviruses, are also likely to be involved in the pathogenesis of T1DM following SARS-CoV-2 infection. This Review addresses the issue of the involvement of SARS-CoV-2 infection in the development of T1DM using evidence from epidemiological, clinical and experimental studies.

中文翻译

环境因素，特别是病毒感染，被认为在1型糖尿病（T1DM）的发病机制中起着重要作用。COVID-19大流行强化了这一假设，因为许多观察性研究和荟萃分析报告称，感染SARS-CoV-2后T1DM的发病率显著增加，并且SARS-CoV-2感染与新发T1DM风险之间存在关联。实验证据表明，人类 β 细胞表达SARS-CoV-2受体，并且SARS-CoV-2可以感染并在 β 细胞中复制，导致这些细胞的结构或功能改变。这些改变包括胰岛素分泌颗粒数量减少、前胰岛素（或胰岛素）分泌受损，以及 β 细胞转分化或去分化。局部或全身性SARS-CoV-2感染引发的炎症环境可能产生一系列信号（如促炎细胞因子），导致 β 细胞改变或凋亡，或旁观者T细胞激活和外周耐受性破坏，从而引发自身免疫。其他机制，如病毒持续存在、分子模拟和人类内源性逆转录病毒的激活，也可能参与SARS-CoV-2感染后T1DM的发病机制。本综述通过流行病学、临床和实验研究的证据，探讨了SARS-CoV-2感染在T1DM发展中的作用。

核心词汇

pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制	infection /ɪnˈfɛkʃən/ : 感染
insulin /ˈɪnsjʊlɪn/ : 胰岛素	secretion /sɪˈkriːʃən/ : 分泌
transdifferentiation /ˌtrænsdɪˈfərənʃiˈeɪʃən/ : 转分化	dedifferentiation /ˌdiːdɪˈfərənʃiˈeɪʃən/ : 去分化
apoptosis /ˌæpəˈtəʊsɪs/ : 细胞凋亡	autoimmunity /ˌɔːtəɪˈmjʊːnɪti/ : 自身免疫
molecular mimicry /məˈlɛkjʊlər ˈmɪmɪkri/ : 分子模拟	epidemiological /ˌɛpɪˌdiːmiəˈlɒdʒɪkəl/ : 流行病学的

Exercise in adults with type 1 diabetes mellitus

成人1型糖尿病患者的运动

期刊: Nature Reviews Endocrinology

日期: 2023

JCR Q1

中科院升级版 医学1区

IF 40.0

Abstract

Regular physical activity improves cardiometabolic and musculoskeletal health, helps with weight management, improves cognitive and psychosocial functioning, and is associated with reduced mortality related to cancer and **diabetes mellitus**. However, turnover rates of **glucose** in the blood increase dramatically during exercise, which often results in either **hypoglycaemia** or **hyperglycaemia** as well as increased **glycaemic variability** in individuals with type 1 **diabetes mellitus** (T1DM). A complex **neuroendocrine** response to an acute exercise session helps to maintain circulating levels of **glucose** in a fairly tight range in healthy individuals, while several abnormal physiological processes and limitations of **insulin** therapy limit the capacity of people with T1DM to exercise in a normoglycaemic state. Knowledge of the acute and chronic effects of exercise and regular physical activity is critical for the formulation of clinical strategies for the management of **insulin** and nutrition for active patients with T1DM. Emerging diabetes-related technologies, such as continuous **glucose** monitors, **automated insulin delivery** systems and the administration of solubilized **glucagon**, are demonstrating efficacy for preserving **glucose homeostasis** during and after exercise in this population of patients. This Review highlights the beneficial effects of regular exercise and details the complex endocrine and metabolic responses to different types of exercise for adults with T1DM. An overview of basic clinical strategies for the preservation of **glucose homeostasis** using emerging technologies is also provided.

中文翻译

规律的身体活动能改善心脏代谢与肌肉骨骼健康，有助于体重管理，提升认知与心理社会功能，并降低癌症和糖尿病相关死亡率。然而，运动期间血糖周转率急剧上升，常导致1型糖尿病患者出现低血糖或高血糖，并增加血糖波动。健康个体通过复杂的神经内分泌反应能在运动期间将血糖维持在相对稳定范围，而1型糖尿病患者因生理过程异常及胰岛素治疗局限，难以在正常血糖状态下进行运动。掌握运动及规律身体活动的急慢性效应，对制定1型糖尿病活跃患者的胰岛素与营养管理临床策略至关重要。新兴糖尿病相关技术——如连续血糖监测仪、自动化胰岛素输注系统和可溶性胰高血糖素给药——已在该患者群体中展现出维持运动期间及运动后血糖稳态的有效性。本综述重点阐述规律运动对成人1型糖尿病患者的益处，详细解析不同类型运动引发的复杂内分泌与代谢反应，并概述运用新兴技术维持血糖稳态的基础临床策略。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtiːz məˈleɪtəs/ : 糖尿病**hyperglycaemia** /ˌhaɪpəˈɡlaɪːsiːmiə/ : 高血糖症**insulin** /ˈɪnsjʊlɪn/ : 胰岛素**neuroendocrine** /ˌnjuːəˈendəkrɪn/ : 神经内分泌**homeostasis** /ˌhəʊmiəʊˈsteɪsɪs/ : 稳态**automated insulin delivery** /ˈɔːtəmeɪtɪd ˈɪnsjʊlɪn dɪˈlɪvəri/ : 自动胰岛素输注**hypoglycaemia** /ˌhaɪpəˈɡlaɪːsiːmiə/ : 低血糖症**glucose** /ˈɡluːkəʊs/ : 葡萄糖**glycaemic variability** /ˌɡlaɪːsiːmɪk ˌveəriəˈbeɪləti/ : 血糖变异性**glucagon** /ˈɡluːkəɡɒn/ : 胰高血糖素**continuous glucose monitor** /kənˈtɪnjuəs ˈɡluːkəʊs ˈmɒnɪtə/ : 连续血糖监测仪

Persistent coxsackievirus B infection and pathogenesis of type 1 diabetes mellitus

持续性柯萨奇病毒B感染与1型糖尿病的发病机制

期刊: Nature Reviews Endocrinology 日期: 2022 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

Enteroviruses are believed to trigger or accelerate **islet autoimmunity** in genetically susceptible individuals, thereby resulting in loss of functional insulin-producing β -cells and type 1 diabetes mellitus (T1DM). Although enteroviruses are primarily involved in acute and lytic infections in vitro and in vivo, they can also establish a **persistent infection**. Prospective **epidemiological studies** have strongly associated the persistence of enteroviruses, especially **coxsackievirus B** (CVB), with the appearance of islet **autoantibodies** and an increased risk of T1DM. CVB can persist in pancreatic ductal and β -cells, which leads to structural or functional alterations of these cells, and to a **chronic inflammatory response** that promotes recruitment and activation of pre-existing **autoreactive T cells** and β -cell autoimmune destruction. CVB persistence in other sites, such as the intestine, blood cells and **thymus**, has been described; these sites could serve as a reservoir for infection or reinfection of the pancreas, and this persistence could have a role in the disturbance of **tolerance** to β -cells. This Review addresses the involvement of persistent **enterovirus** infection in triggering **islet autoimmunity** and T1DM, as well as current strategies to control **enterovirus** infections for preventing or reducing the risk of T1DM onset.

中文翻译

肠道病毒被认为在遗传易感个体中触发或加速胰岛自身免疫，从而导致功能性胰岛素产生 β 细胞的丧失和1型糖尿病（T1DM）的发生。尽管肠道病毒在体外和体内主要引起急性和溶解性感染，但它们也能建立持续性感染。前瞻性流行病学研究强烈表明，肠道病毒（尤其是柯萨奇病毒B型，CVB）的持续存在与胰岛自身抗体的出现及T1DM风险增加密切相关。CVB可在胰腺导管细胞和 β 细胞中持续存在，导致这些细胞的结构或功能改变，并引发慢性炎症反应，促进预先存在的自身反应性T细胞的招募和激活，进而导致 β 细胞的自身免疫性破坏。已有研究描述了CVB在其他部位（如肠道、血细胞和胸腺）的持续存在；这些部位可能作为感染或再感染胰腺的储存库，这种持续存在可能在破坏对 β 细胞的耐受性中发挥作用。本综述探讨了持续性肠道病毒感染在触发胰岛自身免疫和T1DM中的作用，以及当前控制肠道病毒感染以预防或降低T1DM发病风险的策略。

核心词汇

enterovirus /ˈɛntərəʊˌvaɪrəs/ : 肠道病毒	islet autoimmunity /ˈaɪlɪt ˌɔːtuɪˈmjuːnɪti/ : 胰岛自身免疫
coxsackievirus B /kɒkˈsækiˌvaɪrəs biː/ : 柯萨奇病毒B型	pancreatic ductal cells /ˌpæŋkriˈætkəl ˈdʌktəl sɛlz/ : 胰腺导管细胞
autoantibodies /ˌɔːtuːˈæntɪˌbɒdiz/ : 自身抗体	persistent infection /pərˈsɪstənt ɪnˈfɛkʃən/ : 持续性感染
chronic inflammatory response /ˈkrɒnɪk ɪnˈflæmətɔːri riˈspɒns/ : 慢性炎症反应	autoreactive T cells /ˌɔːtuəriˈæktɪv tiː sɛlz/ : 自身反应性T细胞
thymus /ˈθaɪməs/ : 胸腺	tolerance /ˈtɒlərəns/ : 免疫耐受
epidemiological studies /ˌɛpɪˌdɪːmiəˈlɒdʒɪkəl ˈstʌdiz/ : 流行病学研究	

A perspective on treating type 1 diabetes mellitus before insulin is needed

胰岛素需求前治疗1型糖尿病的视角

期刊: Nature Reviews Endocrinology 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

Type 1 **diabetes mellitus** (T1DM) is a progressive **autoimmune disease** that starts long before a **clinical diagnosis** is made. The American Diabetes Association recognizes three stages: stage 1 (**normoglycaemic** and positive for **autoantibodies** to β -cell antigens); stage 2 (asymptomatic with **dysglycaemia**); and stage 3, which is defined by glucose levels consistent with the definition of **diabetes mellitus**. This Perspective focuses on the management of the proportion of individuals with early stage 3 T1DM who do not immediately require **insulin**; a stage we propose should be termed stage 3a. To date, this period of non-**insulin**-dependent T1DM has been largely unrecognized. Importantly, it represents a window of opportunity for intervention, as remaining at this stage might delay the need for **insulin** by months or years. Extending the **insulin**-free period and/or avoiding unnecessary **insulin therapy** are important goals, as there is no risk of **hypoglycaemia** during this period and the adherence burden on patients of **glycaemic** monitoring and daily adjustments for diet and exercise is substantially reduced. Recognizing the pressing need for guidance on adequate management of children and adults with stage 3a T1DM, we present our perspective on the subject, which needs to be tested in formal and adequately powered **clinical trials**.

中文翻译

1型糖尿病是一种进展性自身免疫性疾病，其发病远早于临床诊断。美国糖尿病协会将其分为三个阶段：第一阶段（血糖正常但 β 细胞抗原自身抗体阳性）；第二阶段（无症状但血糖异常）；第三阶段则根据血糖水平符合糖尿病的定义来判断。本文重点关注早期第三阶段1型糖尿病患者中那些不立即需要胰岛素治疗的人群的管理，我们建议将这一阶段称为3a期。迄今为止，这一非胰岛素依赖性的1型糖尿病阶段在很大程度上尚未被充分认识。重要的是，它代表了一个干预的窗口期，因为维持在这一阶段可能会将胰岛素需求推迟数月甚至数年。延长无胰岛素治疗期和/或避免不必要的胰岛素治疗是重要的目标，因为在此期间没有低血糖风险，并且患者在进行血糖监测以及日常饮食和运动调整方面的依从负担显著减轻。鉴于迫切需要针对儿童和成人3a期1型糖尿病患者的适当管理指南，我们提出了对此主题的看法，这需要在正式且具有足够统计效力的临床试验中进行验证。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈleɪtəs/ : 糖尿病	autoimmune disease /ˌɔːtəʊˈmjuːn dɪˈziːz/ : 自身免疫性疾病
autoantibodies /ˌɔːtəʊˈæntɪˌbɒdiz/ : 自身抗体	normoglycaemic /ˌnɔːrməʊɡlaɪˈsiːmɪk/ : 血糖正常
dysglycaemia /dɪsɡlaɪˈsiːmiə/ : 血糖异常	glycaemic /ɡlaɪˈsiːmɪk/ : 血糖的
hypoglycaemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖	insulin /ˈɪnsjʊlɪn/ : 胰岛素
insulin therapy /ˈɪnsjʊlɪn ˈθerəpi/ : 胰岛素治疗	clinical diagnosis /ˈklɪnɪkəl ˌdaɪəɡˈnoʊsɪs/ : 临床诊断
clinical trials /ˈklɪnɪkəl ˈtraɪəlz/ : 临床试验	

Harnessing cellular therapeutics for type 1 diabetes mellitus: Progress, challenges, and the road ahead

利用细胞疗法治疗1型糖尿病：进展、挑战与未来之路

期刊: Nature Reviews Endocrinology 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 42.3

Abstract

Type 1 **diabetes mellitus** (T1DM) is a growing global health concern that affects approximately 8.5 million individuals worldwide. T1DM is characterized by an **autoimmune** destruction of pancreatic β cells, leading to a disruption in **glucose homeostasis**. Therapeutic intervention for T1DM requires a complex regimen of **glycaemic monitoring** and the administration of exogenous **insulin** to regulate blood glucose levels. Advances in continuous glucose monitoring and algorithm-driven **insulin** delivery devices have improved the quality of life of patients. Despite this, mimicking islet function and complex physiological feedback remains challenging. Pancreatic **islet transplantation** represents a potential functional cure for T1DM but is hindered by donor scarcity, variability in harvested cells, aggressive immunosuppressive regimens and suboptimal clinical outcomes. Current research is directed towards generating alternative cell sources, improving **transplantation** methods, and enhancing cell survival without chronic **immunosuppression**. This Review maps the progress in cell replacement therapies for T1DM and outlines the remaining challenges and future directions. We explore the state-of-the-art strategies for generating replenishable β cells, cell delivery technologies and local targeted **immune modulation**. Finally, we highlight relevant **animal models** and the regulatory aspects for advancing these technologies towards clinical deployment.

中文翻译

1型糖尿病（T1DM）已成为日益严峻的全球健康问题，全球约850万人受其影响。该疾病以胰腺 β 细胞遭受自身免疫性破坏为特征，导致葡萄糖稳态失衡。针对T1DM的治疗干预需通过复杂的血糖监测方案与外源性胰岛素给药来调节血糖水平。持续葡萄糖监测技术与算法驱动的胰岛素输注设备的发展已显著改善患者生活质量。尽管如此，模拟胰岛功能及复杂生理反馈机制仍面临挑战。胰岛移植作为T1DM的潜在功能性治愈方案，却受限于供体稀缺、获取细胞质量不均、强效免疫抑制方案及临床疗效欠佳等因素。当前研究重点聚焦于开发替代性细胞来源、优化移植方法、在避免长期免疫抑制的前提下提升细胞存活率。本综述系统梳理T1DM细胞替代疗法的进展，阐明现存挑战与未来方向，深入探讨可再生 β 细胞的前沿制备策略、细胞递送技术及局部靶向免疫调节方案，最后重点阐述相关动物模型及推动这些技术向临床转化所需的监管考量。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈlaɪtəs/ : 糖尿病	autoimmune /ˌɔːtəʊɪˈmjuːn/ : 自身免疫
glucose homeostasis /ˈgluːkəʊs ˌhəʊmiəˈsteɪsɪs/ : 葡萄糖稳态	insulin /ˈɪnsəlɪn/ : 胰岛素
glycaemic monitoring /ˈɡlaɪˈsiːmɪk ˈmɒnɪtərɪŋ/ : 血糖监测	islet transplantation /ˈaɪlət ˌtrænsplænˈteɪʃən/ : 胰岛移植
immunosuppression /ˌɪmjʊnəʊsəˈpreʃən/ : 免疫抑制	cell replacement therapy /sɛl rɪˈpleɪsmənt ˈθerəpi/ : 细胞替代疗法
transplantation /ˌtrænsplænˈteɪʃən/ : 移植	immune modulation /ɪˈmjuːn ˌmɒdjʊˈleɪʃən/ : 免疫调节
animal models /ˈæniməl ˈmɒdəlz/ : 动物模型	

Type 1 Diabetes: Etiology, Immunology, and Therapeutic Strategies

1型糖尿病：病因学、免疫学与治疗策略

期刊: Physiological Reviews 日期: 2011

Abstract

Type 1 diabetes (T1D) is a chronic **autoimmune disease** in which destruction or damaging of the beta-cells in the **islets of Langerhans** results in **insulin deficiency** and **hyperglycemia**. We only know for sure that **autoimmunity** is the predominant effector mechanism of T1D, but may not be its primary cause. T1D precipitates in genetically susceptible individuals, very likely as a result of an environmental trigger. Current genetic data point towards the following genes as susceptibility genes: HLA, insulin, PTPN22, IL2Ra, and CTLA4. Epidemiological and other studies suggest a triggering role for enteroviruses, while other microorganisms might provide protection. Efficacious prevention of T1D will require detection of the earliest events in the process. So far, **autoantibodies** are most widely used as serum **biomarker**, but T-cell readouts and metabolome studies might strengthen and bring forward diagnosis. Current preventive clinical trials mostly focus on environmental triggers. Therapeutic trials test the efficacy of antigen-specific and antigen-nonspecific **immune interventions**, but also include restoration of the affected beta-cell mass by **islet transplantation**, **neogenesis** and **regeneration**, and combinations thereof. In this comprehensive review, we explain the genetic, environmental, and immunological data underlying the prevention and intervention strategies to constrain T1D.

中文翻译

1型糖尿病（T1D）是一种慢性自身免疫性疾病，其特征是胰岛β细胞被破坏或损伤，导致胰岛素缺乏和高血糖。目前我们仅能确定自身免疫是T1D的主要效应机制，但可能并非其根本原因。T1D在具有遗传易感性的个体中发作，很可能由环境因素触发。现有遗传学数据指出以下基因为易感基因：HLA、胰岛素、PTPN22、IL2Ra和CTLA4。流行病学及其他研究表明肠道病毒可能起触发作用，而其他微生物或许具有保护作用。有效预防T1D需要探测疾病进程中的最早事件。目前自身抗体是最广泛使用的血清生物标志物，但T细胞检测和代谢组学研究可能强化并提前诊断。当前预防性临床试验主要聚焦于环境触发因素，治疗性试验则测试抗原特异性和非特异性免疫干预措施的有效性，同时探索通过胰岛移植、新生和再生技术恢复受损β细胞群及其组合法。本综述系统阐述了制约T1D发展的预防与干预策略所依据的遗传学、环境及免疫学数据基础。

核心词汇

autoimmune disease /ˌɔːtəʊˈmjuːn dɪˈziːz/ : 自身免疫性疾病	insulin deficiency /ˈɪnsjʊlɪn dɪˈfɪʃənsi/ : 胰岛素缺乏
hyperglycemia /ˌhaɪpəˈɡliːsɪˈmiə/ : 高血糖	islets of Langerhans /ˈaɪləts əv ˈlɑːŋərˌhɑːns/ : 胰岛
autoimmunity /ˌɔːtəʊˈmjuːnɪti/ : 自身免疫	autoantibodies /ˌɔːtəʊˈæntɪˌbɒdiz/ : 自身抗体
biomarker /ˈbaɪəʊˌmɑːrkər/ : 生物标志物	islet transplantation /ˈaɪlət ˌtrænsˌplænˈteɪʃən/ : 胰岛移植
neogenesis /ˌniːoʊˈdʒenɪsɪs/ : 新生	regeneration /ˌrɪˌdʒɛnəˈreɪʃən/ : 再生
immune interventions /ɪˈmjuːn ˌɪntərˈvenʃənz/ : 免疫干预	

Systematic immune cell dysregulation and molecular subtypes revealed by single-cell RNA-seq of subjects with type 1 diabetes

通过单细胞RNA测序揭示1型糖尿病患者的系统性免疫细胞失调与分子亚型

期刊: Genome Medicine 日期: 2024 JCR Q1 中科院升级版 生物学1区 IF 11.2

Abstract

Abstract Background Type 1 **diabetes mellitus** (T1DM) is a prototypic endocrine **autoimmune disease** resulting from an immune-mediated destruction of pancreatic insulin-secreting β cells. A comprehensive **immune cell** phenotype evaluation in T1DM has not been performed thus far at the single-cell level. **Methods** In this cross-sectional analysis, we generated a single-cell **transcriptomic** dataset of **peripheral blood mononuclear cells** (PBMCs) from 46 manifest T1DM (stage 3) cases and 31 matched controls. **Results** We surprisingly detected profound alterations in circulatory immune cells (1784 **dysregulated genes** in 13 **immune cell** types), far exceeding the count in the comparator systemic **autoimmune disease** SLE. Genes upregulated in T1DM were involved in WNT signaling, interferon signaling and migration of T/NK cells, **antigen presentation** by B cells, and **monocyte activation**. A significant fraction of these differentially expressed genes were also altered in T1DM **pancreatic islets**. We used the single-cell data to construct a T1DM metagene z-score (TMZ score) that distinguished cases and controls and classified patients into **molecular subtypes**. This score correlated with known prognostic immune markers of T1DM, as well as with drug response in clinical trials. **Conclusions** Our study reveals a surprisingly strong systemic dimension at the level of **immune cell** network in T1DM, defines disease-relevant **molecular subtypes**, and has the potential to guide non-invasive test development and **patient stratification**.

中文翻译

摘要 背景 1型糖尿病 (T1DM) 是一种典型的内分泌自身免疫性疾病，由免疫介导的胰腺胰岛素分泌β细胞破坏引起。迄今为止，尚未在单细胞水平对T1DM进行全面免疫细胞表型评估。**方法** 在这项横断面分析中，我们生成了46例显性T1DM (第3阶段) 患者和31例匹配对照的外周血单个核细胞 (PBMC) 单细胞转录组数据集。**结果** 我们意外地检测到循环免疫细胞的深刻改变 (13种免疫细胞类型中1784个基因失调)，远超对照系统性自身免疫病SLE的基因数量。T1DM中上调的基因涉及WNT信号通路、干扰素信号通路、T/NK细胞迁移、B细胞抗原呈递以及单核细胞活化。这些差异表达基因中有相当一部分在T1DM胰岛中也发生了改变。我们利用单细胞数据构建了T1DM元基因z评分 (TMZ评分)，该评分能区分病例与对照，并将患者分为分子亚型。该评分与已知的T1DM预后免疫标志物以及临床试验中的药物反应相关。**结论** 我们的研究揭示了T1DM在免疫细胞网络水平上惊人的强烈系统性维度，定义了疾病相关的分子亚型，并有望指导无创检测方法的开发和患者分层。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈlaɪtəs/ : 糖尿病	autoimmune disease /ˌɔːtəʊˈɪmjʊn dɪˈziːz/ : 自身免疫性疾病
pancreatic islets /ˌpæŋkriˈætsk ˈaɪləts/ : 胰岛	immune cell /ɪˈmjʊn sel/ : 免疫细胞
peripheral blood mononuclear cells /pəˈrɪfərəl bləd ˌmɒnəʊˈnjuːkliə sels/ : 外周血单个核细胞	transcriptomic /ˌtrænskɪptˈtɒmɪk/ : 转录组学
dysregulated genes /dɪsˈrɛɡjuleɪtɪd dʒiːnz/ : 失调基因	antigen presentation /ˈæntɪdʒən ˌpreʒənˈteɪʃən/ : 抗原呈递
monocyte activation /ˈmɒnəsəɪt ˌæktɪˈveɪʃən/ : 单核细胞活化	molecular subtypes /məˈlekjələr ˈsʌbtaɪps/ : 分子亚型
prognostic markers /prɒɡˈnɒstɪk ˈmɑːkəz/ : 预后标志物	patient stratification /ˈpeɪʃənt ˌstrætɪfɪˈkeɪʃən/ : 患者分层

Type 1 diabetes mellitus prevention: Present and future

1型糖尿病预防：现状与未来

期刊: Nature Reviews Endocrinology 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 42.3

Abstract

Type 1 **diabetes mellitus** (T1DM) is a chronic disease with an increasing global incidence. The mortality associated with T1DM **complications** emphasizes the urgency of developing therapeutic strategies to prevent or delay the onset of T1DM. Historically, T1DM was solely described as a T cell-mediated disease. However, the role of beta-cells as active participants in the immune-mediated damage is now well appreciated. Indeed, heterogeneity, vulnerability to stressors and the ability of beta-cells to act as antigen-presenting cells has altered the perspective of what is necessary for effective disease prevention and ongoing beta-cell preservation. Currently, teplizumab, an Fc-**receptor** non-binding humanized CD3-specific **monoclonal antibody**, is the only therapy approved by the FDA for the delay of T1DM onset. The **intravenous administration**, generalized **immunosuppression** and **adverse effects** mean that the transition to routine clinical practice is not without challenges. However, teplizumab could lead to the development of more accessible therapies. In this Review, we explore current and potential therapeutics for T1DM prevention. We offer alternative approaches, such as targeting the **receptor** for advanced **glycation** end products (RAGE). RAGE is a pattern recognition **receptor** that engages a wide range of ligands, including advanced **glycation** end products (AGEs; a family of molecules that includes the well described marker of long-term **glucose** concentrations, HbA(1c)).

中文翻译

1型糖尿病（T1DM）是一种全球发病率持续上升的慢性疾病。与T1DM并发症相关的死亡率凸显了开发治疗策略以预防或延缓T1DM发病的紧迫性。历史上，T1DM仅被描述为一种T细胞介导的疾病。然而，目前学界已充分认识到β细胞在免疫介导损伤中的主动参与作用。事实上，β细胞的异质性、对应激源的易感性以及其作为抗原呈递细胞的能力，已改变了人们对有效预防疾病和持续保护β细胞所需条件的认知。目前，替普利珠单抗（一种非Fc受体结合的人源化CD3特异性单克隆抗体）是唯一获FDA批准用于延缓T1DM发病的疗法。其静脉给药方式、全身性免疫抑制效应及不良反应意味着该疗法向常规临床实践的转化仍面临挑战。但替普利珠单抗可能推动更易获取疗法的开发。本综述探讨了当前及潜在的T1DM预防性治疗策略，并提出替代方案，例如靶向晚期糖基化终末产物受体（RAGE）。RAGE是一种模式识别受体，可与多种配体结合，包括晚期糖基化终末产物（AGEs；该分子家族包含长期血糖浓度的经典标志物HbA(1c)）。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtiːz məˈlaɪtəs/ : 糖尿病	complications /ˌkɒmplɪˈkeɪʃnz/ : 并发症
monoclonal antibody /ˌmɒnəˈklaʊnəl ˈæntɪbɒdi/ : 单克隆抗体	immunosuppression /ˌɪmjʊnəʊsəˈpreʃən/ : 免疫抑制
adverse effects /ˈædvɜːs ɪˈfɛkts/ : 不良反应	intravenous administration /ˌɪntreˈviːnəs ədˌmɪniˈstreɪʃən/ : 静脉给药
receptor /ˈrɛːptə/ : 受体	glycation /ˈɡlaɪˈkeɪʃən/ : 糖化
glucose /ˈɡluːkəʊs/ : 葡萄糖	

Pancreatic islet-specific engineered T_{regs} exhibit robust antigen-specific and bystander immune suppression in type 1 diabetes models

胰岛特异性工程化T_{regs}在1型糖尿病模型中展现出强大的抗原特异性及旁观者免疫抑制效应

期刊: Science Translational Medicine 日期: 2022 JCR Q1 中科院升级版 医学1区 IF 14.6

Abstract

adoptive transfer of regulatory T cells (T_{regs}) is therapeutic in type 1 diabetes (T1D) mouse models. T_{regs} that are specific for pancreatic islets are more potent than polyclonal T_{regs} in preventing disease. However, the frequency of antigen-specific natural T_{regs} is extremely low, and ex vivo expansion may destabilize T_{regs}, leading to an effector phenotype. Here, we generated durable, antigen-specific engineered T_{regs} (EngT_{regs}) from primary human CD4⁺ T cells by combining FOXP3 homology-directed repair editing and lentiviral T cell receptor (TCR) delivery. Using TCRs derived from clonally expanded CD4⁺ T cells isolated from patients with T1D, we generated islet-specific EngT_{regs} that suppressed effector T cell (T_{eff}) proliferation and cytokine production. EngT_{regs} suppressed T_{effs} recognizing the same islet antigen in addition to bystander T_{effs} recognizing other islet antigens through production of soluble mediators and both direct and indirect mechanisms. Adoptively transferred murine islet-specific EngT_{regs} homed to the pancreas and blocked diabetes triggered by islet-specific T_{effs} or diabetogenic polyclonal T_{effs} in recipient mice. These data demonstrate the potential of antigen-specific EngT_{regs} as a targeted therapy for preventing T1D.

中文翻译

调节性T细胞 (Tregs) 的过继转移在1型糖尿病 (T1D) 小鼠模型中具有治疗作用。针对胰岛的特异性Tregs在预防疾病方面比多克隆Tregs更有效。然而, 抗原特异性天然Tregs的频率极低, 且体外扩增可能破坏Tregs的稳定性, 导致其向效应表型转化。本研究通过结合FOXP3同源定向修复编辑和慢病毒T细胞受体 (TCR) 递送技术, 从原代人CD4⁺ T细胞中构建了持久、抗原特异性的工程化Tregs (EngTregs)。利用从T1D患者体内分离的克隆扩增CD4⁺ T细胞中提取的TCR, 我们成功制备了胰岛特异性EngTregs, 这些细胞能有效抑制效应T细胞 (Teffs) 的增殖和细胞因子产生。EngTregs通过可溶性介质的产生以及直接和间接机制, 不仅抑制了识别相同胰岛抗原的Teffs, 还抑制了识别其他胰岛抗原的旁观者Teffs。过继转移的小鼠胰岛特异性EngTregs能够归巢至胰腺, 并在受体小鼠中阻断由胰岛特异性Teffs或致糖尿病多克隆Teffs引发的糖尿病。这些数据证明了抗原特异性EngTregs作为靶向疗法预防T1D的潜力。

核心词汇

regulatory T cells /ˈrɛɡjʊlətɔːri tiː sɛlz/ : 调节性T细胞	pancreatic islets /ˌpæŋkriˈætik ˈaɪləts/ : 胰岛
engineered T cells /ˌɛndʒɪˈnɪəd tiː sɛlz/ : 工程化T细胞	T cell receptor /tiː sɛl rɪˈseptə/ : T细胞受体
effector T cells /ɪˈfɛktə tiː sɛlz/ : 效应T细胞	proliferation /prəˌlɪfəˈreɪʃən/ : 增殖
cytokine /ˈsaɪtəkain/ : 细胞因子	adoptive transfer /əˈdɒptɪv ˈtrænsfɜː/ : 过继性转移
autoimmune disease /ˌɔːtəʊˈɪmjʊn dɪˈziːz/ : 自身免疫性疾病	immunotherapy /ˌɪmjʊnəʊˈθerəpi/ : 免疫治疗

Teplizumab induces persistent changes in the antigen-specific repertoire in individuals at risk for type 1 diabetes

Teplizumab诱导1型糖尿病风险个体抗原特异性库的持久变化

期刊: Journal of Clinical Investigation 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 13.6

Abstract

BACKGROUND Teplizumab, a non-FcR-binding anti-CD3 mAb, is approved to delay progression of type 1 diabetes (T1D) in at-risk patients. Previous investigations described the immediate effects of the 14-day treatment, but longer-term effects of the drug remain unknown. **METHODS** With an extended analysis of study participants, we found that 36% were undiagnosed or remained free of clinical diabetes after 5 years, suggesting **operational tolerance**. Using single-cell RNA sequencing, we compared the phenotypes, **transcriptome**, and repertoire of peripheral blood CD8+ T cells including **autoreactive T cells** from study participants before and after teplizumab and features of responders and non-responders. **RESULTS** At 3 months, there were transcriptional signatures of cell activation in CD4+ and CD8+ T cells including signaling that was reversed at 18 months. At that time, there was reduced expression of genes in T cell receptor and activation pathways in clinical responders. In CD8+ T cells, we found increased expression of genes associated with **exhaustion** and immune regulation with teplizumab treatment. These transcriptional features were further confirmed in an independent cohort. Pseudotime analysis showed differentiation of CD8+ exhausted and **memory cells** with teplizumab treatment. IL7R expression was reduced, and patients with lower expression of CD127 had longer diabetes-free intervals. In addition, the frequency of autoantigen-reactive CD8+ T cells, which expanded in the **placebo** group over 18 months, did not increase in the teplizumab group. **CONCLUSION** These findings indicate that teplizumab promotes **operational tolerance** in T1D, involving activation followed by **exhaustion** and regulation, and prevents expansion of **autoreactive T cells**. **TRIAL REGISTRATION** ClinicalTrials.gov NCT01030861. **FUNDING** National Institute of Diabetes and Digestive and Kidney Diseases/NIH, Juvenile Diabetes Research Foundation.

中文翻译

背景: Teplizumab是一种非FcR结合的抗CD3单克隆抗体, 已被批准用于延缓高危患者1型糖尿病 (T1D) 的进展。先前的研究描述了14天治疗的即时效果, 但该药物的长期效果仍不清楚。**方法:** 通过对研究参与者的扩展分析, 我们发现36%的参与者在5年后未被诊断或未出现临床糖尿病, 这表明了操作性耐受。使用单细胞RNA测序, 我们比较了研究参与者在teplizumab治疗前后外周血CD8+ T细胞 (包括自身反应性T细胞) 的表型、转录组和库, 以及应答者和非应答者的特征。**结果:** 在3个月时, CD4+和CD8+ T细胞中出现了细胞活化的转录特征, 包括在18个月时逆转的信号传导。此时, 临床应答者中T细胞受体和活化通路相关基因的表达减少。在CD8+ T细胞中, 我们发现teplizumab治疗后与耗竭和免疫调节相关的基因表达增加。这些转录特征在独立队列中得到了进一步证实。伪时间分析显示, teplizumab治疗后CD8+耗竭细胞和记忆细胞发生分化。IL7R表达减少, CD127表达较低的患者无糖尿病间隔更长。此外, 在安慰剂组中, 自身抗原反应性CD8+ T细胞的频率在18个月内增加, 而在teplizumab组中未增加。**结论:** 这些发现表明, teplizumab促进了T1D的操作性耐受, 涉及活化后的耗竭和调节, 并防止了自身反应性T细胞的扩增。试验注册: ClinicalTrials.gov NCT01030861。资助: 国家糖尿病、消化和肾脏疾病研究所/美国国立卫生研究院, 青少年糖尿病研究基金会。

核心词汇

autoimmune disease /ˌɔːtəʊˈmjuːn dɪˈziːz/ : 自身免疫性疾病	autoreactive T cells /ˌɔːtəʊriˈæktɪv tiː sɛlz/ : 自身反应性T细胞
exhaustion /ɪɡˈzɔːstʃən/ : 耗竭	memory cells /ˈmeməri sɛlz/ : 记忆细胞
operational tolerance /ˌɒpəˈreɪʃənəl ˈtɒlərəns/ : 功能性耐受	immunotherapy /ˌɪmjʊnəʊˈθerəpi/ : 免疫治疗
monoclonal antibody /ˌmɒnəˈklaʊnl ˈæntɪˌbɒdi/ : 单克隆抗体	transcriptome /trænˈskɪptəʊm/ : 转录组
clinical trial /ˈklɪnɪkl ˈtraɪəl/ : 临床试验	placebo /ˈpləːsiːboʊ/ : 安慰剂

SGLT2 inhibitors and GLP-1 receptor agonists: Established and emerging indications

SGLT2抑制剂与GLP-1受体激动剂：已确立及新兴适应症

期刊: The Lancet 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

SGLT2 inhibitors and GLP-1 receptor agonists are used in patients with type 2 diabetes as glucose lowering therapies, with additional benefits of **weight loss** and **blood pressure reduction**. Data from cardiovascular outcome trials have highlighted that these drugs confer protection against major **cardiovascular disease** in those with established atherosclerotic **cardiovascular disease**, reduce the risk of admission to hospital for **heart failure**, and reduce cardiovascular and all-cause **mortality**. Ongoing research using hard renal endpoints such as **end stage kidney disease** rather than surrogate markers might clarify the **renoprotective** benefits of both agents. When used for glucose lowering, SGLT2 inhibitors are most effective if the estimated **glomerular filtration rate** is more than 60 ml per min per 1.73m² at initiation and should be avoided where there is a risk of diabetic ketoacidosis. GLP-1 receptor agonists are contraindicated in those with a history of medullary thyroid cancer and used with caution in patients with a history of pancreatitis of a known cause. These drugs are now second-line, or even arguably first-line, glucose lowering therapies in patients with cardiorenal disease, irrespective of glycaemic control. If an SGLT2 inhibitor or GLP-1 receptor agonist is considered suitable in patients with type 2 diabetes, treatment should be prioritised according to existing evidence: GLP-1 receptor agonists should be considered in patients at a high risk of, or with established, **cardiovascular disease** and SGLT2 inhibitors considered for patients with **heart failure** (with reduced ejection fraction) or chronic kidney disease (with or without established **cardiovascular disease**). There is now compelling data on the benefits of these drugs for a range of other clinical indications even without type 2 diabetes, including for GLP-1 receptor agonists in patients with obesity and overweight with weight-related comorbidities.

中文翻译

SGLT2抑制剂和GLP-1受体激动剂用于2型糖尿病患者的降糖治疗，具有减重和降压的额外益处。心血管结局试验数据表明，这些药物能为已确诊动脉粥样硬化性心血管疾病的患者提供主要心血管疾病保护，降低因心力衰竭住院的风险，并减少心血管及全因死亡率。目前正在开展以终末期肾病等硬性肾脏终点（而非替代标志物）为指标的研究，可能进一步明确两种药物的肾脏保护作用。用于降糖时，若起始估算肾小球滤过率>60 ml/min/1.73m²，SGLT2抑制剂效果最佳，但需避免用于存在糖尿病酮症酸中毒风险的患者。有甲状腺髓样癌病史者禁用GLP-1受体激动剂，已知病因的胰腺炎病史患者需谨慎使用。无论血糖控制情况如何，这些药物现已成为心肾疾病患者的二线（甚至可视为一线）降糖疗法。若SGLT2抑制剂或GLP-1受体激动剂适用于2型糖尿病患者，应根据现有证据优先选择治疗方案：心血管疾病高风险或已确诊患者应考虑GLP-1受体激动剂，心力衰竭（射血分数降低）或慢性肾脏病（伴或不伴已确诊心血管疾病）患者应考虑SGLT2抑制剂。目前已有确凿数据证实，即使未患2型糖尿病，这些药物对多种其他临床适应症同样具有获益，包括GLP-1受体激动剂用于肥胖或超重合并体重相关并发症的患者。

核心词汇

cardiovascular disease /ˌkɑːrdiəʊˈvæskjələr dɪˈziːz/ : 心血管疾病

heart failure /hɑːt ˈfeɪljər/ : 心力衰竭

glomerular filtration rate /glɒˈmerjələr fɪlˈtreɪʃən reɪt/ : 肾小球滤过率

glucose lowering therapy /ˈgluːkəʊs ˈləʊərɪŋ ˈθerəpi/ : 降糖治疗

weight loss /weɪt lɒs/ : 减重

surrogate marker /ˈsɜːrəget ˈmɑːrkər/ : 替代标志物

atherosclerosis /ˌæθərəʊskləˈrəʊsɪs/ : 动脉粥样硬化

end stage kidney disease /end steɪdʒ ˈkɪdni dɪˈziːz/ : 终末期肾病

mortality /mɔːrˈtæləti/ : 死亡率

blood pressure reduction /blʌd ˈpreʃər rɪˈdʌkʃən/ : 降压

renal endpoint /ˈriːnəl ˈendpɔɪnt/ : 肾脏终点

renoprotective /ˈriːnəʊprəˈtektɪv/ : 肾脏保护性

Single-cell chromatin accessibility identifies pancreatic islet cell type- and state-specific regulatory programs of diabetes risk

单细胞染色质可及性识别胰岛细胞类型及状态特异性糖尿病风险调控程序

期刊: Nature Genetics 日期: 2021 JCR Q1 中科院升级版 生物学1区 IF 29.0

Abstract

Single-nucleus assay for transposase-accessible chromatin using sequencing (snATAC-seq) creates new opportunities to dissect cell type-specific mechanisms of complex diseases. Since pancreatic islets are central to type 2 diabetes (T2D), we profiled 15,298 islet cells by using combinatorial barcoding snATAC-seq and identified 12 clusters, including multiple alpha, beta and delta cell states. We cataloged 228,873 accessible chromatin sites and identified transcription factors underlying lineage- and state-specific regulation. We observed state-specific enrichment of fasting glucose and T2D genome-wide association studies for beta cells and enrichment for other endocrine cell types. At T2D signals localized to islet-accessible chromatin, we prioritized variants with predicted regulatory function and co-accessibility with target genes. A causal T2D variant rs231361 at the KCNQ1 locus had predicted effects on a beta cell enhancer co-accessible with INS and genome editing in embryonic stem cell-derived beta cells affected INS levels. Together our findings demonstrate the power of single-cell epigenomics for interpreting complex disease genetics.

中文翻译

单核转座酶可及染色质测序 (snATAC-seq) 为解析复杂疾病的细胞类型特异性机制提供了新机遇。鉴于胰岛在2型糖尿病 (T2D) 中的核心作用，我们通过组合条形码snATAC-seq技术对15,298个胰岛细胞进行分析，识别出12个细胞簇，包括多种α细胞、β细胞和δ细胞状态。我们系统鉴定了228,873个可及染色质区域，并揭示了谱系与状态特异性调控背后的转录因子。我们观察到β细胞中空腹血糖与T2D全基因组关联研究存在状态特异性富集现象，其他内分泌细胞类型亦呈现富集特征。在定位于胰岛可及染色质的T2D信号区域，我们通过预测调控功能及与靶基因的共可及性分析，对遗传变异进行了优先级排序。KCNQ1基因座的致病性T2D变异rs231361被预测可影响β细胞增强子功能，该增强子与INS基因存在共可及性，胚胎干细胞来源β细胞的基因组编辑实验证实其对INS水平具有调控作用。本研究共同证明了单细胞表观基因组学在解析复杂疾病遗传机制方面的强大能力。

核心词汇

pancreatic islets /ˌpæŋkriˈætɪk ˈaɪlɪts/ : 胰岛	beta cells /ˈbeɪtə sɛlz/ : β细胞
alpha cells /ˈælfə sɛlz/ : α细胞	delta cells /ˈdeltə sɛlz/ : δ细胞
endocrine cell types /ˈɛndəʊkraɪn sɛl taɪps/ : 内分泌细胞类型	accessible chromatin /əkˈsɛsəbəl ˈkroumətɪn/ : 可及染色质
transcription factors /trænˈskɪpʃən ˈfæktərz/ : 转录因子	variant /ˈvɛəriənt/ : 变异
enhancer /ɪnˈhænsər/ : 增强子	

Single-cell RNA sequencing combined with single-cell proteomics identifies the metabolic adaptation of islet cell subpopulations to high-fat diet in mice

单细胞RNA测序结合单细胞蛋白质组学揭示小鼠胰岛细胞亚群对高脂饮食的代谢适应机制

期刊: Diabetologia 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 10.2

Abstract

AIMS/HYPOTHESIS: Islets have complex **heterogeneity** and subpopulations. Cell surface markers representing alpha, beta and **delta cell** subpopulations are urgently needed for investigations to explore the compositional changes of each **subpopulation** in **obesity** progress and **diabetes** onset, and the adaptation mechanism of **islet metabolism** induced by a high-fat diet (HFD). **METHODS:** Single-cell RNA sequencing (scRNA-seq) was applied to identify alpha, beta and **delta cell subpopulation** markers in an HFD-induced mouse model of **glucose intolerance**. Flow cytometry and immunostaining were used to sort and assess the proportion of each **subpopulation**. Single-cell proteomics was performed on sorted cells, and the functional status of each alpha, beta and **delta cell subpopulation** in **glucose intolerance** was deeply elucidated based on protein expression. **RESULTS:** A total of 33,999 cells were analysed by scRNA-seq and clustered into eight populations, including alpha, beta and delta cells. For alpha cells, scRNA-seq revealed that the Ace2 **CONCLUSIONS/INTERPRETATION:** We identified ACE2, CD81 and GLUT2 as surface markers to distinguish, respectively, alpha, beta and **delta cell** subpopulations with heterogeneous maturation and function. The changes in the proportion and functional status of **islet endocrine** subpopulations reflect the metabolic adaptation of islets to high-fat stress, which weakened the function of alpha cells and enhanced the function of beta and delta cells to bring about **glycaemic homeostasis**. Our findings provide a fundamental resource for exploring the mechanisms maintaining each **islet endocrine subpopulation**'s fate and function in health and disease. **DATA AVAILABILITY:** The scRNA-seq analysis datasets from the current study are available in the Gene Expression Omnibus (GEO) repository under the accession number GSE203376.

中文翻译

目的/假设: 胰岛具有复杂的异质性和亚群结构。为探究肥胖进程及糖尿病发病过程中各亚群的组成变化, 以及高脂饮食诱导的胰岛代谢适应机制, 亟需能够表征 α 、 β 和 δ 细胞亚群的细胞表面标志物。**方法:** 本研究采用单细胞RNA测序技术, 在高脂饮食诱导的葡萄糖耐受不良小鼠模型中鉴定 α 、 β 和 δ 细胞亚群标志物。通过流式细胞术和免疫染色技术对各亚群进行分选与比例评估。对分选细胞进行单细胞蛋白质组学分析, 基于蛋白质表达水平深入阐明葡萄糖耐受不良状态下各 α 、 β 和 δ 细胞亚群的功能状态。**结果:** 通过单细胞RNA测序分析33,999个细胞, 聚类形成包括 α 、 β 和 δ 细胞在内的八个细胞群。针对 α 细胞, 单细胞RNA测序显示Ace2基因具有显著表达特征。**结论/阐释:** 本研究确定ACE2、CD81和GLUT2可作为区分具有异质性成熟度与功能状态的 α 、 β 和 δ 细胞亚群的表面标志物。胰岛内分泌亚群比例与功能状态的改变反映了胰岛对高脂应激的代谢适应——该过程削弱 α 细胞功能, 同时增强 β 和 δ 细胞功能以维持血糖稳态。本研究成果为探索健康与疾病状态下维持胰岛各内分泌亚群命运及功能的机制提供了基础资源。**数据可用性:** 本研究单细胞RNA测序分析数据集已存储于基因表达综合数据库, 登录号为GSE203376。

核心词汇

islet /'aɪlət/ : 胰岛

beta cell /'beɪtə sɛl/ : β 细胞

obesity /ou'bi:seɪti/ : 肥胖症

glucose intolerance /'glu:kəʊs ɪn'tɒlərəns/ : 葡萄糖不耐受

endocrine /'endəkrɪn/ : 内分泌

subpopulation /,sʌb'pɒpjʊ'leɪʃən/ : 亚群

alpha cell /'ælfə sɛl/ : α 细胞

delta cell /'deltə sɛl/ : δ 细胞

diabetes /,daɪə'bi:ti:z/ : 糖尿病

metabolism /mə'tæbə,lɪzəm/ : 代谢

heterogeneity /,hetərə'dʒə'ni:əti/ : 异质性

glycaemic homeostasis /glai'si:mɪk ,həʊmɪə'steɪsɪs/ : 血糖稳态

Artesunate improves glucose and lipid metabolism in db/db mice by regulating the metabolic profile and the MAPK/PI3K/akt signalling pathway

青蒿琥酯通过调节代谢谱和MAPK/PI3K/Akt信号通路改善db/db小鼠的糖脂代谢

期刊: Phytomedicine 日期: 2024

JCR Q1

中科院升级版 医学1区

IF 8.3

Abstract

BACKGROUND: **diabetes** is a **metabolic disorder** characterized by chronic **hyperglycaemia**. Chronic metabolic abnormalities and long-term **hyperglycaemia** may result in a wide range of acute and chronic consequences. Previous studies have demonstrated that artesunate(ART) has antidiabetic, anti-inflammatory, antiatherosclerotic, and other beneficial effects, but the specific regulatory mechanism is not completely clear. **AIM:** This study investigated the effects of ART on metabolic disorders in type 2 **diabetes** mellitus (T2DM) model db/db mice and explored the underlying mechanisms involved. **METHODS:** C57BL/KsJ-db/db mice were used to identify the targets and molecular mechanism of ART. Metabolomic methods were used to evaluate the efficacy of ART in improving T2DM-related metabolic disorders. Network pharmacology and transcriptomic sequencing were used to analyse the targets and pathways of ART in T2DM. Finally, molecular biology experiments were performed to verify the key targets and pathways selected by network pharmacology and transcriptomic analyses. **RESULTS:** After a 7-week ART intervention (160 mg/kg), the glucose and **lipid metabolism** levels of the db/db mice improved. Additionally, the **oxidative stress** indices, namely, the MDA and SOD levels, significantly improved ($p<0.01$). Linoleic acid and glycerophospholipid metabolism, **amino acid metabolism**, **bile acid synthesis**, and **purine metabolism** disorders in db/db mice were partially corrected after ART treatment. Network pharmacology analysis identified important targets of ART for the treatment of metabolic disorders in T2DM. These targets are involved in key signalling pathways, including the highest scores observed for the PI3K/Akt signalling pathway. Transcriptomic analysis revealed that ART could activate the MAPK signalling pathway and two key gene targets, HGK and GADD45. **immunoblotting** revealed that ART increases p-PI3K, p-AKT, Glut2, and IRS1 protein expression and suppresses the phosphorylation of p38, ERK1/2, and JNK, returning HGK and GADD45 to their preartesunate levels. **CONCLUSION:** Treatment of db/db mice with 160 mg/kg ART for 7 weeks significantly reduced **fasting blood glucose** and lipid levels. It also improved metabolic imbalances in amino acids, lipids, purines, and bile acids, thereby improving metabolic disorders. These effects are achieved by activating the PI3K/AKT pathway and inhibiting the MAPK pathway, thus demonstrating the efficacy of the drug.

中文翻译

背景: 糖尿病是一种以慢性高血糖为特征的代谢性疾病。慢性代谢异常和长期高血糖可能导致一系列急性和慢性并发症。先前的研究表明, 青蒿琥酯 (ART) 具有抗糖尿病、抗炎、抗动脉粥样硬化等多种有益作用, 但其具体调控机制尚不完全明确。**目的:** 本研究探讨了ART对2型糖尿病 (T2DM) 模型db/db小鼠代谢紊乱的影响, 并探索其潜在机制。**方法:** 使用C57BL/KsJ-db/db小鼠来鉴定ART的作用靶点和分子机制。采用代谢组学方法评估ART改善T2DM相关代谢紊乱的效果。通过网络药理学和转录组测序分析ART在T2DM中的作用靶点和通路。最后, 通过分子生物学实验验证网络药理学和转录组分析筛选出的关键靶点和通路。**结果:** 经过7周的ART干预 (160 mg/kg), db/db小鼠的糖脂代谢水平得到改善。此外, 氧化应激指标MDA和SOD水平显著改善 ($p<0.01$)。ART治疗后, db/db小鼠的亚油酸和甘油磷脂代谢、氨基酸代谢、胆汁酸合成以及嘌呤代谢紊乱得到部分纠正。网络药理学分析确定了ART治疗T2DM代谢紊乱的重要靶点, 这些靶点涉及关键信号通路, 其中PI3K/Akt信号通路得分最高。转录组分析显示, ART能够激活MAPK信号通路以及两个关键基因靶点HGK和GADD45。免疫印迹实验表明, ART增加了p-PI3K、p-AKT、Glut2和IRS1蛋白的表达, 并抑制了p38、ERK1/2和JNK的磷酸化, 使HGK和GADD45恢复到治疗前水平。**结论:** 使用160 mg/kg ART治疗db/db小鼠7周, 显著降低了空腹血糖和血脂水平, 并改善了氨基酸、脂质、嘌呤和胆汁酸的代谢失衡, 从而改善了代谢紊乱。这些效果通过激活PI3K/AKT通路和抑制MAPK通路实现, 证明了该药物的有效性。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病

metabolic disorder /ˌmetəˈbɒlɪk dɪsˈɔːdə/ : 代谢紊乱

lipid metabolism /ˈlɪpɪd məˈtæbəlaɪzəm/ : 脂质代谢

amino acid metabolism /əˈmiːnəʊ ˈæsɪd məˈtæbəlaɪzəm/ : 氨基酸代谢

purine metabolism /ˈpjʊəriːn məˈtæbəlaɪzəm/ : 嘌呤代谢

hyperglycaemia /ˌhaɪpəˈɡleɪˈsiːmiə/ : 高血糖

glucose metabolism /ˈɡluːkəʊs məˈtæbəlaɪzəm/ : 葡萄糖代谢

oxidative stress /ˈɒksɪdətɪv stres/ : 氧化应激

bile acid synthesis /baɪl ˈæsɪd ˈsɪnθəsɪs/ : 胆汁酸合成

fasting blood glucose /ˈfɑːstɪŋ blʌd ˈɡluːkəʊs/ : 空腹血糖

metabolomics /məˌtæbəˈlɒmɪks/ : 代谢组学

transcriptomics /trænˌskɪpˈtɒmɪks/ : 转录组学

immunoblotting /ˌɪmjʊnəʊˈblɒtɪŋ/ : 免疫印迹

云梦山-侵权必究

Epigenetics of type 2 diabetes mellitus and weight change — a tool for precision medicine?

2型糖尿病与体重变化的表观遗传学——精准医学的工具？

期刊: Nature Reviews Endocrinology 日期: 2022 JCR Q1 中科院升级版 医学1区 IF 42.3

Abstract

Pioneering studies performed over the past few decades demonstrate links between **epigenetics** and type 2 diabetes mellitus (T2DM), the **metabolic disorder** with the most rapidly increasing prevalence in the world. Importantly, these studies identified epigenetic modifications, including altered **DNA methylation**, in **pancreatic islets**, **adipose tissue**, **skeletal muscle** and the **liver** from individuals with T2DM. As non-genetic factors that affect the risk of T2DM, such as **obesity**, unhealthy diet, physical inactivity, ageing and the **intrauterine environment**, have been associated with epigenetic modifications in healthy individuals, **epigenetics** probably also contributes to T2DM development. In addition, genetic factors associated with T2DM and **obesity** affect the **epigenome** in human tissues. Notably, causal mediation analyses found **DNA methylation** to be a potential mediator of genetic associations with metabolic traits and disease. In the past few years, translational studies have identified blood-based epigenetic markers that might be further developed and used for **precision medicine** to help patients with T2DM receive optimal therapy and to identify patients at risk of **complications**. This Review focuses on epigenetic mechanisms in the development of T2DM and the regulation of body weight in humans, with a special focus on **precision medicine**.

中文翻译

过去几十年的开创性研究表明，表观遗传学与2型糖尿病之间存在联系，后者是全球患病率增长最快的代谢性疾病。重要的是，这些研究在2型糖尿病患者的胰岛、脂肪组织、骨骼肌和肝脏中发现了表观遗传修饰，包括DNA甲基化的改变。由于影响2型糖尿病风险的非遗传因素，如肥胖、不健康饮食、缺乏运动、衰老和宫内环境，已在健康个体中与表观遗传修饰相关联，表观遗传学很可能也参与了2型糖尿病的发展。此外，与2型糖尿病和肥胖相关的遗传因素会影响人体组织的表观基因组。值得注意的是，因果中介分析发现DNA甲基化可能是遗传与代谢特征及疾病关联的潜在中介。过去几年，转化研究已发现基于血液的表观遗传标记，这些标记可能被进一步开发并用于精准医疗，以帮助2型糖尿病患者获得最佳治疗，并识别有并发症风险的患者。本综述重点关注表观遗传机制在2型糖尿病发展及人体体重调节中的作用，特别聚焦于精准医疗的应用。

核心词汇

epigenetics /ˌepɪdʒəˈnetɪks/ : 表观遗传学	metabolic disorder /ˌmetəˈbɒlɪk dɪsˈɔːdə/ : 代谢性疾病
DNA methylation /ˌdiːɛnˈeɪˌmɛθɪˈleɪʃən/ : DNA甲基化	pancreatic islets /ˌpæŋkriˈæɪtɪk ˈaɪləts/ : 胰岛
adipose tissue /ˈædɪpəʊs ˈtɪʃuː/ : 脂肪组织	skeletal muscle /ˈskeɪtlɪ ˈmʌsl/ : 骨骼肌
liver /ˈlɪvə/ : 肝脏	obesity /əʊˈbiːsɪti/ : 肥胖症
intrauterine environment /ˌɪntrəˈjuːtəraɪn ɪnˈvaɪrənment/ : 宫内环境	epigenome /ˌepɪˈdʒiːnəʊm/ : 表观基因组
precision medicine /prɪˈsɪʒən ˈmedɪsɪn/ : 精准医学	complications /ˌkɒmplɪˈkeɪʃənz/ : 并发症

Hyocholic acid species improve glucose homeostasis through a distinct TGR5 and FXR signaling mechanism

Hyocholic acid species improve glucose homeostasis through a distinct TGR5 and FXR signaling mechanism

期刊: Cell Metabolism 日期: 2021 JCR Q1 中科院升级版 生物学1区 IF 30.9

Abstract

Hyocholic acid (HCA) and its derivatives are found in trace amounts in human blood but constitute approximately 76% of the **bile acid** (BA) pool in pigs, a species known for its exceptional resistance to type 2 **diabetes**. Here, we show that BA depletion in pigs suppressed **secretion** of glucagon-like peptide-1 (GLP-1) and increased **blood glucose** levels. HCA administration in diabetic mouse models improved serum fasting GLP-1 **secretion** and **glucose homeostasis** to a greater extent than tauroursodeoxycholic acid. HCA upregulated GLP-1 production and **secretion** in **enteroendocrine cells** via simultaneously activating G-protein-coupled BA **receptor**, TGR5, and inhibiting farnesoid X **receptor** (FXR), a unique mechanism that is not found in other BA species. We verified the findings in TGR5 knockout, intestinal FXR activation, and GLP-1 **receptor** inhibition mouse models. Finally, we confirmed in a clinical **cohort**, that lower serum concentrations of HCA species were associated with **diabetes** and closely related to **glycemic** markers.

中文翻译

猪胆酸（HCA）及其衍生物在人体血液中仅微量存在，但在猪的胆汁酸（BA）池中占比高达约76%，而猪这一物种以对2型糖尿病具有卓越抵抗力而闻名。本研究揭示，猪体内胆汁酸耗竭会抑制胰高血糖素样肽-1（GLP-1）的分泌并导致血糖水平升高。在糖尿病小鼠模型中，HCA给药对改善血清空腹GLP-1分泌和葡萄糖稳态的效果优于牛磺熊去氧胆酸。HCA通过同时激活G蛋白偶联胆汁酸受体TGR5和抑制法尼醇X受体（FXR），上调肠内分泌细胞中GLP-1的生成与分泌——这一独特机制在其他胆汁酸种类中尚未发现。我们在TGR5敲除、肠道FXR激活及GLP-1受体抑制的小鼠模型中验证了上述发现。最后，通过临床队列研究证实，血清HCA浓度降低与糖尿病发病相关，且与血糖标志物水平密切相关。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病	bile acid /baɪl ˈæsɪd/ : 胆汁酸
glucose /ˈɡluːkʊs/ : 葡萄糖	homeostasis /ˌhoʊmiəˈsteɪsɪs/ : 稳态
enteroendocrine cells /ˌɛntərəʊˈɛndəkrɪn sɛlz/ : 肠内分泌细胞	receptor /rɪˈseptər/ : 受体
secretion /sɪˈkriːʃən/ : 分泌	blood glucose /blʌd ˈɡluːkʊs/ : 血糖
metabolism /məˈtæbəlaɪzəm/ : 代谢	cohort /ˈkoʊhɔːrt/ : 队列
glycemic /ɡlaɪˈsiːmɪk/ : 血糖的	

Dissection of type 2 diabetes: A genetic perspective

2型糖尿病的剖析：遗传学视角

期刊: The Lancet Diabetes & Endocrinology 日期: 2025

Abstract

diabetes is a leading cause of global mortality and disability, and its economic burden is substantial. This Review focuses on type 2 **diabetes**, which makes up 90-95% of all **diabetes** cases. Type 2 **diabetes** involves a progressive loss of **insulin secretion** often alongside **insulin resistance** and **metabolic syndrome**. Although **obesity** and a sedentary lifestyle are considerable contributors, research over the last 25 years has shown that type 2 **diabetes** develops on a predisposing **genetic** background, with family and twin studies indicating considerable **heritability** (ie, 31-72%). This Review explores type 2 **diabetes** from a **genetic** perspective, highlighting insights into its **pathophysiology** and the implications for **precision medicine**. More specifically, the traditional understanding of type 2 **diabetes** genetics has focused on a dichotomy between **monogenic** and **polygenic** forms. However, emerging evidence suggests a continuum that includes **monogenic**, **oligogenic**, and **polygenic** contributions, revealing their complementary roles in type 2 **diabetes pathophysiology**. Recent **genetic** studies provide deeper insights into disease mechanisms and pave the way for **precision medicine** approaches that could transform type 2 **diabetes** management. Additionally, the effect of environmental factors on type 2 **diabetes**, particularly from **epigenetic** modifications, adds another layer of complexity to understanding and addressing this multifaceted disease.

中文翻译

糖尿病是全球死亡和残疾的主要原因之一，其经济负担沉重。本综述聚焦于2型糖尿病，该类型占有所有糖尿病病例的90-95%。2型糖尿病涉及胰岛素分泌的进行性丧失，常伴随胰岛素抵抗和代谢综合征。尽管肥胖和久坐生活方式是重要诱因，但过去25年的研究表明，2型糖尿病是在遗传易感背景下发生的，家族及双胞胎研究显示其具有显著遗传性（即31-72%）。本文从遗传学视角探讨2型糖尿病，深入解析其病理生理机制及其对精准医疗的启示。具体而言，传统对2型糖尿病遗传学的理解主要集中于单基因与多基因形式的二分法。然而，新证据表明存在包含单基因、寡基因和多基因贡献的连续谱系，揭示了它们在2型糖尿病病理生理中的互补作用。近期遗传学研究为疾病机制提供了更深刻的见解，并为可能改变2型糖尿病管理模式的精准医疗方法铺平道路。此外，环境因素（特别是表观遗传修饰）对2型糖尿病的影响，为理解和应对这一复杂疾病增添了新的维度。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病	insulin secretion /ˈɪnsjʊlɪn sɪˈkriːʃən/ : 胰岛素分泌
insulin resistance /ˈɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗	metabolic syndrome /ˌmetəˈbɒlɪk ˈsɪndrəʊm/ : 代谢综合征
obesity /əʊˈbiːsəti/ : 肥胖症	genetic /dʒɪˈnetɪk/ : 遗传的
heritability /ˌherɪtəˈbɪləti/ : 遗传度	monogenic /ˌmɒnəʊˈdʒenɪk/ : 单基因的
polygenic /ˌpɒliˈdʒenɪk/ : 多基因的	oligogenic /ˌɒlɪɡəʊˈdʒenɪk/ : 寡基因的
pathophysiology /ˌpæθəʊˈfɪziːɒlədʒi/ : 病理生理学	precision medicine /prɪˈsɪʒən ˈmedɪsɪn/ : 精准医学
epigenetic /ˌepɪdʒɪˈnetɪk/ : 表观遗传的	

Hypofibrinolysis in type 2 diabetes and its clinical implications: From mechanisms to pharmacological modulation

2型糖尿病中的低纤溶状态及其临床意义：从机制到药物调控

期刊: Cardiovascular Diabetology 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 10.6

Abstract

AbstractA **prothrombotic state** is a typical feature of type 2 diabetes mellitus (T2DM). Apart from increased platelet reactivity, **endothelial dysfunction**, **hyperfibrinogenemia**, and **hypofibrinolysis** are observed in T2DM. A variety of poorly elucidated mechanisms behind impaired **fibrinolysis** in this disease have been reported, indicating complex associations between platelet activation, fibrin formation and clot structure, and **fibrinolysis** inhibitors, in particular, elevated **plasminogen** antigen inhibitor-1 levels which are closely associated with obesity. Abnormal fibrin clot structure is of paramount importance for relative resistance to plasmin-mediated lysis in T2DM. Enhanced **thrombin** generation, a proinflammatory state, increased release of neutrophil extracellular traps, elevated complement C3, along with posttranslational modifications of **fibrinogen** and **plasminogen** have been regarded to contribute to altered clot structure and impaired **fibrinolysis** in T2DM. Antidiabetic agents such as metformin and insulin, as well as antithrombotic agents, including **anticoagulants**, have been reported to improve fibrin properties and accelerate **fibrinolysis** in T2DM. Notably, recent evidence shows that **hypofibrinolysis**, assessed in plasma-based assays, has a predictive value in terms of **cardiovascular events** and cardiovascular mortality in T2DM patients. This review presents the current data on the mechanisms underlying arterial and venous **thrombotic complications** in T2DM patients, with an emphasis on **hypofibrinolysis** and its impact on clinical outcomes. We also discuss potential modulators of **fibrinolysis** in the search for optimal therapy in diabetic patients.

中文翻译

摘要：促血栓状态是2型糖尿病（T2DM）的典型特征。除了血小板反应性增强外，T2DM患者还表现出内皮功能障碍、高纤维蛋白原血症和低纤溶活性。已有研究报道了该疾病中纤溶受损背后多种尚未充分阐明的机制，表明血小板活化、纤维蛋白形成与凝块结构以及纤溶抑制剂之间存在复杂的关联，特别是与肥胖密切相关的纤溶酶原激活物抑制剂-1水平升高。异常的纤维蛋白凝块结构对T2DM中纤溶酶介导的溶解相对抵抗至关重要。增强的凝血酶生成、促炎状态、中性粒细胞胞外陷阱释放增加、补体C3升高，以及纤维蛋白原和纤溶酶原的翻译后修饰，都被认为与T2DM中凝块结构改变和纤溶受损有关。据报道，二甲双胍和胰岛素等降糖药物，以及包括抗凝剂在内的抗血栓药物，可以改善T2DM患者的纤维蛋白特性并加速纤溶。值得注意的是，最近的证据表明，基于血浆检测评估的低纤溶活性对T2DM患者的心血管事件和心血管死亡率具有预测价值。本综述介绍了当前关于T2DM患者动脉和静脉血栓并发症机制的数据，重点探讨低纤溶活性及其对临床结局的影响。我们还讨论了纤溶的潜在调节剂，以寻求糖尿病患者的最佳治疗方案。

核心词汇

prothrombotic state /prəʊθrɒmˈbɒtɪk steɪt/ : 促血栓状态

endothelial dysfunction /ˌendəʊˈθiːliəl dɪsˈfʌŋkʃən/ : 内皮功能障碍

hyperfibrinogenemia /ˌhaɪpəˈfaɪbrɪnədʒəˈniːmiə/ : 高纤维蛋白原血症

hypofibrinolysis /ˌhaɪpəʊˈfaɪbrɪˈnɒlɪsɪs/ : 低纤溶状态

fibrinolysis /ˌfaɪbrɪˈnɒlɪsɪs/ : 纤维蛋白溶解

plasminogen /plæzˈmɪnədʒən/ : 纤溶酶原

fibrinogen /ˈfaɪˈbrɪnədʒən/ : 纤维蛋白原

thrombin /ˈθrɒmbɪn/ : 凝血酶

anticoagulants /ˌæntɪkəʊˈæɡjələnts/ : 抗凝剂

cardiovascular events /ˌkɑːrdɪəʊˈvæskjələr ɪˈvents/ : 心血管事件

thrombotic complications /θrɒmˈbɒtɪk ˌkɒmplɪˈkeɪʃnz/ : 血栓性并发症

Human mesenchymal stem cell derived exosomes alleviate type 2 diabetes mellitus by reversing peripheral insulin resistance and relieving β -cell destruction

人类间充质干细胞来源的外泌体通过逆转外周胰岛素抵抗和缓解 β 细胞破坏来缓解2型糖尿病

期刊: ACS Nano 日期: 2018 JCR Q1 中科院升级版 材料科学1区 IF 16.0

Abstract

Exosomes are nanosized extracellular vesicles (EVs) that show great promise in tissue regeneration and injury repair as **mesenchymal stem cell** (MSC). MSC has been shown to alleviate **diabetes mellitus** (DM) in both animal models and clinical trials. In this study, we aimed to investigate whether exosomes from human umbilical cord MSC (hucMSC-ex) have a therapeutic effect on type 2 DM (T2DM). We established a rat model of T2DM using a high-fat diet and streptozotocin (STZ). We found that the **intravenous injection** of hucMSC-ex reduced blood glucose levels as a main paracrine approach of MSC. HucMSC-ex partially reversed **insulin resistance** in T2DM indirectly to accelerate **glucose metabolism**. HucMSC-ex restored the phosphorylation (tyrosine site) of the **insulin receptor** substrate 1 and **protein kinase B** in T2DM, promoted expression and membrane translocation of glucose transporter 4 in muscle, and increased storage of **glycogen** in the liver to maintain glucose homeostasis. HucMSC-ex inhibited STZ-induced β -cell **apoptosis** to restore the insulin-secreting function of T2DM. Taken together, exosomes from hucMSC can alleviate T2DM by reversing peripheral **insulin resistance** and relieving β -cell destruction, providing an alternative approach for T2DM treatment.

中文翻译

外泌体是纳米级的细胞外囊泡，作为间充质干细胞在组织再生和损伤修复方面展现出巨大潜力。间充质干细胞在动物模型和临床试验中均显示出缓解糖尿病的作用。本研究旨在探讨人脐带间充质干细胞来源的外泌体是否对2型糖尿病具有治疗作用。我们通过高脂饮食联合链脲佐菌素诱导建立了2型糖尿病大鼠模型。研究发现，静脉注射人脐带间充质干细胞外泌体作为间充质干细胞的主要旁分泌途径，能够有效降低血糖水平。该外泌体通过间接逆转2型糖尿病胰岛素抵抗状态，加速葡萄糖代谢过程。其作用机制包括：恢复2型糖尿病模型中胰岛素受体底物1酪氨酸位点及蛋白激酶B的磷酸化水平，促进肌肉组织中葡萄糖转运蛋白4的表达与膜转位，增加肝脏糖原储存以维持葡萄糖稳态。同时，人脐带间充质干细胞外泌体能够抑制链脲佐菌素诱导的 β 细胞凋亡，恢复2型糖尿病的胰岛素分泌功能。综上所述，人脐带间充质干细胞外泌体可通过逆转外周胰岛素抵抗和减轻 β 细胞损伤来缓解2型糖尿病，为2型糖尿病治疗提供了新的潜在策略。

核心词汇

exosome /'ɛksəˌsəʊm/ : 外泌体

mesenchymal stem cell /mɛˈsɛŋkɪməl stɛm sɛl/ : 间充质干细胞

insulin resistance /'ɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗

insulin receptor /'ɪnsjʊlɪn rɪˈseptər/ : 胰岛素受体

glycogen /'glʌɪkəʊdʒən/ : 糖原

intravenous injection /ˌɪntrəˈviːnəs ɪnˈdʒɛkʃən/ : 静脉注射

extracellular vesicle /ˌɛkstrəˈsɛljʊlər ˈvesɪkəl/ : 细胞外囊泡

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈlaɪtəs/ : 糖尿病

glucose metabolism /'gluːkəʊs məˈtæbəˌlɪzəm/ : 葡萄糖代谢

protein kinase B /ˈprəʊtɪːn ˈkaɪneɪz biː/ : 蛋白激酶B

apoptosis /ˌæpəˈtəʊsɪs/ : 细胞凋亡

Subtypes of Type 2 Diabetes Determined From Clinical Parameters

2型糖尿病亚型由临床参数确定

期刊: Diabetes 日期: 2020 JCR Q1 中科院升级版 医学2区 IF 7.5

Abstract

Type 2 **diabetes** (T2D) is defined by a single metabolite, **glucose**, but is increasingly recognized as a highly heterogeneous disease, including individuals with varying clinical characteristics, disease progression, drug response, and risk of complications. Identification of subtypes with differing risk profiles and disease etiologies at diagnosis could open up avenues for **personalized medicine** and allow clinical resources to be focused to the patients who would be most likely to develop diabetic complications, thereby both improving patient health and reducing costs for the health sector. More homogeneous populations also offer increased power in experimental, genetic, and clinical studies. Clinical parameters are easily available and reflect relevant disease pathways, including the effects of both genetic and environmental exposures. We used six clinical parameters (GAD autoantibodies, age at **diabetes** onset, HbA1c, BMI, and measures of **insulin resistance** and **insulin secretion**) to cluster adult-onset **diabetes** patients into five subtypes. These subtypes have been robustly reproduced in several populations and associated with different risks of complications, comorbidities, genetics, and response to treatment. Importantly, the group with severe insulin-deficient **diabetes** (SIDD) had increased risk of **retinopathy** and **neuropathy**, whereas the severe insulin-resistant **diabetes** (SIRD) group had the highest risk for **diabetic kidney disease** (DKD) and **fatty liver**, emphasizing the importance of **insulin resistance** for DKD and **hepatosteatorsis** in T2D. In conclusion, we believe that subclassification using these highly relevant parameters could provide a framework for **personalized medicine** in **diabetes**.

中文翻译

2型糖尿病 (T2D) 虽以单一代谢物——葡萄糖为诊断依据，但日益被认识到是一种高度异质性疾病，患者的临床特征、疾病进展、药物反应及并发症风险各不相同。在诊断时识别具有不同风险特征和疾病病因的亚型，可为个体化医疗开辟新途径，并使临床资源能够集中用于最可能发生糖尿病并发症的患者，从而既改善患者健康，又降低医疗部门的成本。更同质化的群体还能提高实验、遗传和临床研究的效力。临床参数易于获取，且反映了相关疾病通路，包括遗传和环境暴露的影响。我们使用六项临床参数（GAD自身抗体、糖尿病发病年龄、HbA1c、BMI、胰岛素抵抗和胰岛素分泌指标）将成年发病的糖尿病患者分为五个亚型。这些亚型已在多个群体中得到稳健复现，并与不同的并发症风险、合并症、遗传特征及治疗反应相关联。重要的是，严重胰岛素缺乏型糖尿病（SIDD）患者群体患视网膜病变和神经病变的风险增加，而严重胰岛素抵抗型糖尿病（SIRD）群体患糖尿病肾病（DKD）和脂肪肝的风险最高，这强调了胰岛素抵抗在T2D患者发生DKD和肝脂肪变性中的重要性。总之，我们认为使用这些高度相关的参数进行亚型分类，可为糖尿病的个体化医疗提供一个框架。

核心词汇

diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病	glucose /ˈɡluːkʊs/ : 葡萄糖
insulin resistance /ˈɪnsəlɪn rɪˈzɪstəns/ : 胰岛素抵抗	insulin secretion /ˈɪnsəlɪn sɪˈkriːʃən/ : 胰岛素分泌
retinopathy /ˌretɪˈnɒpəθi/ : 视网膜病变	neuropathy /njuˈrɒpəθi/ : 神经病变
diabetic kidney disease /ˌdaɪəˈbetɪk ˈkɪdni dɪˈziːz/ : 糖尿病肾病	fatty liver /ˈfæti ˈlɪvər/ : 脂肪肝
hepatosteatorsis /ˌhepəˈtɒstɪːˈæʊtəʊsɪs/ : 肝脂肪变性	comorbidity /ˌkɒmɔːrˈbɪdəti/ : 共病
personalized medicine /ˈpɜːrsənəlaɪzd ˈmedɪsɪn/ : 个性化医疗	

A novel GLP-1 and FGF21 dual agonist has therapeutic potential for diabetes and non-alcoholic steatohepatitis

一种新型GLP-1和FGF21双重激动剂对糖尿病和非酒精性脂肪性肝炎具有治疗潜力

期刊: EBioMedicine 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 10.8

Abstract

BACKGROUND: **fibroblast growth factor** 21 (FGF21) has become a promising therapeutic target for metabolic diseases such as type 2 diabetes (T2D), **obesity** and non-alcoholic steatohepatitis. However, the clinical application of natural FGF21 molecule is limited because of its instability in vitro and short half-life in vivo. To improve FGF21's therapeutic property, we screened high receptor binding FGF21 analogs and made FGF21-Fc-GLP-1 dual-targeted constructs to investigate their activity in a number of experiments. **METHODS:** Utilizing phage display high-throughput screening we identified mutations that could improve β -Klotho binding property of FGF21. IgG4 Fc was fused to FGF21 variants to extend the in vivo half-life. We further explored the potential synergistic actions of FGF21 with the **incretin** glucagon-like peptide-1 (GLP-1) by generating GLP-1-Fc-FGF21 dual agonists. **FINDINGS:** Two Fc-FGF21 variants showed enhanced β -Klotho binding affinity in vitro as well as improved **glucose lowering** effect in vivo. One of the dual agonists, GLP-1-Fc-FGF21 D1, provided potent and sustained **glucose lowering** effect in diabetic mice models. It also demonstrated superior **weight loss** effect to GLP-1 or FGF21 alone. Moreover, GLP-1-Fc-FGF21 D1 exhibited strong anti-NASH effect in the high-fat diet-induced ob/ob model as it improved **liver function**, serum and hepatic **lipid profile** and reduced NAFLD activity score with an efficacy superior to either FGF21 or GLP-1 analogs alone. **INTERPRETATION:** This novel GLP-1/FGF21 dual **agonist** is worth clinical development for the treatment of T2D, **obesity** and NASH. **FUNDING:** HEC Pharm R&D Co., Ltd, National natural science fund of China.

中文翻译

背景: 成纤维细胞生长因子21 (FGF21) 已成为治疗2型糖尿病 (T2D)、肥胖和非酒精性脂肪性肝炎等代谢性疾病的有前景的靶点。然而,天然FGF21分子在体外不稳定、体内半衰期短,限制了其临床应用。为改善FGF21的治疗特性,我们筛选了高受体结合力的FGF21类似物,并构建了FGF21-Fc-GLP-1双靶向分子,通过系列实验探究其活性。**方法:** 利用噬菌体展示高通量筛选技术,我们鉴定出能提升FGF21与 β -Klotho结合能力的突变位点。将IgG4 Fc与FGF21变体融合以延长体内半衰期。我们进一步通过构建GLP-1-Fc-FGF21双激动剂,探索FGF21与肠促胰岛素胰高血糖素样肽-1 (GLP-1) 的潜在协同作用。**结果:** 两种Fc-FGF21变体在体外显示出增强的 β -Klotho结合亲和力,并在体内表现出改善的降糖效果。其中一种双激动剂GLP-1-Fc-FGF21 D1在糖尿病小鼠模型中展现出强效且持久的降糖作用,其减重效果也优于单独使用GLP-1或FGF21。此外,在高脂饮食诱导的ob/ob模型中, GLP-1-Fc-FGF21 D1表现出强大的抗NASH作用,能改善肝功能、血清及肝脏脂质谱,降低NAFLD活动度评分,其疗效优于单独使用FGF21或GLP-1类似物。**结论:** 这种新型GLP-1/FGF21双激动剂值得临床开发,用于治疗T2D、肥胖及NASH。资助方: 东阳光药业研发有限公司,国家自然科学基金。

核心词汇

fibroblast growth factor /'faɪbrɒs,blæst gruθ 'fæktər/ : 成纤维细胞生长因子

incretin /ɪn 'kri:tɪn/ : 肠促胰岛素

glucose lowering /'glu:kəʊs 'ləʊərɪŋ/ : 降血糖

liver function /'lɪvər 'fʌŋkʃən/ : 肝功能

diabetic mice model /,daɪə'betɪk maɪs 'mɒdəl/ : 糖尿病小鼠模型

obesity /ou'bi:səti/ : 肥胖症

agonist /'æɡənɪst/ : 激动剂

weight loss /weɪt lɒs/ : 减重

lipid profile /'lɪpɪd 'prəʊfaɪl/ : 血脂谱

Use of glucagon-like peptide-1 receptor agonists in patients with type 2 diabetes and cardiovascular disease

胰高血糖素样肽-1受体激动剂在2型糖尿病及心血管疾病患者中的应用

期刊: JAMA Cardiology 日期: 2020 JCR Q1 中科院升级版 医学1区 IF 14.1

Abstract

IMPORTANCE: Recent **randomized clinical trials** have demonstrated that glucagon-like peptide-1 receptor agonists (GLP-1RAs) reduce **cardiovascular events** in at-risk individuals with type 2 diabetes. Despite these findings, GLP-1RAs are underused in eligible patients, particularly by **cardiologists**. **OBSERVATIONS:** To date, **randomized clinical trials** of albiglutide, dulaglutide, liraglutide, and injectable semaglutide have reported favorable cardiovascular outcomes. Most recently approved for clinical use, oral semaglutide has a favorable safety profile and is currently undergoing regulatory evaluation and further study for cardiovascular outcomes. Professional society guidelines now recommend GLP-1RA therapy for cardiovascular risk mitigation in patients with type 2 diabetes and established **atherosclerotic cardiovascular disease** (ASCVD) or multiple ASCVD risk factors, independent of glucose control or background antihyperglycemic therapy (other diabetes medications being used). Additional conditions suitable for GLP-1RA therapy include **obesity** and advanced **chronic kidney disease** (estimated **glomerular filtration rate** <30 mL/min/1.73 m²), for which cardiovascular risk-reducing options are limited. Out-of-pocket costs and secondary advantages (eg, weight loss) may inform shared decision-making discussions regarding potential therapies. GLP-1RA therapy has a favorable safety profile. Its most common adverse effect is **gastrointestinal upset**, which typically wanes during the early weeks of therapy and may be mitigated by starting at the lowest dose and escalating as tolerated. Depending on baseline glycemic control, **sulfonylureas** and **insulin** may need to be decreased before GLP-1RA initiation; without concurrent use of **insulin** or **sulfonylureas**, GLP-1RAs are not associated with **hypoglycemia**. Multidisciplinary follow-up and collaborative care with primary care physicians and/or **endocrinologists** are important. **CONCLUSIONS AND RELEVANCE:** Findings from this review suggest that GLP-1RAs are safe, are well tolerated, and improve cardiovascular outcomes, largely independent of their antihyperglycemic properties, but they remain underused by **cardiologists**. This review provides a practical resource for **cardiologists** for initiating GLP-1RAs and managing the therapy in patients with type 2 diabetes and established ASCVD or high risk for ASCVD.

中文翻译

重要性: 近期随机临床试验表明, 胰高血糖素样肽-1受体激动剂 (GLP-1RAs) 能降低2型糖尿病高危人群的心血管事件风险。尽管有这些发现, GLP-1RAs在符合条件的患者中使用不足, 尤其是在心脏病专家中。观察结果: 迄今为止, 关于阿必鲁肽、度拉糖肽、利拉鲁肽和注射用司美格鲁肽的随机临床试验均报告了良好的心血管结局。最新获批临床使用的口服司美格鲁肽安全性良好, 目前正在接受监管评估和进一步的心血管结局研究。专业学会指南现推荐使用GLP-1RA治疗来降低2型糖尿病合并已确诊动脉粥样硬化性心血管疾病 (ASCVD) 或多个ASCVD危险因素患者的心血管风险, 这与血糖控制或背景降糖治疗 (正在使用的其他糖尿病药物) 无关。适合GLP-1RA治疗的其他情况包括肥胖和晚期慢性肾病 (估计肾小球滤过率 <30 mL/min/1.73 m²), 这些情况中心血管风险降低的选择有限。自付费用和次要优势 (如减轻体重) 可能影响关于潜在治疗的共同决策讨论。GLP-1RA治疗安全性良好。其最常见的不良反应是胃肠道不适, 通常在治疗初期几周内逐渐减轻, 可通过从最低剂量开始并根据耐受性逐渐增加剂量来缓解。根据基线血糖控制情况, 在开始GLP-1RA治疗前可能需要减少磺脲类药物和胰岛素的用量; 在不与胰岛素或磺脲类药物同时使用的情况下, GLP-1RAs不会引起低血糖。多学科随访以及与初级保健医生和/或内分泌科医生的协作护理非常重要。结论与相关性: 本综述的研究结果表明, GLP-1RAs安全、耐受性良好, 并能改善心血管结局, 这很大程度上与其降糖特性无关, 但心脏病专家对其使用仍然不足。本综述为心脏病专家在2型糖尿病合并已确诊ASCVD或ASCVD高风险患者中启动GLP-1RAs治疗和管理该疗法提供了实用资源。

核心词汇

cardiovascular events /ˌkɑːrdiəʊˈvæskjələr ɪˈvents/ : 心血管事件

obesity /ouˈbiːsəti/ : 肥胖症

glomerular filtration rate /ɡləˈmɜːljələr fɪlˈtreɪʃən reɪt/ : 肾小球滤过率

hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖症

cardiologists /ˌkɑːrdiˈɒlədʒɪsts/ : 心脏病学家

atherosclerotic cardiovascular disease /ˌæθərəʊskləˈrɒtɪk ˌkɑːrdiəʊˈvæskjələr dɪˈziːz/ : 动脉粥样硬化性心血管疾病

chronic kidney disease /ˈkrɒnɪk ˈkɪdni dɪˈziːz/ : 慢性肾脏病

gastrointestinal upset /ˌɡæstrəʊɪnˈtestɪnəl ˈʌpset/ : 胃肠道不适

randomized clinical trials /ˈrændəmaɪzd ˈklɪnɪkəl ˈtraɪəlz/ : 随机临床试验

endocrinologists /ˌendəʊkrɪˈnɒlədʒɪsts/ : 内分泌学家

insulin /ˈɪnsjʊlɪn/ : 胰岛素

sulfonylureas /ˌsʌlfəniˈjʊəriəz/ : 磺脲类药物

云梦山-侵权必究

Combining glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium-glucose cotransporter-2 inhibitors (SGLT2is) in patients with type 2 diabetes mellitus (T2DM)

联合应用胰高血糖素样肽-1受体激动剂 (GLP-1RAs) 与钠-葡萄糖协同转运蛋白2抑制剂 (SGLT2is) 治疗2型糖尿病 (T2DM) 患者

期刊: Cardiovascular Diabetology 日期: 2023

JCR Q1

中科院升级版 医学1区

IF 10.6

Abstract

Abstract Due to their cardiovascular protective effect, glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium-glucose cotransporter-2 inhibitors (SGLT2is) represent breakthrough therapies for type 2 diabetes mellitus (T2DM). In this review article, we discuss the mechanistic and clinical synergies that make the combined use of GLP-1RAs and SGLT2is appealing in patients with T2DM. Overall, the presented cumulative evidence supports the benefits of GLP-1RA plus SGLT2i combination therapy on metabolic-cardiovascular-renal disease in patients with T2DM, with a low hypoglycemia risk. Accordingly, we encourage the adoption of GLP-1RA plus SGLT2i combination therapy in patients with T2DM and established atherosclerotic cardiovascular disease (ASCVD) or multiple risk factors for ASCVD (i.e., age ≥ 55 years, overweight/obesity, dyslipidemia, hypertension, current tobacco use, left ventricular hypertrophy, and/or proteinuria). Regarding renal effects, the evidence of SGLT2is in preventing kidney failure is more abundant than for GLP-1RAs, which showed a beneficial effect on albuminuria but not on hard kidney endpoints. Hence, in case of persistent albuminuria and/or uncontrolled metabolic risks (i.e., inadequate glycemic control, hypertension, overweight/obesity) on SGLT2i therapy, GLP-1RAs should be considered as the preferential add-on therapy in T2DM patients with chronic kidney disease. Despite the potential clinical benefits of GLP-1RA plus SGLT2i combination therapy in patients with T2DM, several factors may delay this combination to become a common practice soon, such as reimbursement and costs associated with polypharmacy. Altogether, when administering GLP-1RA plus SGLT2i combination therapy, it is important to adopt an individualized approach to therapy taking into account individual preferences, costs and coverage, toxicity profile, consideration of kidney function and glucose-lowering efficacy, desire for weight loss, and comorbidities.

中文翻译

摘要: 胰高血糖素样肽-1受体激动剂 (GLP-1RAs) 和钠-葡萄糖协同转运蛋白2抑制剂 (SGLT2is) 因其心血管保护作用, 已成为2型糖尿病 (T2DM) 治疗领域的突破性疗法。本文综述了GLP-1RAs与SGLT2is联合应用的机制协同性与临床优势。现有证据表明, GLP-1RA与SGLT2i联合治疗对T2DM患者的代谢-心血管-肾脏疾病具有显著获益, 且低血糖风险较低。因此, 我们建议对已确诊动脉粥样硬化性心血管疾病 (ASCVD) 或具有多重ASCVD危险因素 (即年龄 ≥ 55 岁、超重/肥胖、血脂异常、高血压、当前吸烟、左心室肥厚和/或蛋白尿) 的T2DM患者采用GLP-1RA与SGLT2i联合治疗方案。在肾脏保护方面, SGLT2is预防肾衰竭的证据较GLP-1RAs更为充分, 后者虽对蛋白尿有改善作用, 但对硬性肾脏终点事件的影响尚未明确。因此, 对于接受SGLT2i治疗后仍存在持续性蛋白尿和/或代谢风险控制不佳 (即血糖控制不达标、高血压、超重/肥胖) 的慢性肾脏病T2DM患者, 应优先考虑加用GLP-1RAs。尽管GLP-1RA与SGLT2i联合治疗对T2DM患者具有潜在临床获益, 但报销政策及多药治疗成本等因素可能延缓该联合方案的普及。总体而言, 实施GLP-1RA与SGLT2i联合治疗时, 需综合考虑患者个体偏好、治疗成本与医保覆盖、药物毒性特征、肾功能状况、降糖疗效、减重需求及合并症等因素, 采取个体化治疗策略。

核心词汇

atherosclerotic cardiovascular disease /ˌæθərəʊskləˈrɒtɪk ˌkɑːrdiəʊˈvæskjələr dɪˈziːz/ : 动脉粥样硬化性心血管疾病

dyslipidemia /dɪsˌlɪpɪˈdiːmiə/ : 血脂异常

obesity /ouˈbiːsəti/ : 肥胖

albuminuria /ælˌbjuːmɪˈnjuəriə/ : 蛋白尿

kidney failure /ˈkɪdni ˈfeɪljər/ : 肾衰竭

proteinuria /ˌprəʊtiːnˈjuəriə/ : 蛋白尿

glycemic control /ˈɡlaɪˈsiːmɪk kənˈtrəʊl/ : 血糖控制

hypertension /ˌhaɪpərˈtɛnʃən/ : 高血压

overweight /ˌoʊvərweɪt/ : 超重

hypoglycemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖

chronic kidney disease /ˈkrɒnɪk ˈkɪdni dɪˈziːz/ : 慢性肾脏病

left ventricular hypertrophy /left venˈtrɪkjələr haɪˈpɜːrtroʊfi/ : 左心室肥厚

combination therapy /ˌkɒmbɪˈneɪʃən ˈθerəpi/ : 联合治疗

comorbidities /kəʊmɔːrˈbɪdɪtiz/ : 合并症

Glucagon-like peptide 1 receptor agonists: Cardiovascular benefits and mechanisms of action

胰高血糖素样肽-1受体激动剂：心血管益处与作用机制

期刊: Nature Reviews Cardiology 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 44.2

Abstract

Type 2 **diabetes mellitus** (T2DM) and **obesity** are metabolic disorders characterized by excess **cardiovascular** risk. Glucagon-like peptide 1 (GLP1) **receptor** (GLP1R) agonists reduce body weight, **glycaemia**, **blood pressure**, postprandial lipaemia and **inflammation** - actions that could contribute to the reduction of **cardiovascular** events. **cardiovascular** outcome trials (CVOTs) have demonstrated that GLP1R agonists reduce the rates of major adverse **cardiovascular** events in patients with T2DM. Separate phase III CVOTs of GLP1R agonists are currently being conducted in people living with **heart failure** with preserved **ejection fraction** and in those with **obesity**. Mechanistically, GLP1R is expressed at low levels in the heart and **vasculature**, raising the possibility that GLP1 might have both direct and indirect actions on the **cardiovascular** system. In this Review, we summarize the data from CVOTs of GLP1R agonists in patients with T2DM and describe the actions of GLP1R agonists on the heart and blood vessels. We also assess the potential mechanisms that contribute to the reduction in major adverse **cardiovascular** events in individuals treated with GLP1R agonists and highlight the emerging **cardiovascular** biology of novel GLP1-based multi-agonists currently in development. Understanding how GLP1R signalling protects the heart and blood vessels will optimize the **therapeutic** use and development of next-generation GLP1-based therapies with improved **cardiovascular** safety.

中文翻译

2型糖尿病（T2DM）与肥胖症是以心血管风险增高为特征的代谢性疾病。胰高血糖素样肽-1（GLP1）受体（GLP1R）激动剂能够降低体重、血糖、血压、餐后血脂及炎症水平——这些作用可能有助于减少心血管事件的发生。心血管结局试验（CVOTs）已证实，GLP1R激动剂可降低T2DM患者主要不良心血管事件的发生率。目前，针对射血分数保留型心力衰竭患者及肥胖人群的GLP1R激动剂独立三期心血管结局试验正在进行中。从机制上看，GLP1R在心脏及血管系统中表达水平较低，这提示GLP1可能对心血管系统同时具有直接与间接作用。本文综述了GLP1R激动剂在T2DM患者中的心血管结局试验数据，阐述了其对心脏和血管的作用机制，评估了该类药物降低主要不良心血管事件的潜在机理，并重点探讨了基于GLP1的新型多靶点激动剂在研发中展现的心血管生物学新进展。深入理解GLP1R信号通路如何保护心脏与血管系统，将有助于优化现有疗法的临床应用，并推动具有更佳心血管安全性的新一代GLP1靶向疗法的研究进程。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈlaɪtəs/ : 糖尿病

cardiovascular /ˌkɑːrdiəʊˈvæskjələ/ : 心血管的

ejection fraction /ɪˈdʒɛkʃən ˈfrækʃən/ : 射血分数

blood pressure /blʌd ˈpreʃər/ : 血压

agonist /ˈæɡənɪst/ : 激动剂

vasculature /ˈvæskjələtʃər/ : 血管系统

obesity /ouˈbiːsəti/ : 肥胖症

heart failure /haːrt ˈfeɪljər/ : 心力衰竭

glycaemia /ɡlaɪˈsiːmiə/ : 血糖

inflammation /ɪnˈfleɪməʃən/ : 炎症

receptor /rɪˈseptər/ : 受体

therapeutic /ˌθerəˈpiːtɪk/ : 治疗的

Pharmacological treatment of hyperglycemia in type 2 diabetes

2型糖尿病高血糖的药物治疗

期刊: Journal of Clinical Investigation 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 13.6

Abstract

diabetes mellitus is a major public health problem, affecting about 10% of the population. **pharmacotherapy** aims to protect against **microvascular complications**, including **blindness**, end-stage kidney disease, and **amputations**. Landmark clinical trials have demonstrated that intensive **glycemic control** slows progression of **microvascular complications** (**retinopathy**, **nephropathy**, and **neuropathy**). Long-term follow-up has demonstrated that intensive **glycemic control** also decreases risk of **macrovascular disease**, albeit rigorous evidence of macrovascular benefit did not emerge for over a decade. The US FDA's recent requirement for dedicated **cardiovascular outcome trials** ushered in a golden age for understanding the clinical profiles of new type 2 diabetes drugs. Some clinical trials with sodium-glucose cotransporter-2 (SGLT2) inhibitors and glucagon-like peptide 1 (GLP1) receptor agonists reported data demonstrating cardiovascular benefit (decreased risk of major adverse cardiovascular events and hospitalization for **heart failure**) and slower progression of diabetic kidney disease. This Review discusses current guidelines for use of the 12 classes of drugs approved to promote **glycemic control** in patients with type 2 diabetes. The Review also anticipates future developments with potential to improve the standard of care: availability of generic dipeptidylpeptidase-4 (DPP4) inhibitors and SGLT2 inhibitors; **precision medicine** to identify the best drugs for individual patients; and new therapies to protect against chronic complications of diabetes.

中文翻译

糖尿病是一个重大的公共卫生问题，影响约10%的人口。药物治疗旨在预防微血管并发症，包括失明、终末期肾病和截肢。具有里程碑意义的临床试验表明，强化血糖控制可延缓微血管并发症（视网膜病变、肾病和神经病变）的进展。长期随访显示，强化血糖控制也能降低大血管疾病的风险，尽管大血管获益的严格证据在十多年后才出现。美国FDA最近要求进行专门的心血管结局试验，这为理解新型2型糖尿病药物的临床特征开启了一个黄金时代。一些关于钠-葡萄糖协同转运蛋白2 (SGLT2) 抑制剂和胰高血糖素样肽1 (GLP1) 受体激动剂的临床试验报告了显示心血管获益（降低主要不良心血管事件和心力衰竭住院风险）以及延缓糖尿病肾病进展的数据。本综述讨论了目前用于2型糖尿病患者血糖控制的12类药物使用指南。综述还展望了未来可能改善护理标准的发展：通用型二肽基肽酶-4 (DPP4) 抑制剂和SGLT2抑制剂的可用性；精准医学以确定个体患者的最佳药物；以及预防糖尿病慢性并发症的新疗法。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtiːz məˈlaɪtəs/ : 糖尿病

retinopathy /ˌrɛtɪˈnɒpəθi/ : 视网膜病变

neuropathy /njuːˈrɒpəθi/ : 神经病变

cardiovascular outcome trials /ˌkɑːrdiəʊˈvæskjələr ˈaʊtkʌm ˈtraɪəlz/ : 心血管结局试验

pharmacotherapy /ˌfɑːrməˈkəʊθerəpi/ : 药物治疗

blindness /ˈblaɪndnəs/ : 失明

precision medicine /prɪˈsɪʒən ˈmɛdɪsɪn/ : 精准医疗

microvascular complications /ˌmaɪkrəʊˈvæskjələr ˌkɒmplɪˈkeɪʃnz/ : 微血管并发症

nephropathy /nɪˈfrɒpəθi/ : 肾病

macrovascular disease /ˌmækrəʊˈvæskjələr dɪˈziːz/ : 大血管疾病

glycemic control /ˌɡlaɪˈsiːmɪk kənˈtrɒl/ : 血糖控制

amputations /ˌæmpjʊˈteɪʃnz/ : 截肢

heart failure /hɑːrt ˈfeɪljər/ : 心力衰竭

Monogenic diabetes

单基因糖尿病

期刊: Nature Reviews Disease Primers 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 60.6

Abstract

monogenic diabetes includes several clinical conditions generally characterized by early-onset diabetes, such as **neonatal diabetes**, maturity-onset diabetes of the young (MODY) and various diabetes-associated syndromes. However, patients with apparent type 2 diabetes mellitus may actually have **monogenic diabetes**. Indeed, the same **monogenic diabetes** gene can contribute to different forms of diabetes with early or late onset, depending on the functional impact of the variant, and the same pathogenic variant can produce variable diabetes phenotypes, even in the same family. **monogenic diabetes** is mostly caused by impaired function or development of **pancreatic islets**, with defective **insulin secretion** in the absence of **obesity**. The most prevalent form of **monogenic diabetes** is MODY, which may account for 0.5-5% of patients diagnosed with non-autoimmune diabetes but is probably underdiagnosed owing to insufficient genetic testing. Most patients with **neonatal diabetes** or MODY have **autosomal dominant** diabetes. More than 40 subtypes of **monogenic diabetes** have been identified to date, the most prevalent being deficiencies of GCK and HNF1A. **precision medicine** approaches (including specific treatments for **hyperglycaemia**, monitoring associated extra-pancreatic phenotypes and/or following up clinical trajectories, especially during pregnancy) are available for some forms of **monogenic diabetes** (including GCK- and HNF1A-diabetes) and increase patients' quality of life. Next-generation sequencing has made **genetic diagnosis** affordable, enabling effective **genomic medicine** in **monogenic diabetes**.

中文翻译

单基因糖尿病包括多种临床状况，通常以早发糖尿病为特征，如新生儿糖尿病、青年成熟期发病型糖尿病（MODY）以及各种与糖尿病相关的综合征。然而，表面上患有2型糖尿病的患者实际上可能患有单基因糖尿病。事实上，同一单基因糖尿病基因可导致早发或晚发的不同形式糖尿病，这取决于变异的功能影响，而同一致病性变异可能产生不同的糖尿病表型，即使在同一个家庭中也是如此。单基因糖尿病主要由胰岛功能或发育受损引起，在无肥胖情况下胰岛素分泌缺陷。最常见的单基因糖尿病形式是MODY，可能占诊断为非自身免疫性糖尿病患者的0.5-5%，但由于基因检测不足，可能被低估诊断。大多数新生儿糖尿病或MODY患者患有常染色体显性糖尿病。迄今为止，已发现超过40种单基因糖尿病亚型，最常见的是GCK和HNF1A缺陷。针对某些形式的单基因糖尿病（包括GCK和HNF1A糖尿病），精准医学方法（包括针对高血糖的特定治疗、监测相关的胰腺外表型和/或跟踪临床轨迹，特别是在妊娠期间）是可用的，并能提高患者的生活质量。下一代测序技术使基因诊断变得经济实惠，从而在单基因糖尿病中实现了有效的基因组医学。

核心词汇

monogenic diabetes /ˌmɒnəˈdʒɛnɪk daɪəˈbiːtiːz/ : 单基因糖尿病	neonatal diabetes /ˌniːəʊˈneɪtəl daɪəˈbiːtiːz/ : 新生儿糖尿病
pancreatic islets /ˌpæŋkriˈætɪk ˈaɪlɪts/ : 胰岛	insulin secretion /ˈɪnsjʊlɪn sɪˈkriːʃən/ : 胰岛素分泌
obesity /əʊˈbiːsɪti/ : 肥胖	autosomal dominant /ˌɔːtəʊˈsəʊməl ˈdɒmɪnənt/ : 常染色体显性
hyperglycaemia /ˌhaɪpəˈɡlaɪˈsiːmiə/ : 高血糖	genetic diagnosis /dʒɪˈnetɪk ˌdaɪəɡˈnəʊsɪs/ : 基因诊断
genomic medicine /dʒɪˈnəʊmɪk ˈmedɪsɪn/ : 基因组医学	precision medicine /prɪˈsɪʒən ˈmedɪsɪn/ : 精准医学

Pancreas regeneration

胰腺再生

期刊: Nature 日期: 2018 JCR Q1 中科院升级版 综合性期刊1区 IF 48.5

Abstract

The **pancreas** is made from two distinct components: the **exocrine pancreas**, a reservoir of digestive enzymes, and the **endocrine islets**, the source of the vital metabolic hormone **insulin**. Human islets possess limited regenerative ability; loss of islet β -cells in diseases such as type 1 diabetes requires **therapeutic intervention**. The leading strategy for restoration of β -cell mass is through the generation and **transplantation** of new β -cells derived from human **pluripotent stem cells**. Other approaches include stimulating endogenous β -cell **proliferation**, **reprogramming** non- β -cells to β -like cells, and harvesting islets from **genetically engineered** animals. Together these approaches form a rich pipeline of therapeutic development for pancreatic **regeneration**.

中文翻译

胰腺由两个不同的部分组成：外分泌胰腺，作为消化酶的储存库；以及内分泌胰岛，是至关重要的代谢激素胰岛素的来源。人类胰岛的再生能力有限；在诸如1型糖尿病等疾病中，胰岛 β 细胞的丧失需要治疗干预。恢复 β 细胞质量的主要策略是通过从人类多能干细胞中生成并移植新的 β 细胞。其他方法包括刺激内源性 β 细胞增殖、将非 β 细胞重编程为 β 样细胞，以及从基因工程动物中获取胰岛。这些方法共同构成了胰腺再生治疗发展的丰富渠道。

核心词汇

pancreas /'pærŋkriəs/ : 胰腺

endocrine islets /'endəukraɪn 'aɪlɪts/ : 内分泌胰岛

regeneration /rɪ dʒɛnə'reɪʃən/ : 再生

transplantation /trænsplɑ:n'teɪʃən/ : 移植

reprogramming /,ri:'prəʊgræmɪŋ/ : 重编程

therapeutic intervention /,θɛrə'pjʊ:tɪk ,ɪntə'veɪʃən/ : 治疗干预

exocrine pancreas /'eksəukraɪn 'pærŋkriəs/ : 外分泌胰腺

insulin /'ɪnsjʊlɪn/ : 胰岛素

pluripotent stem cells /plu'rɪpə'tent stem sɛlz/ : 多能干细胞

proliferation /prə'lɪfə'reɪʃən/ : 增殖

genetically engineered /dʒɪ'netɪkli ˌɛndʒɪ'niəd/ : 基因工程改造

Pancreatic β cell regeneration as a possible therapy for diabetes

胰腺 β 细胞再生作为糖尿病的潜在疗法

期刊: Cell Metabolism 日期: 2018 JCR Q1 中科院升级版 生物学1区 IF 30.9

Abstract

diabetes is the result of having inadequate supply of functional **insulin**-producing β cells. Two possible approaches for replenishing the β cells are: (1) replacement by transplanting cadaveric **islets** or β cells derived from human embryonic **stem cells**/induced **pluripotent stem cells** and (2) induction of endogenous **regeneration**. This review focuses on endogenous **regeneration**, which can follow two pathways: enhanced **replication** of existing β cells and formation of new β cells from cells not expressing **insulin**, either by conversion from a differentiated cell type (**transdifferentiation**) or **differentiation** from **progenitors** (**neogenesis**). Exciting progress on both pathways suggest that **regeneration** may have **therapeutic** promise.

中文翻译

糖尿病是由于功能性胰岛素分泌 β 细胞供应不足所致。补充 β 细胞有两种可能途径：一是通过移植尸体胰岛或源自人类胚胎干细胞/诱导多能干细胞分化的 β 细胞进行替代；二是诱导内源性再生。本综述聚焦于内源性再生途径，该途径可通过两种方式实现：一是促进现有 β 细胞的增殖复制，二是通过分化细胞类型的转化（转分化）或祖细胞分化（新生）途径，使非胰岛素表达细胞形成新的 β 细胞。两条路径取得的突破性进展均表明，再生疗法具有潜在的治疗前景。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病	insulin /ˈɪnsjʊlɪn/ : 胰岛素
islets /ˈaɪləts/ : 胰岛	stem cells /stem sɛlz/ : 干细胞
pluripotent stem cells /pluˈrɪpətənt stem sɛlz/ : 多能干细胞	transplantation /ˌtrænsplɑːnˈteɪʃən/ : 移植
regeneration /rɪˌdʒɛnəˈreɪʃən/ : 再生	replication /ˌrɛplɪˈkeɪʃən/ : 复制
transdifferentiation /ˌtrænsdɪˌfɛrənʃiˈeɪʃən/ : 转分化	differentiation /ˌdɪfəˌrɛnʃiˈeɪʃən/ : 分化
neogenesis /ˌniːoʊˈdʒɛnɪsɪs/ : 新生	progenitors /prəˈdʒɛnɪtərz/ : 祖细胞
therapeutic /ˌθɛrəˈpjʊːtɪk/ : 治疗的	

The clinical consequences of heterogeneity within and between different diabetes types

不同糖尿病类型内部及之间的异质性临床后果

期刊: Diabetologia 日期: 2020 JCR Q1 中科院升级版 医学1区 IF 10.2

Abstract

Advances in molecular methods and the ability to share large population-based datasets are uncovering heterogeneity within **diabetes** types, and some commonalities between types. Within type 1 **diabetes**, endotypes have been discovered based on demographic (e.g. age at diagnosis, race/ethnicity), genetic, **immunological**, **histopathological**, **metabolic** and/or clinical course characteristics, with implications for disease prediction, prevention, diagnosis and treatment. In type 2 **diabetes**, the relative contributions of **insulin resistance** and **beta cell dysfunction** are heterogeneous and relate to demographics, genetics and clinical characteristics, with substantial interaction from environmental exposures. Investigators have proposed approaches that vary from simple to complex in combining these data to identify type 2 **diabetes** clusters relevant to prognosis and treatment. Advances in **pharmacogenetics** and **pharmacodynamics** are also improving treatment. **monogenic diabetes** is a prime example of how understanding heterogeneity within **diabetes** types can lead to **precision medicine**, since phenotype and treatment are affected by which gene is mutated. Heterogeneity also blurs the classic distinctions between **diabetes** types, and has led to the definition of additional categories, such as **latent autoimmune diabetes in adults**, type 1.5 **diabetes** and ketosis-prone **diabetes**. Furthermore, **monogenic diabetes** shares many features with type 1 and type 2 **diabetes**, which make diagnosis difficult. These challenges to the current classification framework in adult and paediatric **diabetes** require new approaches. The 'palette model' and the 'threshold hypothesis' can be combined to help explain the heterogeneity within and between **diabetes** types. Leveraging such approaches for therapeutic benefit will be an important next step for **precision medicine** in **diabetes**. Graphical abstract.

中文翻译

分子方法的进步以及共享大规模人群数据集的能力正在揭示糖尿病类型内部的异质性以及不同类型之间的某些共性。在1型糖尿病中，基于人口统计学（如诊断年龄、种族/民族）、遗传学、免疫学、组织病理学、代谢和/或临床病程特征，已发现不同的内型，这对疾病的预测、预防、诊断和治疗具有重要意义。在2型糖尿病中，胰岛素抵抗和β细胞功能障碍的相对贡献存在异质性，并与人口统计学、遗传学和临床特征相关，同时受到环境暴露的显著影响。研究人员提出了从简单到复杂的不同方法，结合这些数据来识别与预后和治疗相关的2型糖尿病聚类。药物遗传学和药效学的进展也在改善治疗。单基因糖尿病是理解糖尿病类型内部异质性如何推动精准医疗的典型例子，因为表型和治疗受突变基因的影响。异质性地模糊了糖尿病类型之间的经典区别，并导致了新类别的定义，如成人隐性自身免疫性糖尿病、1.5型糖尿病和酮症倾向糖尿病。此外，单基因糖尿病与1型和2型糖尿病有许多共同特征，这使得诊断变得困难。这些对当前成人和儿童糖尿病分类框架的挑战需要新的方法。“调色板模型”和“阈值假说”可以结合起来，帮助解释糖尿病类型内部和之间的异质性。利用这些方法实现治疗获益将是糖尿病精准医疗下一步的重要方向。图形摘要。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病

insulin resistance /ˈɪnsəlɪn rɪˈzɪstəns/ : 胰岛素抵抗

pharmacogenetics /ˌfɑːrməkoudʒɪˈnetɪks/ : 药物遗传学

monogenic diabetes /ˌmɒnɒuˈdʒenɪk ˈdaɪəˈbiːtiːz/ : 单基因糖尿病

latent autoimmune diabetes in adults /ˈleɪtənt ˌɔːtəʊˈmjuːn ˈdaɪəˈbiːtiːz ɪn ˈædʌlts/ : 成人隐性自身免疫性糖尿病

immunological /ˌɪmjʊnɒlˈɒdʒɪkəl/ : 免疫学的

endotype /ˈendotaɪp/ : 内型

beta cell dysfunction /ˈbeɪtə sɛl dɪsˈfʌŋkʃən/ : β细胞功能障碍

pharmacodynamics /ˌfɑːrməkoudaɪˈnæmɪks/ : 药效动力学

precision medicine /prɪˈsɪʒən ˈmedɪsɪn/ : 精准医学

histopathological /ˌhɪstəʊˌpæθəˈlɒdʒɪkəl/ : 组织病理学的

metabolic /ˌmetəˈbɒlɪk/ : 代谢的

A roadmap to the molecular human linking multiomics with population traits and diabetes subtypes

通往分子人类路线图：多组学与人群特征及糖尿病亚型的关联

期刊: Nature Communications 日期: 2024 JCR Q1 中科院升级版 综合性期刊1区 IF 15.7

Abstract

In-depth multiomic phenotyping provides molecular insights into complex physiological processes and their pathologies. Here, we report on integrating 18 diverse deep molecular phenotyping (omics-) technologies applied to urine, blood, and saliva samples from 391 participants of the multiethnic **diabetes** Qatar Metabolomics Study of **diabetes** (QMDiab). Using 6,304 quantitative molecular traits with 1,221,345 genetic variants, **methylation** at 470,837 DNA CpG sites, and **gene expression** of 57,000 transcripts, we determine (1) within-platform partial correlations, (2) between-platform mutual best correlations, and (3) **genome**-, **epigenome**-, **transcriptome**-, and phenome-wide associations. Combined into a **molecular network** of > 34,000 statistically significant trait-trait links in biofluids, our study portrays "The Molecular Human". We describe the variances explained by each omics in the phenotypes (age, sex, BMI, and **diabetes** state), platform complementarity, and the inherent correlation structures of multiomics data. Further, we construct multi-**molecular network** of **diabetes** subtypes. Finally, we generated an open-access web interface to "The Molecular Human" (<http://comics.metabolomix.com>), providing interactive data exploration and hypotheses generation possibilities.

中文翻译

深入的多组学表型分析为复杂的生理过程及其病理机制提供了分子层面的见解。本文报道了整合18种不同的深度分子表型（组学）技术，应用于卡塔尔糖尿病代谢组学研究（QMDiab）中391名多民族糖尿病参与者的尿液、血液和唾液样本。通过分析6,304个定量分子特征、1,221,345个遗传变异、470,837个DNA CpG位点的甲基化以及57,000个转录本的基因表达，我们确定了（1）平台内的部分相关性，（2）平台间的最佳互相关性，以及（3）基因组、表观基因组、转录组和表型组范围内的关联。这些结果整合成一个包含超过34,000个统计显著特征-特征链接的生物流体分子网络，我们的研究描绘了“分子人类”。我们描述了每种组学在表型（年龄、性别、BMI和糖尿病状态）中解释的方差、平台互补性以及多组学数据的内在相关结构。此外，我们构建了糖尿病亚型的多分子网络。最后，我们创建了一个开放访问的“分子人类”网络界面（<http://comics.metabolomix.com>），提供交互式数据探索和假设生成的可能性。

核心词汇

diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病	phenotype /ˈfiːnətaɪp/ : 表型
genome /ˈdʒiːnəʊm/ : 基因组	epigenome /ˌepɪˈdʒiːnəʊm/ : 表观基因组
transcriptome /trænskrɪptəʊm/ : 转录组	methylation /ˌmeθɪˈleɪʃən/ : 甲基化
gene expression /dʒiːn ɪkˈspreʃən/ : 基因表达	genetic variant /dʒəˈnetɪk ˈveəriənt/ : 遗传变异
biofluid /ˈbaɪəʊˌfluːɪd/ : 生物体液	association /əˌsəʊsiˈeɪʃən/ : 关联
subtype /ˈsʌbtaɪp/ : 亚型	molecular network /məˈlekjələr ˈnetwɜːrk/ : 分子网络

Single-cell atlas of human pancreatic islet and acinar endothelial cells in health and diabetes

健康与糖尿病状态下人类胰岛及腺泡内皮细胞的单细胞图谱

期刊: Nature Communications 日期: 2025 JCR Q1 中科院升级版 综合性期刊1区 IF 15.7

Abstract

Characterization of the vascular **heterogeneity** within the **pancreas** has previously been lacking. Here, we develop strategies to enrich **islet**-specific endothelial cells (ISECs) and **acinar**-specific endothelial cells (ASECs) from three human pancreases and corroborate these findings with three published pancreatic datasets. Single-cell RNA sequencing reveals the unique molecular signatures of ISECs, including structural genes COL13A1, ESM1, PLVAP, UNC5B, and LAMA4, **angiocrine** genes KDR, THBS1, BMPs and CXCR4, and **metabolic** genes ACE, PASK and F2RL3. ASECs display distinct signatures including GPIHBP1, CCL14, CD74, AQP1, KLF4, and KLF2, which may manage the inflammatory and **metabolic** needs of the **exocrine pancreas**. **ligand-receptor** analysis suggests ISECs and ASECs interact with LUM(+) fibroblasts and RGS5(+) pericytes and smooth muscle cells via VEGF-A:VEGFR2, CXCL12:CXCR4, and LIF:LIFR pathways. Comparative expression and immunohistochemistry indicate disruption of endothelial-expressed CD74, ESM1, PLVAP, THBD, VWA1, and VEGF-A cross-talk among vascular and other cell types in diabetes. Thus, our data provide a single-cell vascular atlas of human **pancreas**, enabling deeper understanding of pancreatic pathophysiology in health and disease.

中文翻译

此前，胰腺内血管异质性的特征尚未明确。本研究从三例人类胰腺中富集胰岛特异性内皮细胞（ISECs）和腺泡特异性内皮细胞（ASECs），并结合三个已发表的胰腺数据集验证结果。单细胞RNA测序揭示了ISECs的独特分子特征，包括结构基因COL13A1、ESM1、PLVAP、UNC5B和LAMA4，血管分泌因子基因KDR、THBS1、BMPs和CXCR4，以及代谢基因ACE、PASK和F2RL3。ASECs则呈现独特特征，包括GPIHBP1、CCL14、CD74、AQP1、KLF4和KLF2，这些可能调控外分泌胰腺的炎症与代谢需求。配体-受体分析表明，ISECs和ASECs通过VEGF-A:VEGFR2、CXCL12:CXCR4和LIF:LIFR通路与LUM(+)成纤维细胞及RGS5(+)周细胞、平滑肌细胞相互作用。比较表达与免疫组化结果显示，糖尿病中血管内皮表达的CD74、ESM1、PLVAP、THBD、VWA1及VEGF-A在血管与其他细胞类型间的信号串扰发生紊乱。因此，本研究绘制了人类胰腺单细胞血管图谱，为深入理解健康与疾病状态下的胰腺病理生理学提供了依据。

核心词汇

pancreas /ˈpæŋkriəs/ : 胰腺	islet /ˈaɪlət/ : 胰岛
endothelial cell /ˌɛndəʊˈθi:liəl sɛl/ : 内皮细胞	acinar /ˈæsɪnəri/ : 腺泡
exocrine /ˈɛksəkrɪn/ : 外分泌	angiocrine /ˈændʒiəʊkraɪn/ : 血管分泌
inflammation /ˌɪnfləˈmeɪʃən/ : 炎症	metabolic /ˌmɛtəˈbɒlɪk/ : 代谢的
ligand /ˈlaɪɡənd/ : 配体	receptor /rɪˈseptər/ : 受体
heterogeneity /ˌhɛtərəʊdʒəˈni:ɪti/ : 异质性	

Fam3a-mediated prohormone convertase switch in α -cells regulates pancreatic GLP-1 production in an Nr4a2-Foxa2-dependent manner

Fam3a介导的 α 细胞中激素原转换酶切换以Nr4a2-Foxa2依赖性方式调节胰腺GLP-1生成

期刊: Metabolism 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 11.9

Abstract

BACKGROUND: Fam3a has been demonstrated to regulate pancreatic β -cell function and glucose homeostasis. However, the role and mechanism of Fam3a in regulating α -cell function remain unexplored. METHODS: glucagon and glucagon-like peptide-1 (GLP-1) levels in pancreas and plasma were measured in global Fam3a knockout (Fam3a RESULTS: Compared with wild-type mice, pancreatic and plasma active GLP-1 levels were increased in Fam3a CONCLUSIONS: Fam3a deficiency in α -cells enhances pancreatic GLP-1 production to improve β -cell function via paracrine signaling in an Nr4a2-Foxa2-PC1/3-dependent manner. Our study unveils a novel strategy for reprogramming α -cell proglucagon processing output from glucagon to GLP-1 and deepen the understanding of crosstalk between α -cells and β -cells.

中文翻译

背景：Fam3a已被证实能调节胰腺 β 细胞功能和葡萄糖稳态。然而，Fam3a在调节 α 细胞功能中的作用和机制尚未被探索。方法：在全基因组Fam3a敲除（Fam3a^{-/-}）小鼠中测量胰腺和血浆中的胰高血糖素和胰高血糖素样肽-1（GLP-1）水平。结果：与野生型小鼠相比，Fam3a^{-/-}小鼠的胰腺和血浆活性GLP-1水平升高，而胰高血糖素水平降低。结论： α 细胞中Fam3a的缺乏通过旁分泌信号以Nr4a2-Foxa2-PC1/3依赖的方式增强胰腺GLP-1的产生，从而改善 β 细胞功能。我们的研究揭示了一种将 α 细胞前胰高血糖素处理输出从胰高血糖素重编程为GLP-1的新策略，并加深了对 α 细胞和 β 细胞之间相互作用的理

核心词汇

glucose homeostasis /'glu:kous 'houmiə'steɪsɪs/ : 葡萄糖稳态	glucagon /'glu:kəɡɒn/ : 胰高血糖素
pancreas /'pæŋkriəs/ : 胰腺	knockout /'nɒkaʊt/ : 基因敲除
plasma /'plæzmə/ : 血浆	paracrine signaling /'pærəkraɪn 'sɪɡnəlɪŋ/ : 旁分泌信号传导
proglucagon processing /prɒu'glu:kəɡɒn 'prɒsesɪŋ/ : 胰高血糖素原加工	crosstalk /'krɒstɔ:k/ : 串扰/交互作用
reprogramming /,ri:'prɒɡræmɪŋ/ : 重编程	

Diabetes mellitus and the key role of endoplasmic reticulum stress in pancreatic β cells

糖尿病与内质网应激在胰腺 β 细胞中的关键作用

期刊: Nature Reviews Endocrinology 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

Insufficient **insulin** secretion by pancreatic β cells is central to the **pathogenesis** of **diabetes mellitus**. As **insulin** is synthesized in the **endoplasmic reticulum** (ER), perturbations in ER **homeostasis** lead to **ER stress** and activate the **ER stress** response. Over the past two decades, considerable data have accumulated on the role of β cell **ER stress** in **diabetes mellitus**. Several **monogenic** forms of **diabetes mellitus** are caused by excessive **ER stress**, perturbed **ER stress** response signalling or impaired ER-Golgi protein trafficking. These pathways are now recognized to contribute to β cell failure in both type 1 and type 2 **diabetes mellitus**. This Review considers the role of β cell **ER stress** in common forms of **diabetes mellitus** and examines whether it is a cause or a consequence of these diseases. The strong genetic evidence for a causal role of **ER stress** in 15 **monogenic** forms of **diabetes mellitus** is summarized, and the effects of **ER stress** on human β cell **differentiation**, **function** and **survival** are described. Although definitive proof is lacking that **ER stress** responses can be therapeutically targeted to improve β cell **function** in **diabetes mellitus**, existing and novel treatments that aim to restore ER **homeostasis** are also outlined.

中文翻译

胰腺 β 细胞胰岛素分泌不足是糖尿病发病机制的核心。由于胰岛素在内质网（ER）中合成，内质网稳态的扰动会导致内质网应激并激活内质网应激反应。过去二十年间，关于 β 细胞内质网应激在糖尿病中的作用已积累了相当多的数据。多种单基因型糖尿病是由过度内质网应激、内质网应激反应信号通路紊乱或内质网-高尔基体蛋白质运输受损引起的。目前认为这些通路在1型和2型糖尿病中均导致 β 细胞功能衰竭。本综述探讨了 β 细胞内质网应激在常见类型糖尿病中的作用，并审视其是这些疾病的病因还是后果。文章总结了内质网应激在15种单基因型糖尿病中因果作用的强有力遗传学证据，描述了内质网应激对人类 β 细胞分化、功能和存活的影响。尽管目前缺乏确凿证据表明通过靶向内质网应激反应可改善糖尿病中的 β 细胞功能，但仍概述了旨在恢复内质网稳态的现有及新型治疗方法。

核心词汇

diabetes mellitus /ˌdaɪəˈbiːtiːz məˈlartəs/ : 糖尿病	insulin /ˈɪnsjʊlɪn/ : 胰岛素
endoplasmic reticulum /ˌɛndəʊˈplæzmiːk rɪˈtɪkjʊləm/ : 内质网	ER stress /ˌiːˈɑːr stres/ : 内质网应激
pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制	monogenic /ˌmɒnəʊˈdʒɛnɪk/ : 单基因的
differentiation /ˌdɪfəˈrɛnʃiːeɪʃən/ : 分化	function /ˈfʌŋkʃən/ : 功能
survival /səˈvaɪvəl/ : 存活	homeostasis /ˌhəʊmiəʊˈsteɪsɪs/ : 稳态
therapeutic /ˌθɛrəˈpjʊːtɪk/ : 治疗的	

Inceptor binds to and directs insulin towards lysosomal degradation in β cells

Inceptor结合并引导胰岛素在 β 细胞中向溶酶体降解

期刊: Nature Metabolism 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 20.8

Abstract

AbstractBlunted first-phase **insulin secretion** and **insulin deficiency** are indicators of β cell dysfunction and **diabetes** manifestation. Therefore, insights into molecular mechanisms that regulate insulin homeostasis might provide entry sites to replenish insulin content and restore β cell function. Here, we identify the insulin inhibitory receptor (inceptor; encoded by the gene IIR/ELAPOR1) as an insulin-binding receptor that regulates insulin stores by **lysosomal degradation**. Using human induced pluripotent stem cell (SC)-derived **islets**, we show that IIR knockout (KO) results in enhanced SC β cell **differentiation** and **survival**. Strikingly, extended in vitro culture of IIR KO SC β cells leads to greatly increased insulin content and glucose-stimulated **insulin secretion** (GSIS). We find that inceptor localizes to clathrin-coated vesicles close to the plasma membrane and in the trans-Golgi network as well as in secretory granules, where it acts as a sorting receptor to direct **proinsulin** and insulin towards **lysosomal degradation**. Targeting inceptor using a **monoclonal antibody** increases **proinsulin** and insulin content and improves SC β cell GSIS. Altogether, our findings reveal the basic mechanisms of β cell insulin turnover and identify inceptor as an insulin degradation receptor.

中文翻译

摘要 胰岛素分泌第一相钝化和胰岛素缺乏是 β 细胞功能障碍和糖尿病表现的重要指标。因此，深入探究调控胰岛素稳态的分子机制，可能为补充胰岛素储备、恢复 β 细胞功能提供切入点。本研究首次鉴定出胰岛素抑制受体（inceptor，由IIR/ELAPOR1基因编码）作为胰岛素结合受体，通过溶酶体降解途径调控胰岛素储存。利用人诱导多能干细胞（SC）分化的胰岛类器官，我们发现IIR基因敲除（KO）能显著促进SC β 细胞的分化与存活。值得注意的是，IIR KO SC β 细胞经过体外长期培养后，其胰岛素含量和葡萄糖刺激的胰岛素分泌（GSIS）能力均大幅提升。研究显示，inceptor定位于质膜附近的网格蛋白包被囊泡、高尔基体反面膜囊以及分泌颗粒中，作为分选受体将胰岛素原和胰岛素导向溶酶体降解途径。使用单克隆抗体靶向抑制inceptor功能，可有效增加胰岛素原和胰岛素含量，并显著改善SC β 细胞的GSIS能力。综上所述，本研究揭示了 β 细胞胰岛素周转的基本机制，并确立inceptor作为胰岛素降解受体的关键作用。

核心词汇

insulin secretion /'ɪnsjʊlɪn sɪ'kriːʃən/ : 胰岛素分泌	insulin deficiency /'ɪnsjʊlɪn dɪ'fɪʃənsi/ : 胰岛素缺乏
diabetes /ˌdaɪə'biːtiːz/ : 糖尿病	islets /'aɪləts/ : 胰岛
lysosomal degradation /ˌlaɪsə'souməl ˌdeɪɡreɪ'deɪʃən/ : 溶酶体降解	differentiation /ˌdɪfə'renʃi'eɪʃən/ : 分化
survival /sə'rvaɪvəl/ : 存活	proinsulin /prəʊ'ɪnsjʊlɪn/ : 胰岛素原
monoclonal antibody /ˌmɒnə'kləʊnəl ˌæntɪˌbɒdi/ : 单克隆抗体	

Pancreatic β -cell development and regeneration

胰腺 β 细胞发育与再生

期刊: Cold Spring Harbor Perspectives in Biology 日期: 2022 JCR Q1 中科院升级版 生物学1区 IF 8.4

Abstract

The pancreatic β -cells are essential for regulating glucose homeostasis through the coordinated release of the insulin hormone. dysfunction of the highly specialized β -cells results in diabetes mellitus, a growing global health epidemic. In this review, we describe the development and function of β -cells the emerging concept of heterogeneity within insulin-producing cells, and the potential of other cell types to assume β -cell functionality via transdifferentiation. We also discuss emerging routes to design cells with minimal β -cell properties and human stem cell differentiation efforts that carry the promise to restore normoglycemia in patients suffering from diabetes.

中文翻译

胰腺 β 细胞通过协调释放胰岛素激素对调节葡萄糖稳态至关重要。高度特化的 β 细胞功能失调会导致糖尿病，这一全球健康流行病正日益加剧。本综述阐述了 β 细胞的发育与功能、胰岛素产生细胞异质性的新兴概念，以及其他细胞类型通过转分化获得 β 细胞功能的潜力。我们还讨论了设计具有基本 β 细胞特性细胞的新兴途径，以及人类干细胞分化研究——这些努力有望为糖尿病患者恢复正常血糖水平。

核心词汇

glucose homeostasis /ˈɡluːkəʊs ˌhoʊmiəˈsteɪsɪs/ : 葡萄糖稳态	insulin /ˈɪnsəlɪn/ : 胰岛素
diabetes mellitus /ˌdaɪəˈbiːtiːz məˈleɪtəs/ : 糖尿病	dysfunction /dɪsˈfʌŋkʃən/ : 功能障碍
heterogeneity /ˌhetərədʒəˈniːti/ : 异质性	transdifferentiation /ˌtrænsdɪˈferənʃiˈeɪʃən/ : 转分化
stem cell /stem sɛl/ : 干细胞	differentiation /ˌdɪfəˈrenʃiˈeɪʃən/ : 分化
normoglycemia /ˌnɔːrməʊɡlaɪˈsiːmiə/ : 正常血糖	cell types /sɛl taɪps/ : 细胞类型
functionality /ˌfʌŋkʃəˈnælɪti/ : 功能性	

Glucagon acting at the GLP-1 receptor contributes to β -cell regeneration induced by glucagon receptor antagonism in diabetic mice

胰高血糖素作用于GLP-1受体促进糖尿病小鼠中胰高血糖素受体拮抗作用诱导的 β 细胞再生

期刊: Diabetes 日期: 2023 JCR Q1 中科院升级版 医学2区 IF 7.5

Abstract

Dysfunction of glucagon-secreting α -cells participates in the progression of diabetes, and glucagon receptor (GCGR) antagonism is regarded as a novel strategy for diabetes therapy. GCGR antagonism upregulates glucagon and glucagon-like peptide 1 (GLP-1) secretion and, notably, promotes β -cell regeneration in diabetic mice. Here, we aimed to clarify the role of GLP-1 receptor (GLP-1R) activated by glucagon and/or GLP-1 in the GCGR antagonism-induced β -cell regeneration. We showed that in db/db mice and type 1 diabetic wild-type or Flox/cre mice, GCGR monoclonal antibody (mAb) improved glucose control, upregulated plasma insulin level, and increased β -cell area. Notably, blockage of systemic or pancreatic GLP-1R signaling by exendin 9-39 (Ex9) or Glp1r knockout diminished the above effects of GCGR mAb. Furthermore, glucagon-neutralizing antibody (nAb), which prevents activation of GLP-1R by glucagon, also attenuated the GCGR mAb-induced insulinotropic effect and β -cell regeneration. In cultured primary mouse islets isolated from normal mice and db/db mice, GCGR mAb action to increase insulin release and to upregulate β -cell-specific marker expression was reduced by a glucagon nAb, by the GLP-1R antagonist Ex9, or by a pancreas-specific Glp1r knockout. These findings suggest that activation of GLP-1R by glucagon participates in β -cell regeneration induced by GCGR antagonism in diabetic mice. Article Highlights glucagon receptor (GCGR) antagonism promotes β -cell regeneration in type 1 and type 2 diabetic mice and in euglycemic nonhuman primates. glucagon and glucagon-like peptide 1 (GLP-1) can activate the GLP-1 receptor (GLP-1R), and their levels are upregulated following GCGR antagonism. We investigated whether GLP-1R activated by glucagon and/or GLP-1 contributed to β -cell regeneration induced by GCGR antagonism. We found that blockage of glucagon-GLP-1R signaling attenuated the GCGR monoclonal antibody-induced insulinotropic effect and β -cell regeneration in diabetic mice. Our study reveals a novel mechanism of β -cell regeneration and uncovers the communication between α -cells and β -cells in regulating β -cell mass.

中文翻译

分泌胰高血糖素的 α 细胞功能障碍参与糖尿病的进展,而胰高血糖素受体(GCGR)拮抗被视为糖尿病治疗的新策略。GCGR拮抗可上调胰高血糖素和胰高血糖素样肽-1(GLP-1)的分泌,并显著促进糖尿病小鼠的 β 细胞再生。本研究旨在阐明由胰高血糖素和/或GLP-1激活的GLP-1受体(GLP-1R)在GCGR拮抗诱导的 β 细胞再生中的作用。我们在db/db小鼠及1型糖尿病野生型或Flox/cre小鼠中发现,GCGR单克隆抗体(mAb)能改善血糖控制、提升血浆胰岛素水平并增加 β 细胞面积。值得注意的是,通过exendin 9-39(Ex9)或Glp1r基因敲除阻断全身或胰腺GLP-1R信号通路,会削弱GCGR mAb的上述效应。此外,能阻止胰高血糖素激活GLP-1R的胰高血糖素中和抗体(nAb)也减弱了GCGR mAb诱导的促胰岛素效应和 β 细胞再生。在从正常小鼠和db/db小鼠分离培养的原代胰岛中,GCGR mAb增加胰岛素释放和上调 β 细胞特异性标志物表达的作用,可被胰高血糖素nAb、GLP-1R拮抗剂Ex9或胰腺特异性Glp1r敲除所减弱。这些发现表明,胰高血糖素激活GLP-1R参与了糖尿病小鼠中GCGR拮抗诱导的 β 细胞再生过程。文章要点:胰高血糖素受体(GCGR)拮抗能促进1型和2型糖尿病小鼠以及血糖正常的非人灵长类动物的 β 细胞再生。胰高血糖素和胰高血糖素样肽-1(GLP-1)均可激活GLP-1受体(GLP-1R),且GCGR拮抗后两者水平均上调。我们探究了由胰高血糖素和/或GLP-1激活的GLP-1R是否参与GCGR拮抗诱导的 β 细胞再生。研究发现,阻断胰高血糖素-GLP-1R信号通路可减弱GCGR单克隆抗体在糖尿病小鼠中诱导的促胰岛素效应和 β 细胞再生。本研究揭示了 β 细胞再生的新机制,并阐明了 α 细胞与 β 细胞在调控 β 细胞数量中的交互作用。

核心词汇

diabetes /ˈdaɪəˈbiːtiːz/ : 糖尿病

insulin /ˈɪnsjʊlɪn/ : 胰岛素

antagonism /ænˈtæɡənɪzəm/ : 拮抗作用

antibody /ˈæntɪˌbɒdi/ : 抗体

islet /ˈaɪlət/ : 胰岛

glucagon /ˈɡluːkəɡɒn/ : 胰高血糖素

regeneration /ˌrɛdʒənəˈreɪʃən/ : 再生

receptor /rɪˈseptə/ : 受体

knockout /ˈnɒkəʊt/ : 基因敲除

glucose /ˈɡluːkeɪs/ : 葡萄糖

Gut microbiota-tryptophan metabolism-GLP-1 axis participates in β -cell regeneration induced by dapagliflozin

肠道微生物群-色氨酸代谢-GLP-1轴参与达格列净诱导的 β 细胞再生

期刊: Diabetes 日期: 2024 JCR Q1 中科院升级版 医学2区 IF 7.5

Abstract

Sodium-glucose cotransporter 2 inhibitors, efficacious antidiabetic agents that have cardiovascular and renal benefits, can promote pancreatic β -cell **regeneration** in type 2 diabetic mice. However, the underlying mechanism remains unclear. In this study, we aimed to use multiomics to identify the mediators involved in β -cell **regeneration** induced by dapagliflozin. We showed that dapagliflozin lowered **blood glucose** level, upregulated plasma **insulin** level, and increased **islet** area in db/db mice. Dapagliflozin reshaped **gut microbiota** and modulated microbiotic and plasmatic metabolites related to **tryptophan** metabolism, especially l-**tryptophan**, in the diabetic mice. Notably, l-**tryptophan** upregulated the mRNA level of glucagon-like peptide 1 (GLP-1) production-related gene (Gcg and Pcsk1) expression and promoted GLP-1 secretion in cultured mouse intestinal L cells, and it increased the supernatant **insulin** level in primary human islets, which was eliminated by GPR142 **antagonist**. Transplant of fecal microbiota from dapagliflozin-treated mice, supplementation of l-**tryptophan**, or treatment with dapagliflozin upregulated l-**tryptophan**, GLP-1, and **insulin** or C-peptide levels and promoted β -cell **regeneration** in db/db mice. Addition of exendin 9-39, a GLP-1 **receptor** (GLP-1R) **antagonist**, or pancreatic Glp1r **knockout** diminished these beneficial effects. In summary, treatment with dapagliflozin in type 2 diabetic mice promotes β -cell **regeneration** by upregulating GLP-1 production, which is mediated via **gut microbiota** and **tryptophan** metabolism.

中文翻译

钠-葡萄糖协同转运蛋白2抑制剂作为具有心血管和肾脏保护作用的有效抗糖尿病药物，可促进2型糖尿病小鼠的胰腺 β 细胞再生，但其潜在机制尚不明确。本研究旨在通过多组学方法鉴定达格列净诱导 β 细胞再生的关键介质。结果显示，达格列净可降低db/db小鼠血糖水平、上调血浆胰岛素水平并增加胰岛面积。该药物重塑了糖尿病小鼠的肠道菌群，并调节了与色氨酸代谢相关的微生物及血浆代谢物（尤其是L-色氨酸）。值得注意的是，L-色氨酸可上调培养的小鼠肠道L细胞中胰高血糖素样肽-1（GLP-1）生成相关基因（Gcg和Pcsk1）的mRNA表达并促进GLP-1分泌，同时增加原代人胰岛的上清胰岛素水平，而这一效应可被GPR142拮抗剂消除。移植达格列净处理小鼠的粪便菌群、补充L-色氨酸或使用达格列净治疗，均可上调db/db小鼠的L-色氨酸、GLP-1及胰岛素/C肽水平，并促进 β 细胞再生。而添加GLP-1受体（GLP-1R）拮抗剂exendin 9-39或敲除胰腺Glp1r基因则会消除这些有益效应。综上，达格列净通过肠道菌群和色氨酸代谢介导的GLP-1生成上调，促进2型糖尿病小鼠的 β 细胞再生。

核心词汇

regeneration /rɪˌdʒɛnəˈreɪʃən/ : 再生	blood glucose /blʌd ˈɡluːkəʊs/ : 血糖
insulin /ˈɪnsəlɪn/ : 胰岛素	islet /ˈaɪlət/ : 胰岛
gut microbiota /ɡʌt ˈmaɪkrəʊbaɪˈoʊtə/ : 肠道菌群	metabolite /miˈtæbəlaɪt/ : 代谢物
tryptophan /ˈtrɪptəfæn/ : 色氨酸	transplantation /ˌtrænsplænˈteɪʃən/ : 移植
antagonist /ænˈtæɡənɪst/ : 拮抗剂	receptor /rɪˈseptər/ : 受体
knockout /ˈnɒkəʊt/ : 敲除	

A novel hydrogel loaded with plant exosomes and stem cell exosomes as a new strategy for treating diabetic wounds

一种装载植物外泌体和干细胞外泌体的新型水凝胶作为治疗糖尿病伤口的新策略

期刊: Materials Today Bio 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 10.2

Abstract

diabetic wound healing is constrained by various factors, including chronic inflammation, sustained oxidative stress, impaired angiogenesis, and abnormal wound microenvironments. Exosomes derived from mesenchymal stem cells (MSC-exo) contain a wealth of bioactive substances that play a positive role in promoting diabetic wound healing. Plant-derived exosomes, as a novel therapeutic approach, are continuously being explored. Momordica charantia (MC) has been shown to possess blood glucose-lowering effects, and its exosomes are of significant relevance for treating diabetic wounds. However, direct application of exosomes to wounds faces challenges such as poor stability and short retention time, limiting their therapeutic effectiveness and clinical applicability. Encapsulating exosomes in hydrogels is an effective strategy to preserve their bioactivity. In this study, we fabricated a hydrogel loaded with MSC-exo and MC exosomes (MC-exo) by photopolymerization of methacrylated gelatin (GelMA) and dopamine (MEMC-Gel). The resulting MEMC-Gel exhibited favorable mechanical properties, adhesion, degradability, absorbency, and biocompatibility. In vitro, MEMC-Gel demonstrated the ability to resist inflammation, counter oxidative stress, promote fibroblast migration, support endothelial cell angiogenesis, and regulate macrophage polarization. In a diabetic mouse wound model, MEMC-Gel accelerated wound healing by inhibiting inflammation and oxidative stress, modulating macrophage immune responses and hyperglycemia within the microenvironment, promoting angiogenesis, and enhancing epithelialization. In conclusion, MEMC-Gel is an outstanding hydrogel dressing that synergistically promotes repair by loading MSC-exo and MC-exo, significantly accelerating diabetic wound healing through multiple mechanisms. This multifunctional hydrogel, based on exosomes from two different sources, provides an innovative therapeutic strategy for diabetic wound repair with broad clinical application potential.

中文翻译

糖尿病伤口愈合受到多种因素制约，包括慢性炎症、持续氧化应激、血管生成受损及伤口微环境异常。间充质干细胞来源的外泌体（MSC-exo）富含生物活性物质，对促进糖尿病伤口愈合具有积极作用。植物源性外泌体作为一种新型治疗手段正被持续探索，苦瓜已被证实具有降血糖功效，其外泌体对糖尿病伤口治疗具有重要意义。然而外泌体直接应用于伤口面临稳定性差、滞留时间短等挑战，限制了其治疗效果和临床应用潜力。将外泌体封装于水凝胶中是保持其生物活性的有效策略。本研究通过光聚合甲基丙烯酸酯化明胶（GelMA）与多巴胺，构建了负载MSC-exo与苦瓜外泌体（MC-exo）的水凝胶（MEMC-Gel）。所得MEMC-Gel展现出良好的机械性能、粘附性、可降解性、吸液性与生物相容性。体外实验表明，MEMC-Gel具有抗炎、抗氧化、促进成纤维细胞迁移、支持内皮细胞血管生成及调控巨噬细胞极化的能力。在糖尿病小鼠伤口模型中，MEMC-Gel通过抑制炎症与氧化应激、调控巨噬细胞免疫应答及微环境内高血糖状态、促进血管生成和增强上皮化进程，加速了伤口愈合进程。综上所述，MEMC-Gel是一种通过负载MSC-exo与MC-exo协同促进修复的优异水凝胶敷料，能通过多重机制显著加速糖尿病伤口愈合。这种基于两种不同来源外泌体的多功能水凝胶为糖尿病伤口修复提供了创新治疗策略，具有广阔的临床应用前景。

核心词汇

diabetic wound /ˌdaɪəˈbetɪk wuːnd/ : 糖尿病伤口	inflammation /ˌɪnfləˈmeɪʃən/ : 炎症
oxidative stress /ˌɒksɪdeɪtɪv stress/ : 氧化应激	angiogenesis /ˌændʒiəʊˈdʒenɪsɪs/ : 血管生成
mesenchymal stem cell /meˈsenʃkɪməl stem sɛl/ : 间充质干细胞	exosome /ˈeksəsoʊm/ : 外泌体
hydrogel /ˈhaɪdrədʒəl/ : 水凝胶	fibroblast /ˈfaɪbrəʊblæst/ : 成纤维细胞
endothelial cell /ˌendəʊˈθiːliəl sɛl/ : 内皮细胞	macrophage /ˈmækrəfeɪdʒ/ : 巨噬细胞
epithelialization /ˌɛpɪˈθiːliəˈzeɪʃən/ : 上皮化	biocompatibility /ˌbaɪəʊkəmˈpæteːbɪlɪti/ : 生物相容性

Bioinspired nanovesicles released from injectable hydrogels facilitate diabetic wound healing by regulating macrophage polarization and endothelial cell dysfunction

生物启发纳米囊泡从可注射水凝胶中释放通过调节巨噬细胞极化和内皮细胞功能障碍促进糖尿病伤口愈合

期刊: Journal of Nanobiotechnology 日期: 2023 JCR Q1 中科院升级版 生物学1区 IF 12.6

Abstract

AbstractWound healing is one of the major global health concerns in diabetic patients. Overactivation of proinflammatory M1 macrophages could lead to delayed wound healing in diabetes. 4-octyl itaconate (4OI), a derivative of the metabolite itaconate, has aroused growing interest recently on account of its excellent anti-inflammatory properties. Cell membrane coating is widely regarded as a novel biomimetic strategy to deliver drugs and inherit properties derived from source cells for biomedical applications. Herein, we fused induced pluripotent stem cell-derived endothelial cell (iEC) membrane together with M1 type macrophage membrane to construct a hybrid membrane (iEC-M) camouflaged 4OI nanovesicles (4OI@iEC-M). Furthermore, bioinspired nanovesicles 4OI@iEC-M are incorporated into the injectable, multifunctional gelatin methacryloyl hydrogels for diabetic wound repair and regeneration. In our study, bioinspired nanovesicles could achieve dual-targeted deliver of 4OI into both M1 macrophages and endothelial cells, thereby promoting macrophage polarization and protecting endothelial cells. With the synergistically anti-inflammatory and immunoregulative effects, the bioinspired nanovesicles-loaded hydrogels could facilitate neovascularization and exhibit superior diabetic wound repair and regeneration. Taken together, this study might provide a novel strategy to facilitate diabetic wound healing, thereby reducing limb amputation and mortality of diabetes.

中文翻译

摘要伤口愈合是糖尿病患者全球主要健康问题之一。促炎性M1巨噬细胞的过度激活可能导致糖尿病伤口愈合延迟。4-辛基衣康酸酯（4OI）作为代谢物衣康酸的衍生物，因其卓越的抗炎特性近来引起了广泛关注。细胞膜包覆技术被广泛视为一种新型仿生策略，用于药物递送并继承源细胞特性，适用于生物医学应用。本研究将诱导多能干细胞来源的内皮细胞（iEC）膜与M1型巨噬细胞膜融合，构建了仿生混合膜（iEC-M）包覆的4OI纳米囊泡（4OI@iEC-M）。进一步地，将这种仿生纳米囊泡4OI@iEC-M整合至可注射的多功能甲基丙烯酰化明胶水凝胶中，用于糖尿病伤口修复与再生。研究表明，仿生纳米囊泡能够实现4OI对M1巨噬细胞和内皮细胞的双重靶向递送，从而促进巨噬细胞极化并保护内皮细胞。凭借协同的抗炎与免疫调节作用，负载仿生纳米囊泡的水凝胶能够促进新生血管形成，并展现出优异的糖尿病伤口修复与再生效果。综上所述，本研究可能为促进糖尿病伤口愈合提供新策略，从而降低糖尿病患者的截肢风险与死亡率。

核心词汇

wound healing /wu:nd 'hi:lɪŋ/ : 伤口愈合	diabetes /,daɪə'bi:tɪz/ : 糖尿病
macrophage /'mækroʊfeɪdʒ/ : 巨噬细胞	inflammation /,ɪnflə'meɪʃən/ : 炎症
endothelial cell /,endə'thi:lɪəl sel/ : 内皮细胞	hydrogel /'haɪdrədʒel/ : 水凝胶
neovascularization /,ni:ou,væskjʊləraɪ'zeɪʃən/ : 新生血管形成	amputation /,æmpju'teɪʃən/ : 截肢
regeneration /rɪ,dʒɛnə'reɪʃən/ : 再生	biomimetic /,baɪəʊmɪ'metɪk/ : 仿生
immunoregulation /,ɪmjʊnəv,regju'leɪʃən/ : 免疫调节	polarization /,pəʊləraɪ'zeɪʃən/ : 极化

Diabetic emergencies - ketoacidosis, hyperglycaemic hyperosmolar state and hypoglycaemia

糖尿病急症——酮症酸中毒、高血糖高渗状态与低血糖

期刊: Nature Reviews. Endocrinology 日期: 2016

Abstract

diabetic ketoacidosis (DKA), **hyperglycaemic hyperosmolar state** (HHS) and **hypoglycaemia** are serious complications of **diabetes mellitus** that require prompt recognition, **diagnosis** and treatment. DKA and HHS are characterized by **insulinopaenia** and severe **hyperglycaemia**; clinically, these two conditions differ only by the degree of **dehydration** and the severity of **metabolic acidosis**. The overall **mortality** recorded among children and adults with DKA is <1%. **mortality** among patients with HHS is ~10-fold higher than that associated with DKA. The **prognosis** and outcome of patients with DKA or HHS are determined by the severity of **dehydration**, the presence of comorbidities and age >60 years. The estimated annual cost of hospital treatment for patients experiencing hyperglycaemic crises in the USA exceeds US\$2 billion. **hypoglycaemia** is a frequent and serious adverse effect of antidiabetic therapy that is associated with both immediate and delayed adverse clinical outcomes, as well as increased economic costs. Inpatients who develop **hypoglycaemia** are likely to experience a long duration of hospital stay and increased **mortality**. This Review describes the clinical presentation, precipitating causes, **diagnosis** and acute **management** of these diabetic emergencies, including a discussion of practical strategies for their **prevention**.

中文翻译

糖尿病酮症酸中毒 (DKA)、高血糖高渗状态 (HHS) 和低血糖是糖尿病的严重并发症，需要及时识别、诊断和治疗。DKA和HHS以胰岛素缺乏和严重高血糖为特征；临床上，这两种情况的区别仅在于脱水程度和代谢性酸中毒的严重性。儿童和成人DKA的总体死亡率记录低于1%。HHS患者的死亡率比DKA相关死亡率高出约10倍。DKA或HHS患者的预后和结果取决于脱水严重程度、合并症的存在以及年龄是否超过60岁。在美国，高血糖危象患者住院治疗的年度估计费用超过20亿美元。低血糖是抗糖尿病治疗中常见且严重的不良反应，与即时和延迟的不良临床结果以及经济成本增加相关。发生低血糖的住院患者可能经历较长的住院时间和增加的死亡率。本综述描述了这些糖尿病急症的临床表现、诱发原因、诊断和急性管理，包括讨论其预防的实际策略。

核心词汇

diabetic ketoacidosis /ˌdaɪəˈbetɪk ˌkiːtəʊˈæsiˈdəʊsɪs/ : 糖尿病酮症酸中毒

hypoglycaemia /ˌhaɪpəʊɡlaɪˈsiːmiə/ : 低血糖症

insulinopaenia /ˌɪnsjʊlɪnəʊˈpiːniə/ : 胰岛素缺乏症

dehydration /ˌdiːˈhaɪˈdreɪʃən/ : 脱水

mortality /ˈmɔːrˈtælɪti/ : 死亡率

prognosis /ˈprɒɡˈnoʊsɪs/ : 预后

management /ˈmænɪdʒmənt/ : 管理

hyperglycaemic hyperosmolar state /ˌhaɪpərglaɪˈsiːmɪk ˌhaɪpəˈɒzmələr steɪt/ : 高血糖高渗状态

diabetes mellitus /ˌdaɪəˈbiːtɪz məˈleɪtəs/ : 糖尿病

hyperglycaemia /ˌhaɪpərglaɪˈsiːmiə/ : 高血糖症

metabolic acidosis /ˌmetəˈbɒlɪk ˌæsiˈdəʊsɪs/ : 代谢性酸中毒

comorbidity /ˌkəʊmɔːrˈbɪdɪti/ : 共病

diagnosis /ˌdaɪəɡˈnoʊsɪs/ : 诊断

prevention /prɪˈvenʃən/ : 预防

Management of diabetic neuropathy

糖尿病神经病变的管理

期刊: Metabolism 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 11.9

Abstract

diabetic neuropathy is a **neurodegenerative disorder** that may alter both the somatic and autonomic peripheral nervous systems in the context of **diabetes mellitus** (DM). It is a prevalent and burdensome **chronic complication** of DM, that requires timely management. Optimized **glycemic control** (mainly for type 1 DM), multifactorial intervention (mainly for type 2 DM), with lifestyle intervention/physical exercise, and weight loss represent the basis of management for diabetic distal symmetrical **polyneuropathy**, and should be implemented early in the disease course. Despite better understanding of the pathogenetic mechanisms of diabetic peripheral neuropathy, there is still a stringent need for more pathogenetic-based agents that would significantly modify the natural history of the disease. The paper reviews the available drugs and current recommendations for the management of distal symmetrical **polyneuropathy**, including **pain management**, and for diabetic **autonomic neuropathy**. Evaluation of drug combinations that would perhaps be more efficient in slowing the **progression** of the disease or even reversing it, and that would provide a better **pain management** is still needed.

中文翻译

糖尿病神经病变是一种神经退行性疾病，在糖尿病背景下可能影响体感和自主外周神经系统。作为糖尿病常见且负担沉重的慢性并发症，其需要及时管理。优化血糖控制（主要针对1型糖尿病）、多因素干预（主要针对2型糖尿病），结合生活方式干预/体育锻炼及减重，构成了糖尿病远端对称性多发性神经病变管理的基础，应在病程早期实施。尽管对糖尿病周围神经病变的发病机制有了更深入的理解，但仍迫切需要更多基于发病机制的药物，以显著改变疾病的自然进程。本文综述了可用于远端对称性多发性神经病变（包括疼痛管理）及糖尿病自主神经病变的现有药物及当前治疗建议。仍需评估可能更有效延缓甚至逆转疾病进展，并提供更好疼痛管理的药物组合方案。

核心词汇

diabetic neuropathy /ˌdaɪəˈbetɪk njuːˈrɒpəθi/ : 糖尿病神经病变

peripheral nervous system /pəˈrɪfərəl ˈnɜːrvəs ˈsɪstəm/ : 周围神经系统

polyneuropathy /ˌpɒlɪnjuːˈrɒpəθi/ : 多发性神经病

glycemic control /ˈɡlaɪˈsiːmɪk kənˈtrɒl/ : 血糖控制

pain management /peɪn ˈmænɪdʒmənt/ : 疼痛管理

progression /prəˈɡreʃən/ : 进展

neurodegenerative disorder /ˌnjuːəroʊdɪˈdʒɛnərətɪv dɪsˈɔːrdər/ : 神经退行性疾病

diabetes mellitus /ˌdaɪəˈbiːtiːz məˈlaɪtəs/ : 糖尿病

autonomic neuropathy /ˌɔːtəˈnɒmɪk njuːˈrɒpəθi/ : 自主神经病变

pathogenetic mechanism /ˌpæθəˈdʒɛnɪtɪk ˈmekəˌnɪzəm/ : 发病机制

drug combination /drʌɡ ˌkɒmbɪˈneɪʃən/ : 药物联合治疗

chronic complication /ˈkrɒnɪk ˌkɒmplɪˈkeɪʃən/ : 慢性并发症

Stomach-derived human insulin-secreting organoids restore glucose homeostasis

胃源性人胰岛素分泌类器官恢复葡萄糖稳态

期刊: Nature Cell Biology 日期: 2023 JCR Q1 中科院升级版 生物学1区 IF 19.5

Abstract

Gut stem cells are accessible by **biopsy** and propagate robustly in **culture**, offering an invaluable resource for **autologous** cell therapies. **insulin**-producing cells can be induced in mouse gut, but it has not been possible to generate abundant and durable **insulin**-secreting cells from human gut tissues to evaluate their potential as a **cell therapy** for **diabetes**. Here we describe a **protocol** to differentiate cultured human **gastric** stem cells into **pancreatic islet**-like organoids containing **gastric insulin**-secreting (GINS) cells that resemble beta-cells in molecular hallmarks and function. Sequential activation of the inducing factors NGN3 and PDX1-MAFA led human **gastric** stem cells onto a distinctive **differentiation** path, including a SOX4(High) endocrine and Galanin(High) GINS precursor, before adopting beta-cell identity, at efficiencies close to 70%. GINS organoids acquired glucose-stimulated **insulin** secretion in 10 days and restored glucose homeostasis for over 100 days in diabetic mice after transplantation, providing proof of concept for a promising approach to treat **diabetes**.

中文翻译

肠道干细胞可通过活检获取，并在培养中稳定增殖，为自体细胞疗法提供了宝贵资源。小鼠肠道中可诱导产生胰岛素分泌细胞，但尚未能从人类肠道组织中生成充足且持久的胰岛素分泌细胞，以评估其作为糖尿病细胞疗法的潜力。本文描述了一种将培养的人胃干细胞分化为含胃胰岛素分泌（GINS）细胞的胰岛样类器官的方案，这些细胞在分子特征和功能上均类似β细胞。通过依次激活诱导因子NGN3和PDX1-MAFA，人胃干细胞进入独特的分化路径，包括形成SOX4高表达的内分泌前体细胞和Galanin高表达的GINS前体细胞，最终获得β细胞特性，分化效率接近70%。GINS类器官在10天内获得葡萄糖刺激的胰岛素分泌能力，并在移植至糖尿病小鼠后恢复血糖稳态超过100天，为治疗糖尿病提供了有前景的方法的概念验证。

核心词汇

stem cell /stem sɛl/ : 干细胞

cell therapy /sɛl ˈθɛrəpi/ : 细胞治疗

insulin /ˈɪnsjʊlɪn/ : 胰岛素

pancreatic islet /ˌpæŋkriˈætrɪk ˈaɪlət/ : 胰岛

differentiation /ˌdɪfəˈrɛnʃiˈeɪʃən/ : 分化

culture /ˈkʌltʃə/ : 培养

biopsy /ˈbaɪɒpsi/ : 活检

diabetes /ˌdaɪəˈbiːtiːz/ : 糖尿病

gastric /ˈgæstrɪk/ : 胃的

organoid /ˈɔːɡənɔɪd/ : 类器官

autologous /ɔːˈtɒləɡəs/ : 自体的

protocol /ˈprəʊtəkol/ : 方案/规程

Diabetic kidney disease

糖尿病肾病

期刊: Nature Reviews Disease Primers 日期: 2015 JCR Q1 中科院升级版 医学1区 IF 60.6

Abstract

The **kidney** is arguably the most important target of **microvascular damage** in **diabetes**. A substantial proportion of individuals with **diabetes** will develop **kidney** disease owing to their disease and/or other co-morbidity, including **hypertension** and ageing-related **nephron** loss. The presence and severity of **chronic kidney disease** (CKD) identify individuals who are at increased risk of adverse health outcomes and premature **mortality**. Consequently, preventing and managing CKD in patients with **diabetes** is now a key aim of their overall management. Intensive management of patients with **diabetes** includes controlling **blood glucose** levels and **blood pressure** as well as blockade of the renin-angiotensin-aldosterone system; these approaches will reduce the incidence of **diabetic kidney disease** and slow its **progression**. Indeed, the major decline in the incidence of **diabetic kidney disease** (DKD) over the past 30 years and improved patient prognosis are largely attributable to improved **diabetes** care. However, there remains an unmet need for innovative treatment strategies to prevent, arrest, treat and reverse DKD. In this Primer, we summarize what is now known about the molecular **pathogenesis** of CKD in patients with **diabetes** and the key pathways and targets implicated in its **progression**. In addition, we discuss the current evidence for the prevention and management of DKD as well as the many controversies. Finally, we explore the opportunities to develop new interventions through urgently needed investment in dedicated and focused research. For an illustrated summary of this Primer, visit: <http://go.nature.com/NKHDzg>.

中文翻译

肾脏无疑是糖尿病微血管损伤最重要的靶器官。相当一部分糖尿病患者会因其疾病和/或其他合并症（包括高血压和与年龄相关的肾单位丧失）而发展为肾脏疾病。慢性肾脏病（CKD）的存在及其严重程度，标志着患者面临更高的不良健康结局和过早死亡风险。因此，预防和管理糖尿病患者的CKD已成为其整体治疗的关键目标。对糖尿病患者的强化管理包括控制血糖和血压，以及阻断肾素-血管紧张素-醛固酮系统；这些措施将降低糖尿病肾病的发病率并延缓其进展。事实上，过去30年间糖尿病肾病（DKD）发病率的大幅下降和患者预后的改善，很大程度上归功于糖尿病护理的进步。然而，在预防、阻止、治疗和逆转DKD方面，创新治疗策略的需求仍未得到满足。在本篇《基础指南》中，我们总结了目前对糖尿病患者CKD分子发病机制的认识，以及与其进展相关的关键通路和靶点。此外，我们讨论了当前关于DKD预防和管理的证据以及诸多争议。最后，我们探讨了通过对专门和重点研究的急需投资来开发新干预措施的机会。如需查看本指南的图解摘要，请访问：<http://go.nature.com/NKHDzg>。

核心词汇

kidney /'kɪdni/ : 肾脏

hypertension /,haɪpər'tenʃən/ : 高血压

diabetic kidney disease /,daɪə'betɪk 'kɪdni dɪ'zi:z/ : 糖尿病肾病

nephron /'nefrən/ : 肾单位

blood pressure /blʌd 'prefər/ : 血压

progression /prə'ɡrefən/ : 进展

diabetes /,daɪə'bi:ti:z/ : 糖尿病

chronic kidney disease /'krɒnɪk 'kɪdni dɪ'zi:z/ : 慢性肾脏病

microvascular damage /,maɪkrəv'væskjʊlər 'dæmɪdʒ/ : 微血管损伤

blood glucose /blʌd 'glu:kəʊs/ : 血糖

pathogenesis /,pæθə'dʒenɪsɪs/ : 发病机制

mortality /mɔ:ɪ'tæləti/ : 死亡率

Low-dose metformin targets the lysosomal AMPK pathway through PEN2

低剂量二甲双胍通过PEN2靶向溶酶体AMPK通路

期刊: Nature 日期: 2022 JCR Q1 中科院升级版 综合性期刊1区 IF 48.5

Abstract

Abstract Metformin, the most prescribed antidiabetic medicine, has shown other benefits such as anti-ageing and anticancer effects^{1–4}. For clinical doses of metformin, AMP-activated protein kinase (AMPK) has a major role in its mechanism of action^{4,5}; however, the direct molecular target of metformin remains unknown. Here we show that clinically relevant concentrations of metformin inhibit the lysosomal proton pump v-ATPase, which is a central node for AMPK activation following glucose starvation⁶. We synthesize a photoactive metformin probe and identify PEN2, a subunit of γ -secretase⁷, as a binding partner of metformin with a dissociation constant at micromolar levels. metformin-bound PEN2 forms a complex with ATP6AP1, a subunit of the v-ATPase⁸, which leads to the inhibition of v-ATPase and the activation of AMPK without effects on cellular AMP levels. knockout of PEN2 or re-introduction of a PEN2 mutant that does not bind ATP6AP1 blunts AMPK activation. In vivo, liver-specific knockout of Pen2 abolishes metformin-mediated reduction of hepatic fat content, whereas intestine-specific knockout of Pen2 impairs its glucose-lowering effects. Furthermore, knockdown of pen-2 in *Caenorhabditis elegans* abrogates metformin-induced extension of lifespan. Together, these findings reveal that metformin binds PEN2 and initiates a signalling route that intersects, through ATP6AP1, the lysosomal glucose-sensing pathway for AMPK activation. This ensures that metformin exerts its therapeutic benefits in patients without substantial adverse effects.

中文翻译

摘要 二甲双胍是最常用的抗糖尿病药物，并显示出抗衰老和抗癌等其他益处¹⁻⁴。对于临床剂量的二甲双胍，AMP激活的蛋白激酶（AMPK）在其作用机制中起主要作用^{4,5}；然而，二甲双胍的直接分子靶点仍然未知。本文表明，临床相关浓度的二甲双胍可抑制溶酶体质子泵v-ATPase，这是葡萄糖饥饿后激活AMPK的核心节点⁶。我们合成了一种光活性二甲双胍探针，并鉴定出 γ -分泌酶⁷的亚基PEN2作为二甲双胍的结合伴侣，其解离常数在微摩尔水平。与二甲双胍结合的PEN2与v-ATPase⁸的亚基ATP6AP1形成复合物，从而抑制v-ATPase并激活AMPK，而不影响细胞AMP水平。敲除PEN2或重新引入不结合ATP6AP1的PEN2突变体会削弱AMPK的激活。在体内，肝脏特异性敲除Pen2消除了二甲双胍介导的肝脏脂肪含量降低，而肠道特异性敲除Pen2则损害了其降血糖效果。此外，在秀丽隐杆线虫中敲低pen-2会消除二甲双胍诱导的寿命延长。总之，这些发现揭示了二甲双胍结合PEN2并启动一条信号通路，该通路通过ATP6AP1与溶酶体葡萄糖感应途径相交，从而激活AMPK。这确保了二甲双胍在患者中发挥其治疗益处而不会产生显著的不良反应。

核心词汇

metformin /ˈmetˈfɔːrmin/ : 二甲双胍

glucose /ˈgluːkʊs/ : 葡萄糖

hepatic /hiˈpæɪtɪk/ : 肝脏的

knockout /ˈnɒkɔʊt/ : 敲除

lysosomal /ˌlaɪsəˈsɒməl/ : 溶酶体的

subunit /ˈsʌbjuːnɪt/ : 亚基

signalling route /ˈsɪɡnəlɪŋ ruːt/ : 信号通路

antidiabetic /ˌæntɪˈdaɪəˈbetɪk/ : 抗糖尿病的

lifespan /ˈlaɪfspæn/ : 寿命

fat content /fætˈkɒntent/ : 脂肪含量

mutant /ˈmjuːtənt/ : 突变体

proton pump /ˈpraʊtɒn pʌmp/ : 质子泵

dissociation constant /dɪˌsɒʊsiˈeɪʃənˈkɒnstənt/ : 解离常数

therapeutic benefits /ˌθerəˈpjʊːtɪkˈbenɪfɪts/ : 治疗益处

Metformin: Update on mechanisms of action and repurposing potential

二甲双胍：作用机制与再利用潜力的最新进展

期刊: Nature Reviews Endocrinology 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

Currently, **metformin** is the first-line medication to treat type 2 diabetes mellitus (T2DM) in most guidelines and is used daily by >200 million patients. Surprisingly, the mechanisms underlying its therapeutic action are complex and are still not fully understood. Early evidence highlighted the **liver** as the major organ involved in the effect of **metformin** on reducing blood levels of **glucose**. However, increasing evidence points towards other sites of action that might also have an important role, including the **gastrointestinal tract**, the **gut microbial communities** and the tissue-resident **immune cells**. At the molecular level, it seems that the mechanisms of action vary depending on the dose of **metformin** used and duration of treatment. Initial studies have shown that **metformin** targets hepatic **mitochondria**; however, the identification of a novel target at low concentrations of **metformin** at the **lysosome** surface might reveal a new mechanism of action. Based on the efficacy and safety records in T2DM, attention has been given to the repurposing of **metformin** as part of **adjunct therapy** for the treatment of **cancer**, age-related diseases, **inflammatory diseases** and COVID-19. In this Review, we highlight the latest advances in our understanding of the mechanisms of action of **metformin** and discuss potential emerging novel therapeutic uses.

中文翻译

目前，在大多数指南中，二甲双胍是治疗2型糖尿病的一线药物，每日使用患者超过2亿。令人惊讶的是，其治疗作用的机制复杂且尚未完全明了。早期证据强调肝脏是二甲双胍降低血糖水平的主要作用器官。然而，越来越多的证据指向其他可能也起重要作用的作用部位，包括胃肠道、肠道微生物群落和组织驻留免疫细胞。在分子水平上，其作用机制似乎因二甲双胍的剂量和治疗持续时间而异。初步研究表明，二甲双胍靶向肝线粒体；然而，在低浓度二甲双胍作用下，溶酶体表面新靶点的发现可能揭示了一种新的作用机制。基于其在2型糖尿病中的疗效和安全性记录，人们开始关注将二甲双胍重新用作癌症、年龄相关疾病、炎症性疾病和COVID-19辅助治疗的一部分。在本综述中，我们重点介绍了对二甲双胍作用机制理解的最新进展，并讨论了其潜在的新兴治疗用途。

核心词汇

metformin /met'fɔ:rmɪn/ : 二甲双胍	glucose /'glu:kʊs/ : 葡萄糖
liver /'lɪvə/ : 肝脏	gastrointestinal tract /,gæstrəʊɪn'testɪnəl trækt/ : 胃肠道
gut microbial communities /gʌt maɪ'krəʊbiəl ke'mju:nətiz/ : 肠道微生物群落	immune cells /ɪ'mju:n selz/ : 免疫细胞
mitochondria /,maɪtəu'kɒndrɪə/ : 线粒体	lysosome /'laɪsə'səʊm/ : 溶酶体
cancer /'kænsə/ : 癌症	inflammatory diseases /ɪn'flæmə'tɔ:ri dɪ'zi:zɪz/ : 炎症性疾病
adjunct therapy /'ædʒʌŋkt 'θerəpi/ : 辅助治疗	

Endothelial AGGF1 promotes retinal angiogenesis by coordinating TNFSF12/FN14 signalling

内皮AGGF1通过协调TNFSF12/FN14信号促进视网膜血管生成

期刊: Nature Communications 日期: 2025 JCR Q1 中科院升级版 综合性期刊1区 IF 15.7

Abstract

Abnormal **angiogenesis** is a key process associated with ischaemic retinopathies such as **diabetic retinopathy**, for which the underlying **pathological** mechanisms are still poorly understood. Here, we confirm that angiogenic factor 1 with a G patch and FHA domain (AGGF1) is elevated in the diabetics and induces retinal **angiogenesis**. Mechanistic investigations demonstrate that HIF-1 α directly regulates AGGF1 **expression**. AGGF1 upregulates the **expression** of cell cycle proteins by increasing the binding of tumour necrosis factor ligand superfamily member 12 (TNFSF12) to fibroblast -growth -factor-inducible 14 (FN14, TNFRSF12A). Furthermore, targeting AGGF1 attenuates **pathological neovascularisation** in **ischaemic retinopathy**. Additionally, we discover that sodium-glucose cotransporter 2 inhibitors (SGLT2i) could inhibit the AGGF1 **signalling pathway** early to achieve **therapeutic** effects. Overall, we elucidate the **mechanism** underlying **pathological** retinal **angiogenesis** involved in **endothelial** AGGF1-dependent events and highlight a therapy for the effective treatment of **ischaemic retinopathy**.

中文翻译

异常血管生成是与缺血性视网膜病变（如糖尿病视网膜病变）相关的关键过程，其潜在的病理机制尚不清楚。本文证实，具有G斑块和FHA结构域的血管生成因子1（AGGF1）在糖尿病患者中升高并诱导视网膜血管生成。机制研究表明，HIF-1 α 直接调控AGGF1的表达。AGGF1通过增加肿瘤坏死因子配体超家族成员12（TNFSF12）与成纤维细胞生长因子诱导型14（FN14，TNFRSF12A）的结合，上调细胞周期蛋白的表达。此外，靶向AGGF1可减轻缺血性视网膜病变中的病理性新生血管形成。另外，我们发现钠-葡萄糖协同转运蛋白2抑制剂（SGLT2i）可早期抑制AGGF1信号通路，从而达到治疗效果。总之，我们阐明了涉及内皮AGGF1依赖性事件的病理性视网膜血管生成的机制，并强调了一种有效治疗缺血性视网膜病变的方法。

核心词汇

angiogenesis /ˌændʒiəʊˈdʒɛnɪsɪs/ : 血管生成

diabetic retinopathy /ˌdaɪəˈbɛtɪk ˌrɛtɪˈnɒpəθi/ : 糖尿病视网膜病变

neovascularisation /ˌniːəʊ.væskjʊləraɪˈzeɪʃən/ : 新生血管形成

pathological /ˌpæθəˈlɒdʒɪkəl/ : 病理性的

expression /ɪkˈsprɛʃən/ : 表达

therapeutic /ˌθɛrəˈpjʊːtɪk/ : 治疗的

ischaemic retinopathy /ɪsˈkiːmɪk ˌrɛtɪˈnɒpəθi/ : 缺血性视网膜病变

retina /ˈrɛtɪnə/ : 视网膜

endothelial /ˌɛndəʊˈθiːliəl/ : 内皮的

mechanism /ˈmɛkənɪzəm/ : 机制

inhibitor /ɪnˈhɪbɪtə/ : 抑制剂

signalling pathway /ˈsɪgnəlɪŋ ˈpæθweɪ/ : 信号通路

Diabetic retinopathy

糖尿病视网膜病变

期刊: Nature Reviews Disease Primers 日期: 2016 JCR Q1 中科院升级版 医学1区 IF 60.6

Abstract

diabetic retinopathy (DR) is a common complication of **diabetes mellitus** and is a major cause of vision loss in middle-aged and elderly people. One-third of people with diabetes have DR. Severe stages of DR include proliferative DR, caused by the abnormal growth of new retinal blood vessels, and **diabetic macular oedema**, in which there is exudation and oedema in the central part of the **retina**. DR is strongly associated with a prolonged duration of diabetes, **hyperglycaemia** and **hypertension**. It is traditionally regarded as a **microvascular disease**, but retinal **neurodegeneration** is also involved. Complex interrelated pathophysiological mechanisms triggered by **hyperglycaemia** underlie the development of DR. These mechanisms include genetic and epigenetic factors, increased production of free radicals, advanced glycosylation end products, inflammatory factors and vascular endothelial growth factor (VEGF). Optimal control of blood glucose and blood pressure in individuals with diabetes remains the cornerstone for preventing the development and arresting the progression of DR. Anti-VEGF therapy is currently indicated for **diabetic macular oedema** associated with vision loss, whereas **laser photocoagulation** prevents severe vision loss in eyes with proliferative DR. These measures, together with increasing public awareness and access to regular screening for DR with **retinal photography**, and the development of new treatments to address early disease stages, will lead to better outcomes and prevent blindness for patients with DR.

中文翻译

糖尿病视网膜病变（DR）是糖尿病的常见并发症，也是中老年人视力丧失的主要原因。三分之一的糖尿病患者患有DR。DR的严重阶段包括由视网膜新生血管异常生长引起的增殖性DR，以及糖尿病性黄斑水肿，即视网膜中央部分出现渗出和水肿。DR与糖尿病病程延长、高血糖和高血压密切相关。传统上，DR被视为一种微血管疾病，但视网膜神经退行性变也参与其中。高血糖引发的复杂且相互关联的病理生理机制是DR发展的基础。这些机制包括遗传和表观遗传因素、自由基产生增加、晚期糖基化终末产物、炎症因子和血管内皮生长因子（VEGF）。对糖尿病患者血糖和血压的最佳控制仍然是预防DR发生和阻止其进展的基石。目前，抗VEGF疗法适用于与视力丧失相关的糖尿病性黄斑水肿，而激光光凝术可预防增殖性DR患者的严重视力丧失。这些措施，加上提高公众意识、通过视网膜摄影进行定期筛查的普及，以及针对疾病早期阶段的新疗法的开发，将为DR患者带来更好的预后并预防失明。

核心词汇

diabetic retinopathy /ˌdaɪəˈbetɪk ˌretɪˈnɒpəθi/ : 糖尿病视网膜病变

proliferative retinopathy /prəˈlɪfəretɪv ˌretɪˈnɒpəθi/ : 增殖性视网膜病变

retina /ˈretɪnə/ : 视网膜

hyperglycaemia /ˌhaɪpəˈɡlaɪˈsiːmiə/ : 高血糖

microvascular disease /ˌmaɪkrəʊˈvæskjələ dɪˈziːz/ : 微血管病变

pathophysiology /ˌpæθəʊˈfɪziːˈɒlədʒi/ : 病理生理学

retinal photography /ˈretɪnəl feˈtɒɡrəfi/ : 视网膜摄影

diabetes mellitus /ˌdaɪəˈbiːtiːz məˈlaɪtəs/ : 糖尿病

diabetic macular oedema /ˌdaɪəˈbetɪk ˈmækjʊlə ɪˈdiːmə/ : 糖尿病黄斑水肿

macula /ˈmækjʊlə/ : 黄斑

hypertension /ˌhaɪpəˈtenʃən/ : 高血压

neurodegeneration /ˌnjuərəʊdɪˌdʒɛnəˈreɪʃən/ : 神经退行性变

laser photocoagulation /ˈleɪzə ˌfəʊtəʊˌkeʊæɡjuˈleɪʃən/ : 激光光凝术

Diabetic ketoacidosis

糖尿病酮症酸中毒

期刊: Nature reviews. Disease primers 日期: 2020

Abstract

diabetic ketoacidosis (DKA) is the most common acute **hyperglycaemic emergency** in people with **diabetes mellitus**. A **diagnosis** of DKA is confirmed when all of the three **criteria** are present - 'D', either elevated **blood glucose** levels or a family history of **diabetes mellitus**; 'K', the presence of high urinary or blood ketoacids; and 'A', a high anion gap metabolic **acidosis**. Early **diagnosis** and management are paramount to improve patient outcomes. The mainstays of treatment include restoration of circulating volume, insulin therapy, electrolyte replacement and treatment of any underlying precipitating event. Without optimal treatment, DKA remains a condition with appreciable, although largely preventable, morbidity and mortality. In this Primer, we discuss the epidemiology, pathogenesis, risk factors and **diagnosis** of DKA and provide practical recommendations for the management of DKA in adults and children.

中文翻译

糖尿病酮症酸中毒 (DKA) 是糖尿病患者中最常见的急性高血糖急症。当以下三个标准同时满足时即可确诊DKA——"D"指血糖水平升高或糖尿病家族史；"K"指尿液或血液中酮酸水平升高；"A"指高阴离子间隙代谢性酸中毒。早期诊断和处理对改善患者预后至关重要。治疗核心包括恢复循环血量、胰岛素治疗、电解质补充以及处理任何潜在的诱发事件。若未获得最佳治疗，DKA仍会带来显著（但基本可预防）的发病率和死亡率。本入门指南将探讨DKA的流行病学、发病机制、危险因素和诊断方法，并为成人和儿童DKA的管理提供实用建议。

核心词汇

diabetic ketoacidosis /ˌdaɪəˈbetɪk ˌkiːtəʊˈæsiˈdʊʊsɪs/ : 糖尿病酮症酸中毒	hyperglycaemic /ˌhaɪpəˈɡlaɪˈsiːmɪk/ : 高血糖的
diabetes mellitus /ˌdaɪəˈbiːtiːz məˈleɪtəs/ : 糖尿病	diagnosis /ˌdaɪəˈɡnoʊsɪs/ : 诊断
blood glucose /blʌd ˈɡluːkʊʊs/ : 血糖	ketosis /kiːˈtʊʊsɪs/ : 酮症
acidosis /ˌæsiˈdʊʊsɪs/ : 酸中毒	emergency /ɪˈmɜːrdʒənsɪ/ : 急症
criteria /kraɪˈtɪriə/ : 标准	metabolism /məˈtæbəlaɪzəm/ : 代谢

Hepatic krüppel-like factor 16 (KLF16) targets PPAR α to improve steatohepatitis and insulin resistance

肝Krüppel样因子16 (KLF16) 靶向PPAR α 改善脂肪性肝炎和胰岛素抵抗

期刊: Gut 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 25.8

Abstract

Objective Impaired hepatic fatty acids oxidation results in lipid accumulation and redox imbalance, promoting the development of fatty liver diseases and **insulin resistance**. However, the underlying pathogenic mechanism is poorly understood. Krüppel-like factor 16 (KLF16) is a **transcription factor** that abounds in liver. We explored whether and by what mechanisms KLF16 affects hepatic lipid catabolism to improve hepatosteatosis and **insulin resistance**. Design KLF16 expression was determined in patients with non-alcoholic **fatty liver disease** (NAFLD) and mice models. The role of KLF16 in the regulation of **lipid metabolism** was investigated using **hepatocyte**-specific KLF16-deficient mice fed a high-fat diet (HFD) or using an adenovirus/adenovirus-associated virus to alter KLF16 expression in mouse primary hepatocytes (MPHs) and in vivo livers. RNA-seq, luciferase reporter gene assay and ChIP analysis served to explore the molecular mechanisms involved. Results KLF16 expression was decreased in patients with NAFLD, mice models and oleic acid and palmitic acid (OA and PA) cochallenged hepatocytes. Hepatic KLF16 **knockout** impaired **fatty acid oxidation**, aggravated **mitochondrial stress**, ROS burden, advancing **hepatic steatosis** and **insulin resistance**. Conversely, KLF16 **overexpression** reduced lipid deposition and improved **insulin resistance** via directly binding the promoter of peroxisome proliferator-activated receptor α (PPAR α) to accelerate fatty acids oxidation and attenuate **mitochondrial stress**, **oxidative stress** in db/db and HFD mice. PPAR α deficiency diminished the KLF16-evoked protective effects against lipid deposition in MPHs. Hepatic-specific PPAR α **overexpression** effectively rescued KLF16 deficiency-induced **hepatic steatosis**, altered redox balance and **insulin resistance**. Conclusions These findings prove that a direct KLF16–PPAR α pathway closely links hepatic lipid homeostasis and redox balance, whose dysfunction promotes **insulin resistance** and **hepatic steatosis**.

中文翻译

目的 肝脏脂肪酸氧化受损导致脂质积累和氧化还原失衡，促进脂肪肝病和胰岛素抵抗的发展。然而，其潜在的致病机制尚不清楚。Krüppel样因子16 (KLF16) 是一种在肝脏中丰富的转录因子。我们探讨了KLF16是否以及通过何种机制影响肝脏脂质分解代谢，以改善肝脂肪变性和胰岛素抵抗。设计 在非酒精性脂肪肝病 (NAFLD) 患者和小鼠模型中测定KLF16的表达。通过使用高脂饮食 (HFD) 喂养的肝细胞特异性KLF16缺陷小鼠，或使用腺病毒/腺相关病毒改变小鼠原代肝细胞 (MPHs) 和体内肝脏中KLF16的表达，研究KLF16在脂质代谢调控中的作用。RNA测序、荧光素酶报告基因检测和ChIP分析用于探索相关分子机制。结果 在NAFLD患者、小鼠模型以及油酸和棕榈酸 (OA和PA) 共刺激的肝细胞中，KLF16表达降低。肝脏KLF16敲除损害了脂肪酸氧化，加剧了线粒体应激和ROS负担，促进了肝脏脂肪变性和胰岛素抵抗。相反，KLF16过表达通过直接结合过氧化物酶体增殖物激活受体 α (PPAR α) 的启动子，加速脂肪酸氧化并减轻线粒体应激和氧化应激，从而减少脂质沉积并改善胰岛素抵抗，这在db/db和HFD小鼠中得到验证。PPAR α 缺失削弱了KLF16在MPHs中对脂质沉积的保护作用。肝脏特异性PPAR α 过表达有效挽救了KLF16缺失引起的肝脏脂肪变性、氧化还原平衡改变和胰岛素抵抗。结论 这些发现证明，KLF16–PPAR α 直接通路紧密连接肝脏脂质稳态和氧化还原平衡，其功能障碍促进胰岛素抵抗和肝脏脂肪变性。

核心词汇

fatty liver disease /'fæti 'lɪvər dɪ'zi:z/ : 脂肪肝病

hepatocyte /hi 'pætəsaɪt/ : 肝细胞

oxidative stress /'ɒksɪdətɪv stres/ : 氧化应激

lipid metabolism /'lɪpɪd mɪ'tæbəlaɪzəm/ : 脂质代谢

transcription factor /træn'skrɪpʃən 'fæktər/ : 转录因子

overexpression /'əʊvərɪk'spreʃən/ : 过表达

insulin resistance /'ɪnsjʊlɪn rɪ'zɪstəns/ : 胰岛素抵抗

mitochondrial stress /,maɪtə'kɒndrɪəl stres/ : 线粒体应激

hepatic steatosis /hi 'pætrɪk ,sti:ə'teʊsɪs/ : 肝脂肪变性

fatty acid oxidation /'fæti 'æsaɪd ,ɒksɪ'deɪʃən/ : 脂肪酸氧化

knockout /'nɒkəʊt/ : 敲除

Steatotic liver disease

脂肪肝病

期刊: The Lancet

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JCR Q1

中科院升级版 医学1区

IF 88.5

Abstract

steatotic liver disease is the overarching term for conditions characterised by abnormal lipid accumulation in the liver (liver or hepatic **steatosis**). **steatotic liver disease** encompasses what was previously termed non-alcoholic fatty liver disease (NAFLD), which is now called metabolic dysfunction-associated **steatotic liver disease** (MASLD). Additionally, **steatotic liver disease** includes alcohol-related liver disease (ALD) and MetALD, the new classification for the overlap between MASLD and ALD, and rare causes of liver **steatosis**. **cirrhosis** is globally the 11th leading cause of death, and **steatotic liver disease** has become the leading cause of **cirrhosis** in the EU and USA. **steatotic liver disease** affects around 30% of the global population and is mainly driven by **obesity**, type 2 **diabetes**, and alcohol intake, but only a minor proportion with **steatotic liver disease** progress to **cirrhosis**. The presence and progression of liver **fibrosis** led by hepatic **inflammation** is the main predictor of liver-related death across the entire spectrum of steatotic liver diseases. A combination of recent advancements of widely available biomarkers for early detection of liver **fibrosis** together with considerable advancements in therapeutic interventions offer the possibility to reduce **morbidity** and **mortality** in patients with **steatotic liver disease**. This Seminar covers the recent reclassification of **steatotic liver disease** and how it reflects clinical practice and **prognosis**. For early detection of liver **fibrosis**, we propose a collaborative diagnostic framework between primary care and liver specialists. Lastly, we discuss current best practices for managing **steatotic liver disease**, we explore therapeutic targets across the spectrum of steatotic liver diseases, and we review the pipeline of drugs in development for MASLD.

中文翻译

脂肪性肝病是描述肝脏异常脂质积聚（肝脏或肝脂肪变性）相关疾病的总称。脂肪性肝病涵盖了先前被称为非酒精性脂肪性肝病（NAFLD）的疾病，现称为代谢功能障碍相关脂肪性肝病（MASLD）。此外，脂肪性肝病还包括酒精相关性肝病（ALD）以及MetALD（MASLD与ALD重叠的新分类）和罕见的肝脂肪变性病因。肝硬化在全球范围内是第11大死因，而脂肪性肝病已成为欧盟和美国肝硬化的主要原因。全球约30%的人口受脂肪性肝病影响，其主要驱动因素为肥胖、2型糖尿病和酒精摄入，但仅有少数脂肪性肝病患者会进展为肝硬化。由肝脏炎症引发的肝纤维化的存在和进展是预测所有类型脂肪性肝病相关死亡的主要指标。近期广泛可用的生物标志物在早期检测肝纤维化方面的进展，结合治疗干预手段的显著进步，为降低脂肪性肝病患者的发病率和死亡率提供了可能。本次研讨会涵盖了脂肪性肝病的最新重新分类及其如何反映临床实践和预后。为早期检测肝纤维化，我们提出了初级保健与肝脏专科医生之间的协作诊断框架。最后，我们讨论了当前管理脂肪性肝病的最佳实践，探索了脂肪性肝病谱系中的治疗靶点，并回顾了针对MASLD正在研发的药物管线。

核心词汇

steatotic liver disease /sti:ə'tɒtɪk 'lɪvə dɪ'zi:z/ : 脂肪性肝病	cirrhosis /sɪ'raʊsɪs/ : 肝硬化
fibrosis /faɪ'bɹoʊsɪs/ : 纤维化	inflammation /ˌɪnflə'meɪʃən/ : 炎症
steatosis /sti:ə'toʊsɪs/ : 脂肪变性	biomarker /'baɪəʊ'mɑ:rkə/ : 生物标志物
obesity /ou'bi:seɪti/ : 肥胖症	diabetes /ˌdaɪə'bi:ti:z/ : 糖尿病
morbidity /mɔ:r'brɪdəti/ : 发病率	mortality /mɔ:r'tæləti/ : 死亡率
prognosis /prɒg'nəʊsɪs/ : 预后	therapeutic intervention /ˌθerə'pjʊ:ɪtk ˌɪntər'veɪʃən/ : 治疗干预

Non-alcoholic fatty liver disease

非酒精性脂肪肝病

期刊: The Lancet 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

Non-alcoholic fatty liver disease (NAFLD) has a global prevalence of 25% and is a leading cause of **cirrhosis** and **hepatocellular carcinoma**. NAFLD encompasses a disease continuum from **steatosis** with or without mild **inflammation** (non-alcoholic fatty liver), to non-alcoholic steatohepatitis (NASH), which is characterised by **necroinflammation** and faster **fibrosis** progression than non-alcoholic fatty liver. NAFLD has a bidirectional association with components of the **metabolic syndrome**, and type 2 diabetes increases the risk of **cirrhosis** and related complications. Although the leading causes of death in people with NAFLD are cardiovascular disease and extrahepatic malignancy, advanced liver **fibrosis** is a key **prognostic marker** for liver-related outcomes and overall mortality, and can be assessed with combinations of non-invasive tests. Patients with **cirrhosis** should be screened for **hepatocellular carcinoma** and **oesophageal varices**. There is currently no approved therapy for NAFLD, although several drugs are in advanced stages of development. Because of the complex **pathophysiology** and substantial heterogeneity of disease phenotypes, **combination treatment** is likely to be required for many patients with NAFLD. Healthy lifestyle and weight reduction remain crucial to the prevention and treatment of NAFLD.

中文翻译

非酒精性脂肪性肝病（NAFLD）全球患病率为25%，是导致肝硬化和肝细胞癌的主要原因。NAFLD涵盖从伴有或不伴有轻度炎症的脂肪变性（非酒精性脂肪肝）到非酒精性脂肪性肝炎（NASH）的疾病连续谱，后者以坏死性炎症和比非酒精性脂肪肝更快的纤维化进展为特征。NAFLD与代谢综合征各组分存在双向关联，2型糖尿病会增加肝硬化及相关并发症的风险。尽管NAFLD患者的主要死因是心血管疾病和肝外恶性肿瘤，但晚期肝纤维化是肝脏相关结局和总体死亡率的关键预后标志，可通过非侵入性检测组合进行评估。肝硬化患者应筛查肝细胞癌和食管静脉曲张。目前尚无获批的NAFLD治疗方法，尽管有几种药物处于研发后期阶段。由于复杂的病理生理学和疾病表型的显著异质性，许多NAFLD患者可能需要联合治疗。健康的生活方式和减重对于NAFLD的预防和治疗仍然至关重要。

核心词汇

cirrhosis /sɪˈrʊʊsɪs/ : 肝硬化**steatosis** /stiːəˈtəʊsɪs/ : 脂肪变性**necroinflammation** /ˌnekroʊɪnfləˈmeɪʃən/ : 坏死性炎症**metabolic syndrome** /ˌmetəˈbɒlɪk ˈsɪndrəʊm/ : 代谢综合征**oesophageal varices** /ɪˌsoʊəˈdʒiːəl ˈvæɪsɪːz/ : 食管静脉曲张**combination treatment** /ˌkɒmbɪˈneɪʃən ˈtriːtmənt/ : 联合治疗**hepatocellular carcinoma** /ˌhepəteʊˈseljʊlə ˈkɑːsɪˈneʊmə/ : 肝细胞癌**inflammation** /ˌɪnfləˈmeɪʃən/ : 炎症**fibrosis** /faɪˈbrʊʊsɪs/ : 纤维化**prognostic marker** /prɒɡˈnɒstɪk ˈmɑːkə/ : 预后标志物**pathophysiology** /ˌpæθəʊˈfɪziːɒlədʒi/ : 病理生理学

MAFLD: How is it different from NAFLD?

MAFLD：与NAFLD有何不同？

期刊: Clinical and Molecular Hepatology 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 16.9

Abstract

“metabolic dysfunction-associated fatty liver disease (MAFLD)” is the term suggested in 2020 to refer to fatty liver disease related to systemic metabolic dysregulation. The name change from nonalcoholic fatty liver disease (NAFLD) to MAFLD comes with a simple set of criteria to enable easy diagnosis at the bedside for the general medical community, including primary care physicians. Since the introduction of the term, there have been key areas in which the superiority of MAFLD over the traditional NAFLD terminology has been demonstrated, including for the risk of liver and extrahepatic mortality, disease associations, and for identifying high-risk individuals. Additionally, MAFLD has been adopted by a number of leading pan-national and national societies due to its concise diagnostic criterion, removal of the requirement to exclude concomitant liver diseases, and reduction in the stigma associated with this condition. The current article explores the differences between MAFLD and NAFLD diagnosis, areas of benefit, some potential limitations, and how the MAFLD terminology has opened up new fields of research.

中文翻译

“代谢功能障碍相关脂肪性肝病（MAFLD）”是2020年提出的术语，用于指代与全身代谢失调相关的脂肪性肝病。从非酒精性脂肪性肝病（NAFLD）更名为MAFLD，附带了一套简单的诊断标准，便于包括初级保健医生在内的广大医学界在床边进行诊断。自该术语引入以来，MAFLD在多个关键领域显示出优于传统NAFLD术语的特点，包括肝脏及肝外死亡风险、疾病关联性以及高风险人群识别等方面。此外，由于其诊断标准简洁、无需排除伴随性肝病，并减少了与此病症相关的污名化，MAFLD已被多个领先的跨国及国家级学会采纳。本文探讨了MAFLD与NAFLD诊断的差异、受益领域、一些潜在局限性，以及MAFLD术语如何开辟了新的研究领域。

核心词汇

fatty liver disease /ˈfæti ˈlɪvər dɪˈziːz/ : 脂肪肝病	metabolic dysfunction /ˌmetəˈbɒlɪk dɪsˈfʌŋkʃən/ : 代谢功能障碍
diagnosis /ˌdaɪəɡˈnəʊsɪs/ : 诊断	mortality /mɔːˈtæləti/ : 死亡率
liver /ˈlɪvər/ : 肝脏	extrahepatic /ˌekstreˈhɪˈpætɪk/ : 肝外的
disease association /dɪˈziːz əˈsəʊsiˈeɪʃən/ : 疾病关联	concomitant liver disease /kənˈkɒmɪtənt ˈlɪvər dɪˈziːz/ : 伴随性肝病
stigma /ˈstɪɡmə/ : 病耻感	primary care physician /ˈpraɪməri keə ˈfɪːziʃən/ : 初级保健医生
terminology /ˌtɜːmɪˈnɒlədʒi/ : 术语学	

Hepatic selective insulin resistance at the intersection of insulin signaling and metabolic dysfunction-associated steatotic liver disease

肝脏选择性胰岛素抵抗在胰岛素信号传导与代谢功能障碍相关脂肪性肝病的交汇点

期刊: Cell Metabolism 日期: 2024 JCR Q1 中科院升级版 生物学1区 IF 30.9

Abstract

insulin resistance (IR) is a major pathogenic factor in the progression of MASLD. In the liver, insulin suppresses gluconeogenesis and enhances de novo lipogenesis (DNL). During IR, there is a defect in insulin-mediated suppression of gluconeogenesis, but an unrestrained increase in hepatic lipogenesis persists. The mechanism of increased hepatic steatosis in IR is unclear and remains controversial. The key discrepancy is whether insulin retains its ability to directly regulate hepatic lipogenesis. Blocking insulin/IRS/AKT signaling reduces liver lipid deposition in IR, suggesting insulin can still regulate lipid metabolism; hepatic glucose metabolism that bypasses insulin's action may contribute to lipogenesis; and due to peripheral IR, other tissues are likely to impact liver lipid deposition. We here review the current understanding of insulin's action in governing different aspects of hepatic lipid metabolism under normal and IR states, with the purpose of highlighting the essential issues that remain unsettled.

中文翻译

胰岛素抵抗 (IR) 是MASLD进展中的一个主要致病因素。在肝脏中，胰岛素抑制糖异生并促进新生脂肪生成 (DNL)。在IR状态下，胰岛素介导的糖异生抑制存在缺陷，但肝脏脂肪生成却不受控制地持续增加。IR状态下肝脏脂肪变性增加的机制尚不明确，且存在争议。关键分歧在于胰岛素是否仍保留直接调控肝脏脂肪生成的能力。阻断胰岛素/IRS/AKT信号通路可减少IR状态下的肝脏脂质沉积，表明胰岛素仍能调控脂质代谢；绕过胰岛素作用的肝脏葡萄糖代谢可能促进脂肪生成；由于外周IR，其他组织也可能影响肝脏脂质沉积。本文综述了目前对正常及IR状态下胰岛素调控肝脏脂质代谢不同方面的理解，旨在强调尚未解决的核心问题。

核心词汇

insulin resistance /ˈɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗	gluconeogenesis /ˌɡluːkəʊˈniːoʊˈdʒenɪsɪs/ : 糖异生
lipogenesis /ˌlɪpəʊˈdʒenɪsɪs/ : 脂肪生成	hepatic steatosis /hɪˈpæɪtɪk ˌstiːəˈtoʊsɪs/ : 肝脂肪变性
liver /ˈlɪvər/ : 肝脏	insulin /ˈɪnsjʊlɪn/ : 胰岛素
lipid metabolism /ˈlɪpɪd məˈtæbəˌlɪzəm/ : 脂质代谢	glucose metabolism /ˌɡluːkəʊs məˈtæbəˌlɪzəm/ : 葡萄糖代谢
pathogenic factor /ˌpæθəˈdʒenɪk ˈfæktər/ : 致病因素	progression /prəˈɡreʃən/ : 进展
mechanism /ˈmekəˌnɪzəm/ : 机制	

Pathogenesis and treatment of non-alcoholic steatohepatitis and its fibrosis

非酒精性脂肪性肝炎及其纤维化的发病机制与治疗

期刊: Clinical and Molecular Hepatology 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 16.9

Abstract

The initial presentation of non-alcoholic steatohepatitis (NASH) is **hepatic steatosis**. The dysfunction of **lipid metabolism** within hepatocytes caused by genetic factors, diet, and **insulin resistance** causes lipid accumulation. **lipotoxicity**, **oxidative stress**, mitochondrial dysfunction, and endoplasmic reticulum stress would further contribute to hepatocyte injury and death, leading to **inflammation** and immune dysfunction in the liver. During the healing process, the accumulation of an excessive amount of **fibrosis** might occur while healing. During the development of NASH and liver **fibrosis**, the gut-liver axis, adipose-liver axis, and renin-angiotensin system (RAS) may be dysregulated and impaired. Translocation of bacteria or its end-products entering the liver could activate hepatocytes, Kupffer cells, and **hepatic stellate cells**, exacerbating **hepatic steatosis**, **inflammation**, and **fibrosis**. **bile acids** regulate glucose and **lipid metabolism** through Farnesoid X receptors in the liver and intestine. Increased adipose tissue-derived non-esterified fatty acids would aggravate **hepatic steatosis**. Increased **leptin** also plays a role in hepatic fibrogenesis, and decreased **adiponectin** may contribute to hepatic **insulin resistance**. Moreover, dysregulation of peroxisome proliferator-activated receptors in the liver, adipose, and muscle tissues may impair **lipid metabolism**. In addition, the RAS may contribute to hepatic fatty acid metabolism, **inflammation**, and **fibrosis**. The treatment includes lifestyle modification, **pharmacological therapy**, and non-**pharmacological therapy**. Currently, weight reduction by lifestyle modification or surgery is the most effective therapy. However, vitamin E, pioglitazone, and obeticholic acid have also been suggested. In this review, we will introduce some new **clinical trials** and experimental therapies for the treatment of NASH and related **fibrosis**.

中文翻译

非酒精性脂肪性肝炎 (NASH) 的初始表现为肝脏脂肪变性。由遗传因素、饮食和胰岛素抵抗引起的肝细胞内脂质代谢功能障碍导致脂质积累。脂毒性、氧化应激、线粒体功能障碍和内质网应激会进一步导致肝细胞损伤和死亡，引发肝脏炎症和免疫功能紊乱。在愈合过程中，可能会出现过量的纤维化积累。在NASH和肝纤维化的发展过程中，肠-肝轴、脂肪-肝轴和肾素-血管紧张素系统 (RAS) 可能出现失调和损伤。细菌或其终产物易位进入肝脏可能激活肝细胞、库普弗细胞和肝星状细胞，加剧肝脏脂肪变性、炎症和纤维化。胆汁酸通过肝脏和肠道中的法尼醇X受体调节葡萄糖和脂质代谢。脂肪组织来源的非酯化脂肪酸增加会加重肝脏脂肪变性。瘦素增加也在肝纤维化中发挥作用，而脂联素减少可能导致肝脏胰岛素抵抗。此外，肝脏、脂肪和肌肉组织中过氧化物酶体增殖物激活受体的失调可能损害脂质代谢。另外，RAS可能影响肝脏脂肪酸代谢、炎症和纤维化。治疗包括生活方式调整、药物治疗和非药物治疗。目前，通过生活方式调整或手术减轻体重是最有效的治疗方法。然而，维生素E、吡格列酮和奥贝胆酸也被建议使用。在本综述中，我们将介绍一些治疗NASH及相关纤维化的新临床试验和实验性疗法。

核心词汇

hepatic steatosis /hi'pætk ,sti:ə'teʊsɪs/ : 肝脂肪变性

insulin resistance /'ɪnsjʊlɪn rɪ'zɪstəns/ : 胰岛素抵抗

oxidative stress /'ɒksɪdeɪtɪv stres/ : 氧化应激

fibrosis /faɪ'brəʊsɪs/ : 纤维化

bile acids /baɪl 'æsɪdz/ : 胆汁酸

adiponectin /,ædɪpəʊ'nektɪn/ : 脂联素

clinical trials /'klɪnɪkəl 'traɪəlz/ : 临床试验

lipid metabolism /'lɪpɪd mɪ'tæbəlaɪzəm/ : 脂质代谢

lipotoxicity /,lɪpəʊtɒk'sɪsɪtɪ/ : 脂毒性

inflammation /,ɪnflə'meɪʃən/ : 炎症

hepatic stellate cells /hi'pætk 'steɪlət sɛlz/ : 肝星状细胞

leptin /'leptɪn/ : 瘦素

pharmacological therapy /fɑ:məke'lɒdʒɪkəl 'θerəpi/ : 药物治疗

Metabolic dysfunction-associated steatotic liver disease: Heterogeneous pathomechanisms and effectiveness of metabolism-based treatment

代谢功能障碍相关脂肪性肝病：异质性病理机制及基于代谢治疗的有效性

期刊: The Lancet Diabetes & Endocrinology 日期: 2025

Abstract

The global epidemic of metabolic dysfunction-associated steatotic liver disease (MASLD) is increasing worldwide. People with MASLD can progress to **cirrhosis** and **hepatocellular carcinoma** and are at increased risk of developing type 2 diabetes, **cardiovascular disease**, **chronic kidney disease**, and extrahepatic cancers. Most people with MASLD die from cardiac-related causes. This outcome is attributed to the shared pathogenesis of MASLD and cardiometabolic diseases, involving unhealthy dietary habits, dysfunctional adipose tissue, **insulin resistance**, and subclinical **inflammation**. In addition, the steatotic and inflamed liver affects the vasculature and heart via increased glucose production and release of procoagulant factors, **dyslipidaemia**, and dysregulated release of hepatokines and microRNAs. However, there is substantial heterogeneity in the contributors to the **pathophysiology** of MASLD, which might influence its rate of progression, its relationship with cardiometabolic diseases, and the response to therapy. The most effective non-pharmacological treatment approaches for people with MASLD include **weight loss**. Paradoxically, some effective pharmacological approaches to improve liver health in people with MASLD are associated with no change in bodyweight or even with weight gain, and similar response heterogeneity has been observed for changes in cardiometabolic risk factors. In this Review, we address the heterogeneity of MASLD with respect to its pathogenesis, outcomes, and metabolism-based treatment responses. Although there is currently insufficient evidence for the implementation of **precision medicine** for risk prediction, prevention, and treatment of MASLD, we discuss whether knowledge about this heterogeneity might help achieving this goal in the future.

中文翻译

代谢功能障碍相关脂肪性肝病（MASLD）的全球流行率正在世界范围内上升。MASLD患者可能进展为肝硬化和肝细胞癌，并面临更高的2型糖尿病、心血管疾病、慢性肾病及肝外癌症风险。多数MASLD患者最终死于心脏相关病因，这一结果归因于MASLD与心脏代谢疾病共有的发病机制，涉及不健康的饮食习惯、脂肪组织功能障碍、胰岛素抵抗及亚临床炎症。此外，脂肪变性且存在炎症的肝脏通过增加葡萄糖生成、释放促凝血因子、血脂异常以及肝因子和微小RNA的失调释放，影响血管系统和心脏。然而，MASLD病理生理学的驱动因素存在显著异质性，这可能影响其进展速度、与心脏代谢疾病的关系以及对治疗的反应。对MASLD患者最有效的非药物治疗方法包括减重。矛盾的是，某些改善MASLD患者肝脏健康的有效药物治疗方法并未伴随体重变化甚至可能导致体重增加，心脏代谢风险因素的变化也观察到类似的反应异质性。本综述探讨了MASLD在发病机制、临床结局及基于代谢的治疗反应方面的异质性。尽管目前实施精准医学进行MASLD风险预测、预防和治疗尚缺乏充分证据，但我们讨论了关于这种异质性的认识是否有助于未来实现这一目标。

核心词汇

cirrhosis /sɪˈrɒsɪs/ : 肝硬化	hepatocellular carcinoma /ˌhepəˈtɒʊˈsɛljʊləˌkɑːrsɪˈnɒmə/ : 肝细胞癌
cardiovascular disease /ˌkɑːrdɪəʊˈvæskjʊləˌdɪˈziːz/ : 心血管疾病	chronic kidney disease /ˈkrɒnɪk ˈkɪdniˌdɪˈziːz/ : 慢性肾脏病
insulin resistance /ˈɪnsjʊlɪn rɪˈzɪstəns/ : 胰岛素抵抗	inflammation /ˌɪnˈfleɪməʃən/ : 炎症
dyslipidaemia /dɪsˌlɪpɪˈdiːmiə/ : 血脂异常	pathophysiology /ˌpæθəʊˈfɪzɪˈɒlədʒi/ : 病理生理学
weight loss /weɪt lɒs/ : 减重	precision medicine /prɪˈsɪʒən ˈmedɪsɪn/ : 精准医学

Sexual dimorphism of metabolic dysfunction-associated steatotic liver disease

代谢功能障碍相关脂肪性肝病的性别二态性

期刊: Trends in Molecular Medicine 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 13.8

Abstract

Metabolic dysfunction-associated steatotic liver disease (MASLD) is the most common **chronic liver condition**. MASLD is a sexually dimorphic condition, with its development and progression influenced by **sex chromosomes** and **hormones**. **estrogens** typically protect against, whereas **androgens** promote, MASLD. Therapeutic approaches for a sex-specific **personalized medicine** include **estrogen replacement**, **androgen blockers**, and novel drugs targeting **hormonal pathways**. However, the interactions between hormonal factors and inherited **genetic variation** impacts MASLD risk, necessitating more tailored therapies. Understanding **sex disparities** and the role of **estrogens** could improve MASLD interventions and management, whereas **clinical trials** addressing sex differences are crucial for advancing personalized treatment. This review explores the underappreciated impact of **sexual dimorphism** in MASLD and discusses the potential therapeutic application of sex-related **hormones**.

中文翻译

代谢功能障碍相关脂肪性肝病（MASLD）是最常见的慢性肝脏疾病。MASLD具有性别二态性，其发生和发展受性染色体和激素的影响。雌激素通常对MASLD具有保护作用，而雄激素则促进其发展。针对性别特异性个体化医疗的治疗方法包括雌激素替代疗法、雄激素阻断剂以及靶向激素通路的新药。然而，激素因素与遗传变异之间的相互作用影响MASLD风险，因此需要更精准的治疗方案。理解性别差异及雌激素的作用有助于改进MASLD的干预与管理，而针对性别差异的临床试验对于推进个体化治疗至关重要。本综述探讨了性别二态性在MASLD中未被充分认识的影响，并讨论了性激素潜在的治疗应用。

核心词汇

chronic liver condition /'kronɪk 'lɪvər kən'dɪʃən/ : 慢性肝脏疾病	sexual dimorphism /'sɛkʃʊəl daɪ'mɔːfɪzəm/ : 性别二态性
estrogens /'ɛstrədʒənz/ : 雌激素	androgens /'ændrədʒənz/ : 雄激素
sex chromosomes /sɛks 'krouməsəʊmz/ : 性染色体	hormones /'hɔːrməʊnz/ : 激素
personalized medicine /'pɜːrsənəlaɪzd 'mɛdɪsɪn/ : 个体化医疗	estrogen replacement /'ɛstrədʒən rɪ'pleɪsmənt/ : 雌激素替代疗法
androgen blockers /'ændrədʒən 'blɒkərz/ : 雄激素阻断剂	hormonal pathways /'hɔːr'məʊnəl 'pæθweɪz/ : 激素通路
genetic variation /dʒə'netɪk 'veəri'etʃən/ : 遗传变异	sex disparities /sɛks dɪ'spærɪtɪz/ : 性别差异
clinical trials /'klɪnɪkəl 'traɪəlz/ : 临床试验	

Current and future pharmacological therapies for NAFLD/NASH

当前及未来针对NAFLD/NASH的药理疗法

期刊: Journal of Gastroenterology 日期: 2018 JCR Q1 中科院升级版 医学1区 IF 5.5

Abstract

nonalcoholic fatty liver disease (NAFLD) is the most prevalent liver disease worldwide, and there is no approved **pharmacotherapy**. The efficacy of vitamin E and pioglitazone has been established in **nonalcoholic steatohepatitis** (NASH), a progressive form of NAFLD. GLP-1RA and SGLT2 inhibitors, which are currently approved for use in diabetes, have shown early efficacy in NASH, and also have beneficial cardiovascular or renal effects. Innovative NASH therapies include four main pathways. The first approach is targeting **hepatic fat accumulation**. Medications in this approach include modulation of peroxisome proliferator-activator receptors (e.g., pemafibrate, elafibranor), medications targeting farnesoid X receptor axis [obeticholic acid; OCA], inhibitors of de novo lipogenesis (aramchol, ACC inhibitor), and fibroblast growth factor-21 analogues. A second target is **oxidative stress**, **inflammation**, and **apoptosis**. This class of drug includes **apoptosis** signaling kinase 1 (ASK1) inhibitor and emricasan (an irreversible caspase inhibitor). A third target is intestinal microbiomes and **metabolic endotoxemia**. Several agents are in ongoing trials, including IMMe124, TLR4 antagonist, and solithromycin (macrolide antibiotics). The final target is **hepatic fibrosis**, which is strongly associated with all-cause or liver-related **mortality** in NASH. Antifibrotic agents are a cysteine-cysteine motif chemokine receptor-2/5 antagonist (cenicriviroc; CVC) and galectin 3 antagonist. Among a variety of medications in development, four agents such as OCA, elafibranor, ASK1 inhibitor, and CVC are currently being evaluated in an international phase 3 trial for the treatment of NASH. Within the next few years, the availability of therapeutic options for NASH will hopefully curb the rising trend of NASH-related diseases.

中文翻译

非酒精性脂肪性肝病 (NAFLD) 是全球范围内最普遍的肝脏疾病, 目前尚无获批的药物治疗方案。维生素E和吡格列酮在非酒精性脂肪性肝炎 (NASH, NAFLD的一种进展形式) 中的疗效已得到证实。目前获批用于糖尿病的GLP-1RA和SGLT2抑制剂在NASH中显示出早期疗效, 并对心血管或肾脏具有有益影响。创新的NASH疗法主要包括四种途径。第一种方法是针对肝脏脂肪积累。此类药物包括调节过氧化物酶体增殖物激活受体 (如培马贝特、埃拉贝特)、靶向法尼醇X受体轴的药物 (奥贝胆酸; OCA)、抑制新生脂肪生成的药物 (阿拉姆科尔、ACC抑制剂) 以及成纤维细胞生长因子-21类似物。第二个靶点是氧化应激、炎症和细胞凋亡。这类药物包括凋亡信号调节激酶1 (ASK1) 抑制剂和埃米卡桑 (一种不可逆的caspase抑制剂)。第三个靶点是肠道微生物群和代谢性内毒素血症。目前正在进行试验的药物包括IMMe124、TLR4拮抗剂和索利霉素 (大环内酯类抗生素)。最后一个靶点是肝纤维化, 这与NASH患者的全因死亡率或肝脏相关死亡率密切相关。抗纤维化药物包括半胱氨酸-半胱氨酸基序趋化因子受体-2/5拮抗剂 (塞尼克里韦罗克; CVC) 和半乳糖凝集素3拮抗剂。在众多研发药物中, OCA、埃拉贝特、ASK1抑制剂和CVC等四种药物目前正在国际三期试验中评估其治疗NASH的效果。未来几年内, NASH治疗方案的推出有望遏制NASH相关疾病的上升趋势。

核心词汇

nonalcoholic fatty liver disease /ˌnɒnælkeɪˈhɒlɪk ˈfæti ˈliːvə dɪˈziːz/ : 非酒精性脂肪性肝病

hepatic fibrosis /hiˈpæɪk faɪˈbrəʊsɪs/ : 肝纤维化

inflammation /ˌɪnfləˈmeɪʃən/ : 炎症

hepatic fat accumulation /hiˈpæɪk fæt əˈkjuːmjəˈleɪʃən/ : 肝脏脂肪堆积

metabolic endotoxemia /ˌmetəˈbɒlɪk ˌendəʊtɒkˈsiːmiə/ : 代谢性内毒素血症

antifibrotic agent /ˌæntɪfaɪˈbrɒtɪk ˈeɪdʒənt/ : 抗纤维化药物

mortality /mɔːˈtælɪti/ : 死亡率

nonalcoholic steatohepatitis /ˌnɒnælkeɪˈhɒlɪk ˌstiːətəʊˈhepeɪˈtaɪtɪs/ : 非酒精性脂肪性肝炎

oxidative stress /ˈɒksɪdeɪtɪv stres/ : 氧化应激

apoptosis /ˌæpəpˈtəʊsɪs/ : 细胞凋亡

intestinal microbiome /ɪnˈtestɪnəl ˌmaɪkrəʊˈbaɪəʊm/ : 肠道微生物组

pharmacotherapy /ˌfɑːməkeʊˈθerəpi/ : 药物治疗

clinical trial /ˈklɪnɪkəl ˈtraɪəl/ : 临床试验

Emerging therapeutic approaches for the treatment of NAFLD and type 2 diabetes mellitus

非酒精性脂肪肝病与2型糖尿病治疗的新兴疗法

期刊: Nature Reviews Endocrinology 日期: 2021 JCR Q1 中科院升级版 医学1区 IF 42.3

Abstract

Non-alcoholic fatty liver disease (NAFLD) has emerged as the most prevalent liver disease in the world, yet there are still no approved pharmacological therapies to prevent or treat this condition. NAFLD encompasses a spectrum of severity, ranging from simple **steatosis** to non-alcoholic steatohepatitis (NASH). Although NASH is linked to an increased risk of **hepatocellular carcinoma** and **cirrhosis** and has now become the leading cause of **liver failure**-related **transplantation**, the majority of patients with NASH will ultimately die as a result of complications of type 2 diabetes mellitus (T2DM) and **cardiometabolic diseases**. Importantly, NAFLD is closely linked to **obesity** and tightly interrelated with **insulin resistance** and T2DM. Thus, targeting these interconnected conditions and taking a holistic attitude to the treatment of **metabolic disease** could prove to be a very beneficial approach. This Review will explore the latest relevant literature and discuss the ongoing therapeutic options for NAFLD focused on targeting intermediary metabolism, **insulin resistance** and T2DM to remedy the global health burden of these diseases.

中文翻译

非酒精性脂肪性肝病（NAFLD）已成为全球最常见的肝脏疾病，但目前尚无获批的药物疗法来预防或治疗此病。NAFLD涵盖了一系列严重程度，从简单的脂肪变性到非酒精性脂肪性肝炎（NASH）。尽管NASH与肝细胞癌和肝硬化的风险增加有关，并已成为肝衰竭相关移植的主要原因，但大多数NASH患者最终将死于2型糖尿病（T2DM）和心脏代谢疾病的并发症。重要的是，NAFLD与肥胖密切相关，并与胰岛素抵抗和T2DM紧密相关。因此，针对这些相互关联的疾病，并采取整体态度治疗代谢性疾病，可能被证明是一种非常有益的方法。本综述将探讨最新的相关文献，并讨论针对NAFLD的持续治疗选择，重点关注靶向中间代谢、胰岛素抵抗和T2DM，以减轻这些疾病的全球健康负担。

核心词汇

steatosis /sti:ə'tousɪs/ : 脂肪变性	hepatocellular carcinoma /ˌhepətu'seljʊlər ˌkɑ:rsɪ'noumə/ : 肝细胞癌
cirrhosis /sɪ'rouɪsɪs/ : 肝硬化	transplantation /ˌtrænsplæn'teɪʃən/ : 移植
obesity /ou'bi:səti/ : 肥胖症	insulin resistance /'ɪnsəlɪn rɪ'zɪstəns/ : 胰岛素抵抗
metabolic disease /ˌmetə'bɒlɪk dɪ'zi:z/ : 代谢性疾病	cardiometabolic diseases /ˌkɑ:rdiəvˌmetə'bɒlɪk dɪ'zi:zɪz/ : 心脏代谢性疾病
liver failure /'lɪvər ˈfeɪljər/ : 肝功能衰竭	

Circadian rhythms in the pathogenesis and treatment of fatty liver disease

昼夜节律在脂肪肝疾病发病机制与治疗中的作用

期刊: Gastroenterology 日期: 2020

JCR Q1

中科院升级版 医学1区

IF 25.1

Abstract

circadian clock proteins are endogenous timing mechanisms that control the **transcription** of hundreds of genes. Their integral role in coordinating **metabolism** has led to their scrutiny in a number of diseases, including **nonalcoholic fatty liver disease** (NAFLD). Discoordination between central and peripheral circadian rhythms is a core feature of nearly every genetic, dietary, or environmental model of **metabolic syndrome** and NAFLD. Restricting feeding to a defined daily interval (time-restricted feeding) can synchronize the central and peripheral circadian rhythms, which in turn can prevent or even treat the **metabolic syndrome** and **hepatic steatosis**. Importantly, a number of proteins currently under study as drug targets in NAFLD (sterol regulatory element-binding protein [SREBP], acetyl-CoA carboxylase [ACC], peroxisome proliferator-activator receptors [PPARs], and incretins) are modulated by circadian proteins. Thus, the clock can be used to maximize the benefits and minimize the adverse effects of pharmaceutical agents for NAFLD. The **circadian clock** itself has the potential for use as a target for the **treatment** of NAFLD.

中文翻译

昼夜节律钟蛋白是控制数百个基因转录的内源性计时机制。它们在协调新陈代谢中的核心作用使其在包括非酒精性脂肪肝病 (NAFLD) 在内的多种疾病中受到关注。中枢与外周昼夜节律失调是几乎所有遗传性、饮食性或环境性代谢综合征及NAFLD模型的核心特征。将进食限制在每日特定时段 (限时进食) 可使中枢与外周昼夜节律同步化, 从而预防甚至治疗代谢综合征及肝脏脂肪变性。值得注意的是, 当前作为NAFLD药物靶点研究的多种蛋白 (甾醇调节元件结合蛋白 [SREBP]、乙酰辅酶A羧化酶[ACC]、过氧化物酶体增殖物激活受体[PPARs]及肠促胰岛素) 均受昼夜节律蛋白调控。因此, 可利用生物钟机制最大化NAFLD药物的疗效并最小化其不良反应。昼夜节律钟本身亦具备成为NAFLD治疗靶点的潜力。

核心词汇

circadian clock /sɜːrˈkeɪdiən kloʊk/ : 昼夜节律钟

metabolic syndrome /ˌmetəˈbɒlɪk ˈsɪndroʊm/ : 代谢综合征

transcription /trænˈskɪpʃən/ : 转录

circadian rhythm /sɜːrˈkeɪdiən ˈrɪðəm/ : 昼夜节律

drug target /drʌg ˈtɑːrɡɪt/ : 药物靶点

treatment /ˈtriːtmənt/ : 治疗

nonalcoholic fatty liver disease /ˌnɒnælkəˈhɒlɪk ˈfæti ˈlɪvər dɪˈziːz/ : 非酒精性脂肪性肝病

hepatic steatosis /hɪˈpæɪtɪk ˌstiːəˈtoʊsɪs/ : 肝脂肪变性

metabolism /məˈtæbəlaɪzəm/ : 新陈代谢

synchronization /ˌsɪŋkrənaɪˈzeɪʃən/ : 同步化

pharmaceutical agent /ˌfɑːrməˈsuːtɪkəl ˈeɪdʒənt/ : 药物制剂

Gut microbiota-derived indole compounds attenuate metabolic dysfunction-associated steatotic liver disease by improving fat metabolism and inflammation

肠道微生物群衍生的吲哚化合物通过改善脂肪代谢和炎症减轻代谢功能障碍相关脂肪性肝病

期刊: Gut Microbes 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 11.0

Abstract

Metabolic dysfunction-associated steatotic liver disease (MASLD) is the most common chronic liver disease, and its prevalence has increased worldwide in recent years. Additionally, there is a close relationship between MASLD and gut microbiota-derived metabolites. However, the mechanisms of MASLD and its metabolites are still unclear. We demonstrated decreased indole-3-propionic acid (IPA) and indole-3-acetic acid (IAA) in the feces of patients with hepatic steatosis compared to healthy controls. Here, IPA and IAA administration ameliorated hepatic steatosis and inflammation in an animal model of WD-induced MASLD by suppressing the NF-κB signaling pathway through a reduction in endotoxin levels and inactivation of macrophages.

中文翻译

代谢功能障碍相关脂肪性肝病（MASLD）是最常见的慢性肝病，近年来其患病率在全球范围内持续上升。此外，MASLD与肠道菌群衍生代谢物之间存在密切关联，但其具体作用机制尚不明确。研究发现，与健康对照组相比，肝脂肪变性患者粪便中的吲哚-3-丙酸（IPA）和吲哚-3-乙酸（IAA）水平显著降低。在西方饮食诱导的MASLD动物模型中，补充IPA和IAA通过降低内毒素水平、抑制巨噬细胞活化，进而阻断NF-κB信号通路，有效改善了肝脏脂肪变性和炎症反应。

核心词汇

hepatic steatosis /hiˈpætik stiːəˈtəʊsɪs/ : 肝脂肪变性	inflammation /ˌɪnfləˈmeɪʃən/ : 炎症
gut microbiota /gʌt ˌmaɪkrəʊbaɪˈoʊtə/ : 肠道微生物群	metabolites /miˈtæbəlaɪts/ : 代谢物
feces /ˈfiːsiːz/ : 粪便	endotoxin /ˌendəʊˈtɒksɪn/ : 内毒素
macrophages /ˈmækrəfeɪdʒɪz/ : 巨噬细胞	chronic liver disease /ˈkrɒnɪk ˈlɪvər dɪˈziːz/ : 慢性肝病
animal model /ˈæniməl ˈmɒdəl/ : 动物模型	administration /ədˌmɪnɪˈstreɪʃən/ : 给药
healthy controls /ˈheɪlθi kənˈtrəʊlz/ : 健康对照	

Adaptations in gastrointestinal physiology after sleeve gastrectomy and roux-en-Y gastric bypass

袖状胃切除术与Roux-en-Y胃旁路术后胃肠生理适应性变化

期刊: The Lancet Gastroenterology & Hepatology 日期: 2021

Abstract

Linked to the growing **obesity** epidemic, demand for bariatric and **metabolic surgery** has increased, the most common procedures being **sleeve gastrectomy** and Roux-en-Y gastric bypass. Originally, bariatric procedures were described as purely restrictive, malabsorptive, or combined restrictive-malabsorptive procedures limiting food intake, nutrient absorption, or both. Nowadays, anatomical alterations are known to affect **gastrointestinal physiology**, which in turn affects the digestion and absorption of nutrients and drugs. Therefore, understanding **gastrointestinal physiology** is crucial to prevent postoperative nutritional deficiencies and to optimise postoperative **drug therapy**. Preclinical and clinical research indicates that **sleeve gastrectomy** accelerates liquid and solid **gastric emptying** and small **intestinal transit**, and increases **bile acid** serum levels, whereas its effects on gastrointestinal acidity, gastric and pancreatic secretions, surface area, and colonic transit remain largely unknown. Roux-en-Y gastric bypass diminishes **gastric acid secretion**, accelerates liquid **gastric emptying**, and increases **bile acid** serum levels, but its effects on intestinal pH, solid **gastric emptying**, **intestinal transit** time, gastric enzyme secretions, and surface area remain largely unknown. In this Review, we summarise current knowledge of the effects of these two procedures on **gastrointestinal physiology** and assess the knowledge gaps.

中文翻译

随着肥胖症的日益流行，对减重与代谢手术的需求也随之增加，其中最常见的术式是袖状胃切除术和Roux-en-Y胃旁路术。最初，减重手术被描述为纯粹的限制性、吸收不良性或两者结合的手术，旨在限制食物摄入、营养吸收或两者兼有。如今，人们认识到解剖结构的改变会影响胃肠道生理，进而影响营养物质和药物的消化与吸收。因此，了解胃肠道生理对于预防术后营养缺乏和优化术后药物治疗至关重要。临床前和临床研究表明，袖状胃切除术会加速液体和固体胃排空以及小肠转运，并增加胆汁酸血清水平，而其对胃肠道酸度、胃和胰腺分泌、表面积以及结肠转运的影响在很大程度上仍未知。Roux-en-Y胃旁路术会减少胃酸分泌，加速液体胃排空，并增加胆汁酸血清水平，但其对肠道pH值、固体胃排空、肠道转运时间、胃酶分泌和表面积的影响在很大程度上仍未知。在本综述中，我们总结了目前关于这两种手术对胃肠道生理影响的知识，并评估了知识空白。

核心词汇

obesity /ou'bi:seɪti/ : 肥胖症

metabolic surgery /,metə'bolɪk 'sɜ:rdʒəri/ : 代谢手术

gastrointestinal physiology /,gæstrouɪn'testɪnəl ,fɪzi'ɒlədʒi/ : 胃肠道生理学

intestinal transit /ɪn'testɪnəl 'trænsɪt/ : 肠道转运

gastric acid secretion /'gæstrɪk 'æsaɪd sɪ'kri:fən/ : 胃酸分泌

nutritional deficiency /nju:'tri:fənəl dɪ'fɪʃənsi/ : 营养缺乏

bariatric surgery /,bæri'ætrɪk 'sɜ:rdʒəri/ : 减重手术

sleeve gastrectomy /sli:v gæs'trektəmi/ : 袖状胃切除术

gastric emptying /'gæstrɪk 'emptɪŋ/ : 胃排空

bile acid /baɪl 'æsaɪd/ : 胆汁酸

pancreatic secretion /,pæŋkri'ætrɪk sɪ'kri:fən/ : 胰液分泌

drug therapy /drʌg 'θerəpi/ : 药物治疗

Post sleeve gastrectomy-enriched gut commensal clostridia promotes secondary bile acid increase and weight loss

袖状胃切除术后富集的肠道共生梭菌促进次级胆汁酸增加与体重减轻

期刊: Gut Microbes 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 11.0

Abstract

The **gut microbiome** is altered after **bariatric surgery** and is associated with weight loss. However, the commensal bacteria involved and the underlying mechanism remain to be determined. We performed shotgun **metagenomic sequencing** in obese subjects before and longitudinally after **sleeve gastrectomy** (SG), and found a significant enrichment in microbial species in Clostridia and **bile acid** metabolizing genes after SG treatment. **bile acid** profiling further revealed decreased primary bile acids (PBAs) and increased conjugated secondary bile acids (C-SBAs) after SG. Specifically, glycodeoxycholic acid (GDCA) and taurodeoxycholic acid (TDCA) were increased at different follow-ups after SG, and were associated with the increased abundance of Clostridia and body weight reduction. **fecal microbiome transplantation** with post-SG feces increased SBA levels, and alleviated body weight gain in the recipient mice. Furthermore, both Clostridia-enriched spore-forming bacteria and GDCA supplementation increased the expression of genes responsible for **lipolysis** and **fatty acid oxidation** in **adipose tissue** and reduced **adiposity** via Takeda G-protein-coupled receptor 5 (TGR5) signaling. Our findings reveal post-SG **gut microbiome** and C-SBAs as contributory to SG-induced weight loss, in part via TGR5 signaling, and suggest SBA-producing gut microbes as a potential **therapeutic target** for **obesity** intervention.

中文翻译

肠道微生物群在减重手术后发生改变，并与体重减轻相关。然而，所涉及的共生细菌及其潜在机制仍有待确定。我们对肥胖受试者在袖状胃切除术（SG）前后进行了纵向的鸟枪法宏基因组测序，发现SG治疗后梭菌属微生物种类和胆汁酸代谢基因显著富集。胆汁酸谱进一步显示SG后初级胆汁酸（PBAs）减少，而结合次级胆汁酸（C-SBAs）增加。具体而言，甘氨脱氧胆酸（GDCA）和牛磺脱氧胆酸（TDCA）在SG后的不同随访时间点均有所增加，并与梭菌属丰度的增加和体重减轻相关。使用SG后粪便进行粪便微生物移植可提高SBA水平，并减轻受体小鼠的体重增加。此外，富含梭菌的孢子形成细菌和GDCA补充剂均能增加脂肪组织中负责脂肪分解和脂肪酸氧化的基因表达，并通过Takeda G蛋白偶联受体5（TGR5）信号通路减少脂肪堆积。我们的研究揭示了SG后肠道微生物群和C-SBAs对SG诱导的体重减轻具有促进作用，部分通过TGR5信号通路实现，并提示产生SBA的肠道微生物可能成为肥胖干预的潜在治疗靶点。

核心词汇

bariatric surgery /ˌbæɪrɪˈætrɪk ˈsɜːrdʒəri/ : 减重手术	obesity /ouˈbiːsəti/ : 肥胖症
sleeve gastrectomy /sliːv ɡæsˈtrektəmi/ : 袖状胃切除术	gut microbiome /ɡʌt ˌmaɪkroʊˈbaɪəʊm/ : 肠道微生物组
metagenomic sequencing /ˌmetəˈdʒiːˈnɒmɪk ˈsiːkwənsɪŋ/ : 宏基因组测序	bile acid /baɪl ˈæsɪd/ : 胆汁酸
lipolysis /liˈpɒlɪsɪs/ : 脂肪分解	fatty acid oxidation /ˈfæti ˈæsɪd ˌɒksɪˈdeɪʃən/ : 脂肪酸氧化
adipose tissue /ˈædɪpəʊs ˈtɪʃuː/ : 脂肪组织	adiposity /ˌædɪˈpɒsɪti/ : 肥胖度
fecal microbiome transplantation /ˈfiːkəl ˌmaɪkroʊˈbaɪəʊm ˌtrænsplænˈteɪʃən/ : 粪便微生物移植	therapeutic target /ˌθerəˈpiːtɪk ˈtɑːrɡɪt/ : 治疗靶点

A microbial metabolite remodels the gut-liver axis following bariatric surgery

微生物代谢物重塑减肥手术后的肠肝轴

期刊: Cell Host & Microbe 日期: 2021

Abstract

bariatric surgery is the most effective treatment for type 2 diabetes and is associated with changes in gut metabolites. Previous work uncovered a gut-restricted TGR5 agonist with anti-diabetic properties-cholic acid-7-sulfate (CA7S)-that is elevated following **sleeve gastrectomy** (SG). Here, we elucidate a **microbiome**-dependent pathway by which SG increases CA7S production. We show that a microbial **metabolite**, lithocholic acid (LCA), is increased in murine portal veins post-SG and by activating the vitamin D **receptor**, induces hepatic mSult2A1/hSULT2A expression to drive CA7S production. An SG-induced shift in the **microbiome** increases gut expression of the **bile acid** transporters Asbt and Ostα, which in turn facilitate selective transport of LCA across the gut **epithelium**. Cecal microbiota **transplant** from SG animals is sufficient to recreate the pathway in germ-free (GF) animals. Activation of this gut-liver pathway leads to CA7S synthesis and GLP-1 **secretion**, causally connecting a microbial **metabolite** with the improvement of diabetic phenotypes.

中文翻译

减肥手术是治疗2型糖尿病最有效的方法，并与肠道代谢物的变化相关。先前的研究发现了一种具有抗糖尿病特性的肠道限制性TGR5激动剂——胆酸-7-硫酸盐（CA7S），其在袖状胃切除术后升高。本文揭示了一条微生物组依赖的途径，通过该途径袖状胃切除术增加了CA7S的产生。我们展示了在袖状胃切除术后，小鼠门静脉中的微生物代谢物石胆酸（LCA）增加，并通过激活维生素D受体，诱导肝脏mSult2A1/hSULT2A表达，从而驱动CA7S的产生。袖状胃切除术引起的微生物组变化增加了肠道胆汁酸转运蛋白Asbt和Ostα的表达，进而促进了LCA选择性穿过肠道上皮的运输。从袖状胃切除动物移植盲肠微生物组足以在无菌动物中重建这一途径。激活这一肠-肝途径导致CA7S合成和GLP-1分泌，从而将微生物代谢物与糖尿病表型的改善因果联系起来。

核心词汇

bariatric surgery /ˌbæriˈætrɪk ˈsɜːrdʒəri/ : 减重手术

microbiome /ˌmaɪkroʊˈbaɪoʊm/ : 微生物组

bile acid /baɪl ˈæsɪd/ : 胆汁酸

transporter /ˈtrænsˈpɔːrtər/ : 转运蛋白

transplant /ˈtrænsplænt/ : 移植

phenotype /ˈfiːnətaɪp/ : 表型

sleeve gastrectomy /sliːv ɡæsˈtrektəmi/ : 袖状胃切除术

metabolite /mɪˈtæbəlaɪt/ : 代谢物

receptor /rɪˈseptər/ : 受体

epithelium /ˌepɪˈθiːliəm/ : 上皮组织

secretion /sɪˈkriːʃən/ : 分泌

Bioavailability of orally administered drugs after bariatric surgery

口服药物在减肥手术后的生物利用度

期刊: Current Obesity Reports 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 11.0

Abstract

PURPOSE OF REVIEW: Oral drug **absorption** after **bariatric surgery** is likely to be altered, but the impact of different **bariatric surgery** procedures on individual drugs is not uniform. The aim of this article is to describe factors influencing the **bioavailability** of orally administered drugs after **bariatric surgery** and to provide readers with practical recommendations for drug dosing. We also discuss the medications that may be harmful after **bariatric surgery**. RECENT FINDINGS: The fundamental factors for enteral drug **absorption** are the production of **gastric acid**; the preserved length of the **intestine**, i.e., the size of the **absorption** surface and/or the preserved **enterohepatic circulation**; and the length of common loop where food and drugs are mixed with digestive enzymes and bile acids. Bypassing of metabolizing enzymes or efflux pumps and changes in **intestinal motility** can also play an important role. Significant changes of drug **absorption** early after the anatomic alteration may also be gradually ameliorated due to gradual intestinal adaptation. The most affected drugs are those with low or variable **bioavailability** and those undergoing **enterohepatic circulation**. Attention should also be paid to oral drug formulations, especially in the early postoperative period, when immediate-release and liquid formulations are preferred. The changes in oral **bioavailability** are especially clinically meaningful in patients treated with drugs possessing narrow **therapeutic index** (e.g., oral anticoagulants, levothyroxine, and anticonvulsants) or in acute conditions (e.g., anti-infectives); nevertheless, it may also influence the therapeutic value of chronic therapy (e.g., antidepressants, antihypertensives, antiplatelets, statins, PPIs, contraceptives, and analgesics); therapeutic effect of chronic therapy is further influenced by pharmacokinetic alterations resulting from weight loss. **therapeutic drug monitoring**, periodical clinical evaluation, and adequate dose adjustments are necessary. Due to safety reasons, patients should avoid oral bisphosphonates, regular use of non-steroidal anti-inflammatory drugs, and, if possible, corticosteroids after **bariatric surgery**.

中文翻译

综述目的：减肥手术后口服药物的吸收可能会发生改变，但不同减肥手术对个体药物的影响并不一致。本文旨在描述影响减肥手术后口服药物生物利用度的因素，并为读者提供药物剂量的实用建议。同时，我们还将讨论减肥手术后可能有害的药物。最新发现：影响肠道药物吸收的基本因素包括胃酸分泌、肠道保留长度（即吸收表面积和/或保留的肠肝循环）、以及食物和药物与消化酶及胆汁酸混合的共通道长度。代谢酶或外排泵的旁路以及肠道动力的改变也可能起到重要作用。解剖结构改变后早期药物吸收的显著变化可能因肠道逐渐适应而得到缓解。受影响最大的药物是那些生物利用度低或可变以及经历肠肝循环的药物。还需关注口服药物剂型，特别是在术后早期，应优先选择速释剂型和液体剂型。口服生物利用度的变化对于使用治疗指数窄的药物（如口服抗凝药、左甲状腺素和抗惊厥药）或急性病症（如抗感染药）的患者尤其具有临床意义；此外，它也可能影响慢性治疗（如抗抑郁药、抗高血压药、抗血小板药、他汀类药物、质子泵抑制剂、避孕药和镇痛药）的疗效；慢性治疗的疗效还受到因体重减轻导致的药代动力学改变的影响。因此，治疗药物监测、定期临床评估和适当的剂量调整是必要的。出于安全考虑，患者应避免口服双膦酸盐类药物、常规使用非甾体抗炎药，并尽可能在减肥手术后避免使用皮质类固醇。

核心词汇

bariatric surgery /ˌbæɪrɪˈætrɪk ˈsɜːrdʒəri/ : 减重手术	bioavailability /ˌbaɪəʊəˈveɪləˈbɪləti/ : 生物利用度
absorption /əbˈsɔːrpfən/ : 吸收	intestine /ɪnˈtestɪn/ : 肠道
gastric acid /ˈɡæstrɪk ˈæsɪd/ : 胃酸	enterohepatic circulation /ˌɛntərəʊhɪˈpætrɪk ˈsɜːrkleɪʃən/ : 肠肝循环
intestinal motility /ɪnˈtestɪnəl moʊˈtɪləti/ : 肠道蠕动	therapeutic index /ˌθerəˈpjʊːtrɪk ˈɪndeks/ : 治疗指数
pharmacokinetics /ˌfɑːrməkoukɪˈnetɪks/ : 药代动力学	therapeutic drug monitoring /ˌθerəˈpjʊːtrɪk drʌɡ ˈmɒnɪtərɪŋ/ : 治疗药物监测
formulation /ˌfɔːrmjəˈleɪʃən/ : 剂型	dose adjustment /dous əˈdʒʌstmənt/ : 剂量调整

Fasting: From physiology to pathology

禁食：从生理学到病理学

期刊: Advanced Science 日期: 2023

JCR Q1

中科院升级版 综合性期刊1区

IF 14.1

Abstract

Abstract Overnutrition is a risk factor for various human diseases, including neurodegenerative diseases, metabolic disorders, and cancers. Therefore, targeting overnutrition represents a simple but attractive strategy for the treatment of these increasing public health threats. fasting as a dietary intervention for combating overnutrition has been extensively studied. fasting has been practiced for millennia, but only recently have its roles in the molecular clock, gut microbiome, and tissue homeostasis and function emerged. fasting can slow aging in most species and protect against various human diseases, including neurodegenerative diseases, metabolic disorders, and cancers. These centuried and unfading adventures and explorations suggest that fasting has the potential to delay aging and help prevent and treat diseases while minimizing side effects caused by chronic dietary interventions. In this review, recent animal and human studies concerning the role and underlying mechanism of fasting in physiology and pathology are summarized, the therapeutic potential of fasting is highlighted, and the combination of pharmacological intervention and fasting is discussed as a new treatment regimen for human diseases.

中文翻译

摘要：营养过剩是多种人类疾病的风险因素，包括神经退行性疾病、代谢紊乱和癌症。因此，针对营养过剩成为治疗这些日益增长的公共卫生威胁的一种简单而具有吸引力的策略。禁食作为一种对抗营养过剩的饮食干预手段已被广泛研究。禁食已有数千年的历史，但直到最近，其在分子钟、肠道微生物群以及组织稳态和功能中的作用才逐渐显现。禁食可以延缓大多数物种的衰老，并对抗多种人类疾病，包括神经退行性疾病、代谢紊乱和癌症。这些历经百年而不衰的探索表明，禁食具有延缓衰老、帮助预防和治疗疾病的潜力，同时最大限度地减少慢性饮食干预带来的副作用。本文综述了近期关于禁食在生理和病理中的作用及其潜在机制的动物和人类研究，强调了禁食的治疗潜力，并探讨了药物干预与禁食相结合作为人类疾病新治疗方案的可能性。

核心词汇

overnutrition /ˌouvnɜːnjuːˈtrɪfən/ : 营养过剩

metabolic disorders /ˌmetəˈbɒlɪk dɪsˈɔːdərz/ : 代谢性疾病

fasting /ˈfæstɪŋ/ : 禁食

molecular clock /məˈleɪkjələr klɒk/ : 分子钟

tissue homeostasis /ˈtɪʃuː ˌhoumiːoʊˈsteɪsɪs/ : 组织稳态

physiology /ˈfɪziˈɒlədʒi/ : 生理学

therapeutic potential /ˌθerəˈpiːtɪk peˈtɛnʃəl/ : 治疗潜力

neurodegenerative diseases /ˌnuːrəʊdɪˈdʒɛnərətɪv dɪˈziːzɪz/ : 神经退行性疾病

cancers /ˈkænsərz/ : 癌症

dietary intervention /ˈdaɪətəri ˌɪntərˈvenʃən/ : 饮食干预

gut microbiome /gʌt ˌmaɪkrəʊˈbaɪoʊm/ : 肠道微生物组

aging /ˈeɪdʒɪŋ/ : 衰老

pathology /pəˈθɒlədʒi/ : 病理学

pharmacological intervention /ˌfɑːrməkeˈlɒdʒɪkəl ˌɪntərˈvenʃən/ : 药物干预

Gut microbiota regulates postprandial GLP-1 response via ileal bile acid-TGR5 signaling

肠道菌群通过回肠胆汁酸-TGR5信号调节餐后GLP-1反应

期刊: Gut Microbes 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 11.0

Abstract

The **gut microbiota** interacts with **intestinal epithelial cells** through **microbial metabolites** to regulate the release of gut hormones. We investigated whether the **gut microbiota** affects the postprandial glucagon-like peptide-1 (GLP-1) response using antibiotic-treated mice and germ-free mice. Gut microbiome depletion completely abolished postprandial GLP-1 response in the **circulation** and **ileum** in a lipid tolerance test. Microbiome depletion did not influence the GLP-1 secretory function of primary ileal cells in response to stimulators in vitro, but dramatically changed the postprandial dynamics of endogenous **bile acids**, particularly ω -muricholic acid (ω MCA) and hyocholic acid (HCA). The bile acid **receptor** Takeda G protein-coupled **receptor** 5 (TGR5) but not farnesoid X **receptor** (FXR), participated in the regulation of postprandial GLP-1 response in the **circulation** and **ileum**, and ω MCA or HCA stimulated GLP-1 **secretion** via TGR5. Finally, **fecal microbiota transplantation** or ω MCA and HCA supplementation restored postprandial GLP-1 response. In conclusion, **gut microbiota** is indispensable for maintaining the postprandial GLP-1 response specifically in the **ileum**, and bile acid (ω MCA and HCA)-TGR5 signaling is involved in this process. This study helps to understand the essential interplay between the **gut microbiota** and host in regulating postprandial GLP-1 response and opens the foundation for new therapeutic targets.

中文翻译

肠道菌群通过微生物代谢物与肠道上皮细胞相互作用，调节肠道激素的释放。我们利用抗生素处理小鼠和无菌小鼠研究了肠道菌群是否影响餐后胰高血糖素样肽-1 (GLP-1) 反应。在脂质耐受试验中，肠道微生物群的耗竭完全消除了循环和回肠中的餐后GLP-1反应。微生物群耗竭不影响原代回肠细胞在体外对刺激物的GLP-1分泌功能，但显著改变了内源性胆汁酸，特别是 ω -鼠胆酸 (ω MCA) 和猪胆酸 (HCA) 的餐后动态。胆汁酸受体Takeda G蛋白偶联受体5 (TGR5) 而非法尼醇X受体 (FXR) 参与了循环和回肠中餐后GLP-1反应的调节， ω MCA或HCA通过TGR5刺激GLP-1分泌。最后，粪便微生物群移植或补充 ω MCA和HCA恢复了餐后GLP-1反应。总之，肠道菌群对于维持餐后GLP-1反应（特别是在回肠中）是不可或缺的，而胆汁酸 (ω MCA和HCA) -TGR5信号通路参与了这一过程。这项研究有助于理解肠道菌群与宿主在调节餐后GLP-1反应中的基本相互作用，并为新的治疗靶点奠定了基础。

核心词汇

gut microbiota /gʌt ˌmaɪkroʊbaɪ ˈoʊtə/ : 肠道微生物群

intestinal epithelial cells /ɪn ˈtɛstɪnəl ˌɛpɪ ˈθiːliəl sɛlz/ : 肠上皮细胞

microbial metabolites /maɪ ˈkroʊbiəl mɪ ˈtæbəlaɪts/ : 微生物代谢物

ileum /ˈɪliəm/ : 回肠

bile acids /baɪl ˈæsɪdz/ : 胆汁酸

receptor /rɪ ˈsɛptər/ : 受体

fecal microbiota transplantation /ˈfiːkəl ˌmaɪkroʊbaɪ ˈoʊtə ˌtrænsplæn ˈteɪʃən/ : 粪便微生物群移植

secretion /sɪ ˈkriːʃən/ : 分泌

circulation /ˌsɜːrkjə ˈleɪʃən/ : 循环

Adipocyte-specific *Hnrnpa1* knockout aggravates obesity-induced metabolic dysfunction via upregulation of CCL2

脂肪细胞特异性*Hnrnpa1* 敲除通过上调CCL2加剧肥胖诱导的代谢功能障碍

期刊: Diabetes 日期: 2024 JCR Q1 中科院升级版 医学2区 IF 7.5

Abstract

Heterogeneous nuclear ribonucleoprotein A1 (HNRNPA1) is involved in lipid and glucose metabolism via mRNA processing. However, whether and how HNRNPA1 alters adipocyte function in obesity remain obscure. Here, we found that the obese state downregulated HNRNPA1 expression in white adipose tissue (WAT). The depletion of adipocyte HNRNPA1 promoted markedly increased macrophage infiltration and expression of proinflammatory and fibrosis genes in WAT of obese mice, eventually leading to exacerbated insulin sensitivity, glucose tolerance, and hepatic steatosis. Mechanistically, HNRNPA1 interacted with Ccl2 and regulated its mRNA stability. Intraperitoneal injection of CCL2-CCR2 signaling antagonist improved adipose tissue inflammation and systemic glucose homeostasis. Furthermore, HNRNPA1 expression in human WAT was negatively correlated with BMI, fat percentage, and subcutaneous fat area. Among individuals with 1-year metabolic surgery follow-up, HNRNPA1 expression was positively related to percentage of total weight loss. These findings identify adipocyte HNRNPA1 as a link between adipose tissue inflammation and systemic metabolic homeostasis, which might be a promising therapeutic target for obesity-related disorders.

中文翻译

异质核糖核蛋白A1（HNRNPA1）通过mRNA加工参与脂质和葡萄糖代谢。然而，HNRNPA1是否以及如何改变肥胖状态下脂肪细胞的功能仍不清楚。本研究发现，肥胖状态会下调白色脂肪组织（WAT）中HNRNPA1的表达。脂肪细胞HNRNPA1的缺失显著促进了肥胖小鼠WAT中巨噬细胞浸润以及促炎和纤维化基因的表达，最终导致胰岛素敏感性、葡萄糖耐受性和肝脂肪变性恶化。机制上，HNRNPA1与Ccl2相互作用并调控其mRNA稳定性。腹腔注射CCL2-CCR2信号拮抗剂可改善脂肪组织炎症和全身葡萄糖稳态。此外，人类WAT中HNRNPA1的表达与BMI、体脂率和皮下脂肪面积呈负相关。在为期一年的代谢手术随访人群中，HNRNPA1表达与总体重下降百分比呈正相关。这些发现将脂肪细胞HNRNPA1确定为脂肪组织炎症与全身代谢稳态之间的关键联系，可能成为肥胖相关疾病有前景的治疗靶点。

核心词汇

obesity /ou'bi:seɪ/ : 肥胖症	adipocyte /'ædɪpou,sart/ : 脂肪细胞
white adipose tissue /waɪt 'ædɪ,pous 'tɪʃu:/ : 白色脂肪组织	macrophage infiltration /'mækroʊ,feɪdʒ ,ɪnfɪl'treɪʃən/ : 巨噬细胞浸润
inflammation /,ɪnflə'meɪʃən/ : 炎症	fibrosis /faɪ'brouzɪs/ : 纤维化
insulin sensitivity /'ɪnsəlɪn ,sensɪ'tɪvətɪ/ : 胰岛素敏感性	glucose tolerance /'glu:kous 'tɑ:lərəns/ : 葡萄糖耐量
hepatic steatosis /hɪ'pætɪk ,sti:ə'tousɪs/ : 肝脂肪变性	metabolic homeostasis /,metə'ba:lɪk ,houmi:ou'steɪsɪs/ : 代谢稳态
metabolic surgery /,metə'ba:lɪk 'sɜ:rdʒəri/ : 代谢手术	therapeutic target /,θerə'pjʊ:tk 'tɑ:rgɪt/ : 治疗靶点

Obesity-enriched gut microbe degrades myo-inositol and promotes lipid absorption

肥胖相关肠道微生物降解肌醇并促进脂质吸收

期刊: Cell Host & Microbe 日期: 2024

Abstract

Numerous studies have reported critical roles for the gut microbiota in obesity. However, the specific microbes that causally contribute to obesity and the underlying mechanisms remain undetermined. Here, we conducted shotgun metagenomic sequencing in a Chinese cohort of 631 obese subjects and 374 normal-weight controls and identified a Megamonas-dominated, enterotype-like cluster enriched in obese subjects. Among this cohort, the presence of Megamonas and polygenic risk exhibited an additive impact on obesity. Megamonas rupellensis possessed genes for myo-inositol degradation, as demonstrated in vitro and in vivo, and the addition of myo-inositol effectively inhibited fatty acid absorption in intestinal organoids. Furthermore, mice colonized with M. rupellensis or E. coli heterologously expressing the myo-inositol-degrading iolG gene exhibited enhanced intestinal lipid absorption, thereby leading to obesity. Altogether, our findings uncover roles for M. rupellensis as a myo-inositol degrader that enhances lipid absorption and obesity, suggesting potential strategies for future obesity management.

中文翻译

大量研究报道了肠道菌群在肥胖中的关键作用。然而，导致肥胖的具体微生物及其潜在机制仍未明确。本研究通过对631名中国肥胖受试者和374名正常体重对照者进行宏基因组测序，发现了一个以Megamonas为主的、类似肠型的菌群簇在肥胖受试者中富集。在该队列中，Megamonas的存在与多基因风险对肥胖具有叠加影响。体外和体内实验证实，Megamonas rupellensis具有降解肌醇的基因，而添加肌醇能有效抑制肠道类器官对脂肪酸的吸收。此外，定植了M. rupellensis或异源表达肌醇降解基因iolG的大肠杆菌的小鼠表现出肠道脂质吸收增强，从而导致肥胖。总之，我们的研究揭示了M. rupellensis作为肌醇降解菌在增强脂质吸收和促进肥胖中的作用，为未来肥胖管理提供了潜在策略。

核心词汇

obesity /ou'bi:seɪ/ : 肥胖症	microbiota /,maɪkrəʊbaɪ'əʊtə/ : 微生物群
metagenomic sequencing /,metədʒɪ'nomɪk 'si:kwənsɪŋ/ : 宏基因组测序	enterotype /'entəraʊtaɪp/ : 肠型
polygenic risk /,pɒli'dʒenɪk rɪsk/ : 多基因风险	degradation /,deɡre'deɪʃən/ : 降解
fatty acid /'fæti 'æsɪd/ : 脂肪酸	absorption /əb'sɔ:rpʃən/ : 吸收
intestinal organoids /ɪn'testɪnəl 'ɔ:rgə,nɔɪdz/ : 肠道类器官	colonization /,kɒlənaɪ'zeɪʃən/ : 定植
lipid /'lɪpɪd/ : 脂质	

Resistant starch intake facilitates weight loss in humans by reshaping the gut microbiota

抗性淀粉摄入通过重塑肠道菌群促进人类减肥

期刊: Nature Metabolism 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 20.8

Abstract

AbstractEmerging evidence suggests that modulation of gut microbiota by dietary fibre may offer solutions for metabolic disorders. In a randomized placebo-controlled crossover design trial (ChiCTR-TTRCC-13003333) in 37 participants with overweight or obesity, we test whether resistant starch (RS) as a dietary supplement influences obesity-related outcomes. Here, we show that RS supplementation for 8 weeks can help to achieve weight loss (mean -2.8 kg) and improve insulin resistance in individuals with excess body weight. The benefits of RS are associated with changes in gut microbiota composition. supplementation with Bifidobacterium adolescentis, a species that is markedly associated with the alleviation of obesity in the study participants, protects male mice from diet-induced obesity. Mechanistically, the RS-induced changes in the gut microbiota alter the bile acid profile, reduce inflammation by restoring the intestinal barrier and inhibit lipid absorption. We demonstrate that RS can facilitate weight loss at least partially through B. adolescentis and that the gut microbiota is essential for the action of RS.

中文翻译

摘要新出现的证据表明，通过膳食纤维调节肠道菌群可能为代谢紊乱提供解决方案。在一项针对37名超重或肥胖参与者的随机安慰剂对照交叉设计试验（ChiCTR-TTRCC-13003333）中，我们测试了抗性淀粉（RS）作为膳食补充剂是否影响肥胖相关结果。在此，我们展示为期8周的RS补充有助于实现体重减轻（平均-2.8公斤）并改善超重个体的胰岛素抵抗。RS的益处与肠道菌群组成的变化相关。补充青春双歧杆菌（一种在研究与缓解肥胖显著相关的菌种）可保护雄性小鼠免受饮食诱导的肥胖。从机制上讲，RS引起的肠道菌群变化改变了胆汁酸谱，通过恢复肠道屏障减少炎症，并抑制脂质吸收。我们证明，RS至少部分通过青春双歧杆菌促进体重减轻，并且肠道菌群对RS的作用至关重要。

核心词汇

metabolic disorders /ˌmɛtəˈbɒlɪk dɪsˈɔːdəz/ : 代谢性疾病	obesity /ouˈbiːsəti/ : 肥胖症
insulin resistance /ˈɪnsəlɪn rɪˈzɪstəns/ : 胰岛素抵抗	gut microbiota /gʌt ˌmaɪkrəʊbaɪˈoʊtə/ : 肠道菌群
dietary fibre /ˈdaɪətəri ˈfaɪbər/ : 膳食纤维	resistant starch /rɪˈzɪstənt stætʃ/ : 抗性淀粉
weight loss /weɪt lɒs/ : 体重减轻	inflammation /ˌɪnfləˈmeɪʃən/ : 炎症
intestinal barrier /ɪnˈtɛstɪnəl ˈbæriər/ : 肠道屏障	lipid absorption /ˈlɪpɪd əbˈsɔːrptʃən/ : 脂质吸收
bile acid /baɪl ˈæsɪd/ : 胆汁酸	supplementation /ˌsʌplɪmənˈteɪʃən/ : 补充疗法

The microbiota-gut-brain axis in obesity

微生物群-肠-脑轴在肥胖中的作用

期刊: The Lancet Gastroenterology & Hepatology 日期: 2017

Abstract

Changes in microbial diversity and composition are increasingly associated with several disease states including **obesity** and behavioural disorders. **obesity**-associated **microbiota** alter host energy harvesting, **insulin resistance**, **inflammation**, and fat deposition. Additionally, intestinal **microbiota** can regulate **metabolism**, **adiposity**, **homoeostasis**, and energy balance as well as central appetite and food reward signalling, which together have crucial roles in **obesity**. Moreover, some strains of bacteria and their metabolites might target the brain directly via **vagal stimulation** or indirectly through immune-**neuroendocrine** mechanisms. Therefore, the gut **microbiota** is becoming a target for new anti-**obesity** therapies. Further investigations are needed to elucidate the intricate gut-**microbiota**-host relationship and the potential of gut-**microbiota**-targeted strategies, such as **dietary interventions** and faecal **microbiota transplantation**, as promising **metabolic therapies** that help patients to maintain a healthy weight throughout life.

中文翻译

微生物多样性和组成的变化日益与多种疾病状态相关联，包括肥胖和行为障碍。与肥胖相关的微生物群会改变宿主的能量获取、胰岛素抵抗、炎症反应及脂肪沉积。此外，肠道微生物群能够调节新陈代谢、肥胖程度、内稳态和能量平衡，以及中枢食欲和食物奖赏信号，这些因素共同在肥胖症中扮演关键角色。更有甚者，某些细菌菌株及其代谢产物可能通过迷走神经刺激直接作用于大脑，或通过免疫-神经内分泌机制间接影响。因此，肠道微生物群正成为新型抗肥胖疗法的靶点。需要进一步研究以阐明复杂的肠道-微生物群-宿主关系，以及针对肠道微生物群的策略（如饮食干预和粪便微生物移植）作为有前景的代谢疗法，帮助患者终身维持健康体重的潜力。

核心词汇

obesity /ou'bi:səti/ : 肥胖症

inflammation /,ɪnflə'meɪʃən/ : 炎症

metabolism /mæ'tæbəlaɪzəm/ : 新陈代谢

homoeostasis /,houmiə'steɪsɪs/ : 稳态

neuroendocrine /,njuərou'endəkrɪn/ : 神经内分泌

metabolic therapies /,metə'bɒlɪk 'θerəpi:z/ : 代谢疗法

microbiota /,maɪkrəu'baɪ'əʊtə/ : 微生物群

insulin resistance /'ɪnsəlɪn rɪ'zɪstəns/ : 胰岛素抵抗

adiposity /,ædɪ'pɒsɪti/ : 肥胖度

vagal stimulation /'veɪgəl ,stɪmjʊ'steɪʃən/ : 迷走神经刺激

transplantation /,trænsplæn'teɪʃən/ : 移植

dietary interventions /'daɪətəri ,ɪntər'veɪnfənz/ : 饮食干预

A brain-to-gut signal controls intestinal fat absorption

脑至肠信号调控肠道脂肪吸收

期刊: Nature 日期: 2024 JCR Q1 中科院升级版 综合性期刊1区 IF 48.5

Abstract

Although **fat** is a crucial source of **energy** in diets, excessive **intake** leads to **obesity**. **fat absorption** in the **gut** is prevalingly thought to occur organ-autonomously by **diffusion**(1-3). Whether the process is controlled by the brain-to-**gut** axis, however, remains largely unknown. Here we demonstrate that the dorsal motor nucleus of vagus (DMV) plays a key part in this process. Inactivation of DMV neurons reduces intestinal **fat absorption** and consequently causes weight loss, whereas activation of the DMV increases **fat absorption** and weight gain. Notably, the inactivation of a subpopulation of DMV neurons that project to the jejunum shortens the length of microvilli, thereby reducing **fat absorption**. Moreover, we identify a natural compound, puerarin, that mimics the suppression of the DMV-vagus pathway, which in turn leads to reduced **fat absorption**. Photoaffinity chemical methods and cryogenic electron microscopy of the structure of a GABA(A) receptor-puerarin complex reveal that puerarin binds to an allosteric modulatory site. Notably, conditional Gabra1 knockout in the DMV largely abolishes puerarin-induced intestinal **fat** loss. In summary, we discover that suppression of the DMV-vagus-jejunum axis controls intestinal **fat absorption** by shortening the length of microvilli and illustrate the therapeutic potential of puerarin binding to GABRA1 in **fat** loss.

中文翻译

尽管脂肪是饮食中重要的能量来源，但过量摄入会导致肥胖。目前普遍认为，肠道中的脂肪吸收是通过扩散以器官自主方式进行的（1-3）。然而，这一过程是否受脑-肠轴调控仍鲜为人知。本研究发现，迷走神经背侧运动核（DMV）在此过程中发挥关键作用。抑制DMV神经元会减少肠道脂肪吸收，进而导致体重下降；而激活DMV则会增加脂肪吸收并促进体重增长。值得注意的是，抑制投射至空肠的DMV神经元亚群会缩短微绒毛长度，从而减少脂肪吸收。此外，我们鉴定出一种天然化合物——葛根素，它能模拟抑制DMV-迷走神经通路的作用，进而减少脂肪吸收。通过光亲和化学方法及冷冻电镜解析GABA(A)受体-葛根素复合物结构，发现葛根素结合于变构调节位点。值得注意的是，条件性敲除DMV中的Gabra1基因可基本消除葛根素诱导的肠道脂肪减少效应。综上，本研究发现抑制DMV-迷走神经-空肠轴可通过缩短微绒毛长度调控肠道脂肪吸收，并揭示了葛根素结合GABRA1在减脂中的治疗潜力。

核心词汇

fat /fæt/ : 脂肪	energy /'enərdʒi/ : 能量
diet /'daɪət/ : 饮食	obesity /ou'bi:səti/ : 肥胖症
absorption /əb'sɔ:rpʃən/ : 吸收	gut /gʌt/ : 肠道
intake /'ɪnteɪk/ : 摄入	diffusion /dɪ'fju:ʒən/ : 扩散

Perilipin 5 links mitochondrial uncoupled respiration in brown fat to healthy white fat remodeling and systemic glucose tolerance

Perilipin 5 将棕色脂肪中的线粒体解偶联呼吸与健康白色脂肪重塑及全身葡萄糖耐受性联系起来

期刊: Nature Communications

日期: 2021

JCR Q1

中科院升级版 综合性期刊1区

IF 15.7

Abstract

Abstract Exposure of mice or humans to cold promotes significant changes in **brown adipose tissue** (BAT) with respect to **histology**, lipid content, **gene expression**, and mitochondrial mass and function. Herein we report that the **lipid droplet** coat protein Perilipin 5 (PLIN5) increases markedly in BAT during exposure of mice to cold. To understand the functional significance of cold-induced PLIN5, we created and characterized gain- and loss-of-function mouse models. Enforcing PLIN5 expression in mouse BAT mimics the effects of cold with respect to mitochondrial cristae packing and uncoupled substrate-driven respiration. PLIN5 is necessary for the maintenance of mitochondrial cristae structure and **respiratory function** during cold stress. We further show that promoting PLIN5 function in BAT is associated with healthy remodeling of subcutaneous **white adipose tissue** and improvements in systemic **glucose tolerance** and diet-induced **hepatic steatosis**. These observations will inform future strategies that seek to exploit thermogenic adipose tissue as a **therapeutic target** for type 2 diabetes, **obesity**, and **nonalcoholic fatty liver disease**.

中文翻译

摘要: 将小鼠或人类暴露于寒冷环境中, 会显著改变棕色脂肪组织 (BAT) 的组织结构、脂质含量、基因表达以及线粒体质量和功能。本文报告, 在小鼠暴露于寒冷期间, 棕色脂肪组织中的脂滴包被蛋白Perilipin 5 (PLIN5) 显著增加。为了理解寒冷诱导的PLIN5的功能意义, 我们创建并表征了功能获得和功能缺失的小鼠模型。在小鼠棕色脂肪组织中强制表达PLIN5, 模拟了寒冷对线粒体嵴排列和解偶联底物驱动呼吸的影响。PLIN5对于在寒冷应激期间维持线粒体嵴结构和呼吸功能是必要的。我们进一步表明, 促进棕色脂肪组织中PLIN5的功能与皮下白色脂肪组织的健康重塑、全身葡萄糖耐量的改善以及饮食诱导的肝脂肪变性有关。这些观察结果将为未来寻求利用产热脂肪组织作为2型糖尿病、肥胖和非酒精性脂肪肝病治疗靶点的策略提供信息。

核心词汇

brown adipose tissue /braʊn 'ædəˌpɒs 'tɪʃuː/ : 棕色脂肪组织

mitochondria /ˌmaɪtəˈkɒndrɪə/ : 线粒体

hepatic steatosis /hɪˈpæɪtɪk ˌstiːəˈtəʊsɪs/ : 肝脂肪变性

nonalcoholic fatty liver disease /ˌnɒnælkeɪˈhɒlɪk ˈfæti ˈlɪvər dɪˈziːz/ : 非酒精性脂肪肝病

gene expression /dʒiːn ɪkˈspreʃən/ : 基因表达

respiratory function /ˈrespəreɪtɔːri ˈfʌŋkʃən/ : 呼吸功能

white adipose tissue /waɪt 'ædəˌpɒs 'tɪʃuː/ : 白色脂肪组织

glucose tolerance /ˈgluːkəʊs ˈtɒlərəns/ : 葡萄糖耐量

obesity /ouˈbiːsəti/ : 肥胖症

histology /hɪˈstɒlədʒi/ : 组织学

lipid droplet /ˈlɪpɪd ˈdrɒplɪt/ : 脂滴

therapeutic target /ˌθerəˈpjʊːtɪk ˈtɑːrgɪt/ : 治疗靶点

Brown adipose tissue: An update on recent findings

棕色脂肪组织：最新研究进展

期刊: Current Obesity Reports 日期: 2017 JCR Q1 中科院升级版 医学1区 IF 11.0

Abstract

PURPOSE OF REVIEW: New treatment approaches to **weight loss** and **weight loss** maintenance in humans are critical. Given its potential role in stimulating **energy expenditure**, **brown adipose tissue** (BAT) activation has become a trending topic as an anti-**obesity** treatment. RECENT FINDINGS: Most studies on BAT stimulation have been conducted in rodents and used cold stimulation. To date, few human trials exist that tested the effect of **cold exposure** on BAT. Those studies show that BAT contributes a small amount to overall **energy metabolism** which is unlikely to cause **weight loss**. Nonetheless, improvements in **glucose metabolism** have been demonstrated in humans. While new **pharmacological** approaches demonstrate some contribution of BAT to overall **energy expenditure**, the potential **cardiovascular risk** (increased **heart rate** and **blood pressure** to sustain the extra **energy expenditure**) may preclude their use. There is no convincing evidence yet to indicate that BAT may be a viable pharmaceutical target for body **weight loss** or even **weight loss** maintenance. More research is needed to confirm the relevance of BAT and **beige tissue** to whole-body **energy metabolism** in humans.

中文翻译

综述目的：探索人类减肥及维持体重的新治疗方法至关重要。鉴于棕色脂肪组织（BAT）在刺激能量消耗方面的潜在作用，其激活已成为抗肥胖治疗的热点话题。最新发现：大多数关于BAT刺激的研究在啮齿动物中进行，并采用冷刺激方法。迄今为止，测试寒冷暴露对BAT影响的人体试验较少。这些研究表明，BAT对整体能量代谢的贡献较小，不太可能导致体重减轻。尽管如此，人类研究中已证实其对葡萄糖代谢的改善作用。虽然新的药理学方法显示BAT对整体能量消耗有一定贡献，但潜在的心血管风险（为维持额外能量消耗导致心率和血压升高）可能限制其应用。目前尚无令人信服的证据表明BAT可能成为减肥甚至维持体重的可行药物靶点。需要更多研究来确认BAT和米色组织与人体整体能量代谢的相关性。

核心词汇

brown adipose tissue /braʊn 'ædəˌpɒs 'tɪʃuː/ : 棕色脂肪组织	obesity /ou'biːsəti/ : 肥胖症
weight loss /weɪt lɔːs/ : 体重减轻	energy expenditure /'ɛnədʒi ɪk'spɛndɪtʃər/ : 能量消耗
glucose metabolism /'gluːkəʊs mə'tæbəˌlɪzəm/ : 葡萄糖代谢	cardiovascular risk /ˌkɑːrdiəʊ'væskjələr rɪsk/ : 心血管风险
heart rate /haːrt reɪt/ : 心率	blood pressure /blʌd 'preʃər/ : 血压
pharmacological /ˌfɑːrmə'kɒlədʒɪkəl/ : 药理学的	cold exposure /kəʊld ɪk'spəʊʒər/ : 冷暴露
beige tissue /beɪʒ 'tɪʃuː/ : 米色脂肪组织	energy metabolism /'ɛnədʒi mə'tæbəˌlɪzəm/ : 能量代谢

Rab18 maintains homeostasis of subcutaneous adipose tissue to prevent obesity-induced metabolic disorders

Rab18维持皮下脂肪组织稳态以预防肥胖诱导的代谢紊乱

期刊: Science China Life Sciences 日期: 2024 JCR Q1 中科院升级版 生物学1区 IF 9.5

Abstract

Metabolically healthy **obesity** refers to obese individuals who do not develop **metabolic disorders**. These people store fat in **subcutaneous adipose tissue** (SAT) rather than in **visceral adipose tissue** (VAT). However, the molecules participating in this specific scenario remain elusive. Rab18, a **lipid droplet** (LD)-associated protein, mediates the contact between the **endoplasmic reticulum** (ER) and LDs to facilitate LD growth and maturation. In the present study, we show that the protein level of Rab18 is specifically upregulated in the SAT of obese people and mice. Rab18 adipocyte-specific knockout (Rab18 AKO) mice had a decreased volume ratio of SAT to VAT compared with wildtype mice. When subjected to high-fat diet (HFD), Rab18 AKO mice had increased **ER stress** and **inflammation**, reduced **adiponectin**, and decreased **triacylglycerol** (TAG) accumulation in SAT. In contrast, TAG accumulation in VAT, **brown adipose tissue** (BAT) or liver of Rab18 AKO mice had a moderate increase without **ER stress** stimulation. Rab18 AKO mice developed **insulin resistance** and systematic **inflammation**. Rab18 AKO mice maintained body temperature in response to acute and chronic cold induction with a thermogenic SAT, similar to the counterpart mice. Furthermore, Rab18-deficient 3T3-L1 adipocytes were more prone to palmitate-induced **ER stress**, indicating the involvement of Rab18 in alleviating lipid toxicity. Rab18 AKO mice provide a good animal model to investigate **metabolic disorders** such as impaired SAT. In conclusion, our studies reveal that Rab18 is a key and specific regulator that maintains the proper functions of SAT by alleviating lipid-induced **ER stress**.

中文翻译

代谢健康型肥胖指的是那些未出现代谢紊乱的肥胖个体。这类人群的脂肪主要储存在皮下脂肪组织（SAT）而非内脏脂肪组织（VAT）中。然而，参与这一特定生理过程的分子机制尚不明确。Rab18作为一种脂滴（LD）相关蛋白，介导内质网（ER）与脂滴之间的接触，促进脂滴的生长与成熟。本研究显示，在肥胖人群和小鼠的皮下脂肪组织中，Rab18蛋白水平特异性上调。与野生型小鼠相比，Rab18脂肪细胞特异性敲除（Rab18 AKO）小鼠的皮下脂肪组织与内脏脂肪组织体积比显著降低。在高脂饮食（HFD）条件下，Rab18 AKO小鼠皮下脂肪组织出现内质网应激加剧、炎症反应增强、脂联素水平下降及甘油三酯（TAG）蓄积减少等现象。相反，其内脏脂肪组织、棕色脂肪组织（BAT）或肝脏中的甘油三酯蓄积呈适度增加，且未引发内质网应激。Rab18 AKO小鼠出现胰岛素抵抗和系统性炎症反应。在急性慢性冷刺激下，该模型小鼠通过产热性皮下脂肪组织维持体温的能力与对照组相当。此外，Rab18缺陷的3T3-L1脂肪细胞更易发生棕榈酸诱导的内质网应激，表明Rab18参与缓解脂质毒性作用。Rab18 AKO小鼠为研究皮下脂肪组织功能障碍等代谢紊乱提供了理想的动物模型。综上所述，我们的研究揭示Rab18是通过缓解脂质诱导的内质网应激来维持皮下脂肪组织正常功能的关键特异性调节因子。

核心词汇

obesity /ou'bi:səti/ : 肥胖症	adipose tissue /'ædɪpəʊs 'tɪʃu:/ : 脂肪组织
subcutaneous adipose tissue /,sʌbkju'teɪniəs 'ædɪpəʊs 'tɪʃu:/ : 皮下脂肪组织	visceral adipose tissue /'vɪsərəl 'ædɪpəʊs 'tɪʃu:/ : 内脏脂肪组织
brown adipose tissue /braʊn 'ædɪpəʊs 'tɪʃu:/ : 棕色脂肪组织	lipid droplet /'lɪpɪd 'drɒplɪt/ : 脂滴
endoplasmic reticulum /,endə'plæzmɪk rɪ'tɪkjʊləm/ : 内质网	ER stress /,i:'ɑ:r stres/ : 内质网应激
inflammation /,ɪnflə'meɪʃən/ : 炎症	insulin resistance /'ɪnsəlɪn rɪ'zɪstəns/ : 胰岛素抵抗
adiponectin /,ædɪpəʊ'nektɪn/ : 脂联素	triacylglycerol /traɪ,æsl'glɪsərəl/ : 三酰甘油
metabolic disorders /,metə'bɒlɪk dɪs'ɔ:dəz/ : 代谢紊乱	thermogenesis /θɜ:rməʊ'dʒenɪsɪs/ : 产热作用

Mild autonomous cortisol secretion: Pathophysiology, comorbidities and management approaches

轻度自主皮质醇分泌：病理生理学、合并症及管理方法

期刊: Nature Reviews Endocrinology 日期: 2024 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

The majority of incidentally discovered adrenal tumours are benign adrenocortical adenomas and the prevalence of adrenocortical adenomas is around 1-7% on cross-sectional abdominal imaging. These can be non-functioning adrenal tumours or they can be associated with **autonomous cortisol secretion** on a spectrum that ranges from rare clinically overt adrenal **Cushing syndrome** to the much more prevalent mild **autonomous cortisol secretion** (MACS) without signs of **Cushing syndrome**. MACS is diagnosed (based on an abnormal overnight **dexamethasone suppression test**) in 20-50% of patients with adrenal adenomas. MACS is associated with **cardiovascular morbidity**, frailty, fragility fractures, decreased quality of life and increased mortality. Management of MACS should be individualized based on patient characteristics and includes **adrenalectomy** or conservative follow-up with treatment of associated comorbidities. Identifying patients with MACS who are most likely to benefit from **adrenalectomy** is challenging, as **adrenalectomy** results in improvement of **cardiovascular morbidity** in some, but not all, patients with MACS. Of note, diagnosis and management of patients with bilateral MACS is especially challenging. Current gaps in MACS clinical practice include a lack of specific biomarkers diagnostic of MACS-related health outcomes and a paucity of clinical trials demonstrating the efficacy of **adrenalectomy** on comorbidities associated with MACS. In addition, little evidence exists to demonstrate the efficacy and safety of long-term **medical therapy** in patients with MACS.

中文翻译

大多数偶然发现的肾上腺肿瘤为良性肾上腺皮质腺瘤，在腹部横断面影像学检查中，肾上腺皮质腺瘤的患病率约为1-7%。这些肿瘤可能为非功能性肾上腺肿瘤，也可能与自主皮质醇分泌相关，其范围从罕见的临床明显肾上腺库欣综合征到更为常见的轻度自主皮质醇分泌（MACS），后者不伴有库欣综合征的体征。在肾上腺腺瘤患者中，有20-50%被诊断为MACS（基于异常的过夜地塞米松抑制试验）。MACS与心血管疾病、虚弱、脆性骨折、生活质量下降和死亡率增加相关。MACS的管理应根据患者特点个体化，包括肾上腺切除术或保守随访并治疗相关合并症。识别最可能从肾上腺切除术中获益的MACS患者具有挑战性，因为肾上腺切除术仅对部分而非所有MACS患者的心血管疾病有改善作用。值得注意的是，双侧MACS患者的诊断和管理尤其具有挑战性。当前MACS临床实践中的不足包括缺乏诊断MACS相关健康结局的特异性生物标志物，以及缺乏证明肾上腺切除术对MACS相关合并症疗效的临床试验。此外，关于MACS患者长期药物治疗的有效性和安全性的证据也很少。

核心词汇

adrenal tumour /əˈdriːnəl ˈtjuːmə/ : 肾上腺肿瘤	adrenocortical adenoma /əˈdriːnouˈkɔːrtɪkəl ˌædɪˈnoumə/ : 肾上腺皮质腺瘤
Cushing syndrome /ˈkʊʃɪŋ ˈsɪndroum/ : 库欣综合征	autonomous cortisol secretion /ɔːˈtɑːnəməs ˈkɔːrtɪsɔːl sɪˈkriːʃən/ : 自主性皮质醇分泌
dexamethasone suppression test /ˌdekseˈmethəsoun səˈpreʃən test/ : 地塞米松抑制试验	cardiovascular morbidity /ˌkɑːrdiəvˈvæskjələr mɔːrˈbɪdɪti/ : 心血管疾病发病率
fragility fracture /frəˈdʒɪlɪti ˈfræktʃər/ : 脆性骨折	adrenalectomy /əˈdriːnəˈlektəmi/ : 肾上腺切除术
comorbidity /ˌkɒmɔːrˈbɪdɪti/ : 合并症	biomarker /ˈbaɪəʊˌmɑːrkər/ : 生物标志物
clinical trial /ˈklɪnɪkəl ˈtraɪəl/ : 临床试验	medical therapy /ˈmedɪkəl ˈθerəpi/ : 药物治疗

Congenital adrenal hyperplasia

先天性肾上腺增生症

期刊: The Lancet 日期: 2017 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

congenital adrenal hyperplasia is a group of autosomal recessive disorders encompassing enzyme deficiencies in the adrenal steroidogenesis pathway that lead to impaired **cortisol** biosynthesis. Depending on the type and severity of steroid block, patients can have various alterations in **glucocorticoid**, **mineralocorticoid**, and **sex steroid** production that require **hormone replacement therapy**. Presentations vary from neonatal salt wasting and atypical genitalia, to adult presentation of **hirsutism** and **irregular menses**. Screening of neonates with elevated 17-hydroxyprogesterone concentrations for classic (severe) 21-hydroxylase deficiency, the most common type of **congenital adrenal hyperplasia**, is in place in many countries, however cosyntropin stimulation testing might be needed to confirm the diagnosis or establish non-classic (milder) subtypes. Challenges in the treatment of **congenital adrenal hyperplasia** include avoidance of **glucocorticoid** overtreatment and control of sex hormone imbalances. Long-term complications include abnormal growth and development, adverse effects on bone and the cardiovascular system, and **infertility**. Novel treatments aim to reduce **glucocorticoid** exposure, improve excess hormone control, and mimic physiological hormone patterns.

中文翻译

先天性肾上腺皮质增生症是一组常染色体隐性遗传疾病，涉及肾上腺类固醇合成途径中的酶缺陷，导致皮质醇生物合成受损。根据类固醇阻断的类型和严重程度，患者可能出现糖皮质激素、盐皮质激素和性类固醇生成的不同改变，需要激素替代治疗。临床表现多样，从新生儿期的盐耗失和非典型生殖器，到成人期的多毛症和月经不调。许多国家已对新生儿进行17-羟孕酮浓度升高的筛查，以检测经典的（严重的）21-羟化酶缺乏症（先天性肾上腺皮质增生症最常见的类型），但可能需要促肾上腺皮质激素刺激试验来确诊或确定非经典（较轻的）亚型。治疗先天性肾上腺皮质增生症的挑战包括避免糖皮质激素过度治疗和控制性激素失衡。长期并发症包括生长发育异常、对骨骼和心血管系统的不良影响以及不孕症。新型治疗方法旨在减少糖皮质激素暴露、改善对过量激素的控制并模拟生理激素模式。

核心词汇

congenital adrenal hyperplasia /kənˈdʒenɪtəl əˈdriːnəl ˌhaɪpərˈpleɪzə/ : 先天性肾上腺皮质增生症	enzyme deficiency /ˈenzaim dɪˈfɪjənsi/ : 酶缺乏症
cortisol /ˈkɔːrtɪsɒl/ : 皮质醇	glucocorticoid /ˌɡluːkɔːˈkɔːrtɪkɔɪd/ : 糖皮质激素
mineralocorticoid /ˌmɪnərələʊˈkɔːrtɪkɔɪd/ : 盐皮质激素	sex steroid /sɛks ˈstɛrɔɪd/ : 性类固醇
hormone replacement therapy /ˈhɔːrmoun rɪˈpleɪsmənt ˈθɛrəpi/ : 激素替代疗法	hirsutism /ˈhɜːrsjʊtɪzəm/ : 多毛症
irregular menses /ɪˈrɛɡjʊlə ˈmɛnsɪz/ : 月经不调	cosyntropin stimulation test /kɔːˈsɪntreɪpɪn ˌstɪmjʊˈleɪʃən test/ : 促皮质素兴奋试验
infertility /ˌɪnfərˈtɪlɪti/ : 不孕症	

Pathogenesis and treatment of primary aldosteronism

原发性醛固酮增多症的发病机制与治疗

期刊: Nature Reviews Endocrinology 日期: 2020 JCR Q1 中科院升级版 医学1区 IF 40.0

Abstract

Early diagnosis and appropriate treatment of **primary aldosteronism**, the most frequent cause of **secondary hypertension**, are crucial to prevent deleterious **cardiovascular outcomes**. In the past decade, the discovery of genetic abnormalities responsible for sporadic and familial forms of **primary aldosteronism** has improved the knowledge of the pathogenesis of this disorder. Mutations in genes encoding ion channels and pumps lead to increased cytosolic concentrations of calcium in **zona glomerulosa** cells, which triggers CYP11B2 expression and autonomous **aldosterone** production. Improved understanding of the mechanisms underlying the disease is key to improving **diagnostics** and to developing and implementing targeted treatments. This Review provides an update on the genetic abnormalities associated with sporadic and familial forms of **primary aldosteronism**, their frequency among different populations and the mechanisms explaining excessive **aldosterone** production and **adrenal** nodule development. The possible effects and uses of these findings for improving the **diagnostics** for **primary aldosteronism** are discussed. Furthermore, current treatment options of **primary aldosteronism** are reviewed, with particular attention to the latest studies on blood pressure and **cardiovascular outcomes** following medical or **surgical treatment**. The new perspectives regarding the use of **targeted drug therapy** for **aldosterone**-producing adenomas with specific somatic mutations are also addressed.

中文翻译

原发性醛固酮增多症作为继发性高血压最常见的病因，其早期诊断与恰当治疗对于预防有害心血管结局至关重要。过去十年间，对散发性和家族性原发性醛固酮增多症致病基因异常的发现深化了人们对该疾病发病机制的认识。编码离子通道和泵的基因突变导致肾小球带细胞胞质内钙离子浓度升高，从而触发CYP11B2表达及自主性醛固酮分泌。深入理解疾病机制是改进诊断方法、开发并实施靶向治疗的关键。本综述系统更新了散发性和家族性原发性醛固酮增多症相关基因异常、不同人群中的突变频率，以及醛固酮过量分泌与肾上腺结节形成的机制解析。文中进一步探讨了这些发现对改进原发性醛固酮增多症诊断体系的潜在价值与应用前景，并全面评述了当前治疗方案，重点关注药物治疗与手术治疗后血压控制及心血管结局的最新研究进展。同时针对特定体细胞突变的醛固酮分泌腺瘤，阐述了靶向药物治疗的新视角。

核心词汇

primary aldosteronism /ˈpraɪməri ælˈdɒstərəˌnɪzəm/ : 原发性醛固酮增多症

cardiovascular outcomes /ˌkɑːdiəvˈvæskjələr ˈaʊtˌkɒmz/ : 心血管结局

zona glomerulosa /ˈzəʊnə glɒˌmɛrjuˈləʊsə/ : 球状带

mutation /mjuˈteɪʃən/ : 突变

cytosolic calcium /ˌsaɪtəˈsɒlɪk ˈkælsiəm/ : 胞质钙

diagnostics /ˌdaɪəɡˈnɒstɪks/ : 诊断学

surgical treatment /ˈsɜːrdʒɪkəl ˈtriːtmənt/ : 手术治疗

secondary hypertension /ˈsekəndəri ˌhaɪpərˈtenʃən/ : 继发性高血压

adrenal /əˈdriːnəl/ : 肾上腺

aldosterone /ælˈdɒstəˌroʊn/ : 醛固酮

ion channel /ˈaɪən ˈtʃænəl/ : 离子通道

adenoma /ˌædɪˈnoʊmə/ : 腺瘤

targeted drug therapy /ˈtɑːɡɪtɪd drʌɡ ˈθerəpi/ : 靶向药物治疗

Diagnosis and treatment of primary aldosteronism

原发性醛固酮增多症的诊断与治疗

期刊: The Lancet Diabetes & Endocrinology 日期: 2021

Abstract

primary aldosteronism is a common cause of secondary **hypertension** associated with excess cardiovascular morbidities. **primary aldosteronism** is underdiagnosed because it does not have a specific, easily identifiable feature and clinicians can be poorly aware of the disease. The diagnostic investigation is a multistep process of screening, confirmatory testing, and subtype differentiation of unilateral from bilateral forms for therapeutic management. **adrenal venous sampling** is key for reliable subtype identification, but can be bypassed in patients with specific characteristics. For unilateral disease, surgery offers the possibility of cure, with total laparoscopic unilateral adrenalectomy being the treatment of choice. Bilateral forms are treated mainly with **mineralocorticoid receptor antagonists**. The goals of treatment are to normalise both blood pressure and excessive **aldosterone** production, and the primary aims are to reduce associated comorbidities, improve quality of life, and reduce mortality. Prompt diagnosis of **primary aldosteronism** and the use of targeted treatment strategies mitigate **aldosterone**-specific target organ damage and with appropriate patient management outcomes can be excellent. Advances in **molecular histopathology** challenge the traditional concept of **primary aldosteronism** as a binary disease, caused by either a unilateral **aldosterone**-producing **adenoma** or bilateral **adrenal hyperplasia**. Somatic mutations drive autonomous **aldosterone** production in most adenomas. Many of these same mutations have been identified in nodular lesions adjacent to an **aldosterone**-producing **adenoma** and in patients with bilateral disease. In addition, germline mutations cause rare familial forms of aldosteronism (**familial hyperaldosteronism** types 1-4). Genetic testing for inherited forms in suspected cases of **familial hyperaldosteronism** avoids the burdensome diagnostic investigation in positive patients. In this Review, we discuss advances and future management approaches in the diagnosis of **primary aldosteronism**.

中文翻译

原发性醛固酮增多症是继发性高血压的常见病因，与心血管疾病发病率增加密切相关。由于缺乏特异性且易于识别的特征，加之临床医生对该病的认识不足，原发性醛固酮增多症常被漏诊。其诊断过程包括筛查、确诊试验以及区分单侧与双侧亚型以指导治疗的多步骤流程。肾上腺静脉采血是可靠亚型鉴定的关键手段，但对于具有特定特征的患者可省略此步骤。对于单侧病变，手术可实现治愈可能，其中完全腹腔镜下单侧肾上腺切除术是首选治疗方案。双侧病变则主要采用盐皮质激素受体拮抗剂治疗。治疗目标在于使血压和过量醛固酮分泌恢复正常，核心目的是降低相关并发症发生率、改善生活质量并减少死亡率。及时诊断原发性醛固酮增多症并采用针对性治疗策略可减轻醛固酮特异性靶器官损伤，通过规范的患者管理可获得良好预后。分子组织病理学的进展挑战了传统观念——原发性醛固酮增多症曾被认为是由单侧醛固酮分泌腺瘤或双侧肾上腺增生引起的二元性疾病。体细胞突变驱动了大多数腺瘤的自主性醛固酮分泌，这些相同突变在醛固酮分泌腺瘤邻近的结节性病变及双侧疾病患者中亦被识别。此外，种系突变导致罕见的家族性醛固酮增多症（家族性高醛固酮血症1-4型）。对疑似家族性高醛固酮血症病例进行遗传学检测，可避免阳性患者承受繁琐的诊断流程。本综述将探讨原发性醛固酮增多症诊断领域的最新进展与未来管理策略。

核心词汇

primary aldosteronism /ˈpraɪməri ælˈdɒstərouˌnɪzəm/ : 原发性醛固酮增多症

adrenal /əˈdriːnəl/ : 肾上腺

hyperplasia /ˌhaɪpərˈpleɪzə/ : 增生

laparoscopic adrenalectomy /ˌlæpərəˈskopɪk əˈdriːnəˈlektəmi/ : 腹腔镜肾上腺切除术

aldosterone /ælˈdɒstəroun/ : 醛固酮

germline mutation /ˈdʒɜːrmlaɪn mjuˈteɪʃən/ : 种系突变

molecular histopathology /məˈleɪkjələr ˌhɪstəʊpəˈθɒlədʒi/ : 分子组织病理学

hypertension /ˌhaɪpərˈtɛnʃən/ : 高血压

adenoma /ˌædɪˈnəʊmə/ : 腺瘤

adrenal venous sampling /əˈdriːnəl ˈviːnəs ˈsæmplɪŋ/ : 肾上腺静脉采血

mineralocorticoid receptor antagonists /ˌmɪnərələʊˈkɔːrtɪkɔɪd rɪˈseptər ænˈtæɡənɪsts/ : 盐皮质激素受体拮抗剂

somatic mutation /səʊˈmætɪk mjuˈteɪʃən/ : 体细胞突变

familial hyperaldosteronism /fəˈmɪliəl ˌhaɪpərælˈdɒstərouˌnɪzəm/ : 家族性醛固酮增多症

Phaeochromocytoma

嗜铬细胞瘤

期刊: The Lancet 日期: 2005 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

Phaeochromocytomas are rare neuroendocrine tumours with a highly variable clinical presentation but most commonly presenting with episodes of headaches, sweating, palpitations, and **hypertension**. The serious and potentially lethal cardiovascular complications of these tumours are due to the potent effects of secreted catecholamines. **biochemical testing** for **phaeochromocytoma** is indicated not only in symptomatic patients, but also in patients with adrenal incidentalomas or identified genetic predispositions (eg, multiple endocrine neoplasia type 2, von Hippel-Lindau syndrome, neurofibromatosis type 1, and mutations of the succinate dehydrogenase genes). **imaging** techniques such as CT or MRI and functional ligands such as (123)I-MIBG are used to localise biochemically proven tumours. After the use of appropriate **preoperative treatment** to block the effects of secreted catecholamines, **laparoscopic tumour removal** is the preferred procedure. If removal of **phaeochromocytoma** is timely, **prognosis** is excellent. However, **prognosis** is poor in patients with metastases, which especially occur in patients with large, extra-adrenal tumours.

中文翻译

嗜铬细胞瘤是一种罕见的神经内分泌肿瘤，临床表现多样，但最常见的是阵发性头痛、出汗、心悸和高血压。这些肿瘤严重且可能致命的心血管并发症源于分泌的儿茶酚胺的强大作用。生化检测不仅适用于有症状的患者，也适用于肾上腺偶发瘤或已确定遗传易感性的患者（例如，多发性内分泌腺瘤病2型、von Hippel-Lindau综合征、神经纤维瘤病1型以及琥珀酸脱氢酶基因突变）。影像学技术如CT或MRI以及功能性配体如(123)I-MIBG用于定位经生化证实的肿瘤。在使用适当的术前治疗阻断分泌的儿茶酚胺效应后，腹腔镜肿瘤切除是首选手术。如果及时切除嗜铬细胞瘤，预后良好。然而，对于发生转移的患者，尤其是那些大型肾上腺外肿瘤的患者，预后较差。

核心词汇

phaeochromocytoma /fi:əʊˌkrəʊməʊsaɪˈtəʊmə/ : 嗜铬细胞瘤	neuroendocrine tumour /ˌnju:əʊˈendəʊkraɪn ˈtju:mə/ : 神经内分泌肿瘤
hypertension /ˌhaɪpəˈtenʃən/ : 高血压	catecholamine /ˌkætiˈkəʊləmi:n/ : 儿茶酚胺
adrenal incidentaloma /əˈdri:nəl ˌɪnsɪˈdenteɪləmə/ : 肾上腺偶发瘤	biochemical testing /ˌbaɪəʊˈkemɪkəl ˈtestɪŋ/ : 生化检测
imaging /ˈɪmɪdʒɪŋ/ : 影像学检查	laparoscopic /ˌlæpərəˈskɒpɪk/ : 腹腔镜的
tumour removal /ˈtju:mə riˈmu:vəl/ : 肿瘤切除术	metastasis /məˈtæstəsɪs/ : 转移
prognosis /prɒɡˈnəʊsɪs/ : 预后	preoperative treatment /ˌpri:ˈɒperətɪv ˈtri:tment/ : 术前治疗

Cushing's syndrome

库欣综合征

期刊: The Lancet 日期: 2023 JCR Q1 中科院升级版 医学1区 IF 88.5

Abstract

Endogenous **Cushing's syndrome** results from excess **glucocorticoid secretion**, which leads to a myriad of clinical manifestations, comorbidities, and increased **mortality** despite treatment. Molecular mechanisms and genetic alterations associated with different causes of **Cushing's syndrome** have been described in the last decade. Imaging modalities and **biochemical testing** have evolved; however, both the **diagnosis** and management of **Cushing's syndrome** remain challenging. **surgery** is the preferred treatment for all causes, but **medical therapy** has markedly advanced, with new drug options becoming available. Nevertheless, several comorbidities remain even after patient **remission**, which can affect quality of life. Accurate and timely **diagnosis** and treatment are essential for mitigating chronic complications of excess glucocorticoids and improving patient quality of life. In this Seminar, we aim to update several important aspects of **diagnosis**, complications, and treatment of endogenous **Cushing's syndrome** of all causes.

中文翻译

内源性库欣综合征源于糖皮质激素分泌过多，导致一系列临床表现、合并症，以及即使在治疗后死亡率仍增加的情况。过去十年中，已描述了与库欣综合征不同病因相关的分子机制和遗传改变。影像学检查和生化检测技术有所进步；然而，库欣综合征的诊断和治疗仍然具有挑战性。手术是所有病因的首选治疗方法，但药物治疗已显著进步，新的药物选择不断出现。尽管如此，即使在患者缓解后，仍存在一些影响生活质量的合并症。准确及时的诊断和治疗对于减轻糖皮质激素过多引起的慢性并发症和改善患者生活质量至关重要。在本研讨会中，我们旨在更新所有病因的内源性库欣综合征的诊断、并发症和治疗方面的几个重要方面。

核心词汇

Cushing's syndrome /'kʊʃɪŋz 'sɪndroum/ : 库欣综合征	glucocorticoid /glu:kou'kɔ:rtɪkɔɪd/ : 糖皮质激素
secretion /sɪ'kri:fən/ : 分泌	comorbidity /,koumɔ:r'bɪdəti/ : 共病
mortality /mɔ:r'tæləti/ : 死亡率	molecular mechanism /mə'lekjələr 'mekənɪzəm/ : 分子机制
genetic alteration /dʒə'netɪk ɔ:ltə'reɪʃən/ : 遗传改变	imaging modality /'ɪmɪdʒɪŋ mɒs'dæləti/ : 影像学检查方法
biochemical testing /,baɪəʊ'kemɪkəl 'testɪŋ/ : 生化检测	diagnosis /,daɪəg'nəʊsɪs/ : 诊断
surgery /'sɜ:rdʒəri/ : 外科手术	medical therapy /'medɪkəl 'θerəpi/ : 药物治疗
remission /rɪ'mɪʃən/ : 缓解	complication /,kɑ:mplɪ'keɪʃən/ : 并发症

Identification of the notch ligand DLK1 as an immunotherapeutic target and regulator of tumor cell plasticity and chemoresistance in adrenocortical carcinoma

鉴定DLK1为肾上腺皮质癌免疫治疗靶点及肿瘤细胞可塑性与化疗耐药性调节因子

期刊: Nature Communications 日期: 2025 JCR Q1 中科院升级版 综合性期刊1区 IF 15.7

Abstract

Abstract While immunotherapeutic targeting of cell surface proteins is an increasingly effective **cancer** therapy, identification of new surface proteins, particularly those with biological importance, is critical. Here, we uncover delta-like non-canonical Notch ligand 1 (DLK1) as a cell surface protein with limited normal tissue expression and high expression in multiple refractory adult **metastatic** cancers including **small cell lung cancer** (SCLC) and **adrenocortical carcinoma** (ACC), a rare **cancer** with few effective therapies. In ACC, ADCT-701, a DLK1 targeting antibody-drug conjugate (ADC), shows in vitro and in vivo activity but is overall limited due to high expression and activity of the drug efflux protein ABCB1 (MDR1, P-glycoprotein). In contrast, ADCT-701 induces complete responses in DLK1+ ACC and SCLC in vivo models with low or no ABCB1 expression. Genetic deletion of DLK1 in ACC dramatically downregulates ABCB1 and increases ADC payload and **chemotherapy** sensitivity through NOTCH1-mediated transdifferentiation. This work identifies DLK1 as an immunotherapeutic target that regulates **tumor** cell **plasticity** and **chemoresistance** in ACC and supports an active phase I **clinical trial** targeting DLK1 with an ADC in ACC and neuroendocrine neoplasms (NCT06041516).

中文翻译

摘要 尽管针对细胞表面蛋白的免疫疗法已成为日益有效的癌症治疗手段，但识别新的表面蛋白，尤其是具有生物学重要性的蛋白，仍然至关重要。本研究揭示了δ样非经典Notch配体1 (DLK1) 作为一种细胞表面蛋白，其在正常组织中表达有限，而在多种难治性成人转移性癌症中高表达，包括小细胞肺癌 (SCLC) 和肾上腺皮质癌 (ACC) ——后者是一种罕见且缺乏有效疗法的癌症。在ACC中，靶向DLK1的抗体药物偶联物 (ADC) ADCT-701在体外和体内均显示出活性，但由于药物外排蛋白ABCB1 (MDR1, P-糖蛋白) 的高表达和活性，其总体疗效受限。相比之下，在ABCB1表达低或无表达的DLK1阳性ACC和SCLC体内模型中，ADCT-701能诱导完全缓解。在ACC中基因敲除DLK1可通过NOTCH1介导的转分化显著下调ABCB1，并增强ADC有效载荷和化疗敏感性。这项工作确立了DLK1作为调节ACC肿瘤细胞可塑性和化疗耐药性的免疫治疗靶点，并支持了一项针对ACC和神经内分泌肿瘤的DLK1靶向ADC的I期临床试验 (NCT06041516)。

核心词汇

immunotherapy /ˌɪmjʊnoʊˈθerəpi/ : 免疫治疗	cancer /ˈkænsər/ : 癌症
metastatic /ˌmetəˈstætɪk/ : 转移性	small cell lung cancer /smɔːl sel lʌŋ ˈkænsər/ : 小细胞肺癌
adrenocortical carcinoma /əˌdriːnoʊˈkɔːrtɪkəl ˌkɑːrsɪˈnoʊmə/ : 肾上腺皮质癌	chemotherapy /ˌkiːmoʊˈθerəpi/ : 化疗
chemoresistance /ˌkiːmoʊrɪˈzɪstəns/ : 化疗耐药	neuroendocrine neoplasm /ˌnjuːəroʊˈendəkrɪn ˈniːəpləzəm/ : 神经内分泌肿瘤
clinical trial /ˈklɪnɪkəl ˈtraɪəl/ : 临床试验	tumor /ˈtuːmə/ : 肿瘤
plasticity /plæsˈtɪsɪti/ : 可塑性	

Adrenocortical carcinoma: A clinician's update

肾上腺皮质癌：临床医生最新指南

期刊: Nature Reviews Endocrinology 日期: 2011 JCR Q1 中科院升级版 医学1区 IF 42.3

Abstract

adrenocortical carcinoma is a rare heterogeneous **neoplasm** with an incompletely understood **pathogenesis** and a poor **prognosis**. Previous studies have identified overexpression of insulin-like growth factor 2 (IGF-2) and constitutive activation of β -catenin as key factors involved in the development of **adrenocortical carcinoma**. Most patients present with steroid hormone excess, for example **Cushing syndrome** or **virilization**, or abdominal mass effects, but a growing proportion of patients with **adrenocortical carcinoma** (currently >15%) is initially diagnosed incidentally. No general consensus on the diagnostic and therapeutic measures for **adrenocortical carcinoma** exists, but collaborative efforts, such as international conferences and networks, including the European Network for the Study of Adrenal Tumors (ENSAT), have substantially advanced the field. In patients with suspected **adrenocortical carcinoma**, a thorough **endocrine** and **imaging** work-up is recommended to guide the surgical approach aimed at complete **resection** of the tumor. To establish an adequate basis for treatment decisions, **pathology** reports include the Weiss score to assess **malignancy**, the **resection** status and the Ki67 index. As **recurrence** is frequent, close follow-up initially every 3 months is mandatory. Most patients benefit from **adjuvant** mitotane treatment. In **metastatic** disease, mitotane is the cornerstone of initial treatment, and **cytotoxic** drugs should be added in case of progression. Results of a large phase III trial in advanced **adrenocortical carcinoma** are anticipated for 2011 and will hopefully establish a benchmark therapy. New targeted therapies, for example, IGF-1 receptor inhibitors, are under investigation and may soon improve current treatment options.

中文翻译

肾上腺皮质癌是一种罕见的异质性肿瘤，其发病机制尚未完全明确，且预后较差。先前的研究已确定胰岛素样生长因子2（IGF-2）的过度表达和 β -连环蛋白的持续激活是肾上腺皮质癌发生的关键因素。大多数患者表现为类固醇激素过量，例如库欣综合征或男性化，或腹部肿块效应，但越来越多的肾上腺皮质癌患者（目前超过15%）最初是偶然诊断的。对于肾上腺皮质癌的诊断和治疗措施尚无普遍共识，但通过国际合作努力，如国际会议和网络，包括欧洲肾上腺肿瘤研究网络（ENSAT），已显著推动了该领域的发展。对于疑似肾上腺皮质癌的患者，建议进行全面的内分泌和影像学检查，以指导旨在完全切除肿瘤的手术方法。为了为治疗决策建立充分的基础，病理报告包括评估恶性肿瘤的韦斯评分、切除状态和Ki67指数。由于复发频繁，最初必须每3个月进行一次密切随访。大多数患者受益于辅助米托坦治疗。在转移性疾病中，米托坦是初始治疗的基石，如果病情进展，应加用细胞毒性药物。预计2011年将有一项大型III期试验在晚期肾上腺皮质癌中取得结果，并有望建立基准疗法。新的靶向疗法，例如IGF-1受体抑制剂，正在研究中，可能很快会改善当前的治疗选择。

核心词汇

adrenocortical carcinoma /əˌdriːnouˈkɔːrtɪkəlˌkɑːrsɪˈnoumə/ : 肾上腺皮质癌

pathogenesis /ˌpæθəˈdʒɛnɪsɪs/ : 发病机制

Cushing syndrome /ˈkʊʃɪŋˌsɪndroum/ : 库欣综合征

endocrine /ˈendoukrɪn/ : 内分泌

resection /rɪˈsekʃən/ : 切除术

malignancy /məˈlɪgnənsi/ : 恶性肿瘤

adjuvant /ˈædʒəvənt/ : 辅助治疗

neoplasm /ˈniːouˌplæzəm/ : 肿瘤

prognosis /ˌprɔːɡˈnoʊsɪs/ : 预后

virilization /ˌvɪrɪlɪˈzeɪʃən/ : 男性化

imaging /ˈɪmɪdʒɪŋ/ : 影像学检查

pathology /pəˈθɔːlədʒi/ : 病理学

recurrence /rɪˈkɜːrəns/ : 复发

metastatic /ˌmetəˈstætɪk/ : 转移性

Adrenocortical carcinoma: A practical guide for clinicians

肾上腺皮质癌：临床医生实用指南

期刊: The Lancet Diabetes & Endocrinology 日期: 2025

Abstract

adrenocortical carcinoma is a rare endocrine malignancy. The management of patients with adrenocortical carcinoma is challenging for several reasons, including its heterogeneous but frequently aggressive biological behaviour; tumour-related hormonal excess (eg, Cushing's syndrome or virilisation); the overall paucity of evidence regarding diagnostic investigation and treatment; the approval of only one drug (mitotane); and the scarcity of centres with sufficient experience. In this Review, we present 25 questions on the most important aspects of the clinical management of adult patients with adrenocortical carcinoma that we have frequently asked ourselves over the past 25 years. We offer our personal answers and perspectives, drawing upon published evidence as well as more than 60 years of collective clinical experience and insights from our management of more than 1700 patients across two centres in Germany and Italy.

中文翻译

肾上腺皮质癌是一种罕见的内分泌恶性肿瘤。管理肾上腺皮质癌患者具有挑战性，原因包括其异质性但通常具有侵袭性的生物学行为；肿瘤相关的激素过量（如库欣综合征或男性化）；诊断检查和治疗方面的证据总体不足；仅有一种药物（米托坦）获得批准；以及拥有足够经验的中心稀缺。在本综述中，我们提出了关于成人肾上腺皮质癌临床管理最重要方面的25个问题，这些问题在过去25年中我们经常自问。我们基于已发表的证据，以及我们在德国和意大利两个中心管理超过1700名患者所积累的超过60年的集体临床经验和见解，提供了我们个人的答案和观点。

核心词汇

adrenocortical carcinoma /əˌdriːnouˈkɔːrtɪkəlˌkɑːrsɪˈnoumə/ : 肾上腺皮质癌	endocrine malignancy /ˈendəkrɪn məˈlɪgnənsi/ : 内分泌恶性肿瘤
Cushing's syndrome /ˈkʊʃɪŋz ˈsɪndroum/ : 库欣综合征	virilisation /ˌvɪrɪlɪzˈzeɪʃən/ : 男性化
adrenal /əˈdriːnəl/ : 肾上腺	carcinoma /ˌkɑːrsɪˈnoumə/ : 癌
tumour /ˈtjuːmə/ : 肿瘤	hormonal excess /ˈhɔːrˈmounəl ˈekses/ : 激素过量
diagnostic investigation /ˌdaɪəɡˈnɒstɪk ɪnˌvestɪˈgeɪʃən/ : 诊断性检查	clinical management /ˈklɪnɪkəl ˈmænɪdʒmənt/ : 临床管理
malignancy /məˈlɪgnənsi/ : 恶性肿瘤	aggressive biological behaviour /əˈɡresɪv ˌbaɪəˈlɒdʒɪkəl bɪˈheɪvjə/ : 侵袭性生物学行为

Targeting pediatric adrenocortical carcinoma: Molecular insights and emerging therapeutic strategies

靶向儿童肾上腺皮质癌：分子洞察与新兴治疗策略

期刊: Cancer Treatment Reviews 日期: 2025 JCR Q1 中科院升级版 医学1区 IF 10.5

Abstract

Pediatric **adrenocortical carcinoma** (pACC) is an exceptionally rare and aggressive **malignancy**, accounting for only 0.2-0.3% of childhood cancers. Characterized by significant **endocrine activity** and often associated with genetic syndromes such as Li-Fraumeni syndrome, pACC exhibits distinct clinical and molecular profiles compared to adult **adrenocortical carcinoma** (ACC). Current treatment approaches, largely adapted from adult protocols, center on **surgery** and **chemotherapy**, including mitotane. However, the lack of pediatric-specific data and major clinical trials underscores a pressing need for tailored therapeutic strategies. Advances in **molecular profiling** have unveiled actionable targets, such as alterations in the Wnt/ β -catenin and MAP/ERK pathways, overexpression of IGF2, and **epigenetic dysregulation**. Emerging therapies, including **immune checkpoint inhibitors**, CAR T-cell therapy, and radiopharmaceuticals, hold promise but remain largely untested in pediatric populations. Targeting metabolic vulnerabilities, such as **steroidogenesis** and lipid metabolism, offers additional avenues for therapeutic innovation. Furthermore, improved diagnostic tools like **liquid biopsy** and steroid profiling may enhance disease monitoring and early detection. Despite progress in understanding pACC biology, significant challenges remain in translating these insights into effective treatments. Collaborative efforts, such as the European Cooperative Study Group for Pediatric Rare Tumors (EXPeRT), and the development of pediatric-specific clinical trials are vital for advancing the field. Multidisciplinary care and international research initiatives will be pivotal in addressing the unmet needs of pACC patients. By leveraging molecular insights and fostering global collaboration, the field can move toward **personalized medicine**, improving outcomes and quality of life for children with this challenging disease. Expanding clinical trials, refining diagnostic tools, and integrating novel therapies into treatment regimens will be critical in bridging the gap between pediatric and adult ACC treatment success.

中文翻译

儿童肾上腺皮质癌（pACC）是一种极为罕见且侵袭性强的恶性肿瘤，仅占儿童癌症的0.2-0.3%。该疾病以内分泌活性显著为特征，常与李-佛美尼综合征等遗传综合征相关，其临床和分子特征与成人肾上腺皮质癌（ACC）存在明显差异。当前的治疗方法主要借鉴成人方案，以手术和化疗（包括米托坦）为核心。然而，儿童特异性数据的缺乏以及大型临床试验的不足，凸显了对针对性治疗策略的迫切需求。分子谱分析技术的进步揭示了可干预的治疗靶点，如Wnt/ β -catenin和MAP/ERK通路改变、IGF2过表达以及表观遗传失调。新兴疗法，包括免疫检查点抑制剂、CAR T细胞疗法和放射性药物，展现出潜力，但在儿童群体中大多尚未经过验证。针对代谢脆弱性，如类固醇生成和脂质代谢，为治疗创新提供了更多途径。此外，液体活检和类固醇谱分析等改进的诊断工具可能加强疾病监测和早期发现。尽管对pACC生物学的理解有所进展，但将这些见解转化为有效治疗仍面临重大挑战。欧洲儿童罕见肿瘤合作研究组（EXPeRT）等协作努力，以及儿童特异性临床试验的发展，对推动该领域至关重要。多学科护理和国际研究计划将是满足pACC患者未满足需求的关键。通过利用分子见解和促进全球合作，该领域可朝着个体化医疗迈进，改善患有这一挑战性疾病儿童的治疗效果和生活质量。扩大临床试验、完善诊断工具以及将新型疗法整合到治疗方案中，对于缩小儿童与成人ACC治疗成功之间的差距至关重要。

核心词汇

adrenocortical carcinoma /əˌdriːnɒsˈkɔːrtɪkəl ˌkɑːrsiˈnoʊmə/ : 肾上腺皮质癌	malignancy /məˈlɪgnənsi/ : 恶性肿瘤
endocrine activity /ˈendoukraɪn ækˈtɪvəti/ : 内分泌活性	genetic syndrome /dʒəˈnetɪk ˈsɪndroʊm/ : 遗传综合征
surgery /ˈsɜːrdʒəri/ : 外科手术	chemotherapy /ˌkiːmoʊˈθerəpi/ : 化学治疗
molecular profiling /məˈlekjələr ˈproufaɪlɪŋ/ : 分子谱分析	epigenetic dysregulation /ˌepɪdʒəˈnetɪk dɪsˌregjʊˈleɪʃən/ : 表观遗传失调
immune checkpoint inhibitors /ɪˈmjuːn ˈtʃekpɔɪnt ɪnˈhɪbɪtəz/ : 免疫检查点抑制剂	steroidogenesis /stəˈrɔɪdɒs ˈdʒenɪsɪs/ : 类固醇生成
liquid biopsy /ˈlɪkwɪd ˈbaɪɒpsi/ : 液体活检	personalized medicine /ˈpɜːrsənəlaɪzd ˈmedɪsɪn/ : 个性化医疗